

Bangladesh University of Engineering and Technology

Department of Computer Science & Engineering

MECHA 165 Question Bank

Basic Mechanical Engineering

Level 1 · Term 2 · Theory Course

Year Coverage: 2003–04 to 2023–24 |
Sections: A (A1, A2, A3), B (B1–B5), C (C1–C4)

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Section A1

Sources of Energy: Conventional & Renewable

A1 — Sources of Energy: Conventional & Renewable

- Q1.** With a neat sketch, briefly describe the working principle of a hydroelectric power plant. **(15 marks)** Write short notes on any **three**: (i) OTEC, (ii) Solar Photovoltaics, (iii) Fissionable and Fissile Nuclei, (iv) Geothermal Energy. **(15 marks)** [2023–24, 2021–22]
- Q2.** How does the integration of solar power impact the stability of the national power grid? Explain briefly. **(10 marks)** [2023–24]
- Q3.** (a) With a neat sketch, briefly describe the working principle of a hydroelectric power plant. **(15 marks)**
(b) Write down advantages and disadvantages of fossil fuels. **(10 marks)** [2021–22]
- Q4.** (a) What is the heating value of a fuel? Describe its types. **(10 marks)**
(b) Write short notes on: (i) Fissile and fissionable nuclei, (ii) Geothermal energy. **(10 marks)** [2020–21]
- Q5.** With suitable diagrams or flowcharts, write short notes on: (i) Ocean Thermal Energy Conversion (OTEC), (ii) Hydroelectric energy. **(15 marks)** [2018–19]
- Q6.** (a) Mention the key challenges in the energy sector of Bangladesh. What are your recommendations to tackle those challenges? **(8 marks)**
(b) Write short notes on any **three**: (i) OTEC, (ii) Wind turbine, (iii) Geothermal energy, (iv) Pressurised water nuclear reactor, (v) Bio-diesel. **(15 marks)** [2017–18]
- Q7.** (a) What are the potential renewable energy sources in Bangladesh? Discuss the challenges and future prospects of renewable energy in Bangladesh. **(15 marks)**
(b) Write short notes on any **four**: (i) Solar photovoltaic cell, (ii) Pressurised water nuclear reactor, (iii) Ocean thermal energy conversion (OTEC), (iv) Pumped hydro storage system, (v) CETO technology. **(20 marks)** [2016–17]
- Q8.** Define renewable energy and give examples. With suitable diagrams or flowcharts, write short notes on: (i) Nuclear-energy-powered Pressurised Water Reactor, (ii) Geothermal Energy. **(20 marks)** [2015–16]
- Q9.** Write down the names of potential sources of renewable energy and briefly describe solar thermal energy. **(10 marks)** [2014–15]
- Q10.** Write down the potential sources of renewable energy and briefly explain geothermal energy. **(12 marks)** [2013–14]
- Q11.** Define renewable energy and list the potential sources of renewable and non-renewable energy. **(10 marks)** [2012–13]
- Q12.** (a) What is fossil fuel? What are its forms? How long are current reserves expected to last? **(10 marks)**
(b) What is renewable energy? What renewable sources are used for electric power generation in Bangladesh? **(8 marks)**
(c) Why are biomasses (wood, leaf, rice husk, etc.) considered renewable? **(7 marks)** [2011–12]
- Q13.** (a) Discuss the various concepts of extracting solar energy. **(10 marks)**
(b) Describe briefly the extraction method of crude oil. **(10 marks)** [2010–11]
- Q14.** (a) What are the localised environmental concerns of air pollution? **(7 marks)**
(b) What do you know about LPG and CNG? **(8 marks)** [2010–11]
- Q15.** (a) State the criteria of a source of energy. Hence explain that nuclear energy qualifies as one. **(12 marks)**
(b) Write a short note on Photovoltaic Cell. **(9 marks)** [2008–09]
- Q16.** (a) What environmental hazards are associated with burning fossil fuels? **(10 marks)**
(b) Write short notes on: (i) Geothermal Energy, (ii) Solar Energy. **(20 marks)** [2006–07]
- Q17.** (a) What is meant by a renewable energy source? Give examples. **(5 marks)**
(b) Explain how steam can be produced from water using solar energy. **(10 marks)**
(c) What is OTEC? With a neat sketch, explain the closed-cycle OTEC system. **(15 marks)** [2005–06]
- Q18.** What are the main renewable energy sources? Explain briefly what geothermal energy is. **(10 marks)** [2003–04]

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Section A2

Introduction to IC Engines (Otto & Diesel Cycles)

A2 — Introduction to IC Engines

- Q1.** (a) What is the benefit of compressing the intake air of an IC engine? Name the systems that do so and distinguish among them. **(10 marks)**
 (b) With a neat sketch, describe the working principle of a four-stroke SI engine. **(15 marks)**
 (c) A 1600 cc four-cylinder engine has a bore of 80 mm. Compression ratio = 9. Engine runs at 2500 rpm. Calculate: (i) stroke length, (ii) total clearance volume, (iii) average piston speed. **(10 marks)** [2023–24]
- Q2.** (a) What are the differences between a turbocharged and a supercharged engine? Explain with diagrams. **(10 marks)**
 (b) Draw a valve timing diagram of a 4-stroke SI engine and explain valve overlap and spark advance. **(10 marks)**
 (c) A 3-litre SI V8 engine operates on a four-stroke cycle at 3000 rpm. Compression ratio = 12, connecting rods 18 cm, engine is square. Calculate: (i) cylinder bore, (ii) stroke length, (iii) average piston speed, (iv) clearance volume per cylinder, (v) power strokes per minute. **(10 marks)** [2023–24]
- Q3.** In an air-standard Diesel cycle, temperature and pressure at end of compression = 650°C and 4.5 MPa. Pressure at start of compression = 0.1 MPa. Cut-off ratio = 2.8. Calculate: (i) compression ratio, (ii) temperature at each state, (iii) heat addition and work per kg, (iv) cycle efficiency, (v) MEP. **(15 marks)** [2023–24]
- Q4.** (a) What happens during the intake stroke of an IC engine? Is there any difference between SI and CI engines in this stroke? **(8 marks)**
 (b) Draw the typical valve timing diagram for a four-stroke SI engine. **(7 marks)** [2021–22]
- Q5.** A 450 cc single-cylinder square four-stroke SI engine at 2400 rpm, compression ratio = 10. Calculate: (i) clearance volume, (ii) bore and stroke, (iii) average piston speed, (iv) air standard efficiency, (v) MEP if heat generated at 7.5 kW. **(20 marks)** [2021–22]
- Q6.** Define internal combustion engine. Briefly discuss its classification in terms of cylinder arrangement and ignition method. **(15 marks)** [2020–21]
- Q7.** With a neat sketch, describe the operation of a four-stroke SI engine. **(17 marks)** [2020–21]
- Q8.** A 200 cc single-cylinder square SI engine, compression ratio = 10. Calculate: (i) clearance volume, (ii) bore and stroke, (iii) air standard efficiency, (iv) air standard MEP if heat input = 150×10^{-6} J/cycle. **(18 marks)** [2020–21]
- Q9.** (a) Do diesel or gasoline engines operate at higher compression ratios? Why? **(7.5 marks)**
 (b) What is the typical fuel of an SI engine? Will it be suitable in a CI engine? **(7.5 marks)** [2019–20]
- Q10.** Develop the following expression for cut-off ratio r_c for an air-standard Diesel cycle:
- $$r_c = \frac{q_{in}}{C_p T_1 r^{k-1}} + 1$$
- Where C_p is specific heat capacity, T is temperature, k is specific heat ratio, r is compression ratio, and q_{in} is heat input. **(15 marks)** [2019–20]
- Q11.** In an air-standard Otto cycle, compression begins at 35°C and 0.1 MPa. Heat added = 1.675 MJ/kg, compression ratio = 7. Find: (i) work done per kg of air, (ii) cycle efficiency. **(20 marks)** [2019–20]
 ↪ Similar: Similar in 2013–14 and Comp-1
- Q12.** A 4-litre SI V8 engine on a four-stroke cycle at 4000 rpm. Compression ratio = 10, engine is square. Calculate: cylinder bore, stroke length, average piston speed, clearance volume per cylinder. **(15 marks)** [2019–20]
- Q13.** (a) What is Mean Effective Pressure (MEP)? Briefly explain the 4-stroke SI engine operation with schematic diagram. **(15 marks)**
 (b) Draw a schematic diagram of a cam-operated valve. **(5 marks)**
 (c) In a Diesel cycle: compression begins at 0.1 MPa, 40°C, compression ratio = 15, heat added = 1.675 MJ/kg, $C_v = 0.718$ kJ/kgK. Find: (i) maximum temperature, (ii) work per kg, (iii) cycle efficiency, (iv) temperature at end of isentropic expansion, (v) cut-off ratio. **(15 marks)** [2018–19]
- Q14.** (a) Draw and label the valve timing diagram of a typical 4-stroke SI engine. Explain valve overlap and spark advance. **(10 marks)**
 (b) A car engine: bore $\times$ stroke = 79 mm $\times$ 77 mm, speed = 3000 rpm. Find capacity in cc and mean piston speed. **(6 marks)**
 (c) With a neat sketch, explain how cams control timing of inlet and exhaust valves. **(7 marks)**
 (d) What is the key difference between knocking in CI and SI engines? **(6 marks)**
 (e) Write brief notes on VVTi technology and piston rings. **(6 marks)** [2017–18]
- Q15.** (a) Define Internal Combustion Engine. With P-v and T-s diagrams, describe the working principle of a four-stroke SI cycle. Mention two advantages and disadvantages of four-stroke over two-stroke. **(20 marks)**
 (b) With a block diagram, describe the fuel supply sub-system of an IC engine. **(10 marks)**
 (c) A V12 engine: stroke = 86.8 mm, bore = 87 mm, clearance volume = 0.05 litre/cylinder. Calculate displacement volume and compression ratio. State whether petrol or diesel. **(5 marks)** [2015–16]
- Q16.** (a) Write down the engine sub-systems and draw a schematic diagram of a carburettor. **(8 marks)**

- (b) Draw an actual valve timing diagram for a 4-stroke SI engine. (10 marks)
- (c) Among diesel and gasoline engines, which operates at lower compression ratio? Why? (5 marks)
- (d) A FERRARI FF 6.3 V12 engine: bore $\times$ stroke = 80 mm $\times$ 90 mm, clearance volume = 0.05 litre/cylinder. Find compression ratio and thermal efficiency. If converted to constant-pressure cycle at cut-off ratio 3.0, find new efficiency. (12 marks) [2014–15]
- Q17.** (a) Briefly write down the function of the lubrication system for an automotive SI engine. (5 marks)
- (b) Draw an actual indicator diagram for a 4-stroke SI engine. (10 marks)
- (c) An ideal Otto cycle: compression ratio = 8, start of compression $P = 100$ kPa, $T = 20^\circ\text{C}$, heat added = 850 kJ/kg. Determine: (i) maximum temperature and pressure, (ii) net work output, (iii) thermal efficiency. (20 marks) [2013–14]
- Q18.** (a) Using simplified block diagrams, identify components of: (i) IC engine lubrication sub-system, (ii) IC engine cooling sub-system. (10 marks)
- (b) Briefly describe the four-stroke spark ignition cycle with P-v and T-s diagrams. (15 marks)
- (c) Draw engine valve timing diagrams for an actual medium-performance engine and an ideal engine. (10 marks) [2012–13]
- Q19.** (a) With necessary sketches, briefly describe the working principle of a four-stroke IC engine. (18 marks)
- (b) A four-stroke IC engine: bore $\times$ stroke = 10 cm $\times$ 10 cm, clearance volume = 100π cc. Find: (i) total volume per cylinder, (ii) compression ratio, (iii) whether SI or CI, (iv) engine displacement for 4 cylinders. (12 marks)
- (c) Draw the actual and ideal thermodynamic cycle of a petrol engine. (5 marks) [2011–12]
- Q20.** (a) With necessary sketches, describe the operating principle of a closed-loop gas turbine. Draw the corresponding thermodynamic cycle. (18 marks)
- (b) With T-S diagrams, explain the merits of adding regeneration, inter-cooling and reheating to a closed-loop gas turbine. (17 marks) [2011–12]
- Q21.** (a) Give a simplistic definition of an Internal Combustion Engine. (6 marks)
- (b) Classify engines on the basis of combustion method. (8 marks)
- (c) Give a brief comparison between spark ignition and compression ignition engines. (9 marks) [2010–11]
- Q22.** Draw the fuel supply system layout of a compression ignition engine with individual fuel pumps for each cylinder. (12 marks) [2010–11]
- Q23.** (a) Explain the importance of engine cooling. (9 marks)
- (b) Draw a neat sketch of a water cooling system for an IC engine. (10 marks)
- (c) Describe the fuel supply system of a spark ignition engine with neat sketch. (16 marks) [2008–09]
- Q24.** (a) Draw a neat sketch of the spark ignition system for a four-cylinder engine. (9 marks)
- (b) Describe the actual valve timing diagram of a four-stroke SI engine. (14 marks)
- (c) Describe the Otto cycle and Dual cycle on p-v and T-s planes. (12 marks) [2008–09]
- Q25.** (a) Define an isentropic process. Which strokes of an ideal Otto cycle are isentropic? (7 marks)
- (b) With necessary sketches, explain the coil-ignition system of a 4-stroke SI engine. (13 marks)
- (c) Draw the actual valve timing diagram of a 4-stroke diesel engine showing all valve openings and closings. (8 marks)
- (d) What do you understand by spark advance? Why is it applied? (7 marks) [2006–07]
- Q26.** (a) What are the basic functions of the lubricating system in an engine? (6 marks)
- (b) Define firing order of a multi-cylinder SI engine. (4 marks) [2006–07]
- Q27.** (a) Explain with necessary sketches the sequence of operation of a 4-stroke SI engine. (12 marks)
- (b) Draw the valve timing diagram of a 4-stroke diesel engine. (8 marks)
- (c) Draw the indicator diagram of a 4-stroke SI engine. (7 marks)
- (d) Why are inlet valves opened earlier and exhaust valves closed later w.r.t. TDC? (8 marks) [2005–06]
- Q28.** (a) An automobile engine is designated as “1000 cc” — what does this mean? (5 marks)
- (b) Draw actual and ideal P-V diagrams of a 4-stroke SI engine. Why is pressure below atmospheric during the suction stroke in the actual cycle? (15 marks)
- (c) Describe with a neat sketch the liquid cooling system of an IC engine. What is the purpose of a thermostat? (15 marks) [2004–05]
- Q29.** (a) Draw a typical ignition circuit for a four-cylinder petrol engine. (10 marks)
- (b) How is the liquid phase of coolant maintained in the cooling circuit? (5 marks)
- (c) Why are four-stroke engines less harmful to the atmosphere than two-stroke engines? (10 marks) [2003–04]

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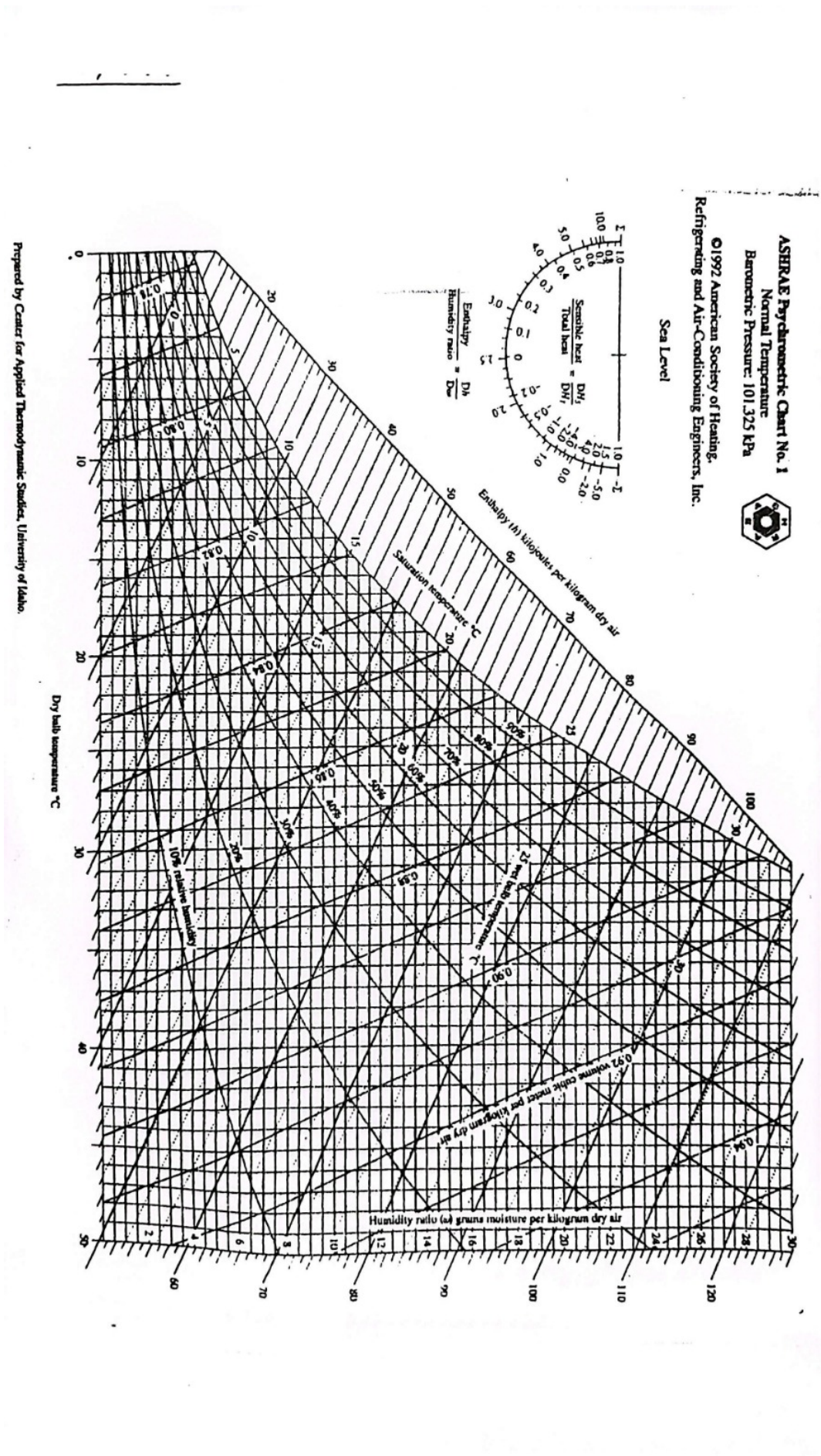
Section A3

Refrigeration & Air Conditioning

A3 — Refrigeration & Air Conditioning

- Q1. (a) On the psychrometric chart, show how the dew-point temperature at a specified state is determined. (5 marks)
- (b) Air at 1 atm, 15°C and 60% RH is heated to 20°C, then humidified to 25°C and 65% RH. Determine: (i) steam added to air, (ii) heat transfer in the heating section. (15 marks)
- (c) Draw a neat sketch of a split-type AC and mention its advantages and disadvantages. (15 marks)

[2023–24]



- Q2. (a) A refrigerator uses R-134a on an ideal vapour compression cycle. Refrigerant evaporates at -22°C and condenses at 28°C . Mass flow rate = 0.05 kg/s . Calculate: (i) rate of heat removal, (ii) power input to compressor, (iii) rate of heat rejection, (iv) COP. (15 marks)
- (b) Define “Refrigerant”. Mention desirable properties of a refrigerant. (10 marks)
- (c) With a neat sketch of components, write a short note on Central Air Conditioning System. (10 marks)

[2023–24]

TABLE A–11
Saturated refrigerant-134a—Temperature table

Temp., $T^{\circ}\text{C}$	Sat. press., P_{sat} kPa	Specific volume, m^3/kg		Internal energy, kJ/kg			Enthalpy, kJ/kg			Entropy, kJ/kg · K		
		Sat. liquid, v_f	Sat. vapor, v_g	Sat. liquid, u_f	Evap., u_{fg}	Sat. vapor, u_g	Sat. liquid, h_f	Evap., h_{fg}	Sat. vapor, h_g	Sat. liquid, s_f	Evap., s_{fg}	Sat. vapor, s_g
-40	51.25	0.0007054	0.36081	-0.036	207.40	207.37	0.000	225.86	225.86	0.00000	0.96866	0.96866
-38	56.86	0.0007083	0.32732	2.475	206.04	208.51	2.515	224.61	227.12	0.01072	0.95511	0.96584
-36	62.95	0.0007112	0.29751	4.992	204.67	209.66	5.037	223.35	228.39	0.02138	0.94176	0.96315
-34	69.56	0.0007142	0.27090	7.517	203.29	210.81	7.566	222.09	229.65	0.03199	0.92859	0.96058
-32	76.71	0.0007172	0.24711	10.05	201.91	211.96	10.10	220.81	230.91	0.04253	0.91560	0.95813
-30	84.43	0.0007203	0.22580	12.59	200.52	213.11	12.65	219.52	232.17	0.05301	0.90278	0.95579
-28	92.76	0.0007234	0.20666	15.13	199.12	214.25	15.20	218.22	233.43	0.06344	0.89012	0.95356
-26	101.73	0.0007265	0.18946	17.69	197.72	215.40	17.76	216.92	234.68	0.07382	0.87762	0.95144
-24	111.37	0.0007297	0.17395	20.25	196.30	216.55	20.33	215.59	235.92	0.08414	0.86527	0.94941
-22	121.72	0.0007329	0.15995	22.82	194.88	217.70	22.91	214.26	237.17	0.09441	0.85307	0.94748
-20	132.82	0.0007362	0.14729	25.39	193.45	218.84	25.49	212.91	238.41	0.10463	0.84101	0.94564
-18	144.69	0.0007396	0.13583	27.98	192.01	219.98	28.09	211.55	239.64	0.11481	0.82908	0.94389
-16	157.38	0.0007430	0.12542	30.57	190.56	221.13	30.69	210.18	240.87	0.12493	0.81729	0.94222
-14	170.93	0.0007464	0.11597	33.17	189.09	222.27	33.30	208.79	242.09	0.13501	0.80561	0.94063
-12	185.37	0.0007499	0.10736	35.78	187.62	223.40	35.92	207.38	243.30	0.14504	0.79406	0.93911
-10	200.74	0.0007535	0.099516	38.40	186.14	224.54	38.55	205.96	244.51	0.15504	0.78263	0.93766
-8	217.08	0.0007571	0.092352	41.03	184.64	225.67	41.19	204.52	245.72	0.16498	0.77130	0.93629
-6	234.44	0.0007608	0.085802	43.66	183.13	226.80	43.84	203.07	246.91	0.17489	0.76008	0.93497
-4	252.85	0.0007646	0.079804	46.31	181.61	227.92	46.50	201.60	248.10	0.18476	0.74896	0.93372
-2	272.36	0.0007684	0.074304	48.96	180.08	229.04	49.17	200.11	249.28	0.19459	0.73794	0.93253
0	293.01	0.0007723	0.069255	51.63	178.53	230.16	51.86	198.60	250.45	0.20439	0.72701	0.93139
2	314.84	0.0007763	0.064612	54.30	176.97	231.27	54.55	197.07	251.61	0.21415	0.71616	0.93031
4	337.90	0.0007804	0.060338	56.99	175.39	232.38	57.25	195.51	252.77	0.22387	0.70540	0.92927
6	362.23	0.0007845	0.056398	59.68	173.80	233.48	59.97	193.94	253.91	0.23356	0.69471	0.92828
8	387.88	0.0007887	0.052762	62.39	172.19	234.58	62.69	192.35	255.04	0.24323	0.68410	0.92733
10	414.89	0.0007930	0.049403	65.10	170.56	235.67	65.43	190.73	256.16	0.25286	0.67356	0.92641
12	443.31	0.0007975	0.046295	67.83	168.92	236.75	68.18	189.09	257.27	0.26246	0.66308	0.92554
14	473.19	0.0008020	0.043417	70.57	167.26	237.83	70.95	187.42	258.37	0.27204	0.65266	0.92470
16	504.58	0.0008066	0.040748	73.32	165.58	238.90	73.73	185.73	259.46	0.28159	0.64230	0.92389
18	537.52	0.0008113	0.038271	76.08	163.88	239.96	76.52	184.01	260.53	0.29112	0.63198	0.92310
20	572.07	0.0008161	0.035969	78.86	162.16	241.02	79.32	182.27	261.59	0.30063	0.62172	0.92234
22	608.27	0.0008210	0.033828	81.64	160.42	242.06	82.14	180.49	262.64	0.31011	0.61149	0.92160
24	646.18	0.0008261	0.031834	84.44	158.65	243.10	84.98	178.69	263.67	0.31958	0.60130	0.92088
26	685.84	0.0008313	0.029976	87.26	156.87	244.12	87.83	176.85	264.68	0.32903	0.59115	0.92018
28	727.31	0.0008366	0.028242	90.09	155.05	245.14	90.69	174.99	265.68	0.33846	0.58102	0.91948
30	770.64	0.0008421	0.026622	92.93	153.22	246.14	93.58	173.08	266.66	0.34789	0.57091	0.91879
32	815.89	0.0008478	0.025108	95.79	151.35	247.14	96.48	171.14	267.62	0.35730	0.56082	0.91811
34	863.11	0.0008536	0.023691	98.66	149.46	248.12	99.40	169.17	268.57	0.36670	0.55074	0.91743
36	912.35	0.0008595	0.022364	101.55	147.54	249.08	102.33	167.16	269.49	0.37609	0.54066	0.91675
38	963.68	0.0008657	0.021119	104.45	145.58	250.04	105.29	165.10	270.39	0.38548	0.53058	0.91606
40	1017.1	0.0008720	0.019952	107.38	143.60	250.97	108.26	163.00	271.27	0.39486	0.52049	0.91536
42	1072.8	0.0008786	0.018855	110.32	141.58	251.89	111.26	160.86	272.12	0.40425	0.51039	0.91464
44	1130.7	0.0008854	0.017824	113.28	139.52	252.80	114.28	158.67	272.95	0.41363	0.50027	0.91391
46	1191.0	0.0008924	0.016853	116.26	137.42	253.68	117.32	156.43	273.75	0.42302	0.49012	0.91315
48	1253.6	0.0008996	0.015939	119.26	135.29	254.55	120.39	154.14	274.53	0.43242	0.47993	0.91236
52	1386.2	0.0009150	0.014265	125.33	130.88	256.21	126.59	149.39	275.98	0.45126	0.45941	0.91067
56	1529.1	0.0009317	0.012771	131.49	126.28	257.77	132.91	144.38	277.30	0.47018	0.43863	0.90880
60	1682.8	0.0009498	0.011434	137.76	121.46	259.22	139.36	139.10	278.46	0.48920	0.41749	0.90669
65	1891.0	0.0009750	0.009950	145.77	115.05	260.82	147.62	132.02	279.64	0.51320	0.39039	0.90359
70	2118.2	0.0010037	0.008642	154.01	108.14	262.15	156.13	124.32	280.46	0.53755	0.36227	0.89982
75	2365.8	0.0010372	0.007480	162.53	100.60	263.13	164.98	115.85	280.82	0.56241	0.33272	0.89512
80	2635.3	0.0010772	0.006436	171.40	92.23	263.63	174.24	106.35	280.59	0.58800	0.30111	0.88912
85	2928.2	0.0011270	0.005486	180.77	82.67	263.44	184.07	95.44	279.51	0.61473	0.26644	0.88117
90	3246.9	0.0011932	0.004599	190.89	71.29	262.18	194.76	82.35	277.11	0.64336	0.22674	0.87010
95	3594.1	0.0012933	0.003726	202.40	56.47	258.87	207.05	65.21	272.26	0.67578	0.17711	0.85289
100	3975.1	0.0015269	0.002630	218.72	29.19	247.91	224.79	33.58	258.37	0.72217	0.08999	0.81215

Source: Tables A–11 through A–13 are generated using the Engineering Equation Solver (EES) software developed by S. A. Klein and F. L. Alvarado. The routine used in calculations is the R134a, which is based on the fundamental equation of state developed by R. Tillner-Roth and H.D. Baehr, “An International Standard Formulation for the Thermodynamic Properties of 1,1,1,2-Tetrafluoroethane (HFC-134a) for temperatures from 170 K to 455 K and Pressures up to 70 MPa,” *J. Phys. Chem, Ref. Data*, Vol. 23, No. 5, 1994. The enthalpy and entropy values of saturated liquid are set to zero at -40°C (and -40°F).

TABLE A–12
Saturated refrigerant-134a—Pressure table

Press., P kPa	Sat. temp., T_{sat} °C	Specific volume, m ³ /kg		Internal energy, kJ/kg			Enthalpy, kJ/kg			Entropy, kJ/kg · K		
		Sat. liquid, v_f	Sat. vapor, v_g	Sat. liquid, u_f	Evap., u_{fg}	Sat. vapor, u_g	Sat. liquid, h_f	Evap., h_{fg}	Sat. vapor, h_g	Sat. liquid, s_f	Evap., s_{fg}	Sat. vapor, s_g
60	-36.95	0.0007098	0.31121	3.798	205.32	209.12	3.841	223.95	227.79	0.01634	0.94807	0.96441
70	-33.87	0.0007144	0.26929	7.680	203.20	210.88	7.730	222.00	229.73	0.03267	0.92775	0.96042
80	-31.13	0.0007185	0.23753	11.15	201.30	212.46	11.21	220.25	231.46	0.04711	0.90999	0.95710
90	-28.65	0.0007223	0.21263	14.31	199.57	213.88	14.37	218.65	233.02	0.06008	0.89419	0.95427
100	-26.37	0.0007259	0.19254	17.21	197.98	215.19	17.28	217.16	234.44	0.07188	0.87995	0.95183
120	-22.32	0.0007324	0.16212	22.40	195.11	217.51	22.49	214.48	236.97	0.09275	0.85503	0.94779
140	-18.77	0.0007383	0.14014	26.98	192.57	219.54	27.08	212.08	239.16	0.11087	0.83368	0.94456
160	-15.60	0.0007437	0.12348	31.09	190.27	221.35	31.21	209.90	241.11	0.12693	0.81496	0.94190
180	-12.73	0.0007487	0.11041	34.83	188.16	222.99	34.97	207.90	242.86	0.14139	0.79826	0.93965
200	-10.09	0.0007533	0.099867	38.28	186.21	224.48	38.43	206.03	244.46	0.15457	0.78316	0.93773
240	-5.38	0.0007620	0.083897	44.48	182.67	227.14	44.66	202.62	247.28	0.17794	0.75664	0.93458
280	-1.25	0.0007699	0.072352	49.97	179.50	229.46	50.18	199.54	249.72	0.19829	0.73381	0.93210
320	2.46	0.0007772	0.063604	54.92	176.61	231.52	55.16	196.71	251.88	0.21637	0.71369	0.93006
360	5.82	0.0007841	0.056738	59.44	173.94	233.38	59.72	194.08	253.81	0.23270	0.69566	0.92836
400	8.91	0.0007907	0.051201	63.62	171.45	235.07	63.94	191.62	255.55	0.24761	0.67929	0.92691
450	12.46	0.0007985	0.045619	68.45	168.54	237.00	68.81	188.71	257.53	0.26465	0.66069	0.92535
500	15.71	0.0008059	0.041118	72.93	165.82	238.75	73.33	185.98	259.30	0.28023	0.64377	0.92400
550	18.73	0.0008130	0.037408	77.10	163.25	240.35	77.54	183.38	260.92	0.29461	0.62821	0.92282
600	21.55	0.0008199	0.034295	81.02	160.81	241.83	81.51	180.90	262.40	0.30799	0.61378	0.92177
650	24.20	0.0008266	0.031646	84.72	158.48	243.20	85.26	178.51	263.77	0.32051	0.60030	0.92081
700	26.69	0.0008331	0.029361	88.24	156.24	244.48	88.82	176.21	265.03	0.33230	0.58763	0.91994
750	29.06	0.0008395	0.027371	91.59	154.08	245.67	92.22	173.98	266.20	0.34345	0.57567	0.91912
800	31.31	0.0008458	0.025621	94.79	152.00	246.79	95.47	171.82	267.29	0.35404	0.56431	0.91835
850	33.45	0.0008520	0.024069	97.87	149.98	247.85	98.60	169.71	268.31	0.36413	0.55349	0.91762
900	35.51	0.0008580	0.022683	100.83	148.01	248.85	101.61	167.66	269.26	0.37377	0.54315	0.91692
950	37.48	0.0008641	0.021438	103.69	146.10	249.79	104.51	165.64	270.15	0.38301	0.53323	0.91624
1000	39.37	0.0008700	0.020313	106.45	144.23	250.68	107.32	163.67	270.99	0.39189	0.52368	0.91558
1200	46.29	0.0008934	0.016715	116.70	137.11	253.81	117.77	156.10	273.87	0.42441	0.48863	0.91303
1400	52.40	0.0009166	0.014107	125.94	130.43	256.37	127.22	148.90	276.12	0.45315	0.45734	0.91050
1600	57.88	0.0009400	0.012123	134.43	124.04	258.47	135.93	141.93	277.86	0.47911	0.42873	0.90784
1800	62.87	0.0009639	0.010559	142.33	117.83	260.17	144.07	135.11	279.17	0.50294	0.40204	0.90498
2000	67.45	0.0009886	0.009288	149.78	111.73	261.51	151.76	128.33	280.09	0.52509	0.37675	0.90184
2500	77.54	0.0010566	0.006936	166.99	96.47	263.45	169.63	111.16	280.79	0.57531	0.31695	0.89226
3000	86.16	0.0011406	0.005275	183.04	80.22	263.26	186.46	92.63	279.09	0.62118	0.25776	0.87894

TABLE A–13
Superheated refrigerant-134a

T °C	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg · K	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg · K	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg · K
	$P = 0.06 \text{ MPa } (T_{sat} = -36.95^\circ \text{C})$				$P = 0.10 \text{ MPa } (T_{sat} = -26.37^\circ \text{C})$				$P = 0.14 \text{ MPa } (T_{sat} = -18.77^\circ \text{C})$			
Sat.	0.31121	209.12	227.79	0.9644	0.19254	215.19	234.44	0.9518	0.14014	219.54	239.16	0.9446
-20	0.33608	220.60	240.76	1.0174	0.19841	219.66	239.50	0.9721				
-10	0.35048	227.55	248.58	1.0477	0.20743	226.75	247.49	1.0030	0.14605	225.91	246.36	0.9724
0	0.36476	234.66	256.54	1.0774	0.21630	233.95	255.58	1.0332	0.15263	233.23	254.60	1.0031
10	0.37893	241.92	264.66	1.1066	0.22506	241.30	263.81	1.0628	0.15908	240.66	262.93	1.0331
20	0.39302	249.35	272.94	1.1353	0.23373	248.79	272.17	1.0918	0.16544	248.22	271.38	1.0624
30	0.40705	256.95	281.37	1.1636	0.24233	256.44	280.68	1.1203	0.17172	255.93	279.97	1.0912
40	0.42102	264.71	289.97	1.1915	0.25088	264.25	289.34	1.1484	0.17794	263.79	288.70	1.1195
50	0.43495	272.64	298.74	1.2191	0.25937	272.22	298.16	1.1762	0.18412	271.79	297.57	1.1474
60	0.44883	280.73	307.66	1.2463	0.26783	280.35	307.13	1.2035	0.19025	279.96	306.59	1.1749
70	0.46269	288.99	316.75	1.2732	0.27626	288.64	316.26	1.2305	0.19635	288.28	315.77	1.2020
80	0.47651	297.41	326.00	1.2997	0.28465	297.08	325.55	1.2572	0.20242	296.75	325.09	1.2288
90	0.49032	306.00	335.42	1.3260	0.29303	305.69	334.99	1.2836	0.20847	305.38	334.57	1.2553
100	0.50410	314.74	344.99	1.3520	0.30138	314.46	344.60	1.3096	0.21449	314.17	344.20	1.2814
	$P = 0.18 \text{ MPa } (T_{sat} = -12.73^\circ \text{C})$				$P = 0.20 \text{ MPa } (T_{sat} = -10.09^\circ \text{C})$				$P = 0.24 \text{ MPa } (T_{sat} = -5.38^\circ \text{C})$			
Sat.	0.11041	222.99	242.86	0.9397	0.09987	224.48	244.46	0.9377	0.08390	227.14	247.28	0.9346
Continued on next page												

T °C	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg · K	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg · K	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg · K
-10	0.11189	225.02	245.16	0.9484	0.09991	224.55	244.54	0.9380				
0	0.11722	232.48	253.58	0.9798	0.10481	232.09	253.05	0.9698	0.08617	231.29	251.97	0.9519
10	0.12240	240.00	262.04	1.0102	0.10955	239.67	261.58	1.0004	0.09026	238.98	260.65	0.9831
20	0.12748	247.64	270.59	1.0399	0.11418	247.35	270.18	1.0303	0.09423	246.74	269.36	1.0134
30	0.13248	255.41	279.25	1.0690	0.11874	255.14	278.89	1.0595	0.09812	254.61	278.16	1.0429
40	0.13741	263.31	288.05	1.0975	0.12322	263.08	287.72	1.0882	0.10193	262.59	287.06	1.0718
50	0.14230	271.36	296.98	1.1256	0.12766	271.15	296.68	1.1163	0.10570	270.71	296.08	1.1001
60	0.14715	279.56	306.05	1.1532	0.13206	279.37	305.78	1.1441	0.10942	278.97	305.23	1.1280
70	0.15196	287.91	315.27	1.1805	0.13641	287.73	315.01	1.1714	0.11310	287.36	314.51	1.1554
80	0.15673	296.42	324.63	1.2074	0.14074	296.25	324.40	1.1983	0.11675	295.91	323.93	1.1825
90	0.16149	305.07	334.14	1.2339	0.14504	304.92	333.93	1.2249	0.12038	304.60	333.49	1.2092
100	0.16622	313.88	343.80	1.2602	0.14933	313.74	343.60	1.2512	0.12398	313.44	343.20	1.2356
$P = 0.28 \text{ MPa } (T_{sat} = -1.25^\circ \text{C})$					$P = 0.32 \text{ MPa } (T_{sat} = 2.46^\circ \text{C})$				$P = 0.40 \text{ MPa } (T_{sat} = 8.91^\circ \text{C})$			
Sat.	0.07235	229.46	249.72	0.9321	0.06360	231.52	251.88	0.9301	0.051201	235.07	255.55	0.9269
0	0.07282	230.44	250.83	0.9362								
10	0.07646	238.27	259.68	0.9680	0.06609	237.54	258.69	0.9544	0.051506	235.97	256.58	0.9305
20	0.07997	246.13	268.52	0.9987	0.06925	245.50	267.66	0.9856	0.054213	244.18	265.86	0.9628
30	0.08338	254.06	277.41	1.0285	0.07231	253.50	276.65	1.0157	0.056796	252.36	275.07	0.9937
40	0.08672	262.10	286.38	1.0576	0.07530	261.60	285.70	1.0451	0.059292	260.58	284.30	1.0236
50	0.09000	270.27	295.47	1.0862	0.07823	269.82	294.85	1.0739	0.061724	268.90	293.59	1.0528
60	0.09324	278.56	304.67	1.1142	0.08111	278.15	304.11	1.1021	0.064104	277.32	302.96	1.0814
70	0.09644	286.99	314.00	1.1418	0.08395	286.62	313.48	1.1298	0.066443	285.86	312.44	1.1094
80	0.09961	295.57	323.46	1.1690	0.08675	295.22	322.98	1.1571	0.068747	294.53	322.02	1.1369
90	0.10275	304.29	333.06	1.1958	0.08953	303.97	332.62	1.1840	0.071023	303.32	331.73	1.1640
100	0.10587	313.15	342.80	1.2222	0.09229	312.86	342.39	1.2105	0.073274	312.26	341.57	1.1907
110	0.10897	322.16	352.68	1.2483	0.09503	321.89	352.30	1.2367	0.075504	321.33	351.53	1.2171
120	0.11205	331.32	362.70	1.2742	0.09775	331.07	362.35	1.2626	0.077717	330.55	361.63	1.2431
130	0.11512	340.63	372.87	1.2997	0.10045	340.39	372.54	1.2882	0.079913	339.90	371.87	1.2688
140	0.11818	350.09	383.18	1.3250	0.10314	349.86	382.87	1.3135	0.082096	349.41	382.24	1.2942
$P = 0.50 \text{ MPa } (T_{sat} = 15.71^\circ \text{C})$					$P = 0.60 \text{ MPa } (T_{sat} = 21.55^\circ \text{C})$				$P = 0.70 \text{ MPa } (T_{sat} = 26.69^\circ \text{C})$			
Sat.	0.041118	238.75	259.30	0.9240	0.034295	241.83	262.40	0.9218	0.029361	244.48	265.03	0.9199
20	0.042115	242.40	263.46	0.9383								
30	0.044338	250.84	273.01	0.9703	0.035984	249.22	270.81	0.9499	0.029966	247.48	268.45	0.9313
40	0.046456	259.26	282.48	1.0011	0.037865	257.86	280.58	0.9816	0.031696	256.39	278.57	0.9641
50	0.048499	267.72	291.96	1.0309	0.039659	266.48	290.28	1.0121	0.033322	265.20	288.53	0.9954
60	0.050485	276.25	301.50	1.0599	0.041389	275.15	299.98	1.0417	0.034875	274.01	298.42	1.0256
70	0.052427	284.89	311.10	1.0883	0.043069	283.89	309.73	1.0705	0.036373	282.87	308.33	1.0549
80	0.054331	293.64	320.80	1.1162	0.044710	292.73	319.55	1.0987	0.037829	291.80	318.28	1.0835
90	0.056205	302.51	330.61	1.1436	0.046318	301.67	329.46	1.1264	0.039250	300.82	328.29	1.1114
100	0.058053	311.50	340.53	1.1705	0.047900	310.73	339.47	1.1536	0.040642	309.95	338.40	1.1389
110	0.059880	320.63	350.57	1.1971	0.049458	319.91	349.59	1.1803	0.042010	319.19	348.60	1.1658
120	0.061687	329.89	360.73	1.2233	0.050997	329.23	359.82	1.2067	0.043358	328.55	358.90	1.1924
130	0.063479	339.29	371.03	1.2491	0.052519	338.67	370.18	1.2327	0.044688	338.04	369.32	1.2186
140	0.065256	348.83	381.46	1.2747	0.054027	348.25	380.66	1.2584	0.046004	347.66	379.86	1.2444
150	0.067021	358.51	392.02	1.2999	0.055522	357.96	391.27	1.2838	0.047306	357.41	390.52	1.2699
160	0.068775	368.33	402.72	1.3249	0.057006	367.81	402.01	1.3088	0.048597	367.29	401.31	1.2951
$P = 0.80 \text{ MPa } (T_{sat} = 31.31^\circ \text{C})$					$P = 0.90 \text{ MPa } (T_{sat} = 35.51^\circ \text{C})$				$P = 1.00 \text{ MPa } (T_{sat} = 39.37^\circ \text{C})$			
Sat.	0.025621	246.79	267.29	0.9183	0.022683	248.85	269.26	0.9169	0.020313	250.68	270.99	0.9156
40	0.027035	254.82	276.45	0.9480	0.023375	253.13	274.17	0.9327	0.020406	251.30	271.71	0.9179
50	0.028547	263.86	286.69	0.9802	0.024809	262.44	284.77	0.9660	0.021796	260.94	282.74	0.9525
60	0.029973	272.83	296.81	1.0110	0.026146	271.60	295.13	0.9976	0.023068	270.32	293.38	0.9850
70	0.031340	281.81	306.88	1.0408	0.027413	280.72	305.39	1.0280	0.024261	279.59	303.85	1.0160
80	0.032659	290.84	316.97	1.0698	0.028630	289.86	315.63	1.0574	0.025398	288.86	314.25	1.0458
90	0.033941	299.95	327.10	1.0981	0.029806	299.06	325.89	1.0860	0.026492	298.15	324.64	1.0748
100	0.035193	309.15	337.30	1.1258	0.030951	308.34	336.19	1.1140	0.027552	307.51	335.06	1.1031
110	0.036420	318.45	347.59	1.1530	0.032068	317.70	346.56	1.1414	0.028584	316.94	345.53	1.1308
120	0.037625	327.87	357.97	1.1798	0.033164	327.18	357.02	1.1684	0.029592	326.47	356.06	1.1580
130	0.038813	337.40	368.45	1.2061	0.034241	336.76	367.58	1.1949	0.030581	336.11	366.69	1.1846
140	0.039985	347.06	379.05	1.2321	0.035302	346.46	378.23	1.2210	0.031554	345.85	377.40	1.2109
150	0.041143	356.85	389.76	1.2577	0.036349	356.28	389.00	1.2467	0.032512	355.71	388.22	1.2368
160	0.042290	366.76	400.59	1.2830	0.037384	366.23	399.88	1.2721	0.033457	365.70	399.15	1.2623
170	0.043427	376.81	411.55	1.3080	0.038408	376.31	410.88	1.2972	0.034392	375.81	410.20	1.2875
180	0.044554	386.99	422.64	1.3327	0.039423	386.52	422.00	1.3221	0.035317	386.04	421.36	1.3124
$P = 1.20 \text{ MPa } (T_{sat} = 46.29^\circ \text{C})$					$P = 1.40 \text{ MPa } (T_{sat} = 52.40^\circ \text{C})$				$P = 1.60 \text{ MPa } (T_{sat} = 57.88^\circ \text{C})$			
Sat.	0.016715	253.81	273.87	0.9130	0.014107	256.37	276.12	0.9105	0.012123	258.47	277.86	0.9078
50	0.017201	257.63	278.27	0.9267								
60	0.018404	267.56	289.64	0.9614	0.015005	264.46	285.47	0.9389	0.012372	260.89	280.69	0.9163

Continued on next page

T °C	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg · K	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg · K	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg · K
70	0.019502	277.21	300.61	0.9938	0.016060	274.62	297.10	0.9733	0.013430	271.76	293.25	0.9535
80	0.020529	286.75	311.39	1.0248	0.017023	284.51	308.34	1.0056	0.014362	282.09	305.07	0.9875
90	0.021506	296.26	322.07	1.0546	0.017923	294.28	319.37	1.0364	0.015215	292.17	316.52	1.0194
100	0.022442	305.80	332.73	1.0836	0.018778	304.01	330.30	1.0661	0.016014	302.14	327.76	1.0500
110	0.023348	315.38	343.40	1.1118	0.019597	313.76	341.19	1.0949	0.016773	312.07	338.91	1.0795
120	0.024228	325.03	354.11	1.1394	0.020388	323.55	352.09	1.1230	0.017500	322.02	350.02	1.1081
130	0.025086	334.77	364.88	1.1664	0.021155	333.41	363.02	1.1504	0.018201	332.00	361.12	1.1360
140	0.025927	344.61	375.72	1.1930	0.021904	343.34	374.01	1.1773	0.018882	342.05	372.26	1.1632
150	0.026753	354.56	386.66	1.2192	0.022636	353.37	385.07	1.2038	0.019545	352.17	383.44	1.1900
160	0.027566	364.61	397.69	1.2449	0.023355	363.51	396.20	1.2298	0.020194	362.38	394.69	1.2163
170	0.028367	374.78	408.82	1.2703	0.024061	373.75	407.43	1.2554	0.020830	372.69	406.02	1.2421
180	0.029158	385.08	420.07	1.2954	0.024757	384.10	418.76	1.2807	0.021456	383.11	417.44	1.2676

Q3. A refrigerator uses R-134a. Enthalpy after expansion valve = 85.26 kJ/kg, condenser rejects 19.1 kW, mass flow rate = 0.1 kg/s. (i) Draw T-S diagram, (ii) determine pressure and temperature at each state, (iii) calculate COP. **(15 marks)** [2023–24]

TABLE A–11
Saturated refrigerant-134a—Temperature table

Temp., T °C	Sat. press., P_{sat} kPa	Specific volume, m ³ /kg		Internal energy, kJ/kg			Enthalpy, kJ/kg			Entropy, kJ/kg · K		
		Sat. liquid, v_f	Sat. vapor, v_g	Sat. liquid, u_f	Evap., u_{fg}	Sat. vapor, u_g	Sat. liquid, h_f	Evap., h_{fg}	Sat. vapor, h_g	Sat. liquid, s_f	Evap., s_{fg}	Sat. vapor, s_g
-40	51.25	0.0007054	0.36081	-0.036	207.40	207.37	0.000	225.86	225.86	0.00000	0.96866	0.96866
-38	56.86	0.0007083	0.32732	2.475	206.04	208.51	2.515	224.61	227.12	0.01072	0.95511	0.96584
-36	62.95	0.0007112	0.29751	4.992	204.67	209.66	5.037	223.35	228.39	0.02138	0.94176	0.96315
-34	69.56	0.0007142	0.27090	7.517	203.29	210.81	7.566	222.09	229.65	0.03199	0.92859	0.96058
-32	76.71	0.0007172	0.24711	10.05	201.91	211.96	10.10	220.81	230.91	0.04253	0.91560	0.95813
-30	84.43	0.0007203	0.22580	12.59	200.52	213.11	12.65	219.52	232.17	0.05301	0.90278	0.95579
-28	92.76	0.0007234	0.20666	15.13	199.12	214.25	15.20	218.22	233.43	0.06344	0.89012	0.95356
-26	101.73	0.0007265	0.18946	17.69	197.72	215.40	17.76	216.92	234.68	0.07382	0.87762	0.95144
-24	111.37	0.0007297	0.17395	20.25	196.30	216.55	20.33	215.59	235.92	0.08414	0.86527	0.94941
-22	121.72	0.0007329	0.15995	22.82	194.88	217.70	22.91	214.26	237.17	0.09441	0.85307	0.94748
-20	132.82	0.0007362	0.14729	25.39	193.45	218.84	25.49	212.91	238.41	0.10463	0.84101	0.94564
-18	144.69	0.0007396	0.13583	27.98	192.01	219.98	28.09	211.55	239.64	0.11481	0.82908	0.94389
-16	157.38	0.0007430	0.12542	30.57	190.56	221.13	30.69	210.18	240.87	0.12493	0.81729	0.94222
-14	170.93	0.0007464	0.11597	33.17	189.09	222.27	33.30	208.79	242.09	0.13501	0.80561	0.94063
-12	185.37	0.0007499	0.10736	35.78	187.62	223.40	35.92	207.38	243.30	0.14504	0.79406	0.93911
-10	200.74	0.0007535	0.099516	38.40	186.14	224.54	38.55	205.96	244.51	0.15504	0.78263	0.93766
-8	217.08	0.0007571	0.092352	41.03	184.64	225.67	41.19	204.52	245.72	0.16498	0.77130	0.93629
-6	234.44	0.0007608	0.085802	43.66	183.13	226.80	43.84	203.07	246.91	0.17489	0.76008	0.93497
-4	252.85	0.0007646	0.079804	46.31	181.61	227.92	46.50	201.60	248.10	0.18476	0.74896	0.93372
-2	272.36	0.0007684	0.074304	48.96	180.08	229.04	49.17	200.11	249.28	0.19459	0.73794	0.93253
0	293.01	0.0007723	0.069255	51.63	178.53	230.16	51.86	198.60	250.45	0.20439	0.72701	0.93139
2	314.84	0.0007763	0.064612	54.30	176.97	231.27	54.55	197.07	251.61	0.21415	0.71616	0.93031
4	337.90	0.0007804	0.060338	56.99	175.39	232.38	57.25	195.51	252.77	0.22387	0.70540	0.92927
6	362.23	0.0007845	0.056398	59.68	173.80	233.48	59.97	193.94	253.91	0.23356	0.69471	0.92828
8	387.88	0.0007887	0.052762	62.39	172.19	234.58	62.69	192.35	255.04	0.24323	0.68410	0.92733
10	414.89	0.0007930	0.049403	65.10	170.56	235.67	65.43	190.73	256.16	0.25286	0.67356	0.92641
12	443.31	0.0007975	0.046295	67.83	168.92	236.75	68.18	189.09	257.27	0.26246	0.66308	0.92554
14	473.19	0.0008020	0.043417	70.57	167.26	237.83	70.95	187.42	258.37	0.27204	0.65266	0.92470
16	504.58	0.0008066	0.040748	73.32	165.58	238.90	73.73	185.73	259.46	0.28159	0.64230	0.92389
18	537.52	0.0008113	0.038271	76.08	163.88	239.96	76.52	184.01	260.53	0.29112	0.63198	0.92310
20	572.07	0.0008161	0.035969	78.86	162.16	241.02	79.32	182.27	261.59	0.30063	0.62172	0.92234
22	608.27	0.0008210	0.033828	81.64	160.42	242.06	82.14	180.49	262.64	0.31011	0.61149	0.92160
24	646.18	0.0008261	0.031834	84.44	158.65	243.10	84.98	178.69	263.67	0.31958	0.60130	0.92088
26	685.84	0.0008313	0.029976	87.26	156.87	244.12	87.83	176.85	264.68	0.32903	0.59115	0.92018
28	727.31	0.0008366	0.028242	90.09	155.05	245.14	90.69	174.99	265.68	0.33846	0.58102	0.91948
30	770.64	0.0008421	0.026622	92.93	153.22	246.14	93.58	173.08	266.66	0.34789	0.57091	0.91879
32	815.89	0.0008478	0.025108	95.79	151.35	247.14	96.48	171.14	267.62	0.35730	0.56082	0.91811
34	863.11	0.0008536	0.023691	98.66	149.46	248.12	99.40	169.17	268.57	0.36670	0.55074	0.91743

Continued on next page

Temp., $T^{\circ}\text{C}$	Sat. press., P_{sat} kPa	Specific volume, m^3/kg		Internal energy, kJ/kg			Enthalpy, kJ/kg			Entropy, kJ/kg · K		
		Sat. liquid, v_f	Sat. vapor, v_g	Sat. liquid, u_f	Evap., u_{fg}	Sat. vapor, u_g	Sat. liquid, h_f	Evap., h_{fg}	Sat. vapor, h_g	Sat. liquid, s_f	Evap., s_{fg}	Sat. vapor, s_g
36	912.35	0.0008595	0.022364	101.55	147.54	249.08	102.33	167.16	269.49	0.37609	0.54066	0.91675
38	963.68	0.0008657	0.021119	104.45	145.58	250.04	105.29	165.10	270.39	0.38548	0.53058	0.91606
40	1017.1	0.0008720	0.019952	107.38	143.60	250.97	108.26	163.00	271.27	0.39486	0.52049	0.91536
42	1072.8	0.0008786	0.018855	110.32	141.58	251.89	111.26	160.86	272.12	0.40425	0.51039	0.91464
44	1130.7	0.0008854	0.017824	113.28	139.52	252.80	114.28	158.67	272.95	0.41363	0.50027	0.91391
46	1191.0	0.0008924	0.016853	116.26	137.42	253.68	117.32	156.43	273.75	0.42302	0.49012	0.91315
48	1253.6	0.0008996	0.015939	119.26	135.29	254.55	120.39	154.14	274.53	0.43242	0.47993	0.91236
52	1386.2	0.0009150	0.014265	125.33	130.88	256.21	126.59	149.39	275.98	0.45126	0.45941	0.91067
56	1529.1	0.0009317	0.012771	131.49	126.28	257.77	132.91	144.38	277.30	0.47018	0.43863	0.90880
60	1682.8	0.0009498	0.011434	137.76	121.46	259.22	139.36	139.10	278.46	0.48920	0.41749	0.90669
65	1891.0	0.0009750	0.009950	145.77	115.05	260.82	147.62	132.02	279.64	0.51320	0.39039	0.90359
70	2118.2	0.0010037	0.008642	154.01	108.14	262.15	156.13	124.32	280.46	0.53755	0.36227	0.89982
75	2365.8	0.0010372	0.007480	162.53	100.60	263.13	164.98	115.85	280.82	0.56241	0.33272	0.89512
80	2635.3	0.0010772	0.006436	171.40	92.23	263.63	174.24	106.35	280.59	0.58800	0.30111	0.88912
85	2928.2	0.0011270	0.005486	180.77	82.67	263.44	184.07	95.44	279.51	0.61473	0.26644	0.88117
90	3246.9	0.0011932	0.004599	190.89	71.29	262.18	194.76	82.35	277.11	0.64336	0.22674	0.87010
95	3594.1	0.0012933	0.003726	202.40	56.47	258.87	207.05	65.21	272.26	0.67578	0.17711	0.85289
100	3975.1	0.0015269	0.002630	218.72	29.19	247.91	224.79	33.58	258.37	0.72217	0.08999	0.81215

Source: Tables A–11 through A–13 are generated using the Engineering Equation Solver (EES) software developed by S. A. Klein and F. L. Alvarado. The routine used in calculations is the R134a, which is based on the fundamental equation of state developed by R. Tillner-Roth and H.D. Baehr, “An International Standard Formulation for the Thermodynamic Properties of 1,1,1,2-Tetrafluoroethane (HFC-134a) for temperatures from 170 K to 455 K and Pressures up to 70 MPa,” *J. Phys. Chem. Ref. Data*, Vol. 23, No. 5, 1994. The enthalpy and entropy values of saturated liquid are set to zero at -40°C (and -40°F).

TABLE A–12
Saturated refrigerant-134a—Pressure table

Press., P kPa	Sat. temp., T_{sat} $^{\circ}\text{C}$	Specific volume, m^3/kg		Internal energy, kJ/kg			Enthalpy, kJ/kg			Entropy, kJ/kg · K		
		Sat. liquid, v_f	Sat. vapor, v_g	Sat. liquid, u_f	Evap., u_{fg}	Sat. vapor, u_g	Sat. liquid, h_f	Evap., h_{fg}	Sat. vapor, h_g	Sat. liquid, s_f	Evap., s_{fg}	Sat. vapor, s_g
60	-36.95	0.0007098	0.31121	3.798	205.32	209.12	3.841	223.95	227.79	0.01634	0.94807	0.96441
70	-33.87	0.0007144	0.26929	7.680	203.20	210.88	7.730	222.00	229.73	0.03267	0.92775	0.96042
80	-31.13	0.0007185	0.23753	11.15	201.30	212.46	11.21	220.25	231.46	0.04711	0.90999	0.95710
90	-28.65	0.0007223	0.21263	14.31	199.57	213.88	14.37	218.65	233.02	0.06008	0.89419	0.95427
100	-26.37	0.0007259	0.19254	17.21	197.98	215.19	17.28	217.16	234.44	0.07188	0.87995	0.95183
120	-22.32	0.0007324	0.16212	22.40	195.11	217.51	22.49	214.48	236.97	0.09275	0.85503	0.94779
140	-18.77	0.0007383	0.14014	26.98	192.57	219.54	27.08	212.08	239.16	0.11087	0.83368	0.94456
160	-15.60	0.0007437	0.12348	31.09	190.27	221.35	31.21	209.90	241.11	0.12693	0.81496	0.94190
180	-12.73	0.0007487	0.11041	34.83	188.16	222.99	34.97	207.90	242.86	0.14139	0.79826	0.93965
200	-10.09	0.0007533	0.099867	38.28	186.21	224.48	38.43	206.03	244.46	0.15457	0.78316	0.93773
240	-5.38	0.0007620	0.083897	44.48	182.67	227.14	44.66	202.62	247.28	0.17794	0.75664	0.93458
280	-1.25	0.0007699	0.072352	49.97	179.50	229.46	50.18	199.54	249.72	0.19829	0.73381	0.93210
320	2.46	0.0007772	0.063604	54.92	176.61	231.52	55.16	196.71	251.88	0.21637	0.71369	0.93006
360	5.82	0.0007841	0.056738	59.44	173.94	233.38	59.72	194.08	253.81	0.23270	0.69566	0.92836
400	8.91	0.0007907	0.051201	63.62	171.45	235.07	63.94	191.62	255.55	0.24761	0.67929	0.92691
450	12.46	0.0007985	0.045619	68.45	168.54	237.00	68.81	188.71	257.53	0.26465	0.66069	0.92535
500	15.71	0.0008059	0.041118	72.93	165.82	238.75	73.33	185.98	259.30	0.28023	0.64377	0.92400
550	18.73	0.0008130	0.037408	77.10	163.25	240.35	77.54	183.38	260.92	0.29461	0.62821	0.92282
600	21.55	0.0008199	0.034295	81.02	160.81	241.83	81.51	180.90	262.40	0.30799	0.61378	0.92177
650	24.20	0.0008266	0.031646	84.72	158.48	243.20	85.26	178.51	263.77	0.32051	0.60030	0.92081
700	26.69	0.0008331	0.029361	88.24	156.24	244.48	88.82	176.21	265.03	0.33230	0.58763	0.91994
750	29.06	0.0008395	0.027371	91.59	154.08	245.67	92.22	173.98	266.20	0.34345	0.57567	0.91912
800	31.31	0.0008458	0.025621	94.79	152.00	246.79	95.47	171.82	267.29	0.35404	0.56431	0.91835
850	33.45	0.0008520	0.024069	97.87	149.98	247.85	98.60	169.71	268.31	0.36413	0.55349	0.91762
900	35.51	0.0008580	0.022683	100.83	148.01	248.85	101.61	167.66	269.26	0.37377	0.54315	0.91692
950	37.48	0.0008641	0.021438	103.69	146.10	249.79	104.51	165.64	270.15	0.38301	0.53323	0.91624
1000	39.37	0.0008700	0.020313	106.45	144.23	250.68	107.32	163.67	270.99	0.39189	0.52368	0.91558
1200	46.29	0.0008934	0.016715	116.70	137.11	253.81	117.77	156.10	273.87	0.42441	0.48863	0.91303
1400	52.40	0.0009166	0.014107	125.94	130.43	256.37	127.22	148.90	276.12	0.45315	0.45734	0.91050

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		<i>Specific volume,</i> m ³ /kg		<i>Internal energy,</i> kJ/kg			<i>Enthalpy,</i> kJ/kg			<i>Entropy,</i> kJ/kg · K		
Press., <i>P</i> kPa	Sat. temp., <i>T</i> _{sat} °C	Sat. liquid, <i>v_f</i>	Sat. vapor, <i>v_g</i>	Sat. liquid, <i>u_f</i>	Evap., <i>u_{fg}</i>	Sat. vapor, <i>u_g</i>	Sat. liquid, <i>h_f</i>	Evap., <i>h_{fg}</i>	Sat. vapor, <i>h_g</i>	Sat. liquid, <i>s_f</i>	Evap., <i>s_{fg}</i>	Sat. vapor, <i>s_g</i>
1600	57.88	0.0009400	0.012123	134.43	124.04	258.47	135.93	141.93	277.86	0.47911	0.42873	0.90784
1800	62.87	0.0009639	0.010559	142.33	117.83	260.17	144.07	135.11	279.17	0.50294	0.40204	0.90498
2000	67.45	0.0009886	0.009288	149.78	111.73	261.51	151.76	128.33	280.09	0.52509	0.37675	0.90184
2500	77.54	0.0010566	0.006936	166.99	96.47	263.45	169.63	111.16	280.79	0.57531	0.31695	0.89226
3000	86.16	0.0011406	0.005275	183.04	80.22	263.26	186.46	92.63	279.09	0.62118	0.25776	0.87894

TABLE A–13
 Superheated refrigerant-134a

<i>T</i> °C	<i>v</i> m ³ /kg	<i>u</i> kJ/kg	<i>h</i> kJ/kg	<i>s</i> kJ/kg · K	<i>v</i> m ³ /kg	<i>u</i> kJ/kg	<i>h</i> kJ/kg	<i>s</i> kJ/kg · K	<i>v</i> m ³ /kg	<i>u</i> kJ/kg	<i>h</i> kJ/kg	<i>s</i> kJ/kg · K
<i>P</i> = 0.06 MPa (<i>T</i> _{sat} = −36.95° C)				<i>P</i> = 0.10 MPa (<i>T</i> _{sat} = −26.37° C)				<i>P</i> = 0.14 MPa (<i>T</i> _{sat} = −18.77° C)				
Sat.	0.31121	209.12	227.79	0.9644	0.19254	215.19	234.44	0.9518	0.14014	219.54	239.16	0.9446
−20	0.33608	220.60	240.76	1.0174	0.19841	219.66	239.50	0.9721				
−10	0.35048	227.55	248.58	1.0477	0.20743	226.75	247.49	1.0030	0.14605	225.91	246.36	0.9724
0	0.36476	234.66	256.54	1.0774	0.21630	233.95	255.58	1.0332	0.15263	233.23	254.60	1.0031
10	0.37893	241.92	264.66	1.1066	0.22506	241.30	263.81	1.0628	0.15908	240.66	262.93	1.0331
20	0.39302	249.35	272.94	1.1353	0.23373	248.79	272.17	1.0918	0.16544	248.22	271.38	1.0624
30	0.40705	256.95	281.37	1.1636	0.24233	256.44	280.68	1.1203	0.17172	255.93	279.97	1.0912
40	0.42102	264.71	289.97	1.1915	0.25088	264.25	289.34	1.1484	0.17794	263.79	288.70	1.1195
50	0.43495	272.64	298.74	1.2191	0.25937	272.22	298.16	1.1762	0.18412	271.79	297.57	1.1474
60	0.44883	280.73	307.66	1.2463	0.26783	280.35	307.13	1.2035	0.19025	279.96	306.59	1.1749
70	0.46269	288.99	316.75	1.2732	0.27626	288.64	316.26	1.2305	0.19635	288.28	315.77	1.2020
80	0.47651	297.41	326.00	1.2997	0.28465	297.08	325.55	1.2572	0.20242	296.75	325.09	1.2288
90	0.49032	306.00	335.42	1.3260	0.29303	305.69	334.99	1.2836	0.20847	305.38	334.57	1.2553
100	0.50410	314.74	344.99	1.3520	0.30138	314.46	344.60	1.3096	0.21449	314.17	344.20	1.2814
<i>P</i> = 0.18 MPa (<i>T</i> _{sat} = −12.73° C)				<i>P</i> = 0.20 MPa (<i>T</i> _{sat} = −10.09° C)				<i>P</i> = 0.24 MPa (<i>T</i> _{sat} = −5.38° C)				
Sat.	0.11041	222.99	242.86	0.9397	0.09987	224.48	244.46	0.9377	0.08390	227.14	247.28	0.9346
−10	0.11189	225.02	245.16	0.9484	0.09991	224.55	244.54	0.9380				
0	0.11722	232.48	253.58	0.9798	0.10481	232.09	253.05	0.9698	0.08617	231.29	251.97	0.9519
10	0.12240	240.00	262.04	1.0102	0.10955	239.67	261.58	1.0004	0.09026	238.98	260.65	0.9831
20	0.12748	247.64	270.59	1.0399	0.11418	247.35	270.18	1.0303	0.09423	246.74	269.36	1.0134
30	0.13248	255.41	279.25	1.0690	0.11874	255.14	278.89	1.0595	0.09812	254.61	278.16	1.0429
40	0.13741	263.31	288.05	1.0975	0.12322	263.08	287.72	1.0882	0.10193	262.59	287.06	1.0718
50	0.14230	271.36	296.98	1.1256	0.12766	271.15	296.68	1.1163	0.10570	270.71	296.08	1.1001
60	0.14715	279.56	306.05	1.1532	0.13206	279.37	305.78	1.1441	0.10942	278.97	305.23	1.1280
70	0.15196	287.91	315.27	1.1805	0.13641	287.73	315.01	1.1714	0.11310	287.36	314.51	1.1554
80	0.15673	296.42	324.63	1.2074	0.14074	296.25	324.40	1.1983	0.11675	295.91	323.93	1.1825
90	0.16149	305.07	334.14	1.2339	0.14504	304.92	333.93	1.2249	0.12038	304.60	333.49	1.2092
100	0.16622	313.88	343.80	1.2602	0.14933	313.74	343.60	1.2512	0.12398	313.44	343.20	1.2356
<i>P</i> = 0.28 MPa (<i>T</i> _{sat} = −1.25° C)				<i>P</i> = 0.32 MPa (<i>T</i> _{sat} = 2.46° C)				<i>P</i> = 0.40 MPa (<i>T</i> _{sat} = 8.91° C)				
Sat.	0.07235	229.46	249.72	0.9321	0.06360	231.52	251.88	0.9301	0.051201	235.07	255.55	0.9269
0	0.07282	230.44	250.83	0.9362								
10	0.07646	238.27	259.68	0.9680	0.06609	237.54	258.69	0.9544	0.051506	235.97	256.58	0.9305
20	0.07997	246.13	268.52	0.9987	0.06925	245.50	267.66	0.9856	0.054213	244.18	265.86	0.9628
30	0.08338	254.06	277.41	1.0285	0.07231	253.50	276.65	1.0157	0.056796	252.36	275.07	0.9937
40	0.08672	262.10	286.38	1.0576	0.07530	261.60	285.70	1.0451	0.059292	260.58	284.30	1.0236
50	0.09000	270.27	295.47	1.0862	0.07823	269.82	294.85	1.0739	0.061724	268.90	293.59	1.0528
60	0.09324	278.56	304.67	1.1142	0.08111	278.15	304.11	1.1021	0.064104	277.32	302.96	1.0814
70	0.09644	286.99	314.00	1.1418	0.08395	286.62	313.48	1.1298	0.066443	285.86	312.44	1.1094
80	0.09961	295.57	323.46	1.1690	0.08675	295.22	322.98	1.1571	0.068747	294.53	322.02	1.1369
90	0.10275	304.29	333.06	1.1958	0.08953	303.97	332.62	1.1840	0.071023	303.32	331.73	1.1640
100	0.10587	313.15	342.80	1.2222	0.09229	312.86	342.39	1.2105	0.073274	312.26	341.57	1.1907
110	0.10897	322.16	352.68	1.2483	0.09503	321.89	352.30	1.2367	0.075504	321.33	351.53	1.2171
120	0.11205	331.32	362.70	1.2742	0.09775	331.07	362.35	1.2626	0.077717	330.55	361.63	1.2431
130	0.11512	340.63	372.87	1.2997	0.10045	340.39	372.54	1.2882	0.079913	339.90	371.87	1.2688
140	0.11818	350.09	383.18	1.3250	0.10314	349.86	382.87	1.3135	0.082096	349.41	382.24	1.2942
<i>P</i> = 0.50 MPa (<i>T</i> _{sat} = 15.71° C)				<i>P</i> = 0.60 MPa (<i>T</i> _{sat} = 21.55° C)				<i>P</i> = 0.70 MPa (<i>T</i> _{sat} = 26.69° C)				

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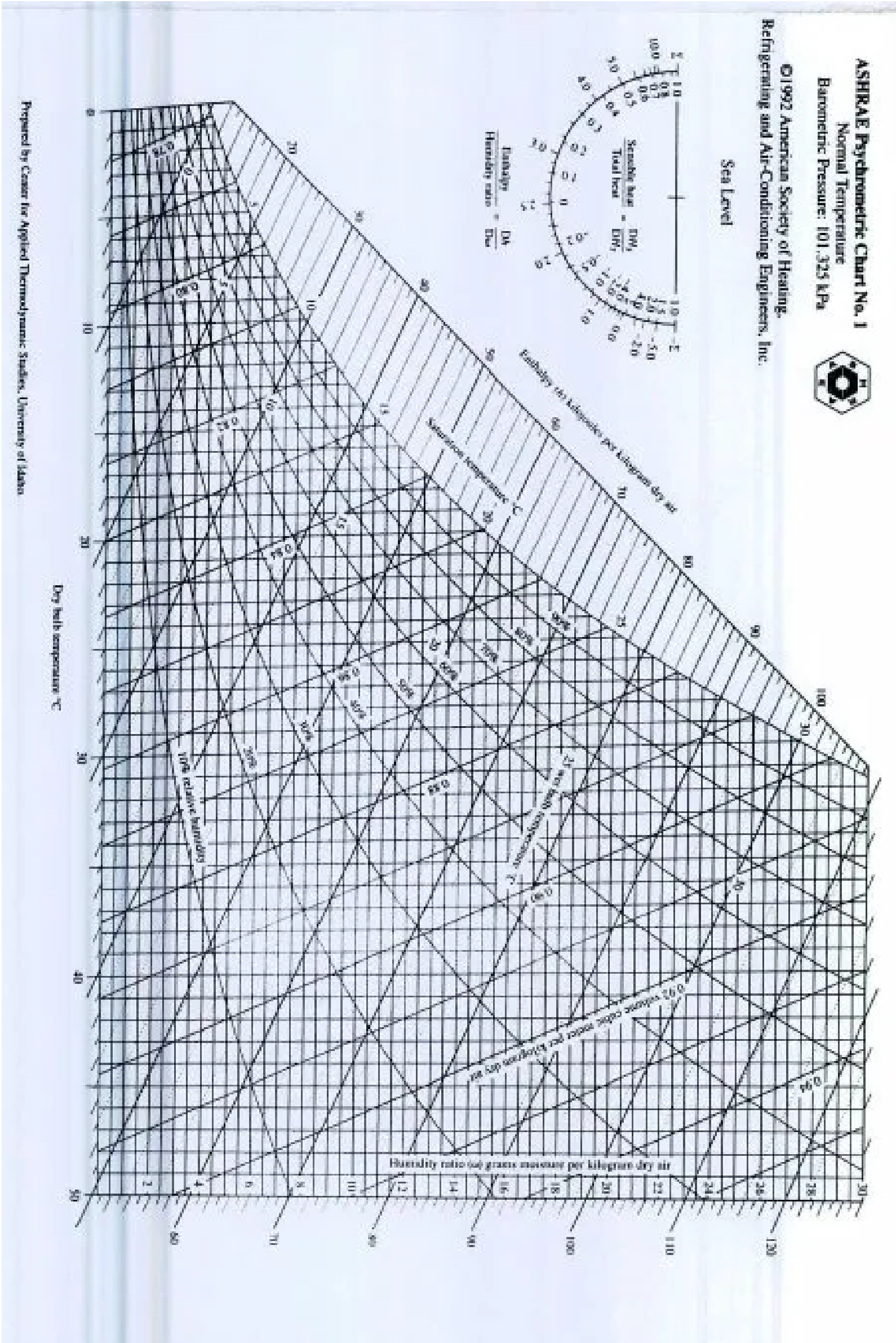
T °C	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg · K	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg · K	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg · K
Sat.	0.041118	238.75	259.30	0.9240	0.034295	241.83	262.40	0.9218	0.029361	244.48	265.03	0.9199
20	0.042115	242.40	263.46	0.9383								
30	0.044338	250.84	273.01	0.9703	0.035984	249.22	270.81	0.9499	0.029966	247.48	268.45	0.9313
40	0.046456	259.26	282.48	1.0011	0.037865	257.86	280.58	0.9816	0.031696	256.39	278.57	0.9641
50	0.048499	267.72	291.96	1.0309	0.039659	266.48	290.28	1.0121	0.033322	265.20	288.53	0.9954
60	0.050485	276.25	301.50	1.0599	0.041389	275.15	299.98	1.0417	0.034875	274.01	298.42	1.0256
70	0.052427	284.89	311.10	1.0883	0.043069	283.89	309.73	1.0705	0.036373	282.87	308.33	1.0549
80	0.054331	293.64	320.80	1.1162	0.044710	292.73	319.55	1.0987	0.037829	291.80	318.28	1.0835
90	0.056205	302.51	330.61	1.1436	0.046318	301.67	329.46	1.1264	0.039250	300.82	328.29	1.1114
100	0.058053	311.50	340.53	1.1705	0.047900	310.73	339.47	1.1536	0.040642	309.95	338.40	1.1389
110	0.059880	320.63	350.57	1.1971	0.049458	319.91	349.59	1.1803	0.042010	319.19	348.60	1.1658
120	0.061687	329.89	360.73	1.2233	0.050997	329.23	359.82	1.2067	0.043358	328.55	358.90	1.1924
130	0.063479	339.29	371.03	1.2491	0.052519	338.67	370.18	1.2327	0.044688	338.04	369.32	1.2186
140	0.065256	348.83	381.46	1.2747	0.054027	348.25	380.66	1.2584	0.046004	347.66	379.86	1.2444
150	0.067021	358.51	392.02	1.2999	0.055522	357.96	391.27	1.2838	0.047306	357.41	390.52	1.2699
160	0.068775	368.33	402.72	1.3249	0.057006	367.81	402.01	1.3088	0.048597	367.29	401.31	1.2951
$P = 0.80 \text{ MPa } (T_{sat} = 31.31^\circ \text{C})$					$P = 0.90 \text{ MPa } (T_{sat} = 35.51^\circ \text{C})$				$P = 1.00 \text{ MPa } (T_{sat} = 39.37^\circ \text{C})$			
Sat.	0.025621	246.79	267.29	0.9183	0.022683	248.85	269.26	0.9169	0.020313	250.68	270.99	0.9156
40	0.027035	254.82	276.45	0.9480	0.023375	253.13	274.17	0.9327	0.020406	251.30	271.71	0.9179
50	0.028547	263.86	286.69	0.9802	0.024809	262.44	284.77	0.9660	0.021796	260.94	282.74	0.9525
60	0.029973	272.83	296.81	1.0110	0.026146	271.60	295.13	0.9976	0.023068	270.32	293.38	0.9850
70	0.031340	281.81	306.88	1.0408	0.027413	280.72	305.39	1.0280	0.024261	279.59	303.85	1.0160
80	0.032659	290.84	316.97	1.0698	0.028630	289.86	315.63	1.0574	0.025398	288.86	314.25	1.0458
90	0.033941	299.95	327.10	1.0981	0.029806	299.06	325.89	1.0860	0.026492	298.15	324.64	1.0748
100	0.035193	309.15	337.30	1.1258	0.030951	308.34	336.19	1.1140	0.027552	307.51	335.06	1.1031
110	0.036420	318.45	347.59	1.1530	0.032068	317.70	346.56	1.1414	0.028584	316.94	345.53	1.1308
120	0.037625	327.87	357.97	1.1798	0.033164	327.18	357.02	1.1684	0.029592	326.47	356.06	1.1580
130	0.038813	337.40	368.45	1.2061	0.034241	336.76	367.58	1.1949	0.030581	336.11	366.69	1.1846
140	0.039985	347.06	379.05	1.2321	0.035302	346.46	378.23	1.2210	0.031554	345.85	377.40	1.2109
150	0.041143	356.85	389.76	1.2577	0.036349	356.28	389.00	1.2467	0.032512	355.71	388.22	1.2368
160	0.042290	366.76	400.59	1.2830	0.037384	366.23	399.88	1.2721	0.033457	365.70	399.15	1.2623
170	0.043427	376.81	411.55	1.3080	0.038408	376.31	410.88	1.2972	0.034392	375.81	410.20	1.2875
180	0.044554	386.99	422.64	1.3327	0.039423	386.52	422.00	1.3221	0.035317	386.04	421.36	1.3124
$P = 1.20 \text{ MPa } (T_{sat} = 46.29^\circ \text{C})$					$P = 1.40 \text{ MPa } (T_{sat} = 52.40^\circ \text{C})$				$P = 1.60 \text{ MPa } (T_{sat} = 57.88^\circ \text{C})$			
Sat.	0.016715	253.81	273.87	0.9130	0.014107	256.37	276.12	0.9105	0.012123	258.47	277.86	0.9078
50	0.017201	257.63	278.27	0.9267								
60	0.018404	267.56	289.64	0.9614	0.015005	264.46	285.47	0.9389	0.012372	260.89	280.69	0.9163
70	0.019502	277.21	300.61	0.9938	0.016060	274.62	297.10	0.9733	0.013430	271.76	293.25	0.9535
80	0.020529	286.75	311.39	1.0248	0.017023	284.51	308.34	1.0056	0.014362	282.09	305.07	0.9875
90	0.021506	296.26	322.07	1.0546	0.017923	294.28	319.37	1.0364	0.015215	292.17	316.52	1.0194
100	0.022442	305.80	332.73	1.0836	0.018778	304.01	330.30	1.0661	0.016014	302.14	327.76	1.0500
110	0.023348	315.38	343.40	1.1118	0.019597	313.76	341.19	1.0949	0.016773	312.07	338.91	1.0795
120	0.024228	325.03	354.11	1.1394	0.020388	323.55	352.09	1.1230	0.017500	322.02	350.02	1.1081
130	0.025086	334.77	364.88	1.1664	0.021155	333.41	363.02	1.1504	0.018201	332.00	361.12	1.1360
140	0.025927	344.61	375.72	1.1930	0.021904	343.34	374.01	1.1773	0.018882	342.05	372.26	1.1632
150	0.026753	354.56	386.66	1.2192	0.022636	353.37	385.07	1.2038	0.019545	352.17	383.44	1.1900
160	0.027566	364.61	397.69	1.2449	0.023355	363.51	396.20	1.2298	0.020194	362.38	394.69	1.2163
170	0.028367	374.78	408.82	1.2703	0.024061	373.75	407.43	1.2554	0.020830	372.69	406.02	1.2421
180	0.029158	385.08	420.07	1.2954	0.024757	384.10	418.76	1.2807	0.021456	383.11	417.44	1.2676

- Q4.** (a) What is meant by a ton of refrigeration? Why should refrigerant have high latent heat of vaporisation? **(10 marks)**
 (b) Using suitable diagrams, briefly explain the working principle of a Conventional All-Air HVAC system. **(20 marks)** [2023–24]
- Q5.** (a) What do you understand by one ton of refrigeration? Convert to SI units. **(6 marks)**
 (b) What does an air handling unit (AHU) do in a central air conditioning system? **(9 marks)** [2021–22]
- Q6.** A refrigerator uses R-134a between 0.10 MPa and 1.10 MPa. Rate of heat removal = 3.5 kW. Calculate mass flow rate of refrigerant. **(10 marks)** [2021–22]

TABLE A–12
Saturated refrigerant-134a—Pressure table

Press., P kPa	Sat. temp., T_{sat} °C	Specific volume, m^3/kg		Internal energy, kJ/kg			Enthalpy, kJ/kg			Entropy, $\text{kJ/kg} \cdot \text{K}$		
		Sat. liquid, v_f	Sat. vapor, v_g	Sat. liquid, u_f	Evap., u_{fg}	Sat. vapor, u_g	Sat. liquid, h_f	Evap., h_{fg}	Sat. vapor, h_g	Sat. liquid, s_f	Evap., s_{fg}	Sat. vapor, s_g
60	-36.95	0.0007098	0.31121	3.798	205.32	209.12	3.841	223.95	227.79	0.01634	0.94807	0.96441
70	-33.87	0.0007144	0.26929	7.680	203.20	210.88	7.730	222.00	229.73	0.03267	0.92775	0.96042
80	-31.13	0.0007185	0.23753	11.15	201.30	212.46	11.21	220.25	231.46	0.04711	0.90999	0.95710
90	-28.65	0.0007223	0.21263	14.31	199.57	213.88	14.37	218.65	233.02	0.06008	0.89419	0.95427
100	-26.37	0.0007259	0.19254	17.21	197.98	215.19	17.28	217.16	234.44	0.07188	0.87995	0.95183
120	-22.32	0.0007324	0.16212	22.40	195.11	217.51	22.49	214.48	236.97	0.09275	0.85503	0.94779
140	-18.77	0.0007383	0.14014	26.98	192.57	219.54	27.08	212.08	239.16	0.11087	0.83368	0.94456
160	-15.60	0.0007437	0.12348	31.09	190.27	221.35	31.21	209.90	241.11	0.12693	0.81496	0.94190
180	-12.73	0.0007487	0.11041	34.83	188.16	222.99	34.97	207.90	242.86	0.14139	0.79826	0.93965
200	-10.09	0.0007533	0.099867	38.28	186.21	224.48	38.43	206.03	244.46	0.15457	0.78316	0.93773
240	-5.38	0.0007620	0.083897	44.48	182.67	227.14	44.66	202.62	247.28	0.17794	0.75664	0.93458
280	-1.25	0.0007699	0.072352	49.97	179.50	229.46	50.18	199.54	249.72	0.19829	0.73381	0.93210
320	2.46	0.0007772	0.063604	54.92	176.61	231.52	55.16	196.71	251.88	0.21637	0.71369	0.93006
360	5.82	0.0007841	0.056738	59.44	173.94	233.38	59.72	194.08	253.81	0.23270	0.69566	0.92836
400	8.91	0.0007907	0.051201	63.62	171.45	235.07	63.94	191.62	255.55	0.24761	0.67929	0.92691
450	12.46	0.0007985	0.045619	68.45	168.54	237.00	68.81	188.71	257.53	0.26465	0.66069	0.92535
500	15.71	0.0008059	0.041118	72.93	165.82	238.75	73.33	185.98	259.30	0.28023	0.64377	0.92400
550	18.73	0.0008130	0.037408	77.10	163.25	240.35	77.54	183.38	260.92	0.29461	0.62821	0.92282
600	21.55	0.0008199	0.034295	81.02	160.81	241.83	81.51	180.90	262.40	0.30799	0.61378	0.92177
650	24.20	0.0008266	0.031646	84.72	158.48	243.20	85.26	178.51	263.77	0.32051	0.60030	0.92081
700	26.69	0.0008331	0.029361	88.24	156.24	244.48	88.82	176.21	265.03	0.33230	0.58763	0.91994
750	29.06	0.0008395	0.027371	91.59	154.08	245.67	92.22	173.98	266.20	0.34345	0.57567	0.91912
800	31.31	0.0008458	0.025621	94.79	152.00	246.79	95.47	171.82	267.29	0.35404	0.56431	0.91835
850	33.45	0.0008520	0.024069	97.87	149.98	247.85	98.60	169.71	268.31	0.36413	0.55349	0.91762
900	35.51	0.0008580	0.022683	100.83	148.01	248.85	101.61	167.66	269.26	0.37377	0.54315	0.91692
950	37.48	0.0008641	0.021438	103.69	146.10	249.79	104.51	165.64	270.15	0.38301	0.53323	0.91624
1000	39.37	0.0008700	0.020313	106.45	144.23	250.68	107.32	163.67	270.99	0.39189	0.52368	0.91558
1200	46.29	0.0008934	0.016715	116.70	137.11	253.81	117.77	156.10	273.87	0.42441	0.48863	0.91303
1400	52.40	0.0009166	0.014107	125.94	130.43	256.37	127.22	148.90	276.12	0.45315	0.45734	0.91050
1600	57.88	0.0009400	0.012123	134.43	124.04	258.47	135.93	141.93	277.86	0.47911	0.42873	0.90784
1800	62.87	0.0009639	0.010559	142.33	117.83	260.17	144.07	135.11	279.17	0.50294	0.40204	0.90498
2000	67.45	0.0009886	0.009288	149.78	111.73	261.51	151.76	128.33	280.09	0.52509	0.37675	0.90184
2500	77.54	0.0010566	0.006936	166.99	96.47	263.45	169.63	111.16	280.79	0.57531	0.31695	0.89226
3000	86.16	0.0011406	0.005275	183.04	80.22	263.26	186.46	92.63	279.09	0.62118	0.25776	0.87894

Q7. Air at 35°C DBT and 60% RH flowing at 2 kg/s is adiabatically mixed with air at 4 kg/s, 20°C WBT and 30°C DBT. Determine for the final mixture: (i) Enthalpy, (ii) Absolute humidity, (iii) Relative humidity, (iv) DBT and WBT, (v) Dew point. **(20 marks)** [2021–22]



Q8. Give examples of the following psychrometric processes and show on a schematic psychrometric chart: (i) Heating with humidification, (ii) Cooling with humidification. **(10 marks)** [2020–21]

Q9. A refrigerator uses R-134a on ideal vapour compression cycle between 0.40 MPa and 1.4 MPa. Mass flow rate = 0.22 kg/s. Determine rate of heat removal from the refrigerated space. **(8 marks)** [2020–21]

TABLE A-12
 Saturated refrigerant-134a—Pressure table

Press., <i>P</i> kPa	Sat. temp., <i>T</i> _{sat} °C	Specific volume, m ³ /kg		Internal energy, kJ/kg			Enthalpy, kJ/kg			Entropy, kJ/kg · K		
		Sat. liquid, <i>v</i> _{<i>f</i>}	Sat. vapor, <i>v</i> _{<i>g</i>}	Sat. liquid, <i>u</i> _{<i>f</i>}	Evap., <i>u</i> _{<i>fg</i>}	Sat. vapor, <i>u</i> _{<i>g</i>}	Sat. liquid, <i>h</i> _{<i>f</i>}	Evap., <i>h</i> _{<i>fg</i>}	Sat. vapor, <i>h</i> _{<i>g</i>}	Sat. liquid, <i>s</i> _{<i>f</i>}	Evap., <i>s</i> _{<i>fg</i>}	Sat. vapor, <i>s</i> _{<i>g</i>}
60	-36.95	0.0007098	0.31121	3.798	205.32	209.12	3.841	223.95	227.79	0.01634	0.94807	0.96441
70	-33.87	0.0007144	0.26929	7.680	203.20	210.88	7.730	222.00	229.73	0.03267	0.92775	0.96042
80	-31.13	0.0007185	0.23753	11.15	201.30	212.46	11.21	220.25	231.46	0.04711	0.90999	0.95710
90	-28.65	0.0007223	0.21263	14.31	199.57	213.88	14.37	218.65	233.02	0.06008	0.89419	0.95427
100	-26.37	0.0007259	0.19254	17.21	197.98	215.19	17.28	217.16	234.44	0.07188	0.87995	0.95183
120	-22.32	0.0007324	0.16212	22.40	195.11	217.51	22.49	214.48	236.97	0.09275	0.85503	0.94779
140	-18.77	0.0007383	0.14014	26.98	192.57	219.54	27.08	212.08	239.16	0.11087	0.83368	0.94456
160	-15.60	0.0007437	0.12348	31.09	190.27	221.35	31.21	209.90	241.11	0.12693	0.81496	0.94190
180	-12.73	0.0007487	0.11041	34.83	188.16	222.99	34.97	207.90	242.86	0.14139	0.79826	0.93965
200	-10.09	0.0007533	0.099867	38.28	186.21	224.48	38.43	206.03	244.46	0.15457	0.78316	0.93773
240	-5.38	0.0007620	0.083897	44.48	182.67	227.14	44.66	202.62	247.28	0.17794	0.75664	0.93458
280	-1.25	0.0007699	0.072352	49.97	179.50	229.46	50.18	199.54	249.72	0.19829	0.73381	0.93210
320	2.46	0.0007772	0.063604	54.92	176.61	231.52	55.16	196.71	251.88	0.21637	0.71369	0.93006
360	5.82	0.0007841	0.056738	59.44	173.94	233.38	59.72	194.08	253.81	0.23270	0.69566	0.92836
400	8.91	0.0007907	0.051201	63.62	171.45	235.07	63.94	191.62	255.55	0.24761	0.67929	0.92691
450	12.46	0.0007985	0.045619	68.45	168.54	237.00	68.81	188.71	257.53	0.26465	0.66069	0.92535
500	15.71	0.0008059	0.041118	72.93	165.82	238.75	73.33	185.98	259.30	0.28023	0.64377	0.92400
550	18.73	0.0008130	0.037408	77.10	163.25	240.35	77.54	183.38	260.92	0.29461	0.62821	0.92282
600	21.55	0.0008199	0.034295	81.02	160.81	241.83	81.51	180.90	262.40	0.30799	0.61378	0.92177
650	24.20	0.0008266	0.031646	84.72	158.48	243.20	85.26	178.51	263.77	0.32051	0.60030	0.92081
700	26.69	0.0008331	0.029361	88.24	156.24	244.48	88.82	176.21	265.03	0.33230	0.58763	0.91994
750	29.06	0.0008395	0.027371	91.59	154.08	245.67	92.22	173.98	266.20	0.34345	0.57567	0.91912
800	31.31	0.0008458	0.025621	94.79	152.00	246.79	95.47	171.82	267.29	0.35404	0.56431	0.91835
850	33.45	0.0008520	0.024069	97.87	149.98	247.85	98.60	169.71	268.31	0.36413	0.55349	0.91762
900	35.51	0.0008580	0.022683	100.83	148.01	248.85	101.61	167.66	269.26	0.37377	0.54315	0.91692
950	37.48	0.0008641	0.021438	103.69	146.10	249.79	104.51	165.64	270.15	0.38301	0.53323	0.91624
1000	39.37	0.0008700	0.020313	106.45	144.23	250.68	107.32	163.67	270.99	0.39189	0.52368	0.91558
1200	46.29	0.0008934	0.016715	116.70	137.11	253.81	117.77	156.10	273.87	0.42441	0.48863	0.91303
1400	52.40	0.0009166	0.014107	125.94	130.43	256.37	127.22	148.90	276.12	0.45315	0.45734	0.91050
1600	57.88	0.0009400	0.012123	134.43	124.04	258.47	135.93	141.93	277.86	0.47911	0.42873	0.90784
1800	62.87	0.0009639	0.010559	142.33	117.83	260.17	144.07	135.11	279.17	0.50294	0.40204	0.90498
2000	67.45	0.0009886	0.009288	149.78	111.73	261.51	151.76	128.33	280.09	0.52509	0.37675	0.90184
2500	77.54	0.0010566	0.006936	166.99	96.47	263.45	169.63	111.16	280.79	0.57531	0.31695	0.89226
3000	86.16	0.0011406	0.005275	183.04	80.22	263.26	186.46	92.63	279.09	0.62118	0.25776	0.87894

- Q10.

How does an absorption refrigeration system differ from a vapour-compression refrigeration system? (10 marks)

[2019–20]
- Q11.

A refrigerator uses R-134a and operates on an ideal vapour compression cycle between 0.14 and 0.8 MPa. Determine COP and power required to service a 150 kW cooling load. (20 marks)

[2019–20]

TABLE A–2

Ideal-gas specific heats of various common gases
(a) At 300 K

Gas	Formula	Gas constant, R	c_p	c_v	k
		$\text{kJ/kg} \cdot \text{K}$	$\text{kJ/kg} \cdot \text{K}$	$\text{kJ/kg} \cdot \text{K}$	
Air	—	0.2870	1.005	0.718	1.400
Argon	Ar	0.2081	0.5203	0.3122	1.667
Butane	C ₄ H ₁₀	0.1433	1.7164	1.5734	1.091
Carbon dioxide	CO ₂	0.1889	0.846	0.657	1.289
Carbon monoxide	CO	0.2968	1.040	0.744	1.400
Ethane	C ₂ H ₆	0.2765	1.7662	1.4897	1.186
Ethylene	C ₂ H ₄	0.2964	1.5482	1.2518	1.237
Helium	He	2.0769	5.1926	3.1156	1.667
Hydrogen	H ₂	4.1240	14.307	10.183	1.405
Methane	CH ₄	0.5182	2.2537	1.7354	1.299
Neon	Ne	0.4119	1.0299	0.6179	1.667
Nitrogen	N ₂	0.2968	1.039	0.743	1.400
Octane	C ₈ H ₁₈	0.0729	1.7113	1.6385	1.044
Oxygen	O ₂	0.2598	0.918	0.658	1.395
Propane	C ₃ H ₈	0.1885	1.6794	1.4909	1.126
Steam	H ₂ O	0.4615	1.8723	1.4108	1.327

Note: The unit $\text{kJ/kg} \cdot \text{K}$ is equivalent to $\text{kJ/kg} \cdot ^\circ \text{C}$.
Source: Chemical and Process Thermodynamics 3/E by Kyle, B. G., © 2000. Adapted by permission of Pearson Education, Inc., Upper Saddle River, NJ.

TABLE A–11
Saturated refrigerant-134a—Temperature table

Temp., $T^\circ \text{C}$	Sat. press., P_{sat} kPa	Specific volume, m^3/kg		Internal energy, kJ/kg			Enthalpy, kJ/kg			Entropy, $\text{kJ/kg} \cdot \text{K}$		
		Sat. liquid, v_f	Sat. vapor, v_g	Sat. liquid, u_f	Evap., u_{fg}	Sat. vapor, u_g	Sat. liquid, h_f	Evap., h_{fg}	Sat. vapor, h_g	Sat. liquid, s_f	Evap., s_{fg}	Sat. vapor, s_g
-40	51.25	0.0007054	0.36081	-0.036	207.40	207.37	0.000	225.86	225.86	0.00000	0.96866	0.96866
-38	56.86	0.0007083	0.32732	2.475	206.04	208.51	2.515	224.61	227.12	0.01072	0.95511	0.96584
-36	62.95	0.0007112	0.29751	4.992	204.67	209.66	5.037	223.35	228.39	0.02138	0.94176	0.96315
-34	69.56	0.0007142	0.27090	7.517	203.29	210.81	7.566	222.09	229.65	0.03199	0.92859	0.96058
-32	76.71	0.0007172	0.24711	10.05	201.91	211.96	10.10	220.81	230.91	0.04253	0.91560	0.95813
-30	84.43	0.0007203	0.22580	12.59	200.52	213.11	12.65	219.52	232.17	0.05301	0.90278	0.95579
-28	92.76	0.0007234	0.20666	15.13	199.12	214.25	15.20	218.22	233.43	0.06344	0.89012	0.95356
-26	101.73	0.0007265	0.18946	17.69	197.72	215.40	17.76	216.92	234.68	0.07382	0.87762	0.95144
-24	111.37	0.0007297	0.17395	20.25	196.30	216.55	20.33	215.59	235.92	0.08414	0.86527	0.94941
-22	121.72	0.0007329	0.15995	22.82	194.88	217.70	22.91	214.26	237.17	0.09441	0.85307	0.94748
-20	132.82	0.0007362	0.14729	25.39	193.45	218.84	25.49	212.91	238.41	0.10463	0.84101	0.94564
-18	144.69	0.0007396	0.13583	27.98	192.01	219.98	28.09	211.55	239.64	0.11481	0.82908	0.94389
-16	157.38	0.0007430	0.12542	30.57	190.56	221.13	30.69	210.18	240.87	0.12493	0.81729	0.94222
-14	170.93	0.0007464	0.11597	33.17	189.09	222.27	33.30	208.79	242.09	0.13501	0.80561	0.94063
-12	185.37	0.0007499	0.10736	35.78	187.62	223.40	35.92	207.38	243.30	0.14504	0.79406	0.93911
-10	200.74	0.0007535	0.099516	38.40	186.14	224.54	38.55	205.96	244.51	0.15504	0.78263	0.93766
-8	217.08	0.0007571	0.092352	41.03	184.64	225.67	41.19	204.52	245.72	0.16498	0.77130	0.93629
-6	234.44	0.0007608	0.085802	43.66	183.13	226.80	43.84	203.07	246.91	0.17489	0.76008	0.93497
-4	252.85	0.0007646	0.079804	46.31	181.61	227.92	46.50	201.60	248.10	0.18476	0.74896	0.93372
-2	272.36	0.0007684	0.074304	48.96	180.08	229.04	49.17	200.11	249.28	0.19459	0.73794	0.93253
0	293.01	0.0007723	0.069255	51.63	178.53	230.16	51.86	198.60	250.45	0.20439	0.72701	0.93139
2	314.84	0.0007763	0.064612	54.30	176.97	231.27	54.55	197.07	251.61	0.21415	0.71616	0.93031
4	337.90	0.0007804	0.060338	56.99	175.39	232.38	57.25	195.51	252.77	0.22387	0.70540	0.92927
6	362.23	0.0007845	0.056398	59.68	173.80	233.48	59.97	193.94	253.91	0.23356	0.69471	0.92828
8	387.88	0.0007887	0.052762	62.39	172.19	234.58	62.69	192.35	255.04	0.24323	0.68410	0.92733
10	414.89	0.0007930	0.049403	65.10	170.56	235.67	65.43	190.73	256.16	0.25286	0.67356	0.92641
12	443.31	0.0007975	0.046295	67.83	168.92	236.75	68.18	189.09	257.27	0.26246	0.66308	0.92554
14	473.19	0.0008020	0.043417	70.57	167.26	237.83	70.95	187.42	258.37	0.27204	0.65266	0.92470
16	504.58	0.0008066	0.040748	73.32	165.58	238.90	73.73	185.73	259.46	0.28159	0.64230	0.92389
18	537.52	0.0008113	0.038271	76.08	163.88	239.96	76.52	184.01	260.53	0.29112	0.63198	0.92310
20	572.07	0.0008161	0.035969	78.86	162.16	241.02	79.32	182.27	261.59	0.30063	0.62172	0.92234
22	608.27	0.0008210	0.033828	81.64	160.42	242.06	82.14	180.49	262.64	0.31011	0.61149	0.92160
24	646.18	0.0008261	0.031834	84.44	158.65	243.10	84.98	178.69	263.67	0.31958	0.60130	0.92088
26	685.84	0.0008313	0.029976	87.26	156.87	244.12	87.83	176.85	264.68	0.32903	0.59115	0.92018
28	727.31	0.0008366	0.028242	90.09	155.05	245.14	90.69	174.99	265.68	0.33846	0.58102	0.91948
30	770.64	0.0008421	0.026622	92.93	153.22	246.14	93.58	173.08	266.66	0.34789	0.57091	0.91879
Continued on next page												

Temp., $T^{\circ}\text{C}$	Sat. press., P_{sat} kPa	Specific volume, m^3/kg		Internal energy, kJ/kg			Enthalpy, kJ/kg			Entropy, kJ/kg · K		
		Sat. liquid, v_f	Sat. vapor, v_g	Sat. liquid, u_f	Evap., u_{fg}	Sat. vapor, u_g	Sat. liquid, h_f	Evap., h_{fg}	Sat. vapor, h_g	Sat. liquid, s_f	Evap., s_{fg}	Sat. vapor, s_g
32	815.89	0.0008478	0.025108	95.79	151.35	247.14	96.48	171.14	267.62	0.35730	0.56082	0.91811
34	863.11	0.0008536	0.023691	98.66	149.46	248.12	99.40	169.17	268.57	0.36670	0.55074	0.91743
36	912.35	0.0008595	0.022364	101.55	147.54	249.08	102.33	167.16	269.49	0.37609	0.54066	0.91675
38	963.68	0.0008657	0.021119	104.45	145.58	250.04	105.29	165.10	270.39	0.38548	0.53058	0.91606
40	1017.1	0.0008720	0.019952	107.38	143.60	250.97	108.26	163.00	271.27	0.39486	0.52049	0.91536
42	1072.8	0.0008786	0.018855	110.32	141.58	251.89	111.26	160.86	272.12	0.40425	0.51039	0.91464
44	1130.7	0.0008854	0.017824	113.28	139.52	252.80	114.28	158.67	272.95	0.41363	0.50027	0.91391
46	1191.0	0.0008924	0.016853	116.26	137.42	253.68	117.32	156.43	273.75	0.42302	0.49012	0.91315
48	1253.6	0.0008996	0.015939	119.26	135.29	254.55	120.39	154.14	274.53	0.43242	0.47993	0.91236
52	1386.2	0.0009150	0.014265	125.33	130.88	256.21	126.59	149.39	275.98	0.45126	0.45941	0.91067
56	1529.1	0.0009317	0.012771	131.49	126.28	257.77	132.91	144.38	277.30	0.47018	0.43863	0.90880
60	1682.8	0.0009498	0.011434	137.76	121.46	259.22	139.36	139.10	278.46	0.48920	0.41749	0.90669
65	1891.0	0.0009750	0.009950	145.77	115.05	260.82	147.62	132.02	279.64	0.51320	0.39039	0.90359
70	2118.2	0.0010037	0.008642	154.01	108.14	262.15	156.13	124.32	280.46	0.53755	0.36227	0.89982
75	2365.8	0.0010372	0.007480	162.53	100.60	263.13	164.98	115.85	280.82	0.56241	0.33272	0.89512
80	2635.3	0.0010772	0.006436	171.40	92.23	263.63	174.24	106.35	280.59	0.58800	0.30111	0.88912
85	2928.2	0.0011270	0.005486	180.77	82.67	263.44	184.07	95.44	279.51	0.61473	0.26644	0.88117
90	3246.9	0.0011932	0.004599	190.89	71.29	262.18	194.76	82.35	277.11	0.64336	0.22674	0.87010
95	3594.1	0.0012933	0.003726	202.40	56.47	258.87	207.05	65.21	272.26	0.67578	0.17711	0.85289
100	3975.1	0.0015269	0.002630	218.72	29.19	247.91	224.79	33.58	258.37	0.72217	0.08999	0.81215

Source: Tables A–11 through A–13 are generated using the Engineering Equation Solver (EES) software developed by S. A. Klein and F. L. Alvarado. The routine used in calculations is the R134a, which is based on the fundamental equation of state developed by R. Tillner-Roth and H.D. Baehr, “An International Standard Formulation for the Thermodynamic Properties of 1,1,1,2-Tetrafluoroethane (HFC-134a) for temperatures from 170 K to 455 K and Pressures up to 70 MPa,” *J. Phys. Chem. Ref. Data*, Vol. 23, No. 5, 1994. The enthalpy and entropy values of saturated liquid are set to zero at -40°C (and -40°F).

TABLE A–12
Saturated refrigerant-134a—Pressure table

Press., P kPa	Sat. temp., T_{sat} $^{\circ}\text{C}$	Specific volume, m^3/kg		Internal energy, kJ/kg			Enthalpy, kJ/kg			Entropy, kJ/kg · K		
		Sat. liquid, v_f	Sat. vapor, v_g	Sat. liquid, u_f	Evap., u_{fg}	Sat. vapor, u_g	Sat. liquid, h_f	Evap., h_{fg}	Sat. vapor, h_g	Sat. liquid, s_f	Evap., s_{fg}	Sat. vapor, s_g
60	-36.95	0.0007098	0.31121	3.798	205.32	209.12	3.841	223.95	227.79	0.01634	0.94807	0.96441
70	-33.87	0.0007144	0.26929	7.680	203.20	210.88	7.730	222.00	229.73	0.03267	0.92775	0.96042
80	-31.13	0.0007185	0.23753	11.15	201.30	212.46	11.21	220.25	231.46	0.04711	0.90999	0.95710
90	-28.65	0.0007223	0.21263	14.31	199.57	213.88	14.37	218.65	233.02	0.06008	0.89419	0.95427
100	-26.37	0.0007259	0.19254	17.21	197.98	215.19	17.28	217.16	234.44	0.07188	0.87995	0.95183
120	-22.32	0.0007324	0.16212	22.40	195.11	217.51	22.49	214.48	236.97	0.09275	0.85503	0.94779
140	-18.77	0.0007383	0.14014	26.98	192.57	219.54	27.08	212.08	239.16	0.11087	0.83368	0.94456
160	-15.60	0.0007437	0.12348	31.09	190.27	221.35	31.21	209.90	241.11	0.12693	0.81496	0.94190
180	-12.73	0.0007487	0.11041	34.83	188.16	222.99	34.97	207.90	242.86	0.14139	0.79826	0.93965
200	-10.09	0.0007533	0.099867	38.28	186.21	224.48	38.43	206.03	244.46	0.15457	0.78316	0.93773
240	-5.38	0.0007620	0.083897	44.48	182.67	227.14	44.66	202.62	247.28	0.17794	0.75664	0.93458
280	-1.25	0.0007699	0.072352	49.97	179.50	229.46	50.18	199.54	249.72	0.19829	0.73381	0.93210
320	2.46	0.0007772	0.063604	54.92	176.61	231.52	55.16	196.71	251.88	0.21637	0.71369	0.93006
360	5.82	0.0007841	0.056738	59.44	173.94	233.38	59.72	194.08	253.81	0.23270	0.69566	0.92836
400	8.91	0.0007907	0.051201	63.62	171.45	235.07	63.94	191.62	255.55	0.24761	0.67929	0.92691
450	12.46	0.0007985	0.045619	68.45	168.54	237.00	68.81	188.71	257.53	0.26465	0.66069	0.92535
500	15.71	0.0008059	0.041118	72.93	165.82	238.75	73.33	185.98	259.30	0.28023	0.64377	0.92400
550	18.73	0.0008130	0.037408	77.10	163.25	240.35	77.54	183.38	260.92	0.29461	0.62821	0.92282
600	21.55	0.0008199	0.034295	81.02	160.81	241.83	81.51	180.90	262.40	0.30799	0.61378	0.92177
650	24.20	0.0008266	0.031646	84.72	158.48	243.20	85.26	178.51	263.77	0.32051	0.60030	0.92081
700	26.69	0.0008331	0.029361	88.24	156.24	244.48	88.82	176.21	265.03	0.33230	0.58763	0.91994
750	29.06	0.0008395	0.027371	91.59	154.08	245.67	92.22	173.98	266.20	0.34345	0.57567	0.91912
800	31.31	0.0008458	0.025621	94.79	152.00	246.79	95.47	171.82	267.29	0.35404	0.56431	0.91835
850	33.45	0.0008520	0.024069	97.87	149.98	247.85	98.60	169.71	268.31	0.36413	0.55349	0.91762
900	35.51	0.0008580	0.022683	100.83	148.01	248.85	101.61	167.66	269.26	0.37377	0.54315	0.91692
950	37.48	0.0008641	0.021438	103.69	146.10	249.79	104.51	165.64	270.15	0.38301	0.53323	0.91624
1000	39.37	0.0008700	0.020313	106.45	144.23	250.68	107.32	163.67	270.99	0.39189	0.52368	0.91558

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		Specific volume, m ³ /kg		Internal energy, kJ/kg			Enthalpy, kJ/kg			Entropy, kJ/kg · K		
		Sat. liquid, v_f	Sat. vapor, v_g	Sat. liquid, u_f	Evap., u_{fg}	Sat. vapor, u_g	Sat. liquid, h_f	Evap., h_{fg}	Sat. vapor, h_g	Sat. liquid, s_f	Evap., s_{fg}	Sat. vapor, s_g
Press., P kPa	Sat. temp., T_{sat} °C											
1200	46.29	0.0008934	0.016715	116.70	137.11	253.81	117.77	156.10	273.87	0.42441	0.48863	0.91303
1400	52.40	0.0009166	0.014107	125.94	130.43	256.37	127.22	148.90	276.12	0.45315	0.45734	0.91050
1600	57.88	0.0009400	0.012123	134.43	124.04	258.47	135.93	141.93	277.86	0.47911	0.42873	0.90784
1800	62.87	0.0009639	0.010559	142.33	117.83	260.17	144.07	135.11	279.17	0.50294	0.40204	0.90498
2000	67.45	0.0009886	0.009288	149.78	111.73	261.51	151.76	128.33	280.09	0.52509	0.37675	0.90184
2500	77.54	0.0010566	0.006936	166.99	96.47	263.45	169.63	111.16	280.79	0.57531	0.31695	0.89226
3000	86.16	0.0011406	0.005275	183.04	80.22	263.26	186.46	92.63	279.09	0.62118	0.25776	0.87894

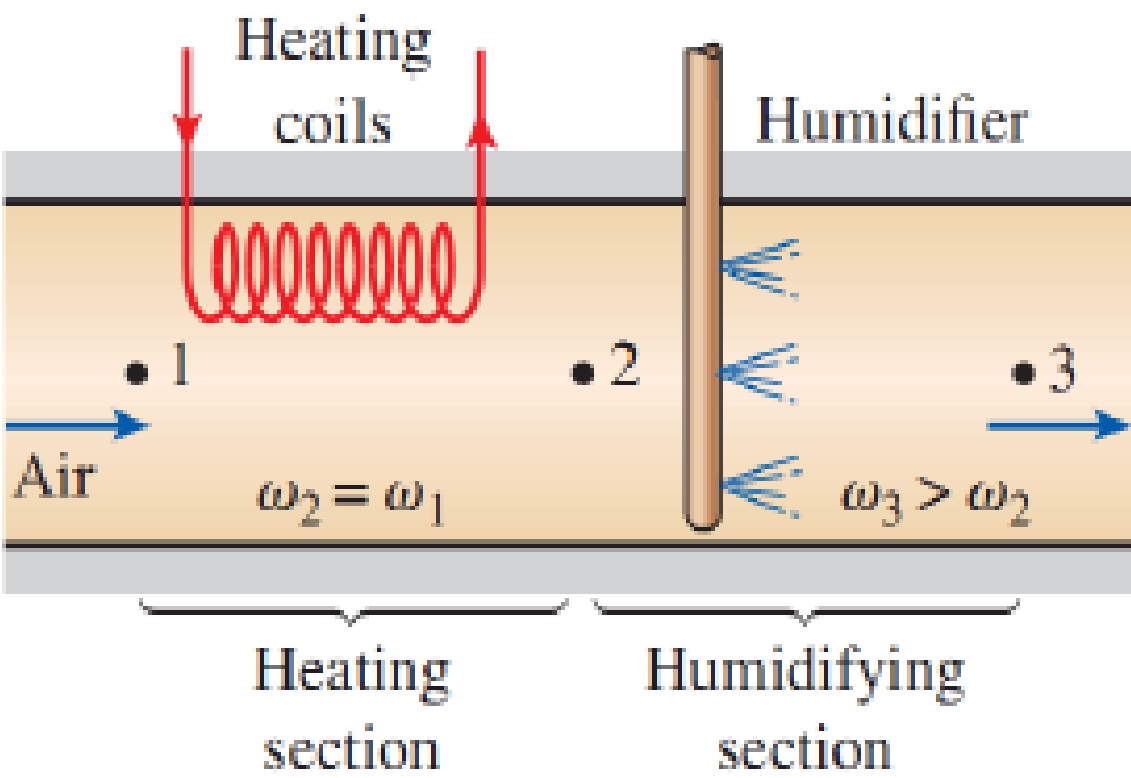
TABLE A–13
 Superheated refrigerant-134a

T °C	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg · K	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg · K	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg · K
$P = 0.06 \text{ MPa } (T_{\text{sat}} = -36.95^\circ \text{C})$					$P = 0.10 \text{ MPa } (T_{\text{sat}} = -26.37^\circ \text{C})$				$P = 0.14 \text{ MPa } (T_{\text{sat}} = -18.77^\circ \text{C})$			
Sat.	0.31121	209.12	227.79	0.9644	0.19254	215.19	234.44	0.9518	0.14014	219.54	239.16	0.9446
-20	0.33608	220.60	240.76	1.0174	0.19841	219.66	239.50	0.9721				
-10	0.35048	227.55	248.58	1.0477	0.20743	226.75	247.49	1.0030	0.14605	225.91	246.36	0.9724
0	0.36476	234.66	256.54	1.0774	0.21630	233.95	255.58	1.0332	0.15263	233.23	254.60	1.0031
10	0.37893	241.92	264.66	1.1066	0.22506	241.30	263.81	1.0628	0.15908	240.66	262.93	1.0331
20	0.39302	249.35	272.94	1.1353	0.23373	248.79	272.17	1.0918	0.16544	248.22	271.38	1.0624
30	0.40705	256.95	281.37	1.1636	0.24233	256.44	280.68	1.1203	0.17172	255.93	279.97	1.0912
40	0.42102	264.71	289.97	1.1915	0.25088	264.25	289.34	1.1484	0.17794	263.79	288.70	1.1195
50	0.43495	272.64	298.74	1.2191	0.25937	272.22	298.16	1.1762	0.18412	271.79	297.57	1.1474
60	0.44883	280.73	307.66	1.2463	0.26783	280.35	307.13	1.2035	0.19025	279.96	306.59	1.1749
70	0.46269	288.99	316.75	1.2732	0.27626	288.64	316.26	1.2305	0.19635	288.28	315.77	1.2020
80	0.47651	297.41	326.00	1.2997	0.28465	297.08	325.55	1.2572	0.20242	296.75	325.09	1.2288
90	0.49032	306.00	335.42	1.3260	0.29303	305.69	334.99	1.2836	0.20847	305.38	334.57	1.2553
100	0.50410	314.74	344.99	1.3520	0.30138	314.46	344.60	1.3096	0.21449	314.17	344.20	1.2814
$P = 0.18 \text{ MPa } (T_{\text{sat}} = -12.73^\circ \text{C})$					$P = 0.20 \text{ MPa } (T_{\text{sat}} = -10.09^\circ \text{C})$				$P = 0.24 \text{ MPa } (T_{\text{sat}} = -5.38^\circ \text{C})$			
Sat.	0.11041	222.99	242.86	0.9397	0.09987	224.48	244.46	0.9377	0.08390	227.14	247.28	0.9346
-10	0.11189	225.02	245.16	0.9484	0.09991	224.55	244.54	0.9380				
0	0.11722	232.48	253.58	0.9798	0.10481	232.09	253.05	0.9698	0.08617	231.29	251.97	0.9519
10	0.12240	240.00	262.04	1.0102	0.10955	239.67	261.58	1.0004	0.09026	238.98	260.65	0.9831
20	0.12748	247.64	270.59	1.0399	0.11418	247.35	270.18	1.0303	0.09423	246.74	269.36	1.0134
30	0.13248	255.41	279.25	1.0690	0.11874	255.14	278.89	1.0595	0.09812	254.61	278.16	1.0429
40	0.13741	263.31	288.05	1.0975	0.12322	263.08	287.72	1.0882	0.10193	262.59	287.06	1.0718
50	0.14230	271.36	296.98	1.1256	0.12766	271.15	296.68	1.1163	0.10570	270.71	296.08	1.1001
60	0.14715	279.56	306.05	1.1532	0.13206	279.37	305.78	1.1441	0.10942	278.97	305.23	1.1280
70	0.15196	287.91	315.27	1.1805	0.13641	287.73	315.01	1.1714	0.11310	287.36	314.51	1.1554
80	0.15673	296.42	324.63	1.2074	0.14074	296.25	324.40	1.1983	0.11675	295.91	323.93	1.1825
90	0.16149	305.07	334.14	1.2339	0.14504	304.92	333.93	1.2249	0.12038	304.60	333.49	1.2092
100	0.16622	313.88	343.80	1.2602	0.14933	313.74	343.60	1.2512	0.12398	313.44	343.20	1.2356
$P = 0.28 \text{ MPa } (T_{\text{sat}} = -1.25^\circ \text{C})$					$P = 0.32 \text{ MPa } (T_{\text{sat}} = 2.46^\circ \text{C})$				$P = 0.40 \text{ MPa } (T_{\text{sat}} = 8.91^\circ \text{C})$			
Sat.	0.07235	229.46	249.72	0.9321	0.06360	231.52	251.88	0.9301	0.051201	235.07	255.55	0.9269
0	0.07282	230.44	250.83	0.9362								
10	0.07646	238.27	259.68	0.9680	0.06609	237.54	258.69	0.9544	0.051506	235.97	256.58	0.9305
20	0.07997	246.13	268.52	0.9987	0.06925	245.50	267.66	0.9856	0.054213	244.18	265.86	0.9628
30	0.08338	254.06	277.41	1.0285	0.07231	253.50	276.65	1.0157	0.056796	252.36	275.07	0.9937
40	0.08672	262.10	286.38	1.0576	0.07530	261.60	285.70	1.0451	0.059292	260.58	284.30	1.0236
50	0.09000	270.27	295.47	1.0862	0.07823	269.82	294.85	1.0739	0.061724	268.90	293.59	1.0528
60	0.09324	278.56	304.67	1.1142	0.08111	278.15	304.11	1.1021	0.064104	277.32	302.96	1.0814
70	0.09644	286.99	314.00	1.1418	0.08395	286.62	313.48	1.1298	0.066443	285.86	312.44	1.1094
80	0.09961	295.57	323.46	1.1690	0.08675	295.22	322.98	1.1571	0.068747	294.53	322.02	1.1369
90	0.10275	304.29	333.06	1.1958	0.08953	303.97	332.62	1.1840	0.071023	303.32	331.73	1.1640
100	0.10587	313.15	342.80	1.2222	0.09229	312.86	342.39	1.2105	0.073274	312.26	341.57	1.1907
110	0.10897	322.16	352.68	1.2483	0.09503	321.89	352.30	1.2367	0.075504	321.33	351.53	1.2171
120	0.11205	331.32	362.70	1.2742	0.09775	331.07	362.35	1.2626	0.077717	330.55	361.63	1.2431
130	0.11512	340.63	372.87	1.2997	0.10045	340.39	372.54	1.2882	0.079913	339.90	371.87	1.2688
140	0.11818	350.09	383.18	1.3250	0.10314	349.86	382.87	1.3135	0.082096	349.41	382.24	1.2942
$P = 0.50 \text{ MPa } (T_{\text{sat}} = 15.71^\circ \text{C})$					$P = 0.60 \text{ MPa } (T_{\text{sat}} = 21.55^\circ \text{C})$				$P = 0.70 \text{ MPa } (T_{\text{sat}} = 26.69^\circ \text{C})$			

Continued on next page

T °C	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg · K	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg · K	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg · K
Sat.	0.041118	238.75	259.30	0.9240	0.034295	241.83	262.40	0.9218	0.029361	244.48	265.03	0.9199
20	0.042115	242.40	263.46	0.9383								
30	0.044338	250.84	273.01	0.9703	0.035984	249.22	270.81	0.9499	0.029966	247.48	268.45	0.9313
40	0.046456	259.26	282.48	1.0011	0.037865	257.86	280.58	0.9816	0.031696	256.39	278.57	0.9641
50	0.048499	267.72	291.96	1.0309	0.039659	266.48	290.28	1.0121	0.033322	265.20	288.53	0.9954
60	0.050485	276.25	301.50	1.0599	0.041389	275.15	299.98	1.0417	0.034875	274.01	298.42	1.0256
70	0.052427	284.89	311.10	1.0883	0.043069	283.89	309.73	1.0705	0.036373	282.87	308.33	1.0549
80	0.054331	293.64	320.80	1.1162	0.044710	292.73	319.55	1.0987	0.037829	291.80	318.28	1.0835
90	0.056205	302.51	330.61	1.1436	0.046318	301.67	329.46	1.1264	0.039250	300.82	328.29	1.1114
100	0.058053	311.50	340.53	1.1705	0.047900	310.73	339.47	1.1536	0.040642	309.95	338.40	1.1389
110	0.059880	320.63	350.57	1.1971	0.049458	319.91	349.59	1.1803	0.042010	319.19	348.60	1.1658
120	0.061687	329.89	360.73	1.2233	0.050997	329.23	359.82	1.2067	0.043358	328.55	358.90	1.1924
130	0.063479	339.29	371.03	1.2491	0.052519	338.67	370.18	1.2327	0.044688	338.04	369.32	1.2186
140	0.065256	348.83	381.46	1.2747	0.054027	348.25	380.66	1.2584	0.046004	347.66	379.86	1.2444
150	0.067021	358.51	392.02	1.2999	0.055522	357.96	391.27	1.2838	0.047306	357.41	390.52	1.2699
160	0.068775	368.33	402.72	1.3249	0.057006	367.81	402.01	1.3088	0.048597	367.29	401.31	1.2951
$P = 0.80 \text{ MPa } (T_{sat} = 31.31^\circ \text{C})$					$P = 0.90 \text{ MPa } (T_{sat} = 35.51^\circ \text{C})$				$P = 1.00 \text{ MPa } (T_{sat} = 39.37^\circ \text{C})$			
Sat.	0.025621	246.79	267.29	0.9183	0.022683	248.85	269.26	0.9169	0.020313	250.68	270.99	0.9156
40	0.027035	254.82	276.45	0.9480	0.023375	253.13	274.17	0.9327	0.020406	251.30	271.71	0.9179
50	0.028547	263.86	286.69	0.9802	0.024809	262.44	284.77	0.9660	0.021796	260.94	282.74	0.9525
60	0.029973	272.83	296.81	1.0110	0.026146	271.60	295.13	0.9976	0.023068	270.32	293.38	0.9850
70	0.031340	281.81	306.88	1.0408	0.027413	280.72	305.39	1.0280	0.024261	279.59	303.85	1.0160
80	0.032659	290.84	316.97	1.0698	0.028630	289.86	315.63	1.0574	0.025398	288.86	314.25	1.0458
90	0.033941	299.95	327.10	1.0981	0.029806	299.06	325.89	1.0860	0.026492	298.15	324.64	1.0748
100	0.035193	309.15	337.30	1.1258	0.030951	308.34	336.19	1.1140	0.027552	307.51	335.06	1.1031
110	0.036420	318.45	347.59	1.1530	0.032068	317.70	346.56	1.1414	0.028584	316.94	345.53	1.1308
120	0.037625	327.87	357.97	1.1798	0.033164	327.18	357.02	1.1684	0.029592	326.47	356.06	1.1580
130	0.038813	337.40	368.45	1.2061	0.034241	336.76	367.58	1.1949	0.030581	336.11	366.69	1.1846
140	0.039985	347.06	379.05	1.2321	0.035302	346.46	378.23	1.2210	0.031554	345.85	377.40	1.2109
150	0.041143	356.85	389.76	1.2577	0.036349	356.28	389.00	1.2467	0.032512	355.71	388.22	1.2368
160	0.042290	366.76	400.59	1.2830	0.037384	366.23	399.88	1.2721	0.033457	365.70	399.15	1.2623
170	0.043427	376.81	411.55	1.3080	0.038408	376.31	410.88	1.2972	0.034392	375.81	410.20	1.2875
180	0.044554	386.99	422.64	1.3327	0.039423	386.52	422.00	1.3221	0.035317	386.04	421.36	1.3124
$P = 1.20 \text{ MPa } (T_{sat} = 46.29^\circ \text{C})$					$P = 1.40 \text{ MPa } (T_{sat} = 52.40^\circ \text{C})$				$P = 1.60 \text{ MPa } (T_{sat} = 57.88^\circ \text{C})$			
Sat.	0.016715	253.81	273.87	0.9130	0.014107	256.37	276.12	0.9105	0.012123	258.47	277.86	0.9078
50	0.017201	257.63	278.27	0.9267								
60	0.018404	267.56	289.64	0.9614	0.015005	264.46	285.47	0.9389	0.012372	260.89	280.69	0.9163
70	0.019502	277.21	300.61	0.9938	0.016060	274.62	297.10	0.9733	0.013430	271.76	293.25	0.9535
80	0.020529	286.75	311.39	1.0248	0.017023	284.51	308.34	1.0056	0.014362	282.09	305.07	0.9875
90	0.021506	296.26	322.07	1.0546	0.017923	294.28	319.37	1.0364	0.015215	292.17	316.52	1.0194
100	0.022442	305.80	332.73	1.0836	0.018778	304.01	330.30	1.0661	0.016014	302.14	327.76	1.0500
110	0.023348	315.38	343.40	1.1118	0.019597	313.76	341.19	1.0949	0.016773	312.07	338.91	1.0795
120	0.024228	325.03	354.11	1.1394	0.020388	323.55	352.09	1.1230	0.017500	322.02	350.02	1.1081
130	0.025086	334.77	364.88	1.1664	0.021155	333.41	363.02	1.1504	0.018201	332.00	361.12	1.1360
140	0.025927	344.61	375.72	1.1930	0.021904	343.34	374.01	1.1773	0.018882	342.05	372.26	1.1632
150	0.026753	354.56	386.66	1.2192	0.022636	353.37	385.07	1.2038	0.019545	352.17	383.44	1.1900
160	0.027566	364.61	397.69	1.2449	0.023355	363.51	396.20	1.2298	0.020194	362.38	394.69	1.2163
170	0.028367	374.78	408.82	1.2703	0.024061	373.75	407.43	1.2554	0.020830	372.69	406.02	1.2421
180	0.029158	385.08	420.07	1.2954	0.024757	384.10	418.76	1.2807	0.021456	383.11	417.44	1.2676

- Q12.** (a) In a fine morning, temperature is 15°C and RH is 100%. What is the wet-bulb temperature? Will dew point be higher or lower than 25°C? **(5 marks)**
 (b) Referring to the psychrometric chart with points 1, 2, 3, which of the following are true: (i) RH at point 2 > point 1, (ii) RH at point 2 > point 3, (iii) RH at point 3 > point 1, (iv) RH at point 3 > point 2. **(5 marks)** [2019–20]



- Q13.** (a) What is COP? Draw the schematic of a Vapour Absorption Refrigeration system and explain each process. **(15 marks)**
(b) Write desirable properties of refrigerants. **(5 marks)**
(c) A refrigerator uses R-134a between 0.14 and 0.8 MPa. Mass flow rate = 0.05 kg/s. Determine: (i) rate of heat removal and power input, (ii) rate of heat rejection, (iii) COP. **(15 marks)** [2018–19]

TABLE A–11
Saturated refrigerant-134a—Temperature table

Temp., <i>T</i> °C	Sat. press., <i>P</i> _{sat} kPa	Specific volume, m ³ /kg		Internal energy, kJ/kg			Enthalpy, kJ/kg			Entropy, kJ/kg · K		
		Sat. liquid, <i>v</i> _{<i>f</i>}	Sat. vapor, <i>v</i> _{<i>g</i>}	Sat. liquid, <i>u</i> _{<i>f</i>}	Evap., <i>u</i> _{<i>fg</i>}	Sat. vapor, <i>u</i> _{<i>g</i>}	Sat. liquid, <i>h</i> _{<i>f</i>}	Evap., <i>h</i> _{<i>fg</i>}	Sat. vapor, <i>h</i> _{<i>g</i>}	Sat. liquid, <i>s</i> _{<i>f</i>}	Evap., <i>s</i> _{<i>fg</i>}	Sat. vapor, <i>s</i> _{<i>g</i>}
-40	51.25	0.0007054	0.36081	-0.036	207.40	207.37	0.000	225.86	225.86	0.00000	0.96866	0.96866
-38	56.86	0.0007083	0.32732	2.475	206.04	208.51	2.515	224.61	227.12	0.01072	0.95511	0.96584
-36	62.95	0.0007112	0.29751	4.992	204.67	209.66	5.037	223.35	228.39	0.02138	0.94176	0.96315
-34	69.56	0.0007142	0.27090	7.517	203.29	210.81	7.566	222.09	229.65	0.03199	0.92859	0.96058
-32	76.71	0.0007172	0.24711	10.05	201.91	211.96	10.10	220.81	230.91	0.04253	0.91560	0.95813
-30	84.43	0.0007203	0.22580	12.59	200.52	213.11	12.65	219.52	232.17	0.05301	0.90278	0.95579
-28	92.76	0.0007234	0.20666	15.13	199.12	214.25	15.20	218.22	233.43	0.06344	0.89012	0.95356
-26	101.73	0.0007265	0.18946	17.69	197.72	215.40	17.76	216.92	234.68	0.07382	0.87762	0.95144
-24	111.37	0.0007297	0.17395	20.25	196.30	216.55	20.33	215.59	235.92	0.08414	0.86527	0.94941
-22	121.72	0.0007329	0.15995	22.82	194.88	217.70	22.91	214.26	237.17	0.09441	0.85307	0.94748
-20	132.82	0.0007362	0.14729	25.39	193.45	218.84	25.49	212.91	238.41	0.10463	0.84101	0.94564
-18	144.69	0.0007396	0.13583	27.98	192.01	219.98	28.09	211.55	239.64	0.11481	0.82908	0.94389
-16	157.38	0.0007430	0.12542	30.57	190.56	221.13	30.69	210.18	240.87	0.12493	0.81729	0.94222
-14	170.93	0.0007464	0.11597	33.17	189.09	222.27	33.30	208.79	242.09	0.13501	0.80561	0.94063
-12	185.37	0.0007499	0.10736	35.78	187.62	223.40	35.92	207.38	243.30	0.14504	0.79406	0.93911
-10	200.74	0.0007535	0.099516	38.40	186.14	224.54	38.55	205.96	244.51	0.15504	0.78263	0.93766
-8	217.08	0.0007571	0.092352	41.03	184.64	225.67	41.19	204.52	245.72	0.16498	0.77130	0.93629
-6	234.44	0.0007608	0.085802	43.66	183.13	226.80	43.84	203.07	246.91	0.17489	0.76008	0.93497
-4	252.85	0.0007646	0.079804	46.31	181.61	227.92	46.50	201.60	248.10	0.18476	0.74896	0.93372
-2	272.36	0.0007684	0.074304	48.96	180.08	229.04	49.17	200.11	249.28	0.19459	0.73794	0.93253
0	293.01	0.0007723	0.069255	51.63	178.53	230.16	51.86	198.60	250.45	0.20439	0.72701	0.93139
2	314.84	0.0007763	0.064612	54.30	176.97	231.27	54.55	197.07	251.61	0.21415	0.71616	0.93031
4	337.90	0.0007804	0.060338	56.99	175.39	232.38	57.25	195.51	252.77	0.22387	0.70540	0.92927
6	362.23	0.0007845	0.056398	59.68	173.80	233.48	59.97	193.94	253.91	0.23356	0.69471	0.92828
8	387.88	0.0007887	0.052762	62.39	172.19	234.58	62.69	192.35	255.04	0.24323	0.68410	0.92733
10	414.89	0.0007930	0.049403	65.10	170.56	235.67	65.43	190.73	256.16	0.25286	0.67356	0.92641
12	443.31	0.0007975	0.046295	67.83	168.92	236.75	68.18	189.09	257.27	0.26246	0.66308	0.92554
14	473.19	0.0008020	0.043417	70.57	167.26	237.83	70.95	187.42	258.37	0.27204	0.65266	0.92470
16	504.58	0.0008066	0.040748	73.32	165.58	238.90	73.73	185.73	259.46	0.28159	0.64230	0.92389
18	537.52	0.0008113	0.038271	76.08	163.88	239.96	76.52	184.01	260.53	0.29112	0.63198	0.92310

TABLE A–12

Saturated refrigerant-134a—Pressure table

Press., <i>P</i> kPa	Sat. temp., <i>T</i> _{sat} °C	Specific volume, m ³ /kg		Internal energy, kJ/kg			Enthalpy, kJ/kg			Entropy, kJ/kg · K		
		Sat. liquid, <i>v</i> _{<i>f</i>}	Sat. vapor, <i>v</i> _{<i>g</i>}	Sat. liquid, <i>u</i> _{<i>f</i>}	Evap., <i>u</i> _{<i>fg</i>}	Sat. vapor, <i>u</i> _{<i>g</i>}	Sat. liquid, <i>h</i> _{<i>f</i>}	Evap., <i>h</i> _{<i>fg</i>}	Sat. vapor, <i>h</i> _{<i>g</i>}	Sat. liquid, <i>s</i> _{<i>f</i>}	Evap., <i>s</i> _{<i>fg</i>}	Sat. vapor, <i>s</i> _{<i>g</i>}
60	-36.95	0.0007098	0.31121	3.798	205.32	209.12	3.841	223.95	227.79	0.01634	0.94807	0.96441
70	-33.87	0.0007144	0.26929	7.680	203.20	210.88	7.730	222.00	229.73	0.03267	0.92775	0.96042
80	-31.13	0.0007185	0.23753	11.15	201.30	212.46	11.21	220.25	231.46	0.04711	0.90999	0.95710
90	-28.65	0.0007223	0.21263	14.31	199.57	213.88	14.37	218.65	233.02	0.06008	0.89419	0.95427
100	-26.37	0.0007259	0.19254	17.21	197.98	215.19	17.28	217.16	234.44	0.07188	0.87995	0.95183
120	-22.32	0.0007324	0.16212	22.40	195.11	217.51	22.49	214.48	236.97	0.09275	0.85503	0.94779
140	-18.77	0.0007383	0.14014	26.98	192.57	219.54	27.08	212.08	239.16	0.11087	0.83368	0.94456
160	-15.60	0.0007437	0.12348	31.09	190.27	221.35	31.21	209.90	241.11	0.12693	0.81496	0.94190
180	-12.73	0.0007487	0.11041	34.83	188.16	222.99	34.97	207.90	242.86	0.14139	0.79826	0.93965
200	-10.09	0.0007533	0.099867	38.28	186.21	224.48	38.43	206.03	244.46	0.15457	0.78316	0.93773
240	-5.38	0.0007620	0.083897	44.48	182.67	227.14	44.66	202.62	247.28	0.17794	0.75664	0.93458
280	-1.25	0.0007699	0.072352	49.97	179.50	229.46	50.18	199.54	249.72	0.19829	0.73381	0.93210
320	2.46	0.0007772	0.063604	54.92	176.61	231.52	55.16	196.71	251.88	0.21637	0.71369	0.93006
360	5.82	0.0007841	0.056738	59.44	173.94	233.38	59.72	194.08	253.81	0.23270	0.69566	0.92836
400	8.91	0.0007907	0.051201	63.62	171.45	235.07	63.94	191.62	255.55	0.24761	0.67929	0.92691
450	12.46	0.0007985	0.045619	68.45	168.54	237.00	68.81	188.71	257.53	0.26465	0.66069	0.92535
500	15.71	0.0008059	0.041118	72.93	165.82	238.75	73.33	185.98	259.30	0.28023	0.64377	0.92400
550	18.73	0.0008130	0.037408	77.10	163.25	240.35	77.54	183.38	260.92	0.29461	0.62821	0.92282
600	21.55	0.0008199	0.034295	81.02	160.81	241.83	81.51	180.90	262.40	0.30799	0.61378	0.92177
650	24.20	0.0008266	0.031646	84.72	158.48	243.20	85.26	178.51	263.77	0.32051	0.60030	0.92081
700	26.69	0.0008331	0.029361	88.24	156.24	244.48	88.82	176.21	265.03	0.33230	0.58763	0.91994
750	29.06	0.0008395	0.027371	91.59	154.08	245.67	92.22	173.98	266.20	0.34345	0.57567	0.91912
800	31.31	0.0008458	0.025621	94.79	152.00	246.79	95.47	171.82	267.29	0.35404	0.56431	0.91835
850	33.45	0.0008520	0.024069	97.87	149.98	247.85	98.60	169.71	268.31	0.36413	0.55349	0.91762
900	35.51	0.0008580	0.022683	100.83	148.01	248.85	101.61	167.66	269.26	0.37377	0.54315	0.91692
950	37.48	0.0008641	0.021438	103.69	146.10	249.79	104.51	165.64	270.15	0.38301	0.53323	0.91624
1000	39.37	0.0008700	0.020313	106.45	144.23	250.68	107.32	163.67	270.99	0.39189	0.52368	0.91558
1200	46.29	0.0008934	0.016715	116.70	137.11	253.81	117.77	156.10	273.87	0.42441	0.48863	0.91303
1400	52.40	0.0009166	0.014107	125.94	130.43	256.37	127.22	148.90	276.12	0.45315	0.45734	0.91050
1600	57.88	0.0009400	0.012123	134.43	124.04	258.47	135.93	141.93	277.86	0.47911	0.42873	0.90784
1800	62.87	0.0009639	0.010559	142.33	117.83	260.17	144.07	135.11	279.17	0.50294	0.40204	0.90498
2000	67.45	0.0009886	0.009288	149.78	111.73	261.51	151.76	128.33	280.09	0.52509	0.37675	0.90184
2500	77.54	0.0010566	0.006936	166.99	96.47	263.45	169.63	111.16	280.79	0.57531	0.31695	0.89226
3000	86.16	0.0011406	0.005275	183.04	80.22	263.26	186.46	92.63	279.09	0.62118	0.25776	0.87894

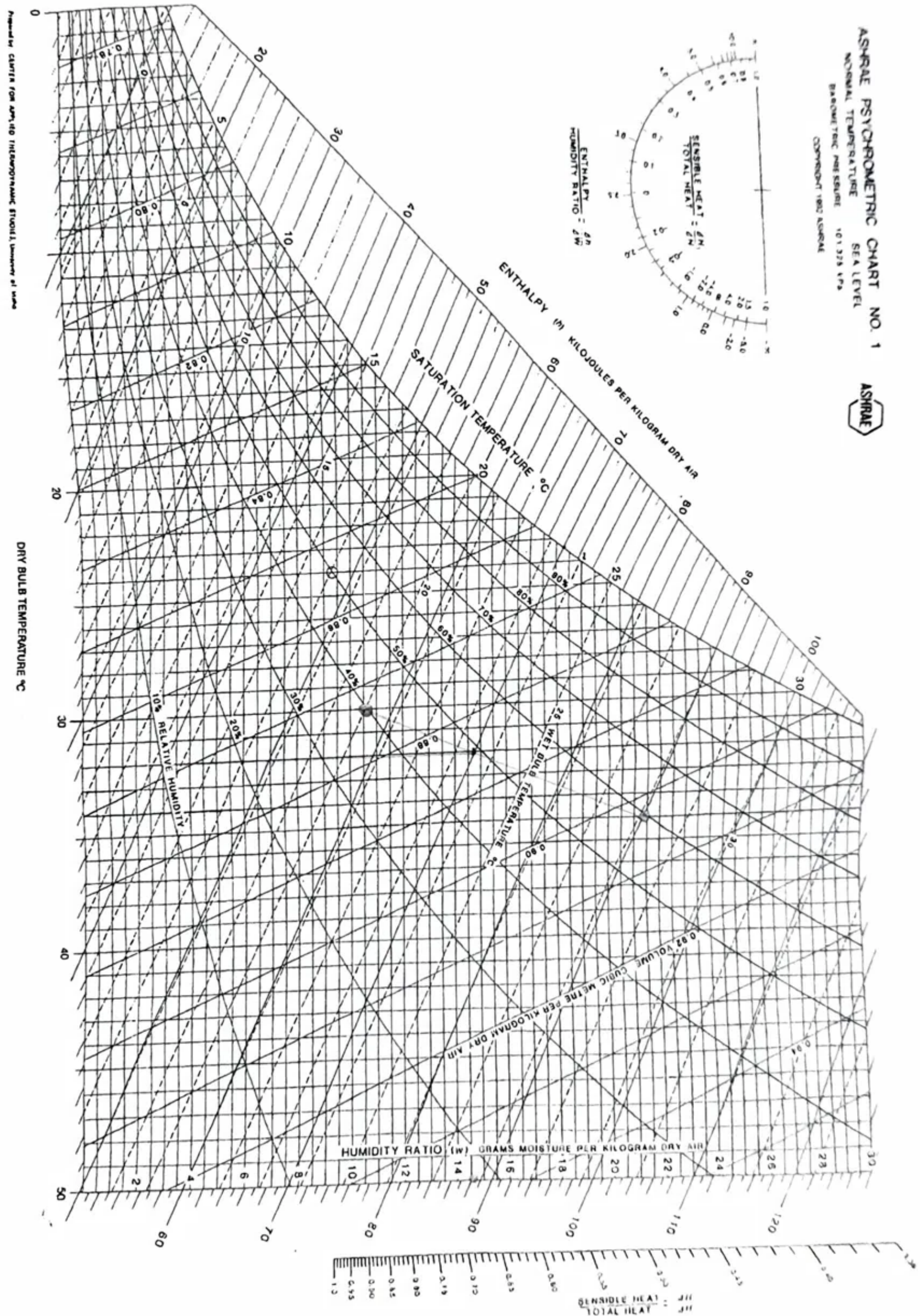
TABLE A–13
Superheated refrigerant-134a

<i>T</i> °C	<i>v</i> m ³ /kg	<i>u</i> kJ/kg	<i>h</i> kJ/kg	<i>s</i> kJ/kg · K	<i>v</i> m ³ /kg	<i>u</i> kJ/kg	<i>h</i> kJ/kg	<i>s</i> kJ/kg · K	<i>v</i> m ³ /kg	<i>u</i> kJ/kg	<i>h</i> kJ/kg	<i>s</i> kJ/kg · K
<i>P</i> = 0.50 MPa (<i>T</i> _{sat} = 15.71° C)					<i>P</i> = 0.60 MPa (<i>T</i> _{sat} = 21.55° C)				<i>P</i> = 0.70 MPa (<i>T</i> _{sat} = 26.69° C)			
Sat.	0.041118	238.75	259.30	0.9240	0.034295	241.83	262.40	0.9218	0.029361	244.48	265.03	0.9199
20	0.042115	242.40	263.46	0.9383								
30	0.044338	250.84	273.01	0.9703	0.035984	249.22	270.81	0.9499	0.029966	247.48	268.45	0.9313
40	0.046456	259.26	282.48	1.0011	0.037865	257.86	280.58	0.9816	0.031696	256.39	278.57	0.9641
50	0.048499	267.72	291.96	1.0309	0.039659	266.48	290.28	1.0121	0.033322	265.20	288.53	0.9954
60	0.050485	276.25	301.50	1.0599	0.041389	275.15	299.98	1.0417	0.034875	274.01	298.42	1.0256
70	0.052427	284.89	311.10	1.0883	0.043069	283.89	309.73	1.0705	0.036373	282.87	308.33	1.0549
80	0.054331	293.64	320.80	1.1162	0.044710	292.73	319.55	1.0987	0.037829	291.80	318.28	1.0835
90	0.056205	302.51	330.61	1.1436	0.046318	301.67	329.46	1.1264	0.039250	300.82	328.29	1.1114
100	0.058053	311.50	340.53	1.1705	0.047900	310.73	339.47	1.1536	0.040642	309.95	338.40	1.1389
110	0.059880	320.63	350.57	1.1971	0.049458	319.91	349.59	1.1803	0.042010	319.19	348.60	1.1658
120	0.061687	329.89	360.73	1.2233	0.050997	329.23	359.82	1.2067	0.043358	328.55	358.90	1.1924
130	0.063479	339.29	371.03	1.2491	0.052519	338.67	370.18	1.2327	0.044688	338.04	369.32	1.2186
140	0.065256	348.83	381.46	1.2747	0.054027	348.25	380.66	1.2584	0.046004	347.66	379.86	1.2444
150	0.067021	358.51	392.02	1.2999	0.055522	357.96	391.27	1.2838	0.047306	357.41	390.52	1.2699

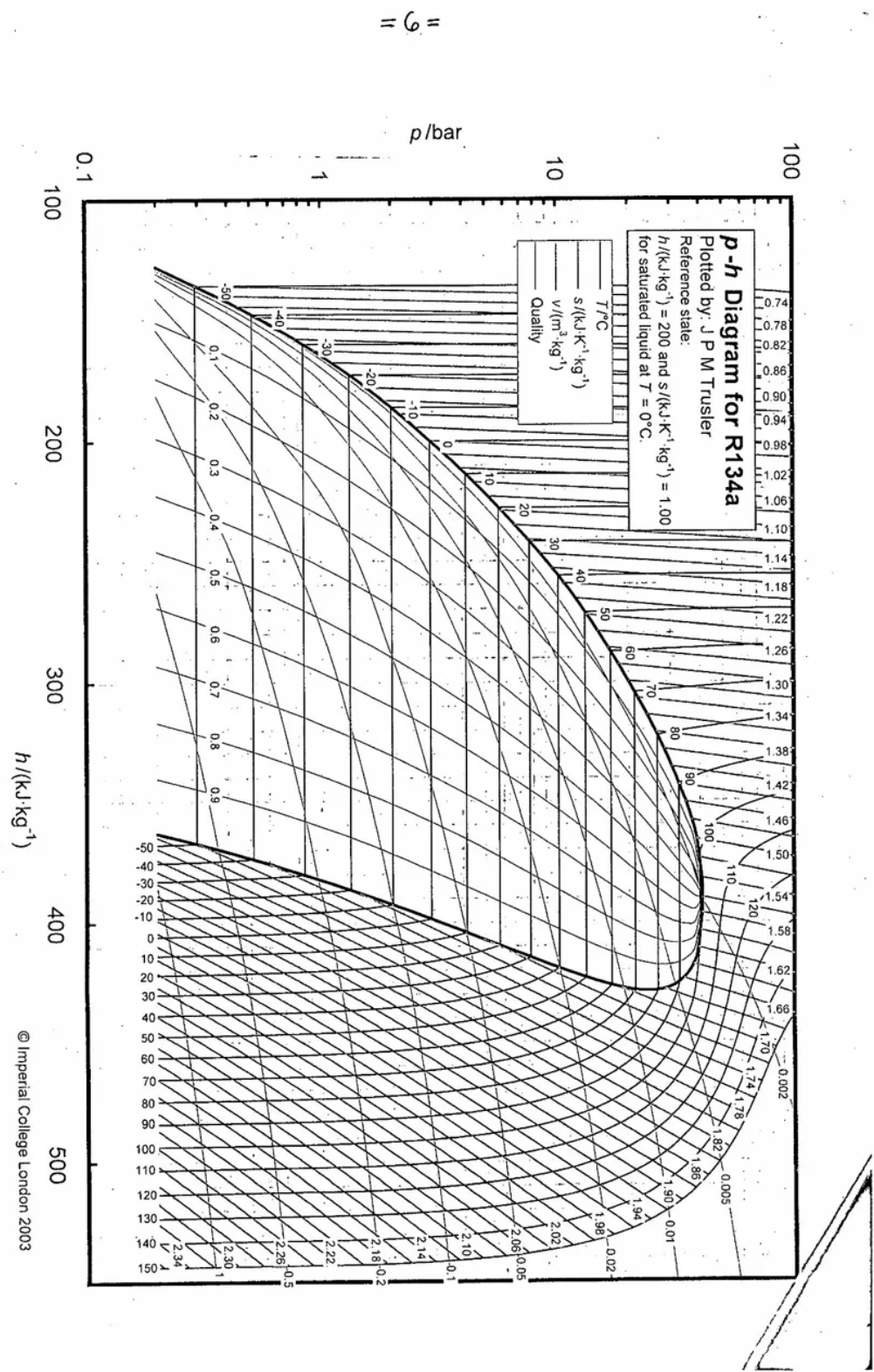
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T °C	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg · K	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg · K	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg · K
160	0.068775	368.33	402.72	1.3249	0.057006	367.81	402.01	1.3088	0.048597	367.29	401.31	1.2951
$P = 0.80 \text{ MPa } (T_{sat} = 31.31^\circ \text{C})$				$P = 0.90 \text{ MPa } (T_{sat} = 35.51^\circ \text{C})$				$P = 1.00 \text{ MPa } (T_{sat} = 39.37^\circ \text{C})$				
Sat.	0.025621	246.79	267.29	0.9183	0.022683	248.85	269.26	0.9169	0.020313	250.68	270.99	0.9156
40	0.027035	254.82	276.45	0.9480	0.023375	253.13	274.17	0.9327	0.020406	251.30	271.71	0.9179
50	0.028547	263.86	286.69	0.9802	0.024809	262.44	284.77	0.9660	0.021796	260.94	282.74	0.9525
60	0.029973	272.83	296.81	1.0110	0.026146	271.60	295.13	0.9976	0.023068	270.32	293.38	0.9850
70	0.031340	281.81	306.88	1.0408	0.027413	280.72	305.39	1.0280	0.024261	279.59	303.85	1.0160
80	0.032659	290.84	316.97	1.0698	0.028630	289.86	315.63	1.0574	0.025398	288.86	314.25	1.0458
90	0.033941	299.95	327.10	1.0981	0.029806	299.06	325.89	1.0860	0.026492	298.15	324.64	1.0748
100	0.035193	309.15	337.30	1.1258	0.030951	308.34	336.19	1.1140	0.027552	307.51	335.06	1.1031
110	0.036420	318.45	347.59	1.1530	0.032068	317.70	346.56	1.1414	0.028584	316.94	345.53	1.1308
120	0.037625	327.87	357.97	1.1798	0.033164	327.18	357.02	1.1684	0.029592	326.47	356.06	1.1580
130	0.038813	337.40	368.45	1.2061	0.034241	336.76	367.58	1.1949	0.030581	336.11	366.69	1.1846
140	0.039985	347.06	379.05	1.2321	0.035302	346.46	378.23	1.2210	0.031554	345.85	377.40	1.2109
150	0.041143	356.85	389.76	1.2577	0.036349	356.28	389.00	1.2467	0.032512	355.71	388.22	1.2368
160	0.042290	366.76	400.59	1.2830	0.037384	366.23	399.88	1.2721	0.033457	365.70	399.15	1.2623
170	0.043427	376.81	411.55	1.3080	0.038408	376.31	410.88	1.2972	0.034392	375.81	410.20	1.2875
180	0.044554	386.99	422.64	1.3327	0.039423	386.52	422.00	1.3221	0.035317	386.04	421.36	1.3124
$P = 1.20 \text{ MPa } (T_{sat} = 46.29^\circ \text{C})$				$P = 1.40 \text{ MPa } (T_{sat} = 52.40^\circ \text{C})$				$P = 1.60 \text{ MPa } (T_{sat} = 57.88^\circ \text{C})$				
Sat.	0.016715	253.81	273.87	0.9130	0.014107	256.37	276.12	0.9105	0.012123	258.47	277.86	0.9078
50	0.017201	257.63	278.27	0.9267								
60	0.018404	267.56	289.64	0.9614	0.015005	264.46	285.47	0.9389	0.012372	260.89	280.69	0.9163
70	0.019502	277.21	300.61	0.9938	0.016060	274.62	297.10	0.9733	0.013430	271.76	293.25	0.9535
80	0.020529	286.75	311.39	1.0248	0.017023	284.51	308.34	1.0056	0.014362	282.09	305.07	0.9875
90	0.021506	296.26	322.07	1.0546	0.017923	294.28	319.37	1.0364	0.015215	292.17	316.52	1.0194
100	0.022442	305.80	332.73	1.0836	0.018778	304.01	330.30	1.0661	0.016014	302.14	327.76	1.0500
110	0.023348	315.38	343.40	1.1118	0.019597	313.76	341.19	1.0949	0.016773	312.07	338.91	1.0795
120	0.024228	325.03	354.11	1.1394	0.020388	323.55	352.09	1.1230	0.017500	322.02	350.02	1.1081
130	0.025086	334.77	364.88	1.1664	0.021155	333.41	363.02	1.1504	0.018201	332.00	361.12	1.1360
140	0.025927	344.61	375.72	1.1930	0.021904	343.34	374.01	1.1773	0.018882	342.05	372.26	1.1632
150	0.026753	354.56	386.66	1.2192	0.022636	353.37	385.07	1.2038	0.019545	352.17	383.44	1.1900
160	0.027566	364.61	397.69	1.2449	0.023355	363.51	396.20	1.2298	0.020194	362.38	394.69	1.2163
170	0.028367	374.78	408.82	1.2703	0.024061	373.75	407.43	1.2554	0.020830	372.69	406.02	1.2421
180	0.029158	385.08	420.07	1.2954	0.024757	384.10	418.76	1.2807	0.021456	383.11	417.44	1.2676

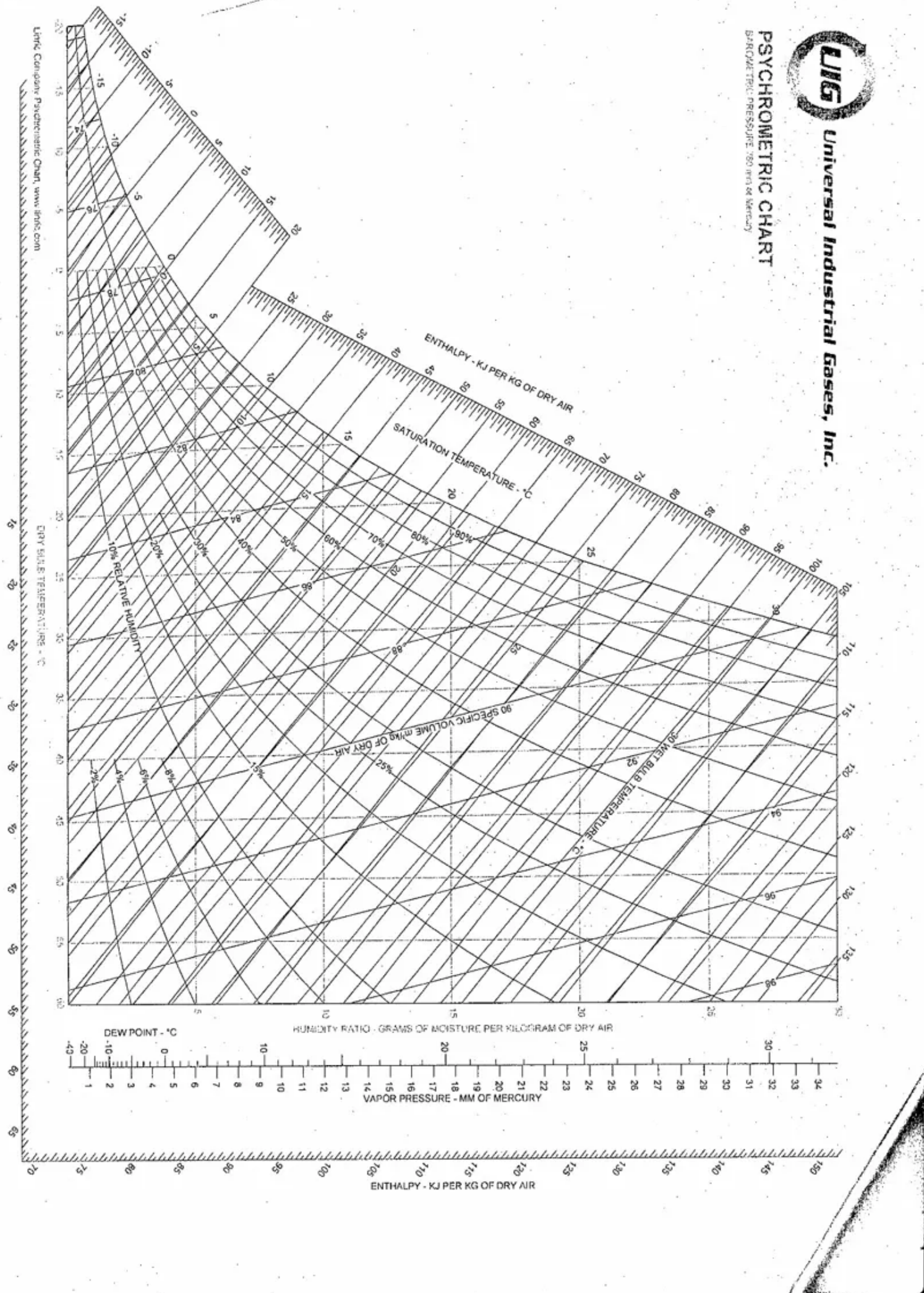
- Q14.** (a) What is Dew Point Temperature? Draw schematic of a chiller-type Air Conditioning System. **(10 marks)**
 (b) Air at 35°C DBT and 60% RH at 2 kg/s mixed with air at 4 kg/s, 20°C WBT and 30°C DBT. Calculate for final mixture: (i) Absolute humidity, (ii) Relative humidity, (iii) DBT and WBT, (iv) Dew point, (v) Enthalpy. **(20 marks)** [2018–19]
 ↪ *Similar: A3-Q12 same numerical*



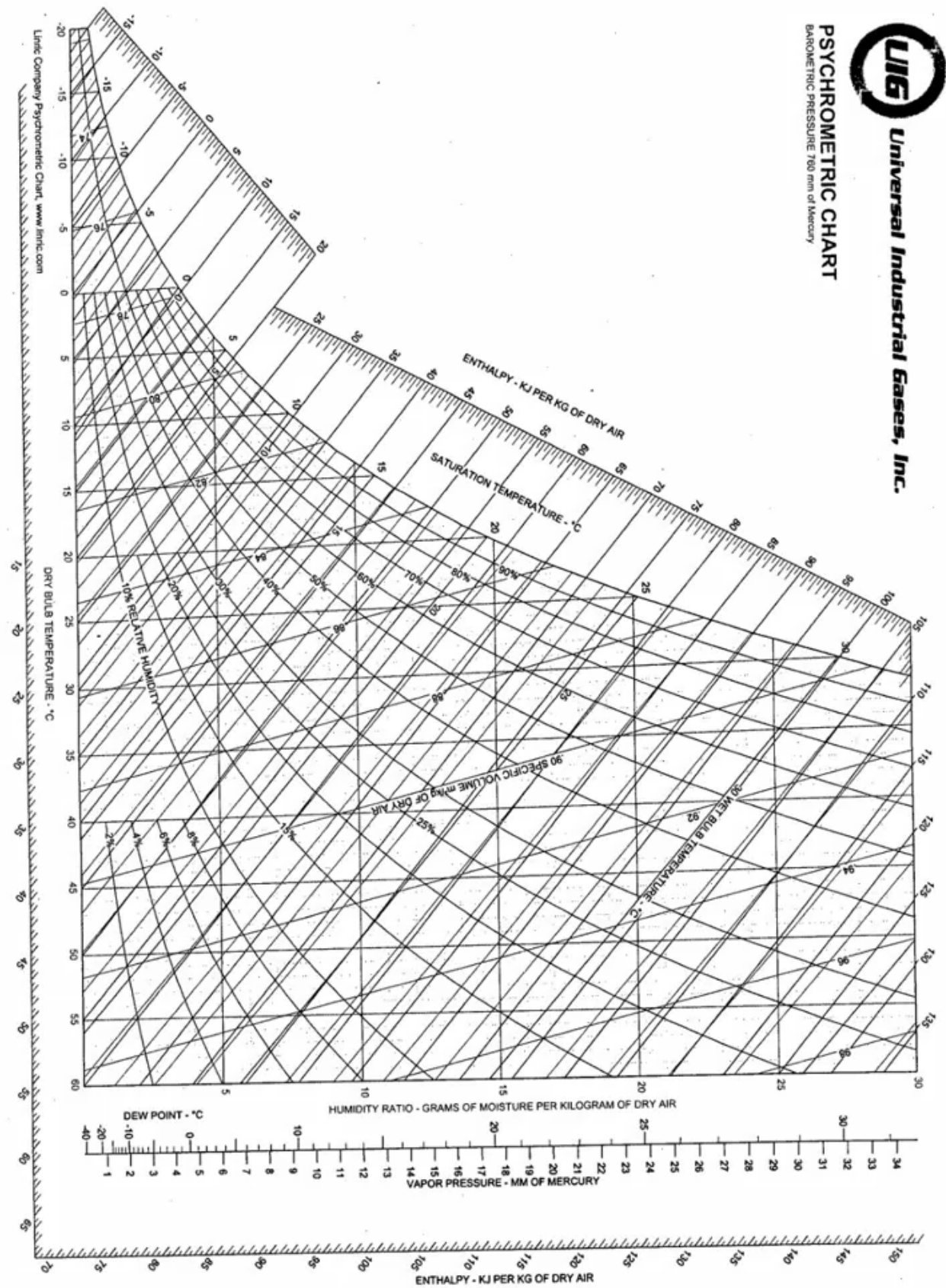
- Q15. (a) Differentiate between vapour compression and vapour absorption refrigeration. (5 marks)
- (b) With necessary sketch, discuss how a central air-conditioning system works. (8 marks)
- (c) What are ODP and GWP? Write chemical formulas of: R22, R600a, R114, R744, R11. (10 marks)
- (d) Tupli makes ice using an R-134a domestic refrigerator. Refrigerant enters compressor as saturated vapour at 1.5 bar, leaves condenser as saturated liquid at 7 bar. Uses 10 kg water/hour at 30°C to make ice at -5°C. ($c_p^{\text{ice}} = 2.1$, $c_p^{\text{water}} = 4.18 \text{ kJ/kgK}$, $L_f = 334 \text{ kJ/kg}$.) Find power input. (12 marks)
- [2017-18]



Q16. An air-conditioning system takes in outdoor air at 10°C and 30% RH at 45 m³/min. It conditions the air to 25°C and 60% RH. Outdoor air is first heated to 22°C then humidified by injecting steam. At 100 kPa, determine the heat supply in the heating stage. **(12 marks)** [2017–18]



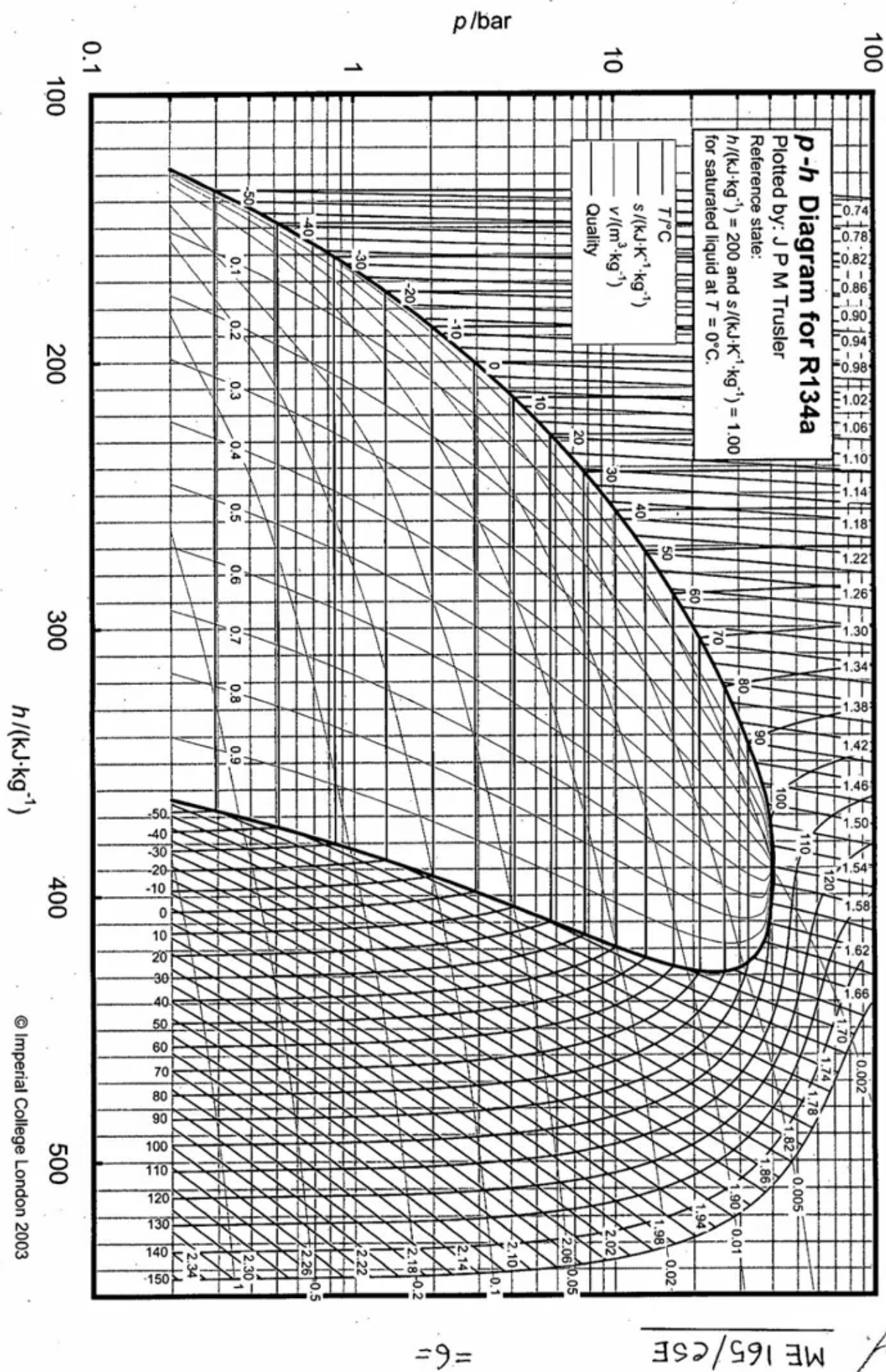
- Q17. (a) Draw the schematic of a conventional air conditioning system. What is the function of a cooling tower? **(10 marks)**
- (b) What factors affect the thermal comfort of humans? **(5 marks)**
- (c) List the key components of a refrigeration system with a suitable sketch. **(7 marks)**
- (d) A fish cold storage converts 5 tonnes of water/hour at 25°C into ice at -4°C using R-134a vapour compression. Evaporator pressure = 1 atm, condenser pressure = 10 atm. Determine: (i) refrigerant mass flow rate, (ii) compressor work, (iii) heat rejected at condenser, (iv) COP. **(13 marks)** [2016-17]



ME 165/CSE

- Q18. (a) What are the functions of a capillary tube in a domestic refrigerator? (5 marks)
 (b) Show the T-v diagram of a pure substance and identify different regions. (5 marks)
 (c) Air enters an evaporative cooler at 1 atm, 35°C and 20% RH. Determine: (i) humidity ratio, enthalpy and specific volume at inlet, (ii) exit temperature, (iii) lowest achievable temperature. (15 marks)

[2016-17]



- Q19. (a) Define refrigeration. What is one ton of refrigeration? With T-s diagrams explain the difference between Reversed Carnot Cycle and Vapour Compression Cycle. (14 marks)
- (b) Mention desirable properties of refrigerants. Write chemical formulas of R-134, R-22, R-718. (6 marks)
- (c) A refrigerator uses R-134a between 0.14 MPa and 0.9 MPa. Mass flow rate = 0.08 kg/s. Determine: (i) rate of heat removed, (ii) rate of heat rejection, (iii) power input, (iv) COP. (15 marks)

[2015–16]

TABLE A–11

Properties of Saturated Refrigerant 134a (Liquid-Vapor): Pressure Table

Press. bar	Temp. °C	Specific Volume m ³ /kg		Internal Energy kJ/kg		Enthalpy kJ/kg			Entropy kJ/kg · K		Press. bar
		Sat. Liquid <i>v_f</i> × 10 ³	Sat. Vapor <i>v_g</i>	Sat. Liquid <i>u_f</i>	Sat. Vapor <i>u_g</i>	Sat. Liquid <i>h_f</i>	Evap. <i>h_{fg}</i>	Sat. Vapor <i>h_g</i>	Sat. Liquid <i>s_f</i>	Sat. Vapor <i>s_g</i>	
0.6	-37.07	0.7097	0.3100	3.41	206.12	3.46	221.27	224.72	0.0147	0.9520	0.6
0.8	-31.21	0.7184	0.2366	10.41	209.46	10.47	217.92	228.39	0.0440	0.9447	0.8
1.0	-26.43	0.7258	0.1917	16.22	212.18	16.29	215.06	231.35	0.0678	0.9395	1.0
1.2	-22.36	0.7323	0.1614	21.23	214.50	21.32	212.54	233.86	0.0879	0.9354	1.2
1.4	-18.80	0.7381	0.1395	25.66	216.52	25.77	210.27	236.04	0.1055	0.9322	1.4
1.6	-15.62	0.7435	0.1229	29.66	218.32	29.78	208.19	237.97	0.1211	0.9295	1.6
1.8	-12.73	0.7485	0.1098	33.31	219.94	33.45	206.26	239.71	0.1352	0.9273	1.8
2.0	-10.09	0.7532	0.0993	36.69	221.43	36.84	204.46	241.30	0.1481	0.9253	2.0
2.4	-5.37	0.7618	0.0834	42.77	224.07	42.95	201.14	244.09	0.1710	0.9222	2.4
2.8	-1.23	0.7697	0.0719	48.18	226.38	48.39	198.13	246.52	0.1911	0.9197	2.8
3.2	2.48	0.7770	0.0632	53.06	228.43	53.31	195.35	248.66	0.2089	0.9177	3.2
3.6	5.84	0.7839	0.0564	57.54	230.28	57.82	192.76	250.58	0.2251	0.9160	3.6
4.0	8.93	0.7904	0.0509	61.69	231.97	62.00	190.32	252.32	0.2399	0.9145	4.0
5.0	15.74	0.8056	0.0409	70.93	235.64	71.33	184.74	256.07	0.2723	0.9117	5.0
6.0	21.58	0.8196	0.0341	78.99	238.74	79.48	179.71	259.19	0.2999	0.9097	6.0
7.0	26.72	0.8328	0.0292	86.19	241.42	86.78	175.07	261.85	0.3242	0.9080	7.0
8.0	31.33	0.8454	0.0255	92.75	243.78	93.42	170.73	264.15	0.3459	0.9066	8.0
9.0	35.53	0.8576	0.0226	98.79	245.88	99.56	166.62	266.18	0.3656	0.9054	9.0
10.0	39.39	0.8695	0.0202	104.42	247.77	105.29	162.68	267.97	0.3838	0.9043	10.0
12.0	46.32	0.8928	0.0166	114.69	251.03	115.76	155.23	270.99	0.4164	0.9023	12.0
14.0	52.43	0.9159	0.0140	123.98	253.74	125.26	148.14	273.40	0.4453	0.9003	14.0
16.0	57.92	0.9392	0.0121	132.52	256.00	134.02	141.31	275.33	0.4714	0.8982	16.0
18.0	62.91	0.9631	0.0105	140.49	257.88	142.22	134.60	276.83	0.4954	0.8959	18.0
20.0	67.49	0.9878	0.0093	148.02	259.41	149.99	127.95	277.94	0.5178	0.8934	20.0
25.0	77.59	1.0562	0.0069	165.48	261.84	168.12	111.06	279.17	0.5687	0.8854	25.0
30.0	86.22	1.1416	0.0053	181.88	262.16	185.30	92.71	278.01	0.6156	0.8735	30.0

Note: Pressure Conversions: 1 bar = 0.1 MPa = 10² kPa.

TABLE A–13
Properties of Superheated Refrigerant 134a

T °C	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg · K	T °C	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg · K
p = 1.4 bar = 0.14 MPa ($T_{\text{sat}} = -18.80^\circ\text{C}$)					p = 1.8 bar = 0.18 MPa ($T_{\text{sat}} = -12.73^\circ\text{C}$)				
Sat.	0.13945	216.52	236.04	0.9322	Sat.	0.10983	219.94	239.71	0.9273
-10	0.14549	223.03	243.40	0.9606	-10	0.11135	222.02	242.06	0.9362
0	0.15219	230.55	251.86	0.9922	0	0.11678	229.67	250.69	0.9684
10	0.15875	238.21	260.43	1.0230	10	0.12207	237.44	259.41	0.9998
20	0.16520	246.01	269.13	1.0532	20	0.12723	245.33	268.23	1.0304
30	0.17155	253.96	277.97	1.0828	30	0.13230	253.36	277.17	1.0604
40	0.17783	262.06	286.96	1.1120	40	0.13730	261.53	286.24	1.0898
50	0.18404	270.32	296.09	1.1407	50	0.14222	269.85	295.45	1.1187
60	0.19020	278.74	305.37	1.1690	60	0.14710	278.31	304.79	1.1472
70	0.19633	287.32	314.80	1.1969	70	0.15193	286.93	314.28	1.1753
80	0.20241	296.06	324.39	1.2244	80	0.15672	295.71	323.92	1.2030
90	0.20846	304.95	334.14	1.2516	90	0.16148	304.63	333.70	1.2303
100	0.21449	314.01	344.04	1.2785	100	0.16622	313.72	343.63	1.2573
p = 8.0 bar = 0.80 MPa ($T_{\text{sat}} = 31.33^\circ\text{C}$)					p = 9.0 bar = 0.90 MPa ($T_{\text{sat}} = 35.53^\circ\text{C}$)				
Sat.	0.02547	243.78	264.15	0.9066	Sat.	0.02255	245.88	266.18	0.9054
40	0.02691	252.13	273.66	0.9374	40	0.02325	250.32	271.25	0.9217
50	0.02846	261.62	284.39	0.9711	50	0.02472	260.09	282.34	0.9566
60	0.02992	271.04	294.98	1.0034	60	0.02609	269.72	293.21	0.9897
70	0.03131	280.45	305.50	1.0345	70	0.02738	279.30	303.94	1.0214
80	0.03264	289.89	316.00	1.0647	80	0.02861	288.87	314.62	1.0521
90	0.03393	299.37	326.52	1.0940	90	0.02880	298.46	325.28	1.0819
100	0.03519	308.93	337.08	1.1227	100	0.03095	308.11	335.96	1.1109
110	0.03642	318.57	347.71	1.1508	110	0.03207	317.82	346.68	1.1392
120	0.03762	328.31	358.40	1.1784	120	0.03316	327.62	357.47	1.1670
130	0.03881	338.14	369.19	1.2055	130	0.03423	337.52	368.33	1.1943
140	0.03997	348.09	380.07	1.2321	140	0.03529	347.51	379.27	1.2211
150	0.04113	358.15	391.05	1.2584	150	0.03633	357.61	390.31	1.2475
160	0.04227	368.32	402.14	1.2843	160	0.03736	367.82	401.44	1.2735
170	0.04340	378.61	413.33	1.3098	170	0.03838	378.14	412.68	1.2992
180	0.04452	389.02	424.63	1.3351	180	0.03939	388.57	424.02	1.3245
p = 14.0 bar = 1.40 MPa ($T_{\text{sat}} = 52.43^\circ\text{C}$)					p = 16.0 bar = 1.60 MPa ($T_{\text{sat}} = 57.92^\circ\text{C}$)				
Sat.	0.01405	253.74	273.40	0.9003	Sat.	0.01208	256.00	275.33	0.8982
60	0.01495	262.17	283.10	0.9297	60	0.01233	258.48	278.20	0.9069
70	0.01603	272.87	295.31	0.9658	70	0.01340	269.89	291.33	0.9457
80	0.01701	283.29	307.10	0.9997	80	0.01435	280.78	303.74	0.9813
90	0.01792	293.55	318.63	1.0319	90	0.01521	291.39	315.72	1.0148
100	0.01878	303.73	330.02	1.0628	100	0.01601	301.84	327.46	1.0467
110	0.01960	313.88	341.32	1.0927	110	0.01677	312.20	339.04	1.0773
120	0.02039	324.05	352.59	1.1218	120	0.01750	322.53	350.53	1.1069
130	0.02115	334.25	363.86	1.1501	130	0.01820	332.87	361.99	1.1357
140	0.02189	344.50	375.15	1.1777	140	0.01887	343.24	373.44	1.1638
150	0.02262	354.82	386.49	1.2048	150	0.01953	353.66	384.91	1.1912
160	0.02333	365.22	397.89	1.2315	160	0.02017	364.15	396.43	1.2181
170	0.02403	375.71	409.36	1.2576	170	0.02080	374.71	407.99	1.2445
180	0.02472	386.29	420.90	1.2843	180	0.02142	385.35	419.62	1.2704
190	0.02541	396.96	432.53	1.3088	190	0.02203	396.08	431.33	1.2960
200	0.02608	407.73	444.24	1.3338	200	0.02263	406.90	443.11	1.3212

Q20. Air at 35°C DBT and 60% RH at 2 kg/s mixed with air at 4 kg/s, 20°C WBT and 30°C DBT. Calculate for final mixture: (i) Absolute humidity, (ii) RH, (iii) DBT and WBT, (iv) Dew point, (v) Enthalpy. **(15 marks)** [2015–16]

↔ Similar: A3-Q12, A3-Q17 identical numerical

ME 165/CSE

=5=

Chart for Q.5b)

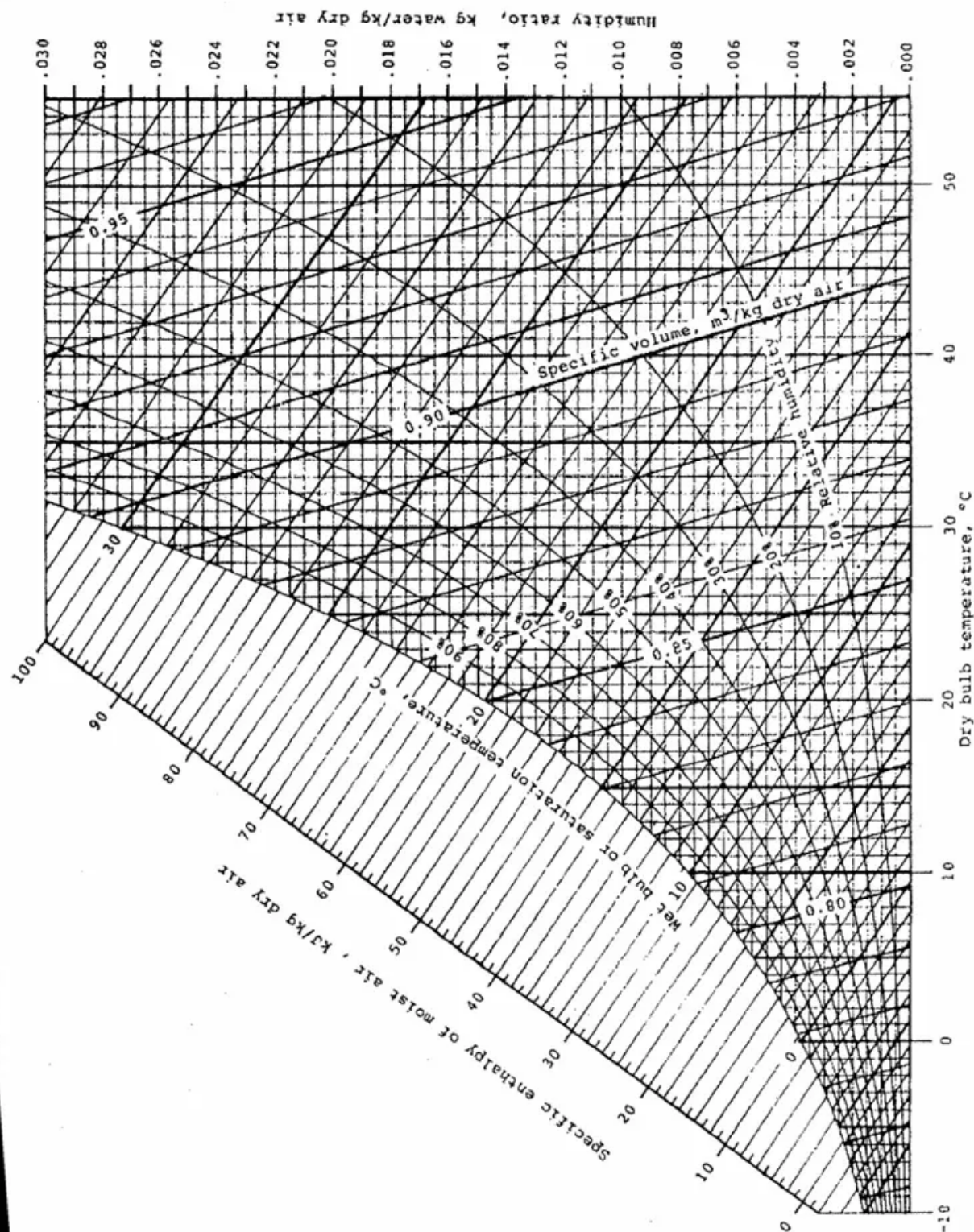
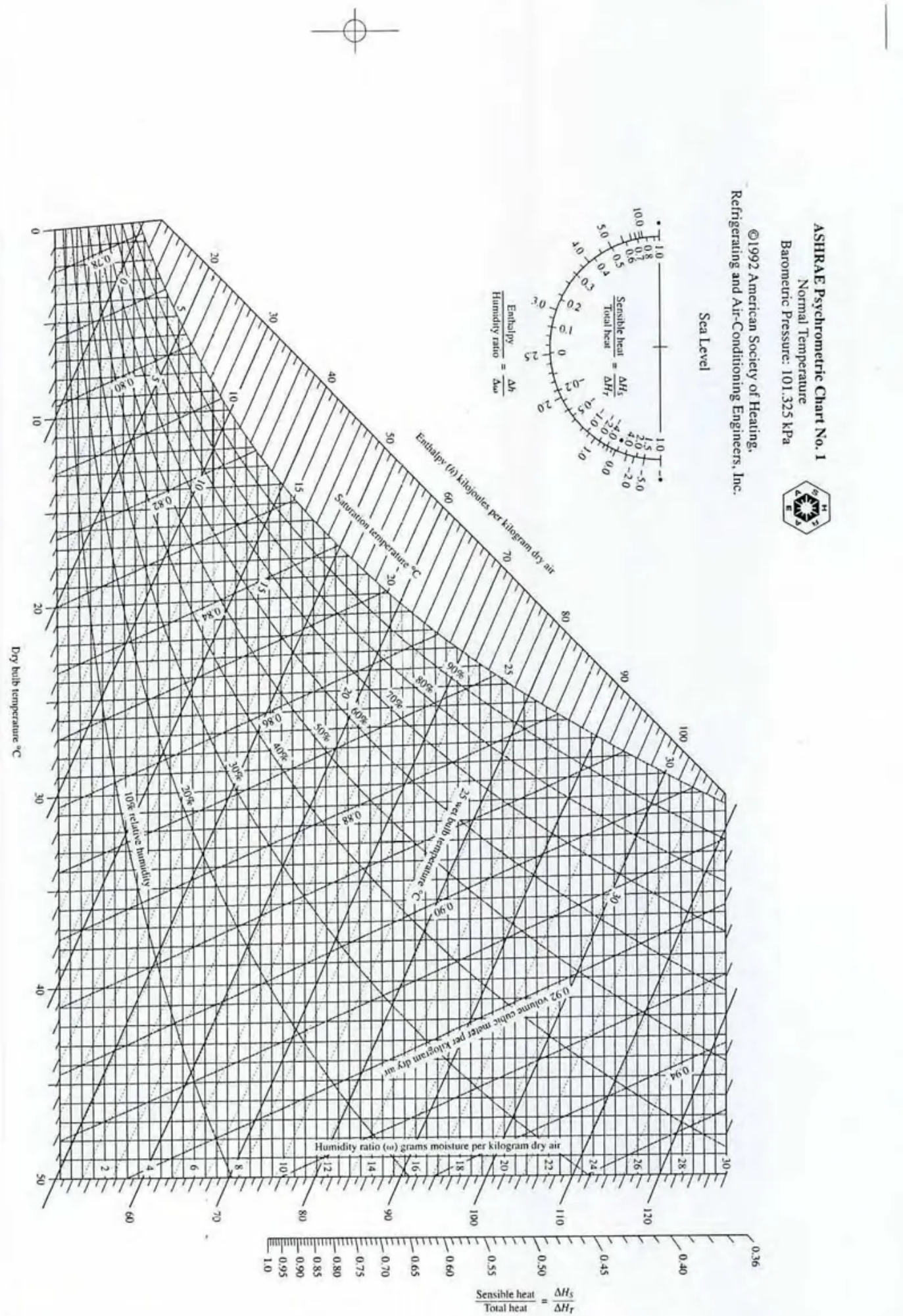


Figure A-9 Psychrometric chart for 1 atm (SI units). Source: Z. Zhang and M. B. Pate, "A Methodology for Implementing a Psychrometric Chart in a Computer Graphics System," *ASHRAE Transactions*, Vol. 94, Pt. 1, 1988.

- Q21.** (a) Write chemical formula of any three: R717, R12, R112, R290. (6 marks)
 (b) Draw schematic and T-s diagram of vapour compression refrigeration with multistage compression. (14 marks)
 (c) A refrigerator: $COP = 4.0$, compressor work = 0.5 kW , mass flow rate = 0.05 kg/s . Determine: (i) rate of heat removal, (ii) rate of heat rejection, (iii) COP as heat pump. (9 marks)
 (d) Define: (i) tonne of refrigeration (TR), (ii) COP. (6 marks) [2014–15]

- Q22.** 2 kg air at 40°C DBT and 50% RH mixed with 1 kg air at 22°C DBT and 17°C WBT. Calculate: (i) temperature, (ii) dew point temperature, (iii) specific humidity, (iv) relative humidity of mixture. (25 marks) [2014–15]



Press., P kPa	Sat. temp., T_{sat} °C	Specific volume, m ³ /kg		Internal energy, kJ/kg			Enthalpy, kJ/kg			Entropy, kJ/kg · K		
		Sat. liquid, v_f	Sat. vapor, v_g	Sat. liquid, u_f	Evap., u_{fg}	Sat. vapor, u_g	Sat. liquid, h_f	Evap., h_{fg}	Sat. vapor, h_g	Sat. liquid, s_f	Evap., s_{fg}	Sat. vapor, s_g
60	-36.95	0.0007098	0.31121	3.798	205.32	209.12	3.841	223.95	227.79	0.01634	0.94807	0.96441
70	-33.87	0.0007144	0.26929	7.680	203.20	210.88	7.730	222.00	229.73	0.03267	0.92775	0.96042
80	-31.13	0.0007185	0.23753	11.15	201.30	212.46	11.21	220.25	231.46	0.04711	0.90999	0.95710
90	-28.65	0.0007223	0.21263	14.31	199.57	213.88	14.37	218.65	233.02	0.06008	0.89419	0.95427
100	-26.37	0.0007259	0.19254	17.21	197.98	215.19	17.28	217.16	234.44	0.07188	0.87995	0.95183
120	-22.32	0.0007324	0.16212	22.40	195.11	217.51	22.49	214.48	236.97	0.09275	0.85503	0.94779
140	-18.77	0.0007383	0.14014	26.98	192.57	219.54	27.08	212.08	239.16	0.11087	0.83368	0.94456
160	-15.60	0.0007437	0.12348	31.09	190.27	221.35	31.21	209.90	241.11	0.12693	0.81496	0.94190
180	-12.73	0.0007487	0.11041	34.83	188.16	222.99	34.97	207.90	242.86	0.14139	0.79826	0.93965
200	-10.09	0.0007533	0.099867	38.28	186.21	224.48	38.43	206.03	244.46	0.15457	0.78316	0.93773
240	-5.38	0.0007620	0.083897	44.48	182.67	227.14	44.66	202.62	247.28	0.17794	0.75664	0.93458
280	-1.25	0.0007699	0.072352	49.97	179.50	229.46	50.18	199.54	249.72	0.19829	0.73381	0.93210
320	2.46	0.0007772	0.063604	54.92	176.61	231.52	55.16	196.71	251.88	0.21637	0.71369	0.93006
360	5.82	0.0007841	0.056738	59.44	173.94	233.38	59.72	194.08	253.81	0.23270	0.69566	0.92836
400	8.91	0.0007907	0.051201	63.62	171.45	235.07	63.94	191.62	255.55	0.24761	0.67929	0.92691
450	12.46	0.0007985	0.045619	68.45	168.54	237.00	68.81	188.71	257.53	0.26465	0.66069	0.92535
500	15.71	0.0008059	0.041118	72.93	165.82	238.75	73.33	185.98	259.30	0.28023	0.64377	0.92400
550	18.73	0.0008130	0.037408	77.10	163.25	240.35	77.54	183.38	260.92	0.29461	0.62821	0.92282
600	21.55	0.0008199	0.034295	81.02	160.81	241.83	81.51	180.90	262.40	0.30799	0.61378	0.92177
650	24.20	0.0008266	0.031646	84.72	158.48	243.20	85.26	178.51	263.77	0.32051	0.60030	0.92081
700	26.69	0.0008331	0.029361	88.24	156.24	244.48	88.82	176.21	265.03	0.33230	0.58763	0.91994
750	29.06	0.0008395	0.027371	91.59	154.08	245.67	92.22	173.98	266.20	0.34345	0.57567	0.91912
800	31.31	0.0008458	0.025621	94.79	152.00	246.79	95.47	171.82	267.29	0.35404	0.56431	0.91835
850	33.45	0.0008520	0.024069	97.87	149.98	247.85	98.60	169.71	268.31	0.36413	0.55349	0.91762
900	35.51	0.0008580	0.022683	100.83	148.01	248.85	101.61	167.66	269.26	0.37377	0.54315	0.91692
950	37.48	0.0008641	0.021438	103.69	146.10	249.79	104.51	165.64	270.15	0.38301	0.53323	0.91624
1000	39.37	0.0008700	0.020313	106.45	144.23	250.68	107.32	163.67	270.99	0.39189	0.52368	0.91558
1200	46.29	0.0008934	0.016715	116.70	137.11	253.81	117.77	156.10	273.87	0.42441	0.48863	0.91303
1400	52.40	0.0009166	0.014107	125.94	130.43	256.37	127.22	148.90	276.12	0.45315	0.45734	0.91050
1600	57.88	0.0009400	0.012123	134.43	124.04	258.47	135.93	141.93	277.86	0.47911	0.42873	0.90784
1800	62.87	0.0009639	0.010559	142.33	117.83	260.17	144.07	135.11	279.17	0.50294	0.40204	0.90498
2000	67.45	0.0009886	0.009288	149.78	111.73	261.51	151.76	128.33	280.09	0.52509	0.37675	0.90184
2500	77.54	0.0010566	0.006936	166.99	96.47	263.45	169.63	111.16	280.79	0.57531	0.31695	0.89226
3000	86.16	0.0011406	0.005275	183.04	80.22	263.26	186.46	92.63	279.09	0.62118	0.25776	0.87894

TABLE A–13
Superheated refrigerant-134a

T °C	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg · K	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg · K	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg · K
	$P = 0.50 \text{ MPa } (T_{sat} = 15.71^\circ \text{C})$				$P = 0.60 \text{ MPa } (T_{sat} = 21.55^\circ \text{C})$				$P = 0.70 \text{ MPa } (T_{sat} = 26.69^\circ \text{C})$			
Sat.	0.041118	238.75	259.30	0.9240	0.034295	241.83	262.40	0.9218	0.029361	244.48	265.03	0.9199
20	0.042115	242.40	263.46	0.9383								
30	0.044338	250.84	273.01	0.9703	0.035984	249.22	270.81	0.9499	0.029966	247.48	268.45	0.9313
40	0.046456	259.26	282.48	1.0011	0.037865	257.86	280.58	0.9816	0.031696	256.39	278.57	0.9641
50	0.048499	267.72	291.96	1.0309	0.039659	266.48	290.28	1.0121	0.033322	265.20	288.53	0.9954
60	0.050485	276.25	301.50	1.0599	0.041389	275.15	299.98	1.0417	0.034875	274.01	298.42	1.0256
70	0.052427	284.89	311.10	1.0883	0.043069	283.89	309.73	1.0705	0.036373	282.87	308.33	1.0549
80	0.054331	293.64	320.80	1.1162	0.044710	292.73	319.55	1.0987	0.037829	291.80	318.28	1.0835
90	0.056205	302.51	330.61	1.1436	0.046318	301.67	329.46	1.1264	0.039250	300.82	328.29	1.1114
100	0.058053	311.50	340.53	1.1705	0.047900	310.73	339.47	1.1536	0.040642	309.95	338.40	1.1389
110	0.059880	320.63	350.57	1.1971	0.049458	319.91	349.59	1.1803	0.042010	319.19	348.60	1.1658
120	0.061687	329.89	360.73	1.2233	0.050997	329.23	359.82	1.2067	0.043358	328.55	358.90	1.1924
130	0.063479	339.29	371.03	1.2491	0.052519	338.67	370.18	1.2327	0.044688	338.04	369.32	1.2186
140	0.065256	348.83	381.46	1.2747	0.054027	348.25	380.66	1.2584	0.046004	347.66	379.86	1.2444
150	0.067021	358.51	392.02	1.2999	0.055522	357.96	391.27	1.2838	0.047306	357.41	390.52	1.2699
160	0.068775	368.33	402.72	1.3249	0.057006	367.81	402.01	1.3088	0.048597	367.29	401.31	1.2951
	$P = 0.80 \text{ MPa } (T_{sat} = 31.31^\circ \text{C})$				$P = 0.90 \text{ MPa } (T_{sat} = 35.51^\circ \text{C})$				$P = 1.00 \text{ MPa } (T_{sat} = 39.37^\circ \text{C})$			
Continued on next page												

T °C	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg · K	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg · K	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg · K
Sat.	0.025621	246.79	267.29	0.9183	0.022683	248.85	269.26	0.9169	0.020313	250.68	270.99	0.9156
40	0.027035	254.82	276.45	0.9480	0.023375	253.13	274.17	0.9327	0.020406	251.30	271.71	0.9179
50	0.028547	263.86	286.69	0.9802	0.024809	262.44	284.77	0.9660	0.021796	260.94	282.74	0.9525
60	0.029973	272.83	296.81	1.0110	0.026146	271.60	295.13	0.9976	0.023068	270.32	293.38	0.9850
70	0.031340	281.81	306.88	1.0408	0.027413	280.72	305.39	1.0280	0.024261	279.59	303.85	1.0160
80	0.032659	290.84	316.97	1.0698	0.028630	289.86	315.63	1.0574	0.025398	288.86	314.25	1.0458
90	0.033941	299.95	327.10	1.0981	0.029806	299.06	325.89	1.0860	0.026492	298.15	324.64	1.0748
100	0.035193	309.15	337.30	1.1258	0.030951	308.34	336.19	1.1140	0.027552	307.51	335.06	1.1031
110	0.036420	318.45	347.59	1.1530	0.032068	317.70	346.56	1.1414	0.028584	316.94	345.53	1.1308
120	0.037625	327.87	357.97	1.1798	0.033164	327.18	357.02	1.1684	0.029592	326.47	356.06	1.1580
130	0.038813	337.40	368.45	1.2061	0.034241	336.76	367.58	1.1949	0.030581	336.11	366.69	1.1846
140	0.039985	347.06	379.05	1.2321	0.035302	346.46	378.23	1.2210	0.031554	345.85	377.40	1.2109
150	0.041143	356.85	389.76	1.2577	0.036349	356.28	389.00	1.2467	0.032512	355.71	388.22	1.2368
160	0.042290	366.76	400.59	1.2830	0.037384	366.23	399.88	1.2721	0.033457	365.70	399.15	1.2623
170	0.043427	376.81	411.55	1.3080	0.038408	376.31	410.88	1.2972	0.034392	375.81	410.20	1.2875
180	0.044554	386.99	422.64	1.3327	0.039423	386.52	422.00	1.3221	0.035317	386.04	421.36	1.3124
$P = 1.20 \text{ MPa } (T_{sat} = 46.29^\circ \text{C})$				$P = 1.40 \text{ MPa } (T_{sat} = 52.40^\circ \text{C})$				$P = 1.60 \text{ MPa } (T_{sat} = 57.88^\circ \text{C})$				
Sat.	0.016715	253.81	273.87	0.9130	0.014107	256.37	276.12	0.9105	0.012123	258.47	277.86	0.9078
50	0.017201	257.63	278.27	0.9267								
60	0.018404	267.56	289.64	0.9614	0.015005	264.46	285.47	0.9389	0.012372	260.89	280.69	0.9163
70	0.019502	277.21	300.61	0.9938	0.016060	274.62	297.10	0.9733	0.013430	271.76	293.25	0.9535
80	0.020529	286.75	311.39	1.0248	0.017023	284.51	308.34	1.0056	0.014362	282.09	305.07	0.9875
90	0.021506	296.26	322.07	1.0546	0.017923	294.28	319.37	1.0364	0.015215	292.17	316.52	1.0194
100	0.022442	305.80	332.73	1.0836	0.018778	304.01	330.30	1.0661	0.016014	302.14	327.76	1.0500
110	0.023348	315.38	343.40	1.1118	0.019597	313.76	341.19	1.0949	0.016773	312.07	338.91	1.0795
120	0.024228	325.03	354.11	1.1394	0.020388	323.55	352.09	1.1230	0.017500	322.02	350.02	1.1081
130	0.025086	334.77	364.88	1.1664	0.021155	333.41	363.02	1.1504	0.018201	332.00	361.12	1.1360
140	0.025927	344.61	375.72	1.1930	0.021904	343.34	374.01	1.1773	0.018882	342.05	372.26	1.1632
150	0.026753	354.56	386.66	1.2192	0.022636	353.37	385.07	1.2038	0.019545	352.17	383.44	1.1900
160	0.027566	364.61	397.69	1.2449	0.023355	363.51	396.20	1.2298	0.020194	362.38	394.69	1.2163
170	0.028367	374.78	408.82	1.2703	0.024061	373.75	407.43	1.2554	0.020830	372.69	406.02	1.2421
180	0.029158	385.08	420.07	1.2954	0.024757	384.10	418.76	1.2807	0.021456	383.11	417.44	1.2676

TABLE A–17
Ideal-gas properties of air

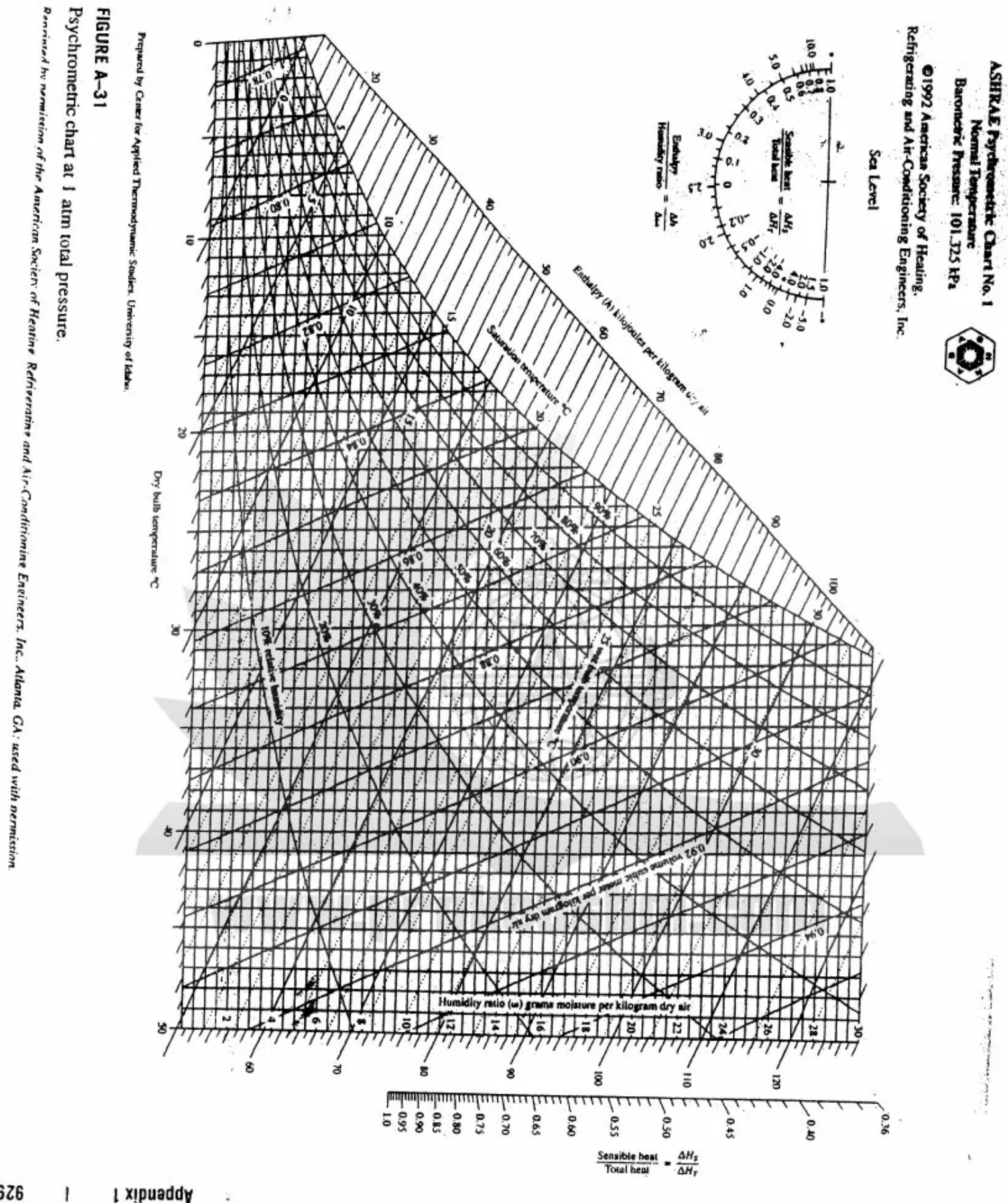
T K	h kJ/kg	P_r	u kJ/kg	v_r	s° kJ/kg·K	T K	h kJ/kg	P_r	u kJ/kg	v_r	s° kJ/kg·K
200	199.97	0.3363	142.56	1707.0	1.29559	580	586.04	14.38	419.55	115.7	2.37348
210	209.97	0.3987	149.69	1512.0	1.34444	590	596.52	15.31	427.15	110.6	2.39140
220	219.97	0.4690	156.82	1346.0	1.39105	600	607.02	16.28	434.78	105.8	2.40902
230	230.02	0.5477	164.00	1205.0	1.43557	610	617.53	17.30	442.42	101.2	2.42644
240	240.02	0.6355	171.13	1084.0	1.47824	620	628.07	18.36	450.09	96.92	2.44356
250	250.05	0.7329	178.28	979.0	1.51917	630	638.63	19.84	457.78	92.84	2.46048
260	260.09	0.8405	185.45	887.8	1.55848	640	649.22	20.64	465.50	88.99	2.47716
270	270.11	0.9590	192.60	808.0	1.59634	650	659.84	21.86	473.25	85.34	2.49364
280	280.13	1.0889	199.75	738.0	1.63279	660	670.47	23.13	481.01	81.89	2.50985
285	285.14	1.1584	203.33	706.1	1.65055	670	681.14	24.46	488.81	78.61	2.52589
290	290.16	1.2311	206.91	676.1	1.66802	680	691.82	25.85	496.62	75.50	2.54175
295	295.17	1.3068	210.49	647.9	1.68515	690	702.52	27.29	504.45	72.56	2.55731
298	298.18	1.3543	212.64	631.9	1.69528	700	713.27	28.80	512.33	69.76	2.57277
300	300.19	1.3860	214.07	621.2	1.70203	710	724.04	30.38	520.23	67.07	2.58810
305	305.22	1.4686	217.67	596.0	1.71865	720	734.82	32.02	528.14	64.53	2.60319
310	310.24	1.5546	221.25	572.3	1.73498	730	745.62	33.72	536.07	62.13	2.61803
315	315.27	1.6442	224.85	549.8	1.75106	740	756.44	35.50	544.02	59.82	2.63280
320	320.29	1.7375	228.42	528.6	1.76690	750	767.29	37.35	551.99	57.63	2.64737
325	325.31	1.8345	232.02	508.4	1.78249	760	778.18	39.27	560.01	55.54	2.66176
330	330.34	1.9352	235.61	489.4	1.79783	780	800.03	43.35	576.12	51.64	2.69013
340	340.42	2.149	242.82	454.1	1.82790	800	821.95	47.75	592.30	48.08	2.71787
350	350.49	2.379	250.02	422.2	1.85708	820	843.98	52.59	608.59	44.84	2.74504
360	360.58	2.626	257.24	393.4	1.88543	840	866.08	57.60	624.95	41.85	2.77170
370	370.67	2.892	264.46	367.2	1.91313	860	888.27	63.09	641.40	39.12	2.79783
380	380.77	3.176	271.69	343.4	1.94001	880	910.56	68.98	657.95	36.61	2.82344
390	390.88	3.481	278.93	321.5	1.96633	900	932.93	75.29	674.58	34.31	2.84856
400	400.98	3.806	286.16	301.6	1.99194	920	955.38	82.05	691.28	32.18	2.87324
410	411.12	4.153	293.43	283.3	2.01699	940	977.92	89.28	708.08	30.22	2.89748

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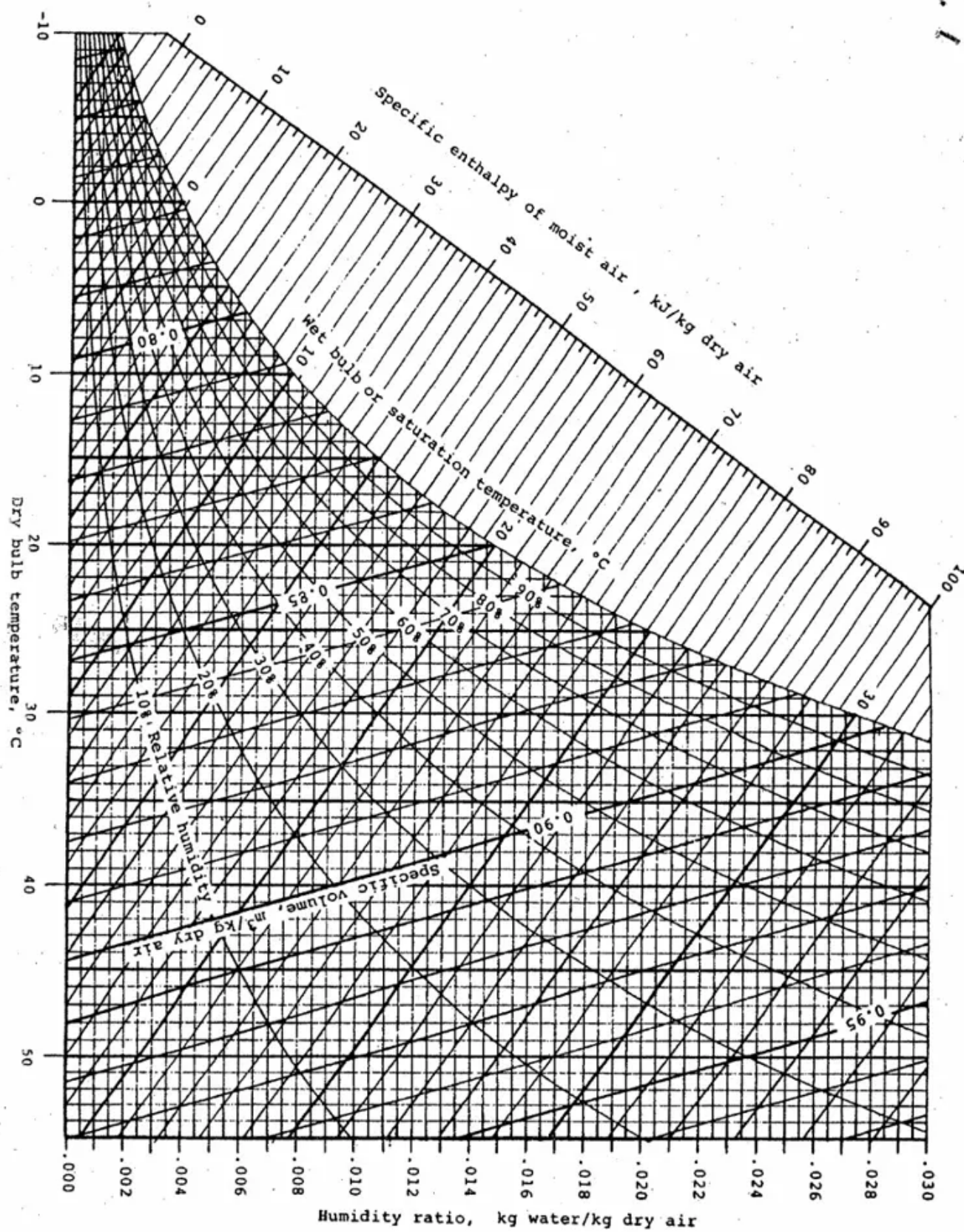
T K	h kJ/kg	P_r	u kJ/kg	v_r	s° kJ/kg·K	T K	h kJ/kg	P_r	u kJ/kg	v_r	s° kJ/kg·K
420	421.26	4.522	300.69	266.6	2.04142	960	1000.55	97.00	725.02	28.40	2.92128
430	431.43	4.915	307.99	251.1	2.06533	980	1023.25	105.2	741.98	26.73	2.94468
440	441.61	5.332	315.30	236.8	2.08870	1000	1046.04	114.0	758.94	25.17	2.96770
450	451.80	5.775	322.62	223.6	2.11161	1020	1068.89	123.4	776.10	23.72	2.99034
460	462.02	6.245	329.97	211.4	2.13407	1040	1091.85	133.3	793.36	23.29	3.01260
470	472.24	6.742	337.32	200.1	2.15604	1060	1114.86	143.9	810.62	21.14	3.03449
480	482.49	7.268	344.70	189.5	2.17760	1080	1137.89	155.2	827.88	19.98	3.05608
490	492.74	7.824	352.08	179.7	2.19876	1100	1161.07	167.1	845.33	18.896	3.07732
500	503.02	8.411	359.49	170.6	2.21952	1120	1184.28	179.7	862.79	17.886	3.09825
510	513.32	9.031	366.92	162.1	2.23993	1140	1207.57	193.1	880.35	16.946	3.11883
520	523.63	9.684	374.36	154.1	2.25997	1160	1230.92	207.2	897.91	16.064	3.13916
530	533.98	10.37	381.84	146.7	2.27967	1180	1254.34	222.2	915.57	15.241	3.15916
540	544.35	11.10	389.34	139.7	2.29906	1200	1277.79	238.0	933.33	14.470	3.17888
550	555.74	11.86	396.86	133.1	2.31809	1220	1301.31	254.7	951.09	13.747	3.19834
560	565.17	12.66	404.42	127.0	2.33685	1240	1324.93	272.3	968.95	13.069	3.21751
570	575.59	13.50	411.97	121.2	2.35531						
1260	1348.55	290.8	986.90	12.435	3.23638	1600	1757.57	791.2	1298.30	5.804	3.52364
1280	1372.24	310.4	1004.76	11.835	3.25510	1620	1782.00	834.1	1316.96	5.574	3.53879
1300	1395.97	330.9	1022.82	11.275	3.27345	1640	1806.46	878.9	1335.72	5.355	3.55381
1320	1419.76	352.5	1040.88	10.747	3.29160	1660	1830.96	925.6	1354.48	5.147	3.56867
1340	1443.60	375.3	1058.94	10.247	3.30959	1680	1855.50	974.2	1373.24	4.949	3.58335
1360	1467.49	399.1	1077.10	9.780	3.32724	1700	1880.1	1025	1392.7	4.761	3.5979
1380	1491.44	424.2	1095.26	9.337	3.34474	1750	1941.6	1161	1439.8	4.328	3.6336
1400	1515.42	450.5	1113.52	8.919	3.36200	1800	2003.3	1310	1487.2	3.994	3.6684
1420	1539.44	478.0	1131.77	8.526	3.37901	1850	2065.3	1475	1534.9	3.601	3.7023
1440	1563.51	506.9	1150.13	8.153	3.39586	1900	2127.4	1655	1582.6	3.295	3.7354
1460	1587.63	537.1	1168.49	7.801	3.41247	1950	2189.7	1852	1630.6	3.022	3.7677
1480	1611.79	568.8	1186.95	7.468	3.42892	2000	2252.1	2068	1678.7	2.776	3.7994
1500	1635.97	601.9	1205.41	7.152	3.44516	2050	2314.6	2303	1726.8	2.555	3.8303
1520	1660.23	636.5	1223.87	6.854	3.46120	2100	2377.7	2559	1775.3	2.356	3.8605
1540	1684.51	672.8	1242.43	6.569	3.47712	2150	2440.3	2837	1823.8	2.175	3.8901
1560	1708.82	710.5	1260.99	6.301	3.49276	2200	2503.2	3138	1872.4	2.012	3.9191
1580	1733.17	750.0	1279.65	6.046	3.50829	2250	2566.4	3464	1921.3	1.864	3.9474

Note: The properties P_r (relative pressure) and v_r (relative specific volume) are dimensionless quantities used in the analysis of isentropic processes, and should not be confused with the properties pressure and specific volume.
Source: Kenneth Wark, *Thermodynamics*, 4th ed. (New York: McGraw-Hill, 1983), pp. 785–86, table A–5. Originally published in J. H. Keenan and J. Kaye, *Gas Tables* (New York: John Wiley & Sons, 1948).

Q24. 1 kg air at 30°C DBT and 40% RH mixed with 1 kg air at 20°C DBT and 15°C WBT. Calculate temperature and humidity of the mixture. **(17 marks)**
[2013–14]



- Q25.** (a) Refrigerant R50 maintained at 0.1 MPa, 80°C. Using ideal gas assumption, determine its density. **(10 marks)**
 (b) An insulated space has air at 30°C DBT and specific volume 0.88 m³/kg. After AC operation at 24°C, 60% RH, with 50 kg initial air — how much water was removed? **(15 marks)** [2012–13]
- Q26.** (a) Define air conditioning and give brief descriptions of the four major AC loads. **(10 marks)**
 (b) Give a concise description of processes in an ideal vapour compression refrigeration cycle with P-h and T-s diagrams. **(15 marks)**
 (c) Write commercial names of: (i) Ammonia, (ii) Methane, (iii) Tetrachloromethane, (iv) Tetrafluoromethane, (v) Dichlorodifluoromethane. **(10 marks)** [2012–13]
- Q27.** (a) Differentiate between natural cooling and refrigeration. **(7 marks)**
 (b) Define ton of refrigeration. **(6 marks)**
 (c) Draw the Vapour Compression Refrigeration system and corresponding P-h diagram. **(10 marks)**
 (d) DBT = 30°C and WBT = 25°C in a room. Determine: (i) humidity ratio, RH, enthalpy, specific volume, dew point; (ii) after AC changes condition to 25°C DBT, 50% RH — new humidity ratio, WBT, enthalpy; (iii) heat and water removed. **(12 marks)** [2011–12]

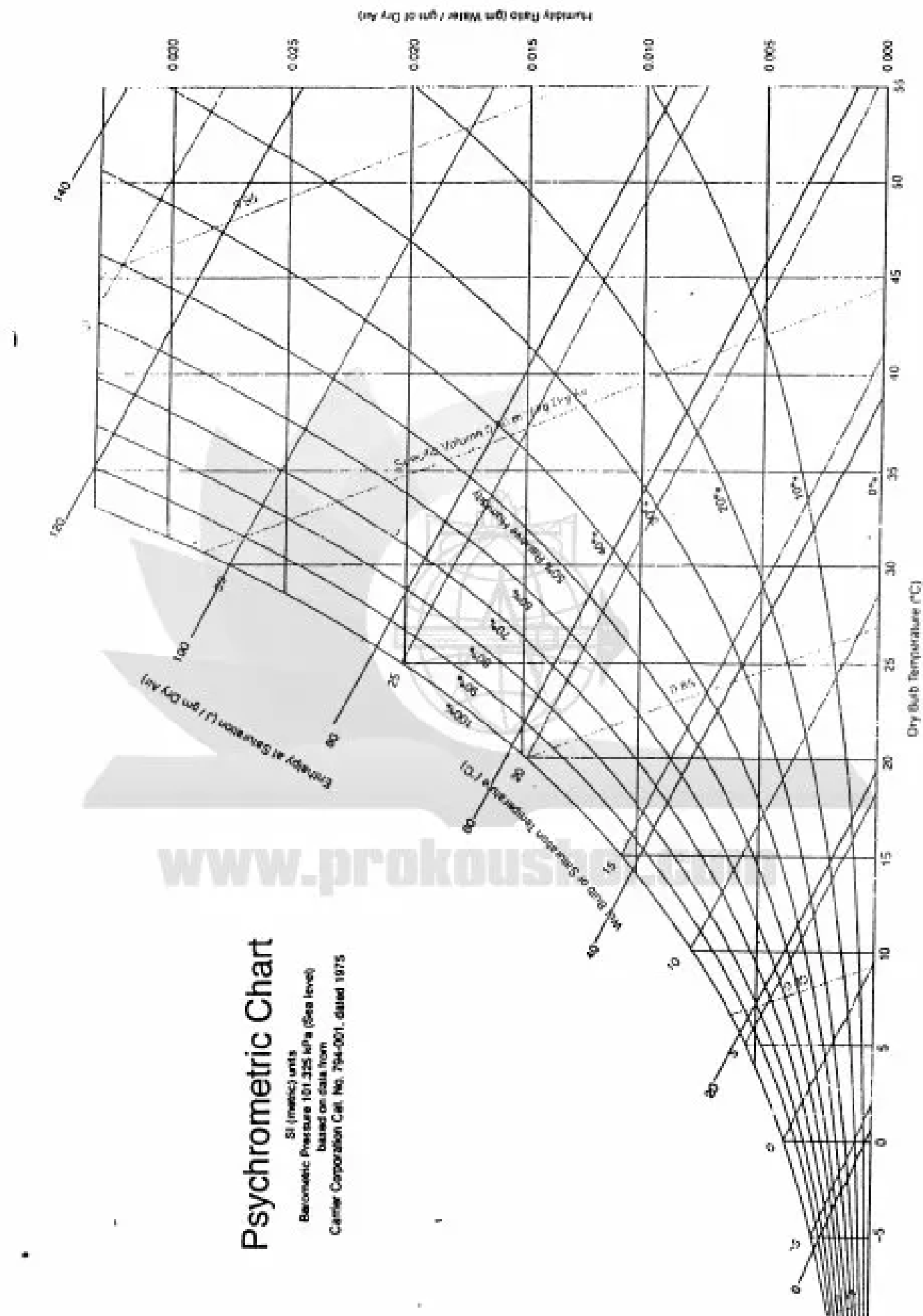


- Q28. (a) Mention desirable properties of a refrigerant. (8 marks)
 (b) What do you know about COP and RT? (8 marks)
 (c) Briefly describe the working principle of a vapour compression refrigeration system. (13 marks)
 (d) Draw the schematic of an all-air type central AC system. (6 marks)

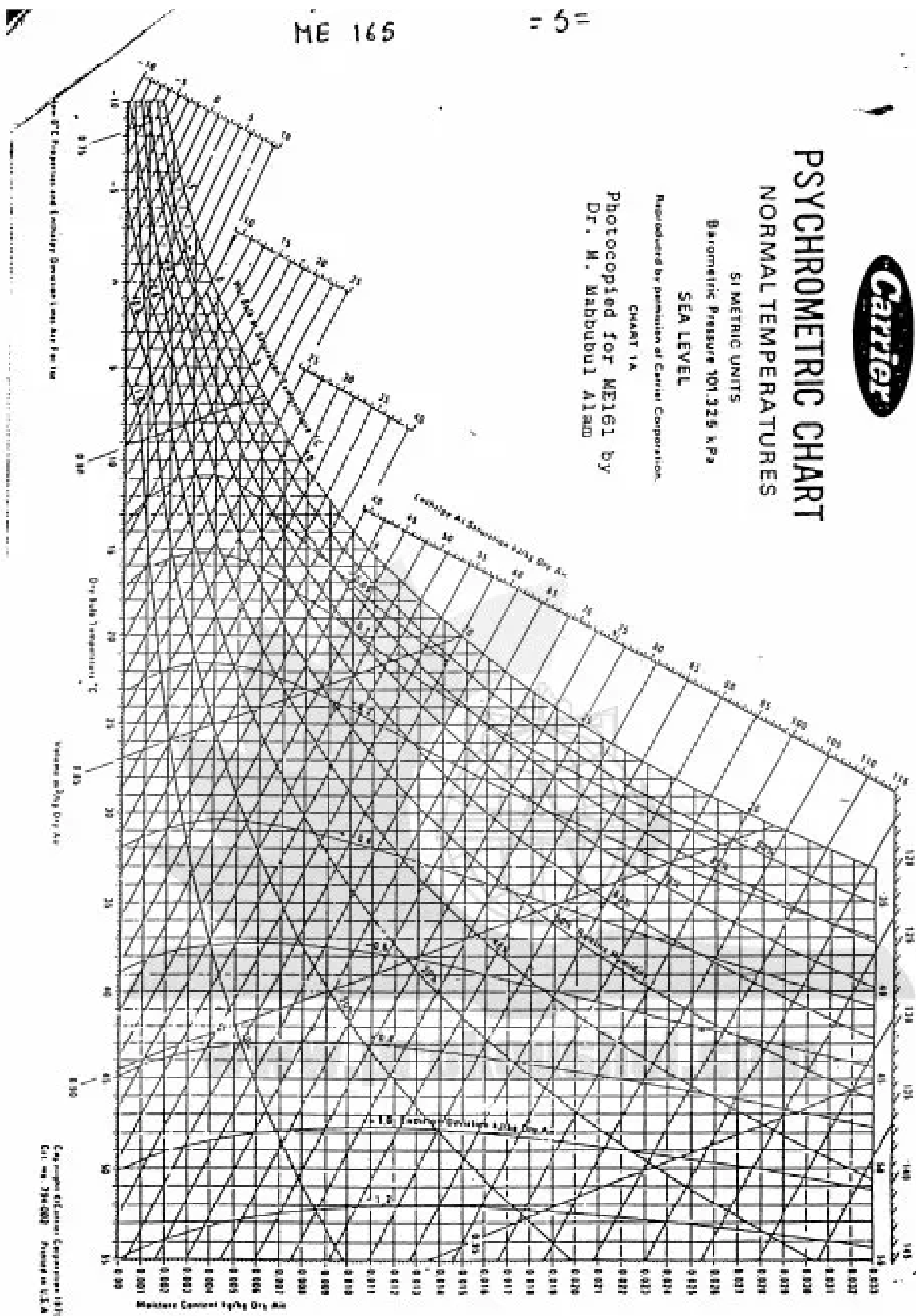
[2010–11]

- Q29. (a) What are the desirable properties of a good refrigerant? Write commercial name, chemical formula and chemical name of R₁₁ and R₁₁₃. (12 marks)

- (b) Describe a winter air-conditioning system with necessary sketch. **(11 marks)**
 (c) In a summer AC system, air is at 32°C WBT, 70% RH before cooling coil and exits at 18°C WBT, 50% RH. Calculate condensate formed and heat removed. **(12 marks)** [2008–09]



- Q30.** (a) Differentiate among refrigerator, heat engine, and heat pump. **(6 marks)**
 (b) What is an azeotrope? Give two examples. **(5 marks)**
 (c) Mention advantages of vapour absorption over vapour compression refrigeration. **(8 marks)**
 (d) In vapour compression refrigeration, why is pressure low in the evaporator and high in the condenser? **(6 marks)**
 (e) With a neat sketch, briefly describe the working principle of a summer AC system. **(10 marks)** [2006–07]
- Q31.** (a) What are the different processes in a standard vapour-compression refrigeration cycle? Show on P-h and T-s diagrams. **(12 marks)**
 (b) What are the desirable properties of a refrigerant? **(5 marks)**
 (c) Draw the simple vapour absorption system with its components. **(8 marks)**
 (d) Show cooling and dehumidification on a psychrometric chart. Draw the window air-conditioning unit. **(10 marks)** [2005–06]
- Q32.** (a) What are the basic components of a vapour compression refrigeration cycle? Show in schematic diagram and T-S / P-h diagrams. **(15 marks)**
 (b) Calculate capacity of a refrigerator that converts 50 kg of water into ice (both at 0°C) in 24 minutes. **(5 marks)**
 (c) A closed room initially at 47°C with 30% RH. After AC: WBT = 15°C, RH = 20%. With 50 kg initial air, find: (i) temperature reduction, (ii) moisture removed. **(15 marks)** [2004–05]



- Q33.** (a) Define coefficient of performance and refrigeration effect. **(6 marks)**
 (b) Write chemical name and formula of refrigerants: R22, R114, R744, R40. **(8 marks)**
 (c) Draw schematic of a refrigeration system with two compressors, one evaporator, and an intercooler. Show the P-h diagram. **(10 marks)**
 (d) Show on a psychrometric chart: (i) Cooling with dehumidification, (ii) Humidification at constant DBT, (iii) Mixing in an air handling unit. **(11 marks)**
 [2003–04]

BUET · Level 1 · Term 2

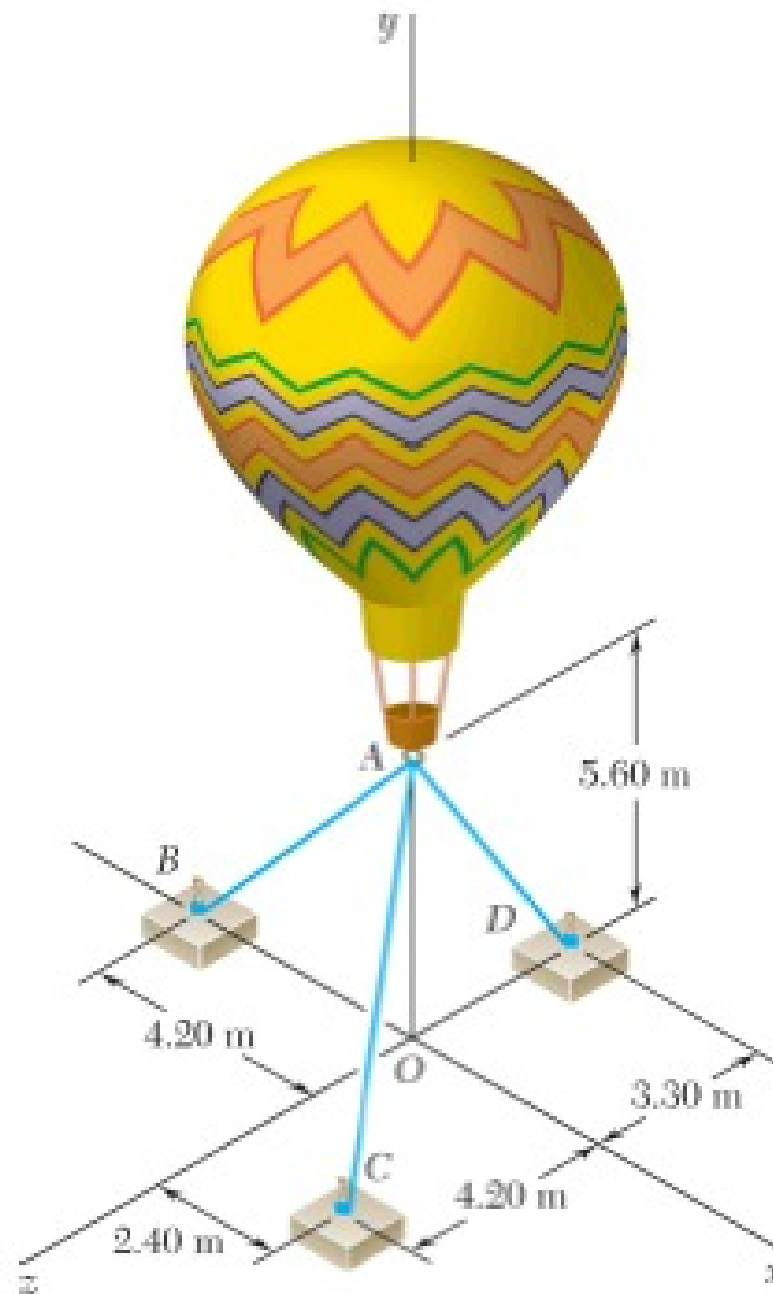
MECHA 165

Section B1

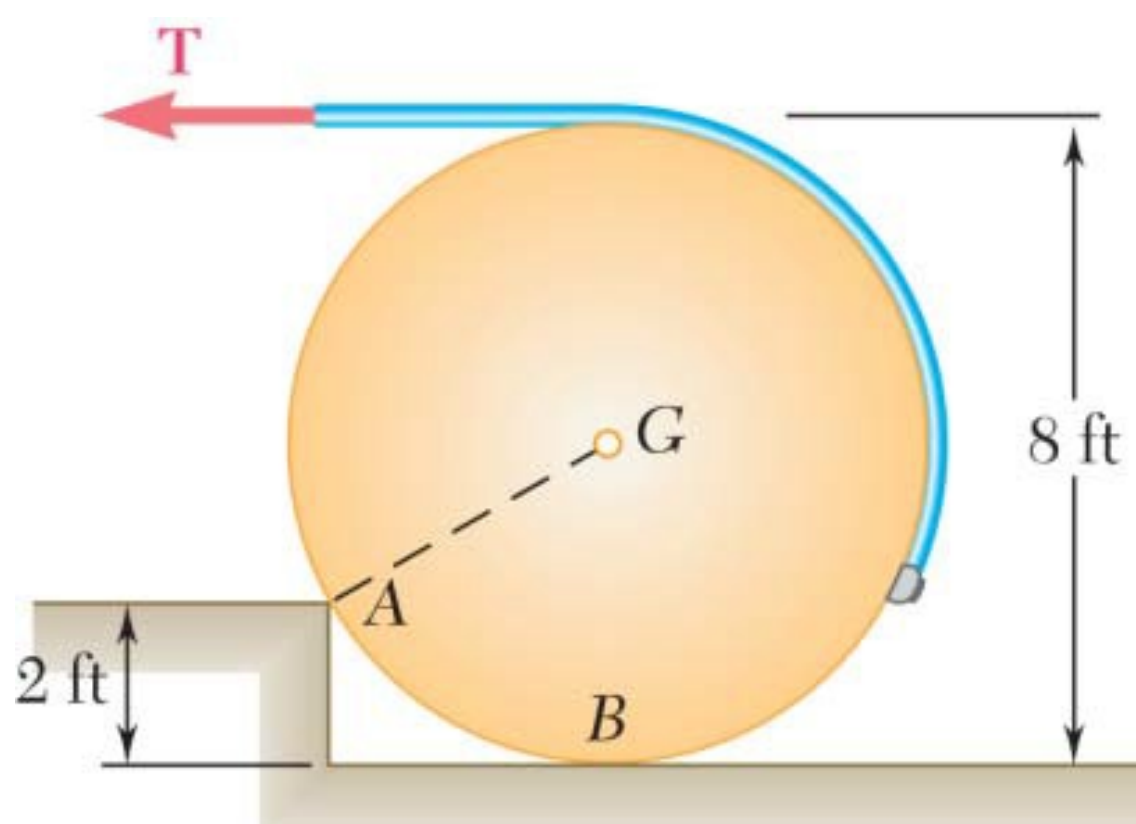
Statics of Particles & Rigid Bodies

B1 — Statics of Particles & Rigid Bodies

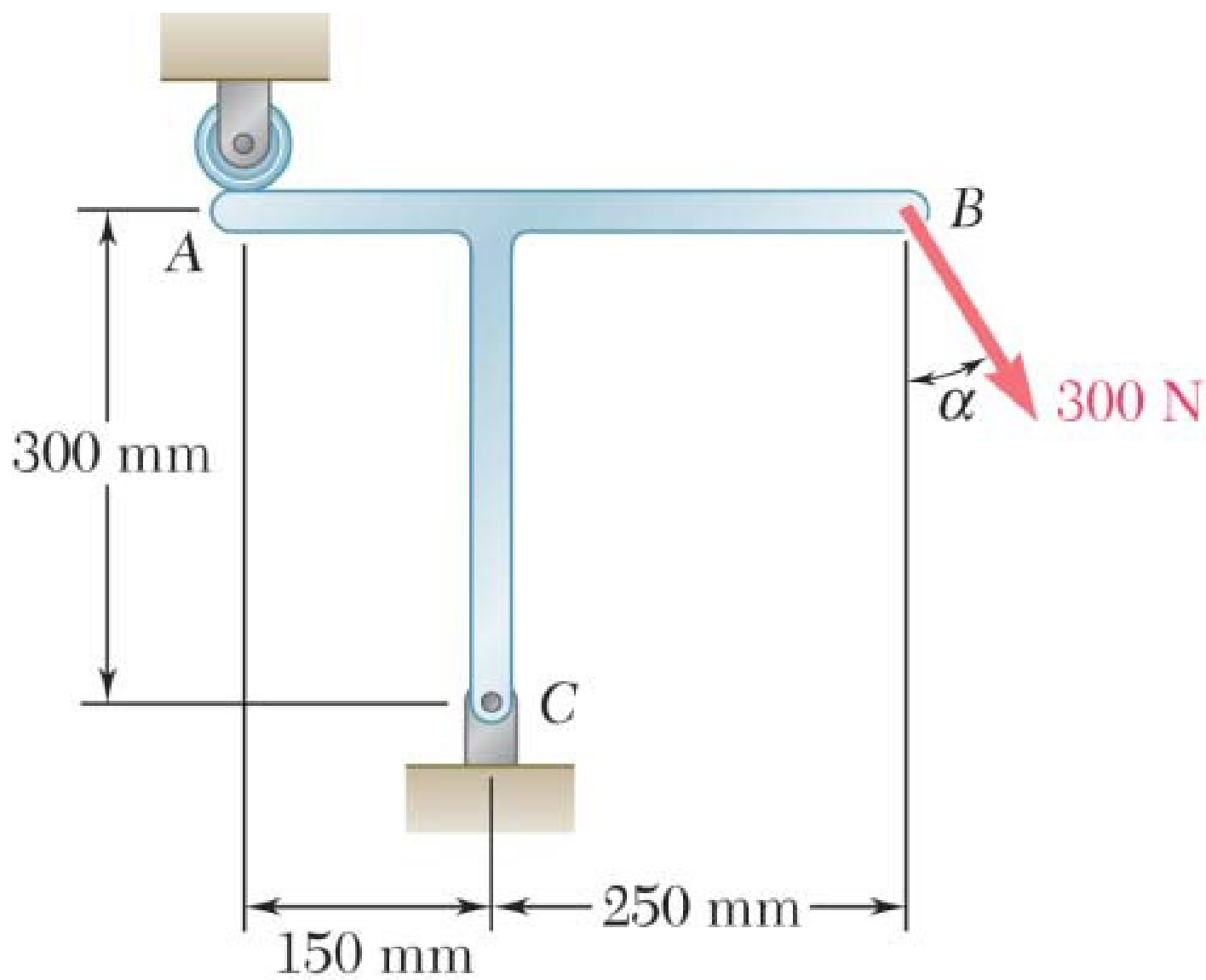
- Q1.** Three cables tether a balloon. Tension in cable AD = 481 N. Determine the vertical force exerted by the balloon at A. **(15 marks)**
 [2023–24 , 2020–21, 2018–19]



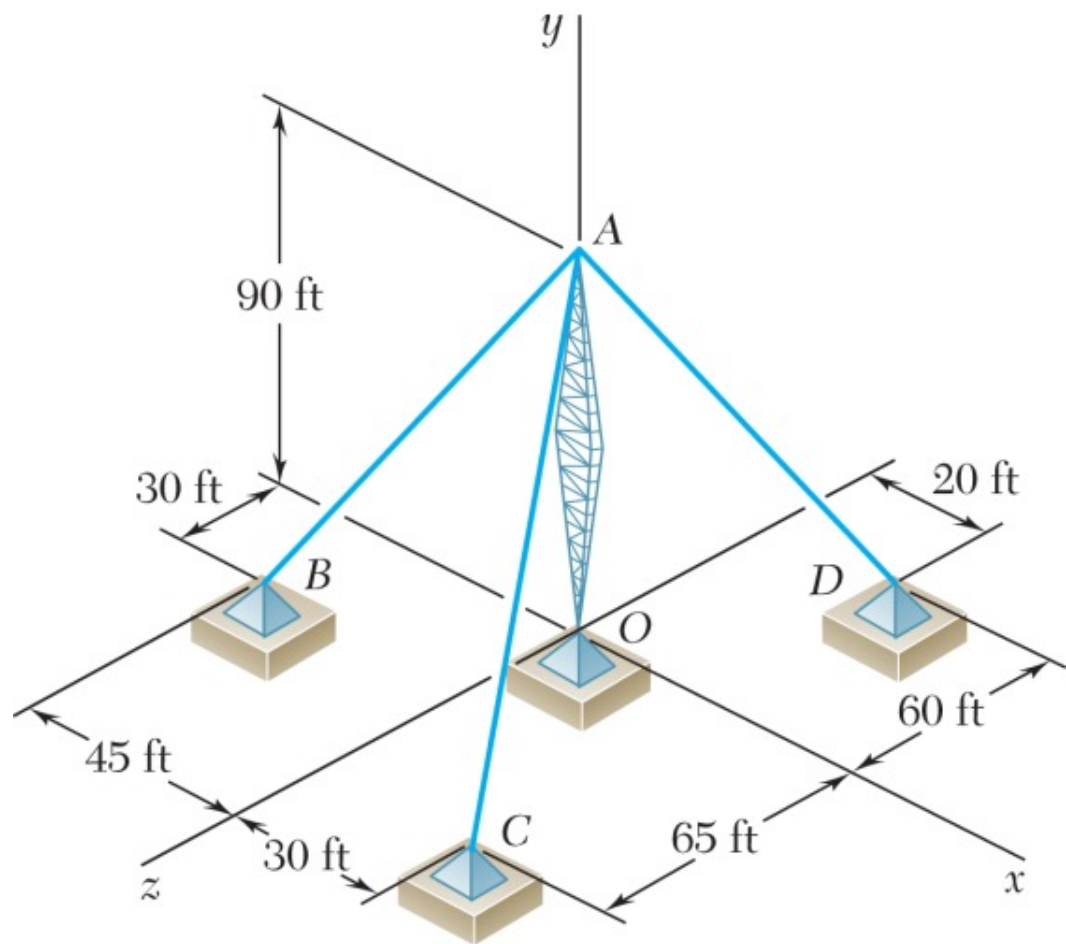
- Q2.** A 500-lb cylindrical tank, 8 ft in diameter, is to be raised over a 2-ft obstruction. A cable is wrapped around the tank and pulled horizontally. Find the required tension in the cable and the reaction at A. **(15 marks)**
 [2023–24]



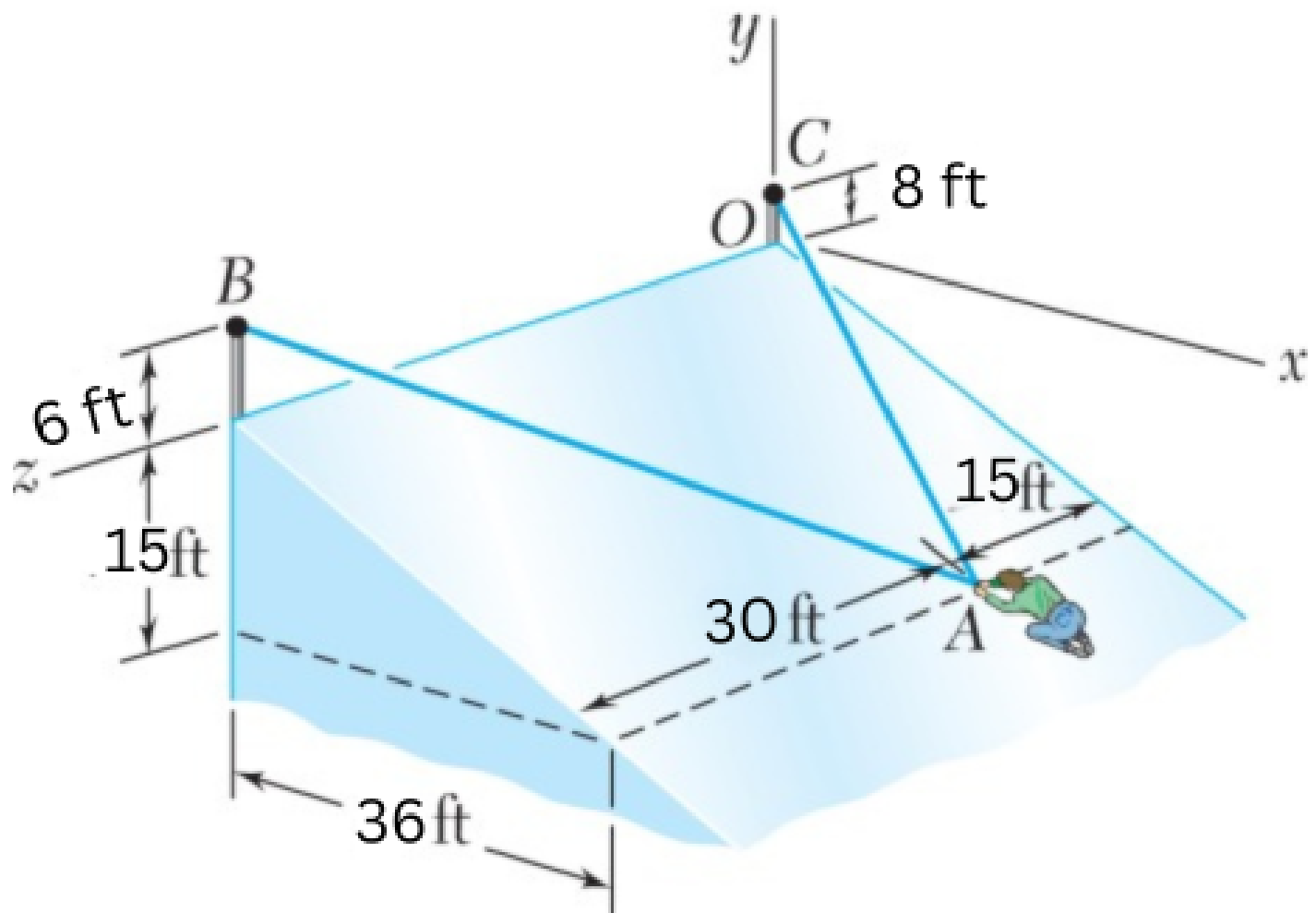
- Q3.** T-shaped bracket supports a 300-N load. Determine reactions at A and C when $\alpha = 60^\circ$. **(15 marks)**
 ↪ Similar: B1-Q20 same bracket, $\alpha = 45^\circ$



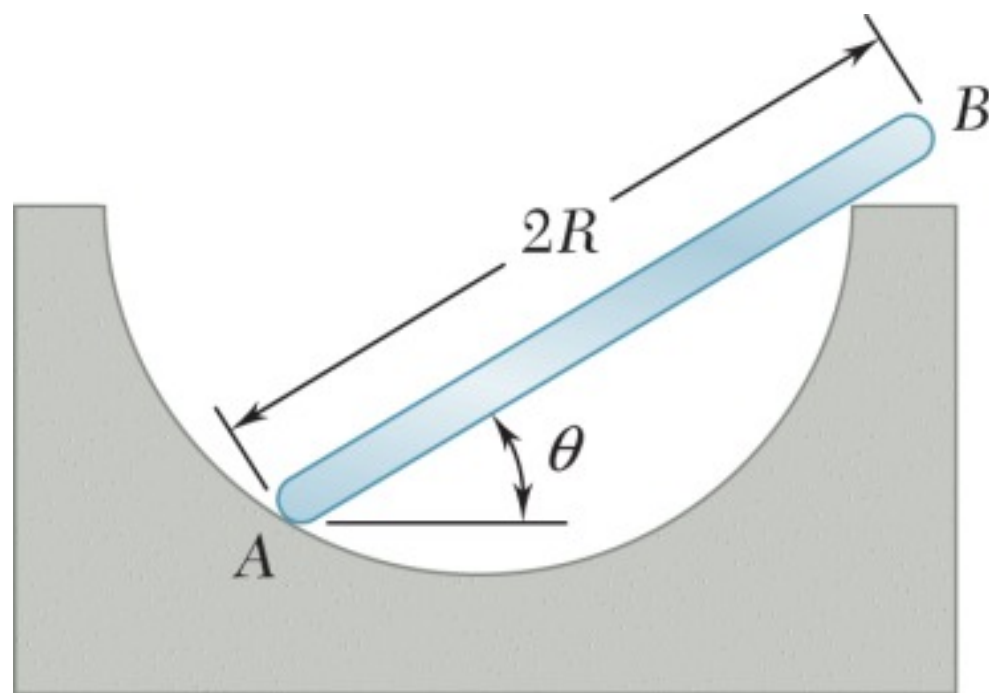
Q4. Transmission tower held by three guy wires at A anchored at B, C, D. Tension in wire AB = 630 lb. Determine vertical force P exerted by tower on pin at A. (15 marks) [2023–24]



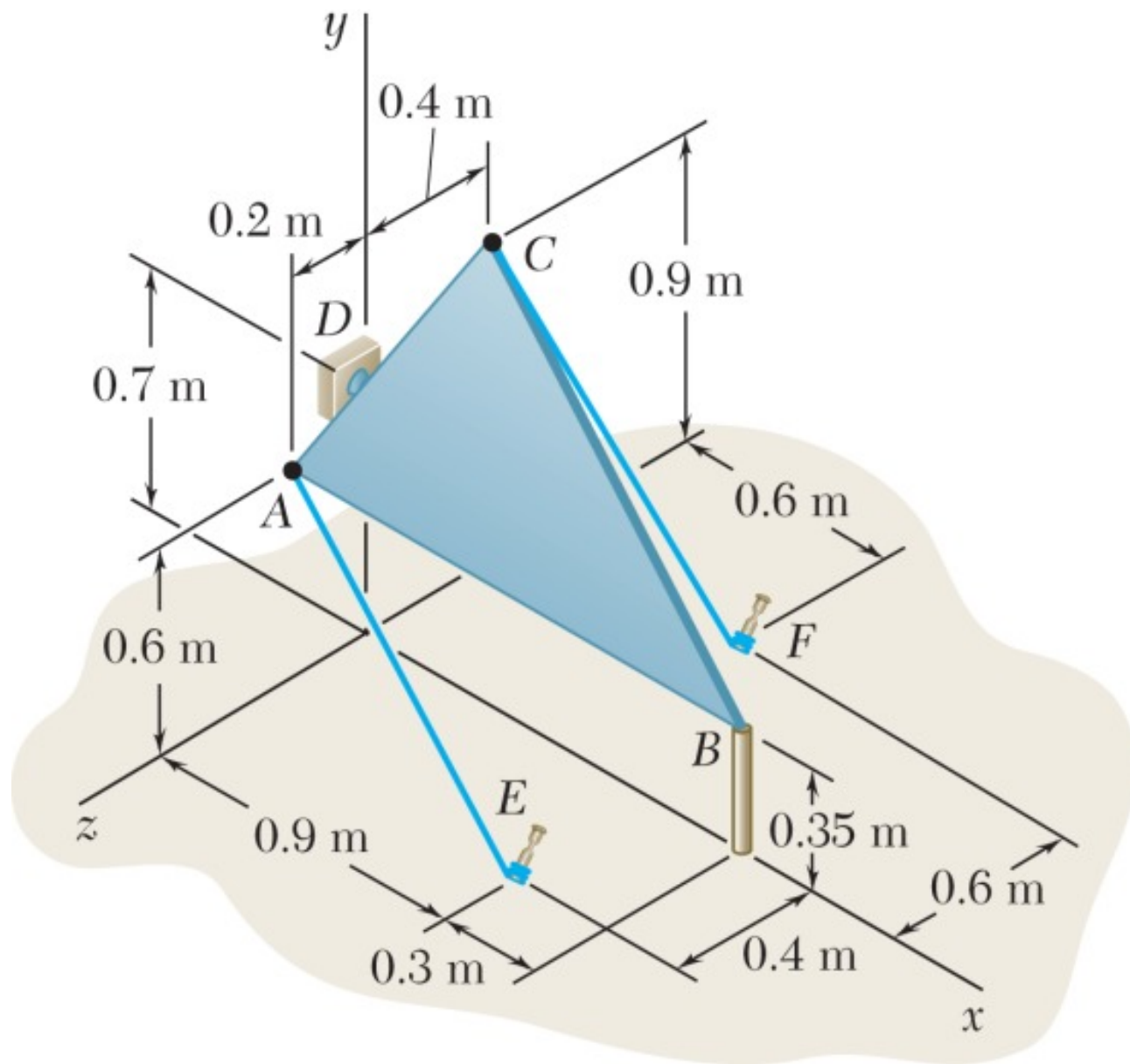
Q5. In trying to move across an icy surface, a 180-lb man uses two ropes AB and AC. A friend pulls at A with $P = -(48 \text{ lb})\hat{k}$. Determine tension in each rope. (17 marks) [2021–22]



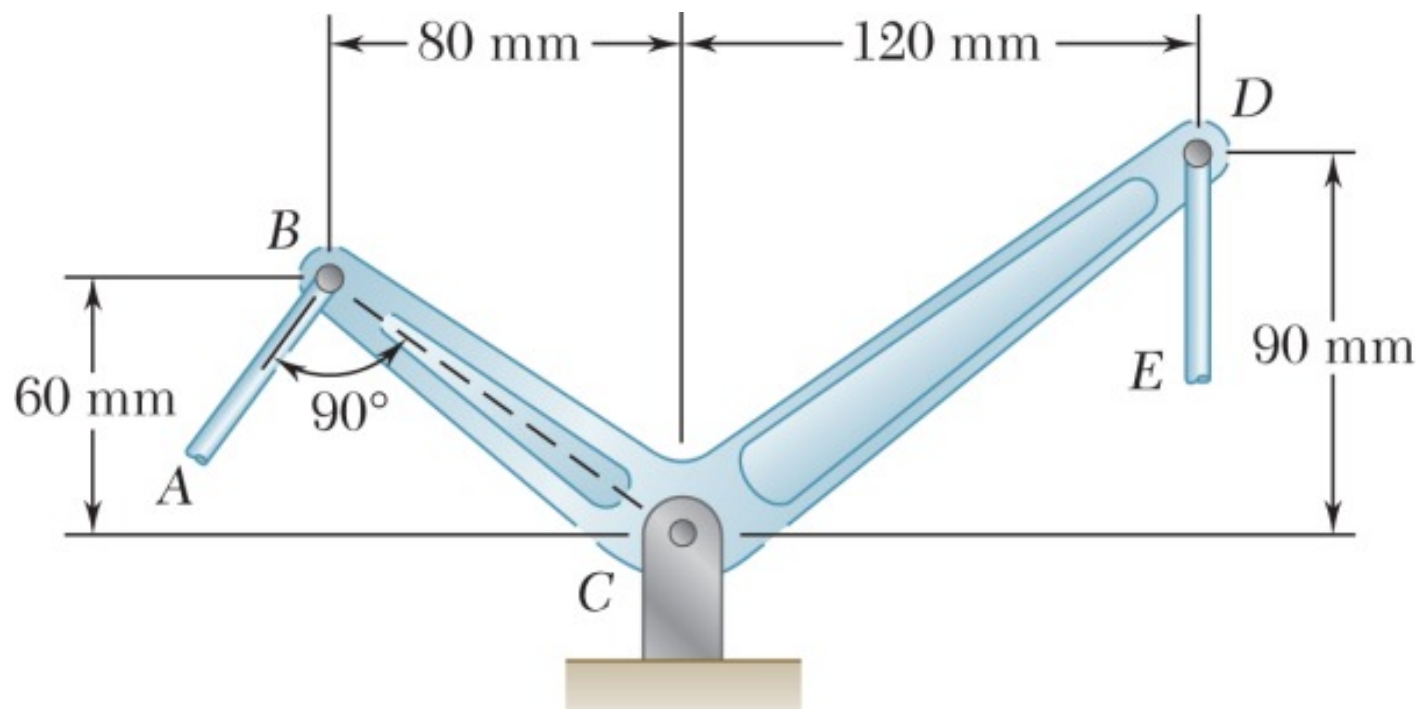
Q6. A uniform rod AB of length $2R$ and weight W rests inside a hemispherical bowl of radius R . Neglecting friction, determine the equilibrium angle θ . **(15 marks)** [2021–22]



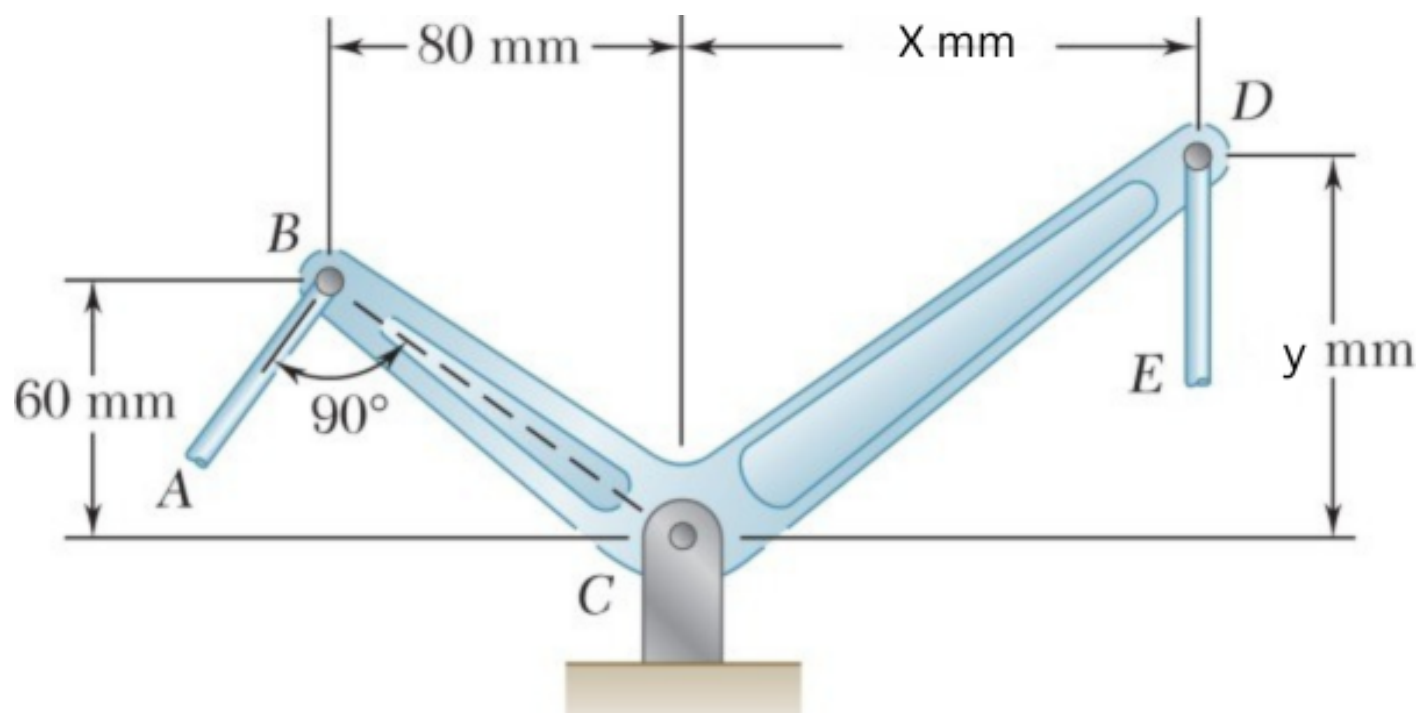
Q7. Triangular plate ABC supported by ball-and-socket joints at B and D, held by cables AE and CF. Force in cable AE = 55 N. Determine force in cable CF. **(18 marks)** [2021–22]



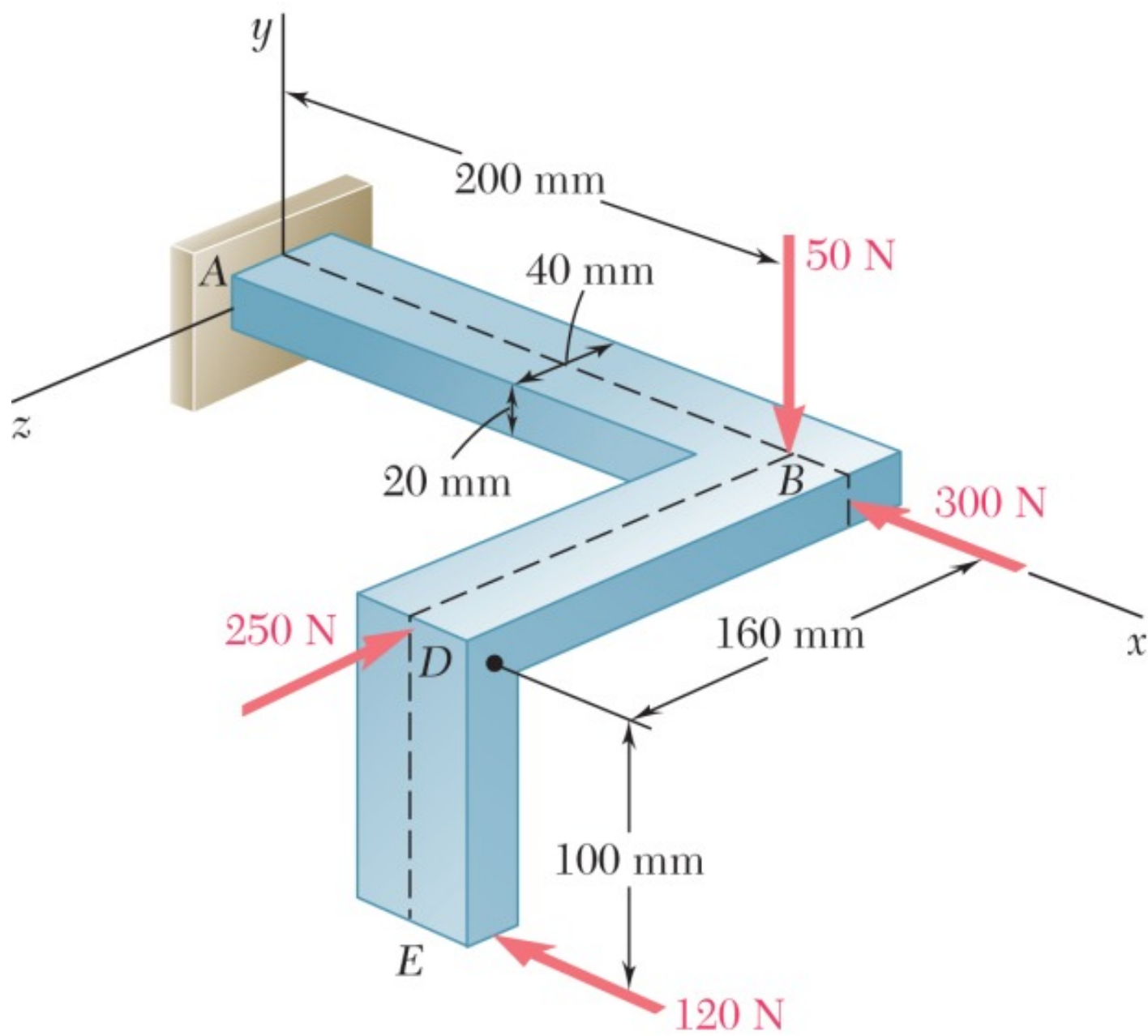
- Q8.** Why is equilibrium of particles different from equilibrium of rigid bodies? Explain briefly with an example. (5 marks) [2019–20, 2020–21]
- Q9.** Two links AB and DE connected by a bell crank. Tension in link AB = 720 N. Determine: (i) tension in link DE, (ii) reaction at C. (14 marks) [2020–21]



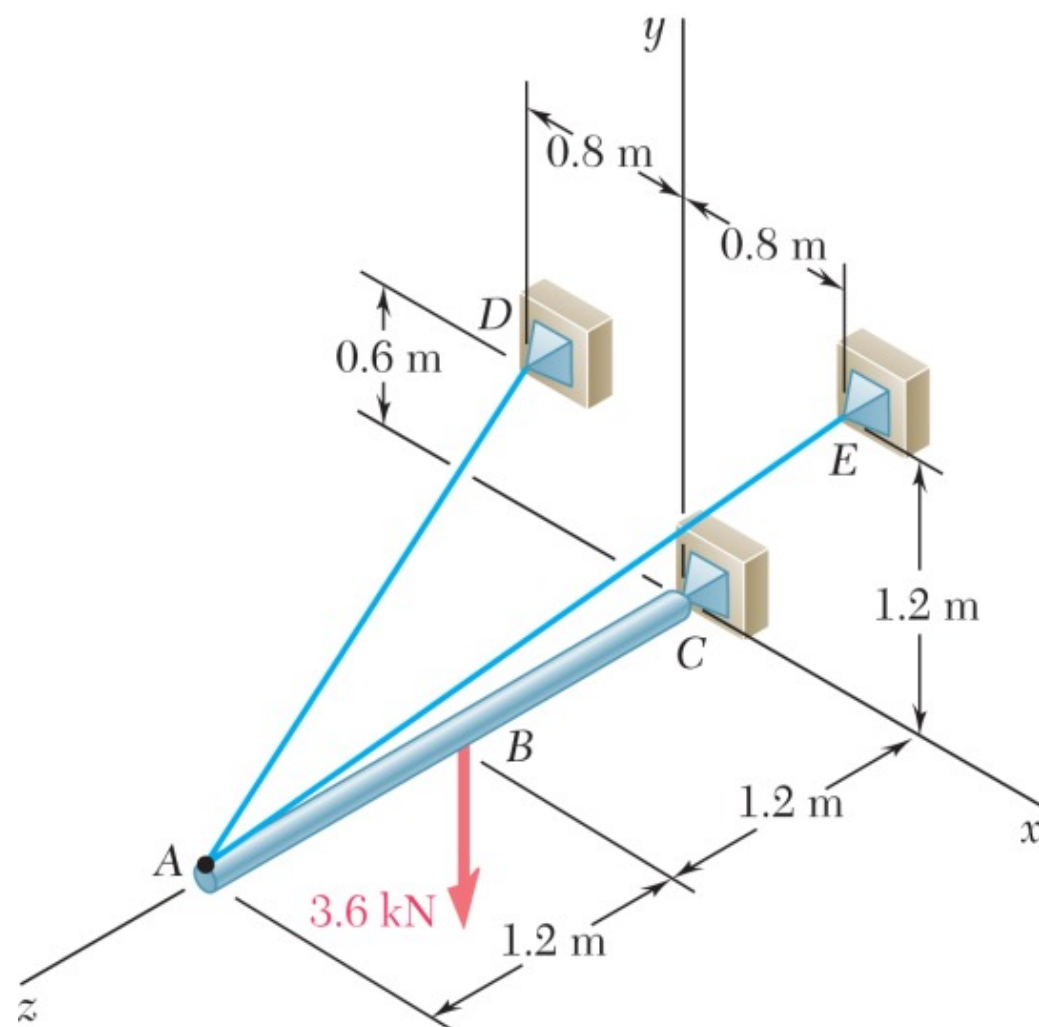
- Q10.** Two links AB and DE are connected by a bell crank. Tension in link AB = P N where $P = 450 + 3 \times (\text{last 3 digits of student ID})$. Determine: (a) tension in link DE (vertical), (b) reaction at C. (25 marks) [2019–20]



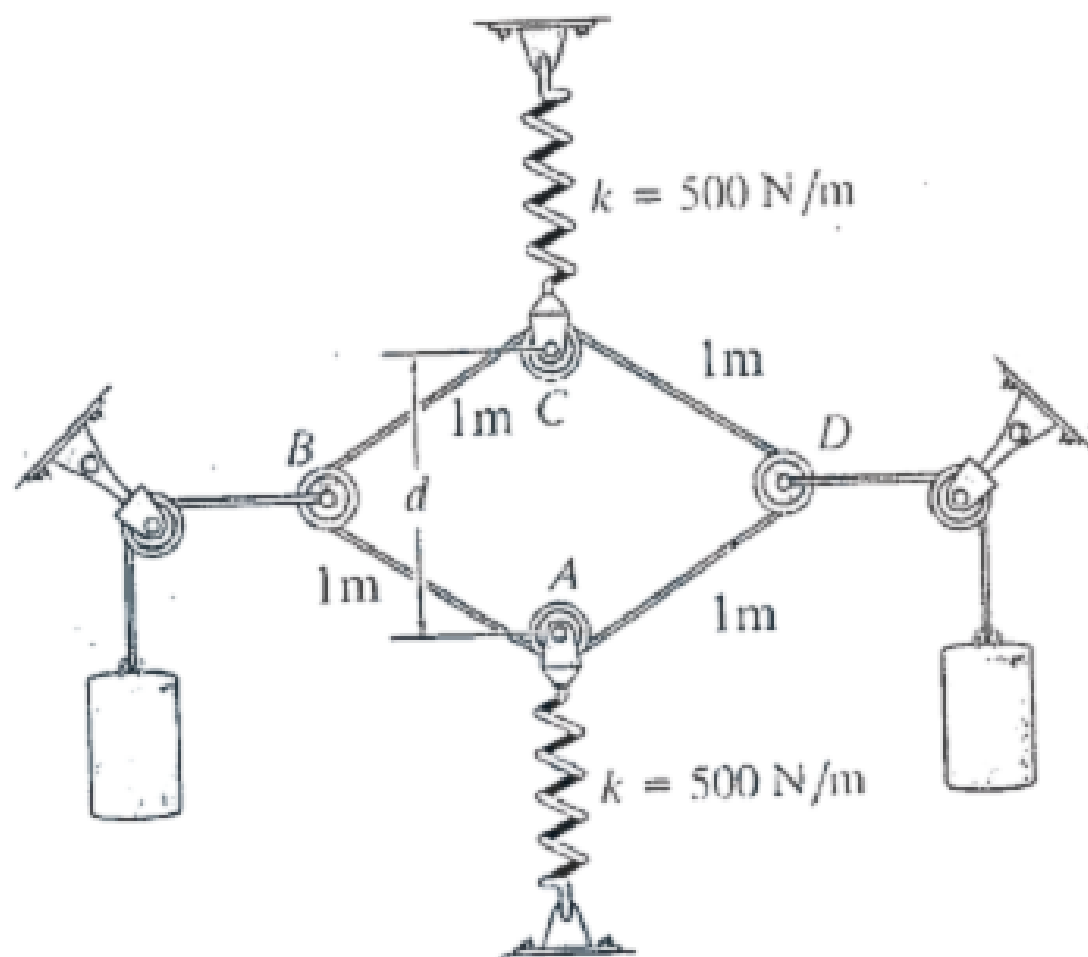
Q11. Four forces applied to machine component ABDE. Replace these forces with an equivalent force-couple system at A. **(20 marks)**
[2016–17, 2018–19]



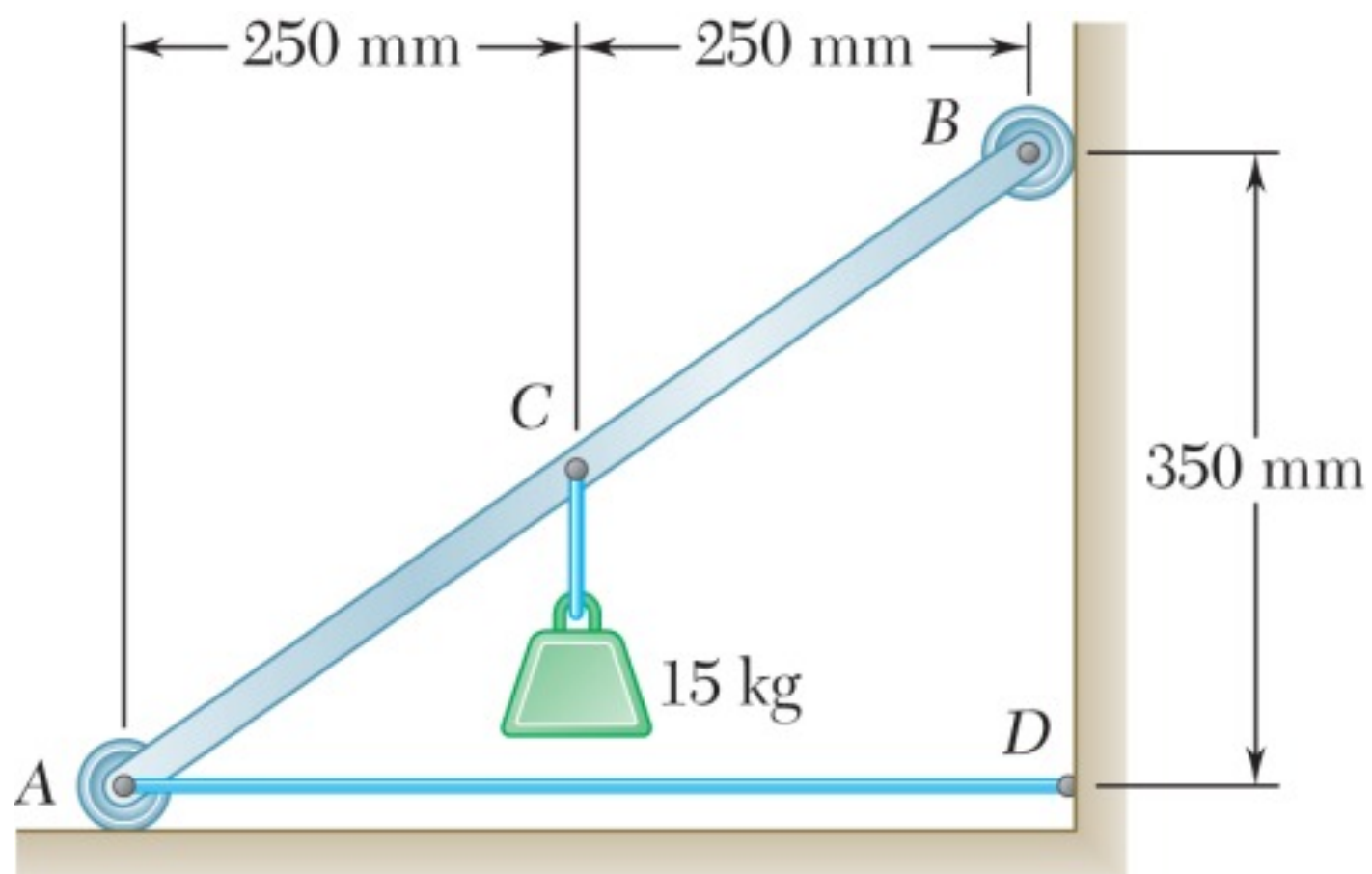
Q12. A 2.4-m boom held by ball-and-socket joint at C and by cables AD and AE. Determine tension in each cable and reaction at C. **(20 marks)**
[2016–17, 2018–19]



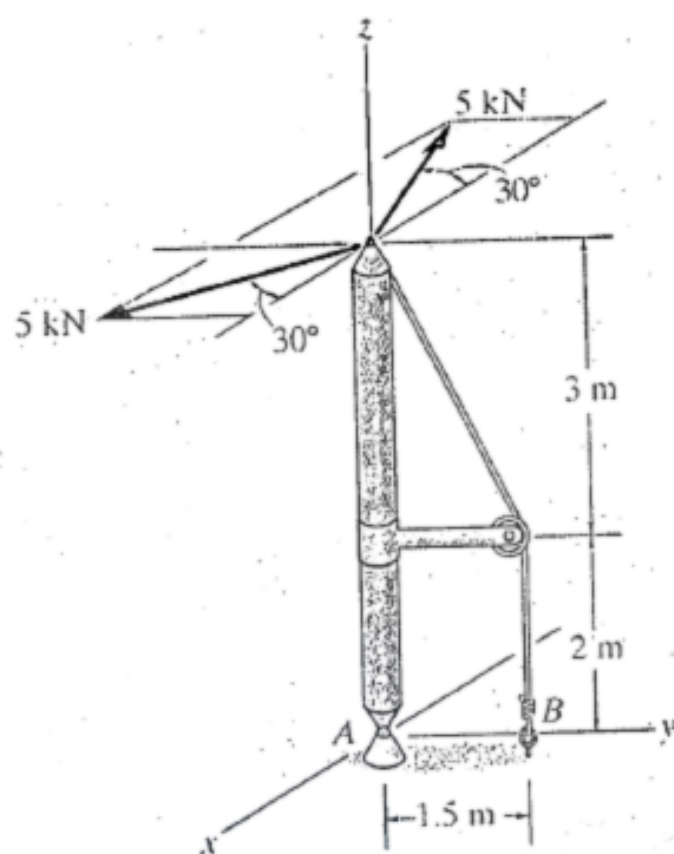
- Q13.** A continuous cable of total length 4 m is wrapped around pulleys at A, B, C, D. Each spring is stretched 300 mm. Springs unstretched when $d = 2$ m. Determine mass of each block. (20 marks) [2017–18]



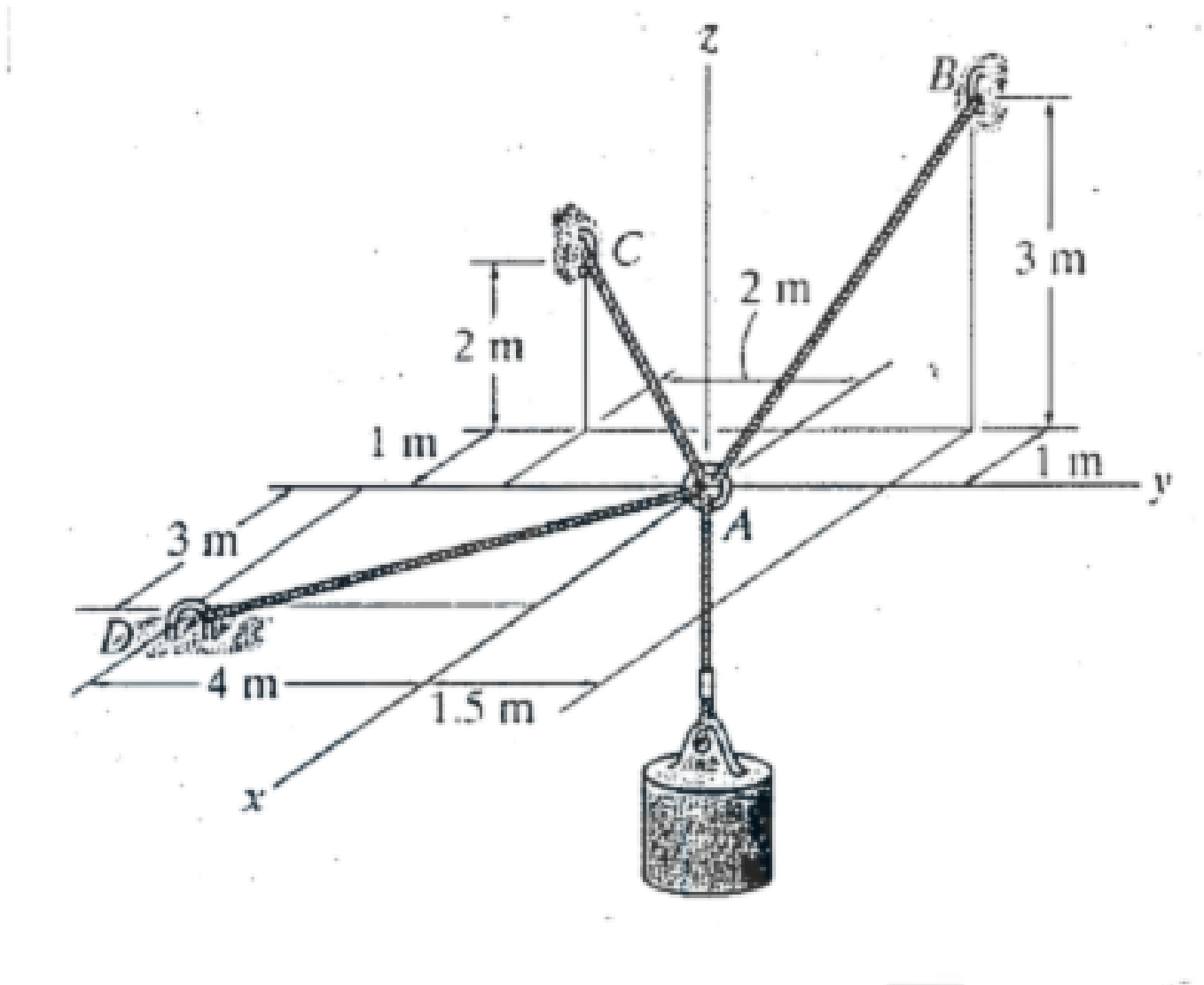
- Q14.** Light bar AB supports a 15-kg block at midpoint C. Rollers at A and B rest against frictionless surfaces; horizontal cable AD attached at A. Determine: (a) tension in cable AD, (b) reactions at A and B. (15 marks) [2017–18]



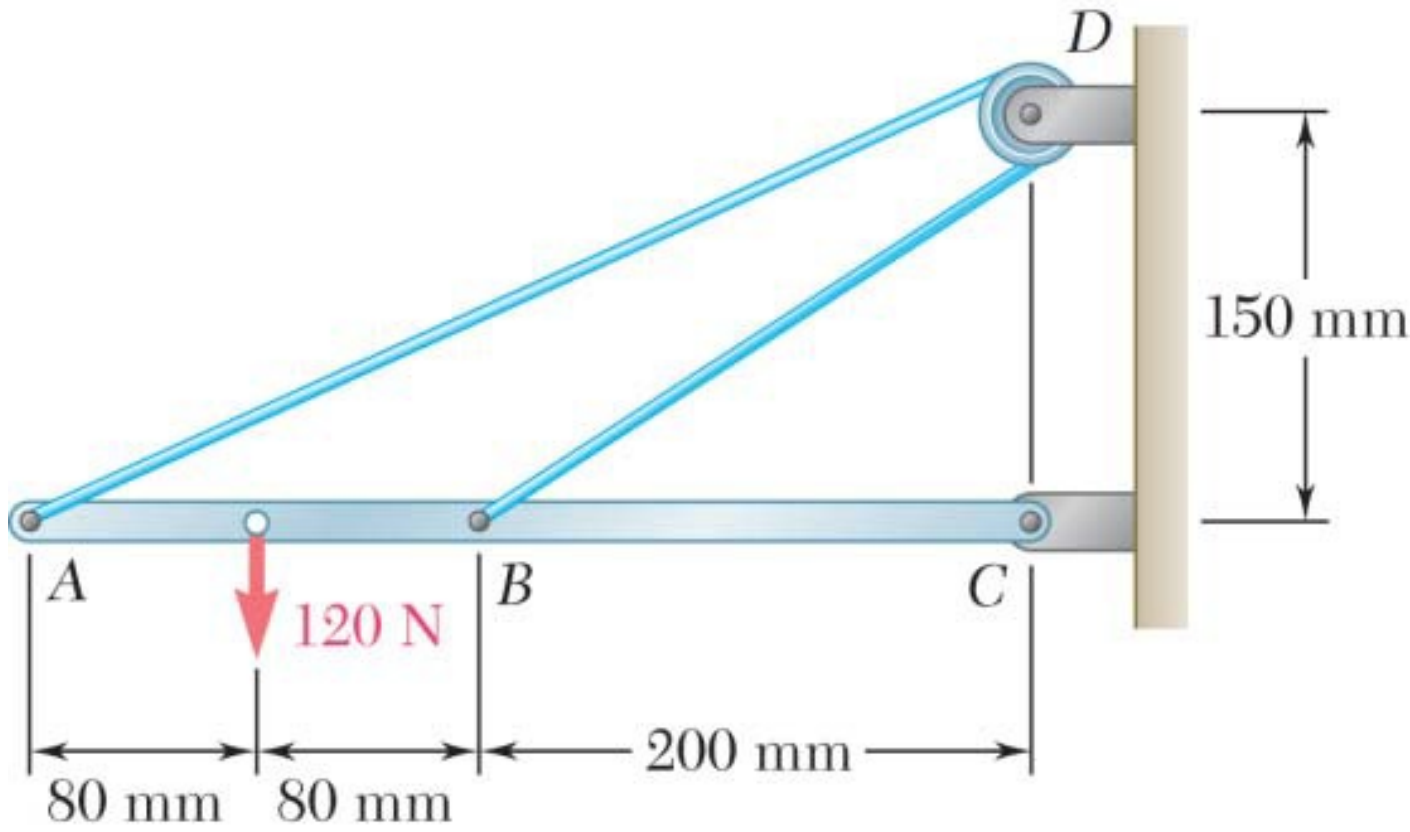
Q15. Boom supported by ball-and-socket joint at A and guy wire at B. Two 5-kN loads lie in a plane parallel to x-y plane. Determine x, y, z components of reaction at A and tension in cable. **(17 marks)** [2017–18]



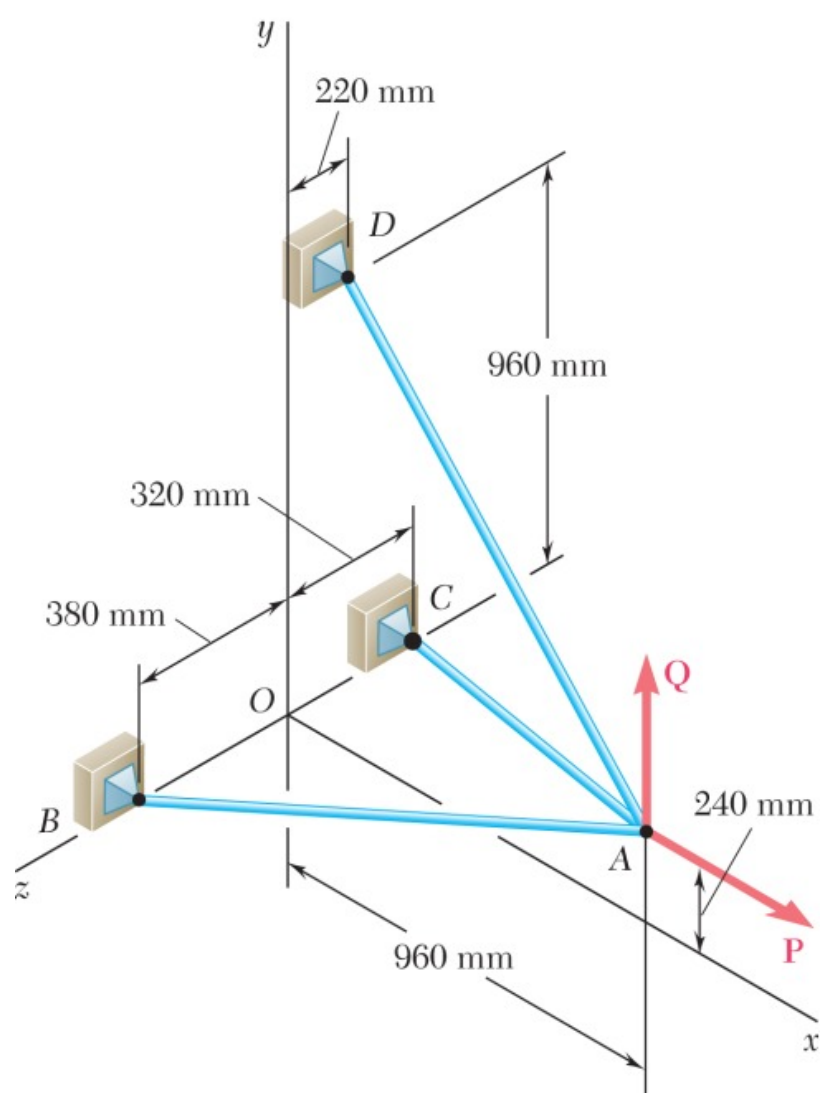
Q16. A cable system maintains a cylinder in equilibrium. Tension in cable AB = 1000 N. Determine the mass of the cylinder. **(15 marks)** [2016–17]



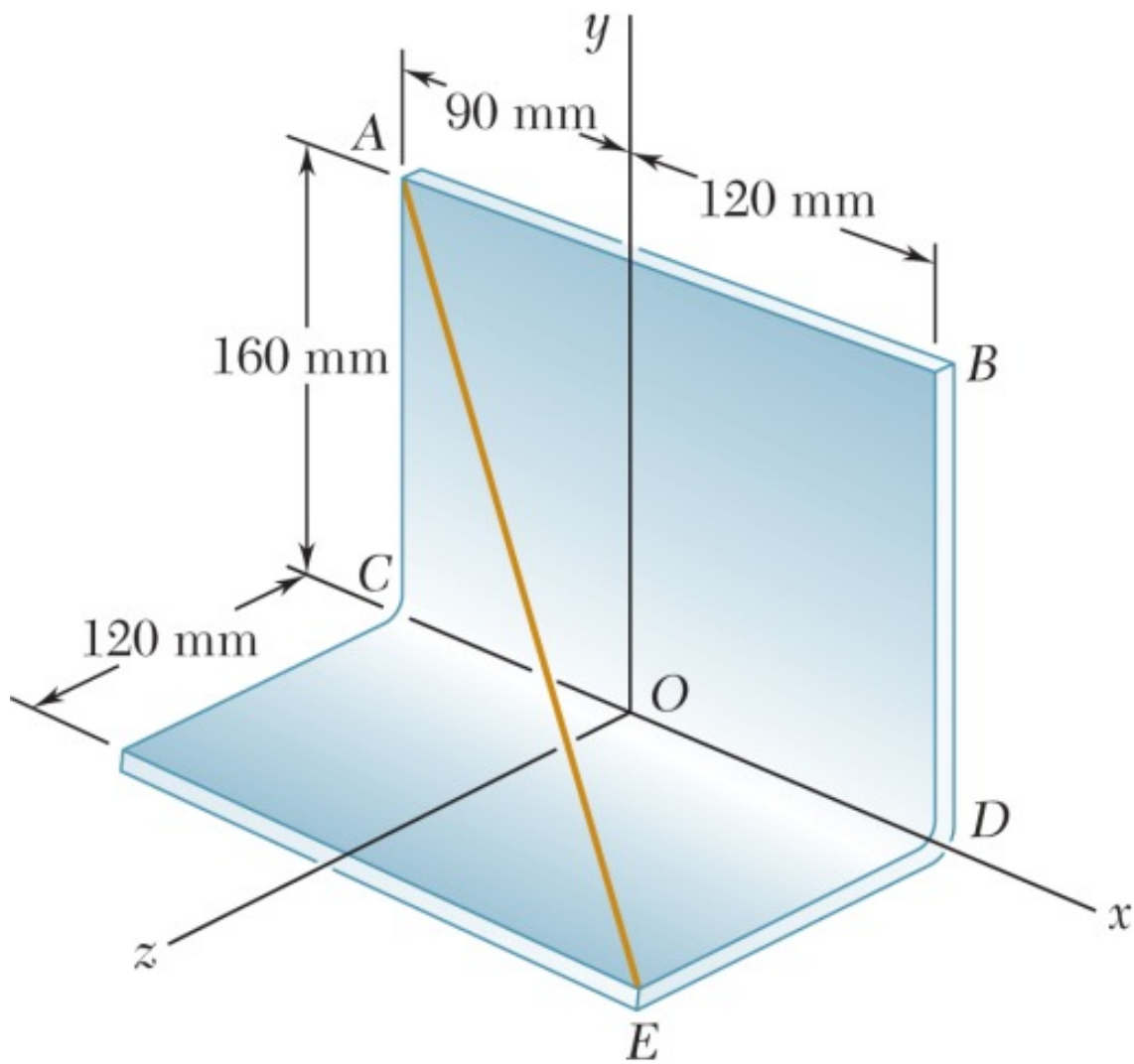
Q17. Neglecting friction and radius of pulley, determine: (i) tension in cable ADB, (ii) reaction at C. (15 marks) [2016–17]



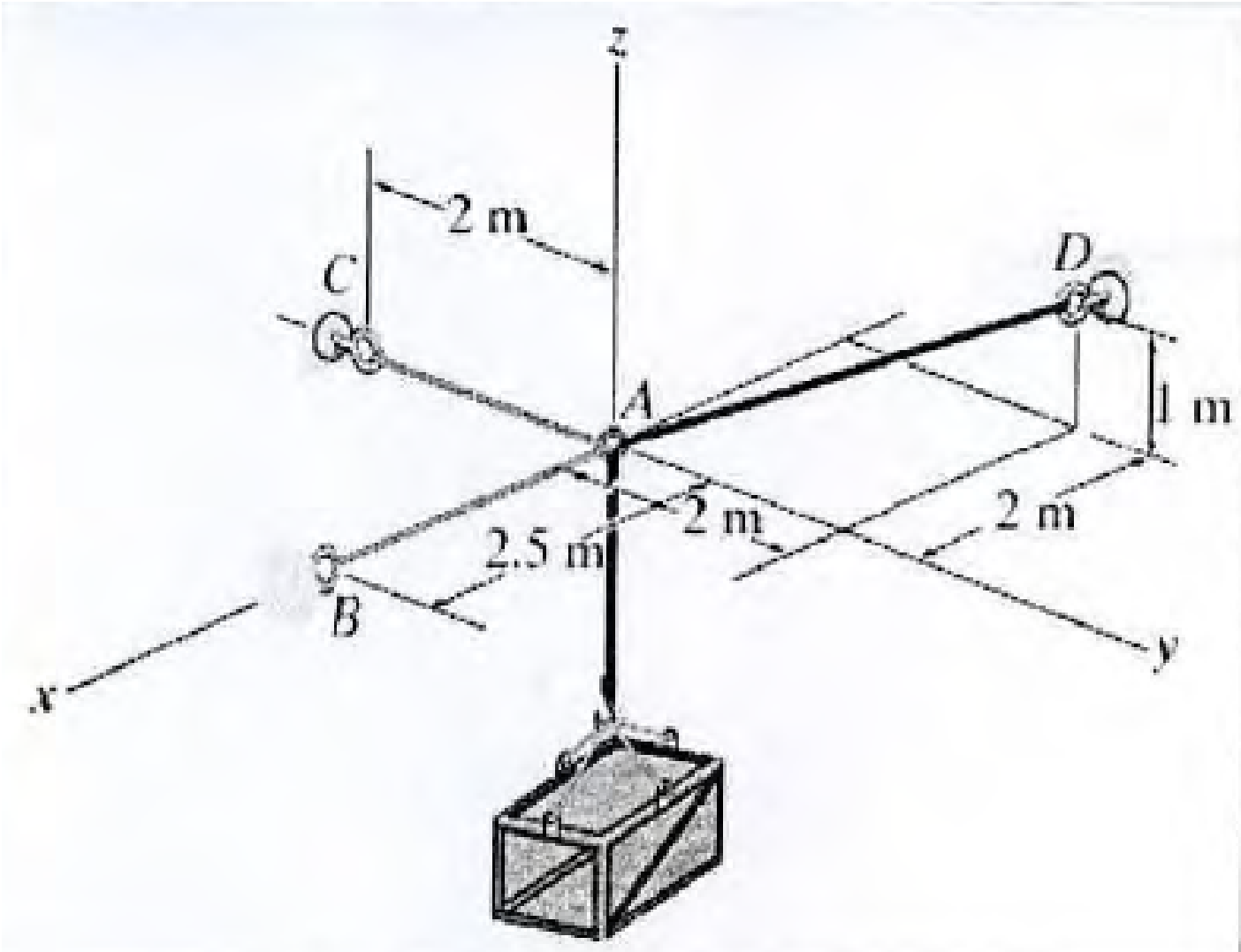
Q18. Three cables connected at A where force P is applied. Find value of P for which tension in cable AD is 305 N. (15 marks) [2015–16]



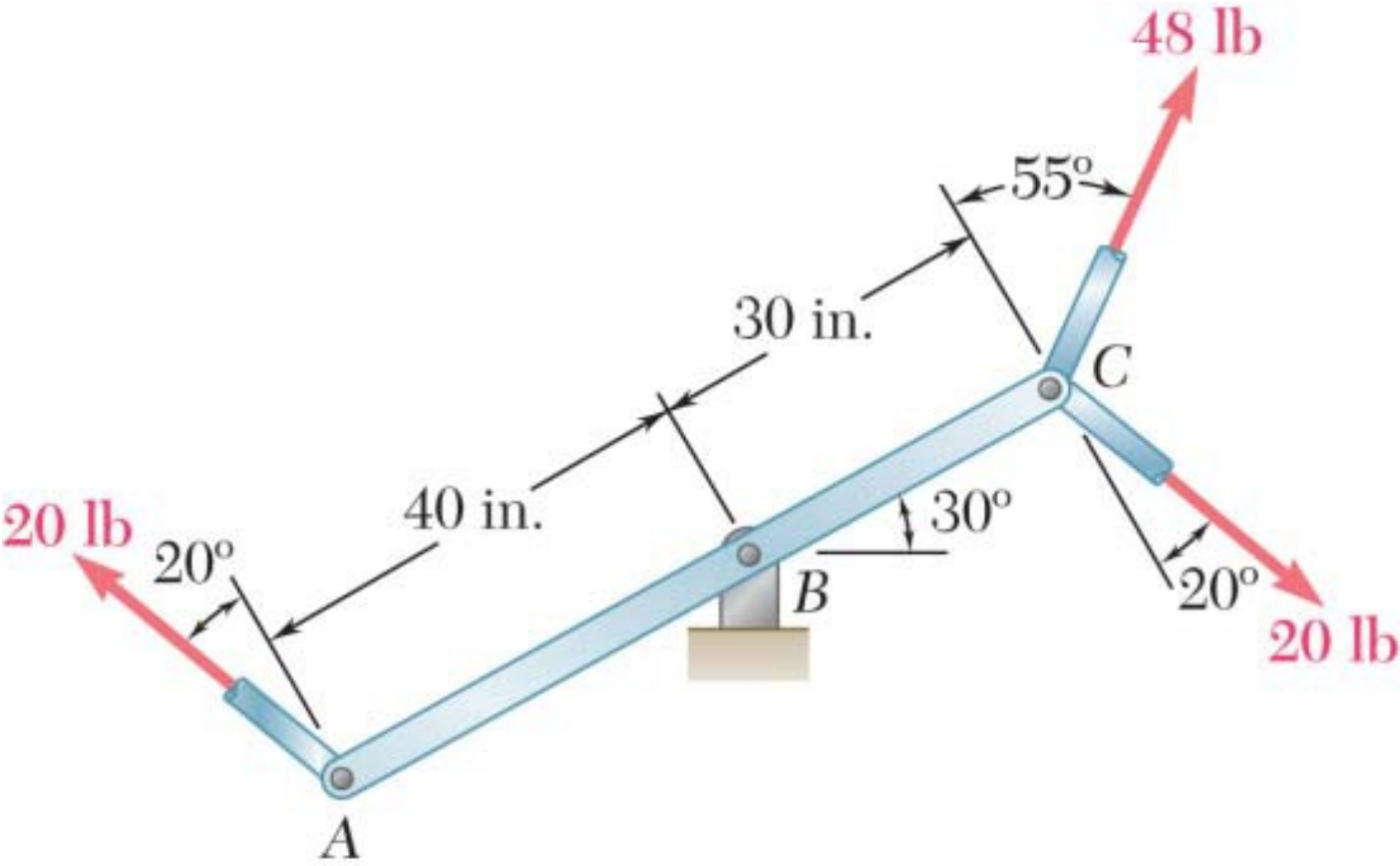
Q19. Wire AE stretched between corners A and E of a bent plate. Tension = 435 N. Determine: (i) moment about O , (ii) moment about each coordinate axis. (20 marks) [2015–16]



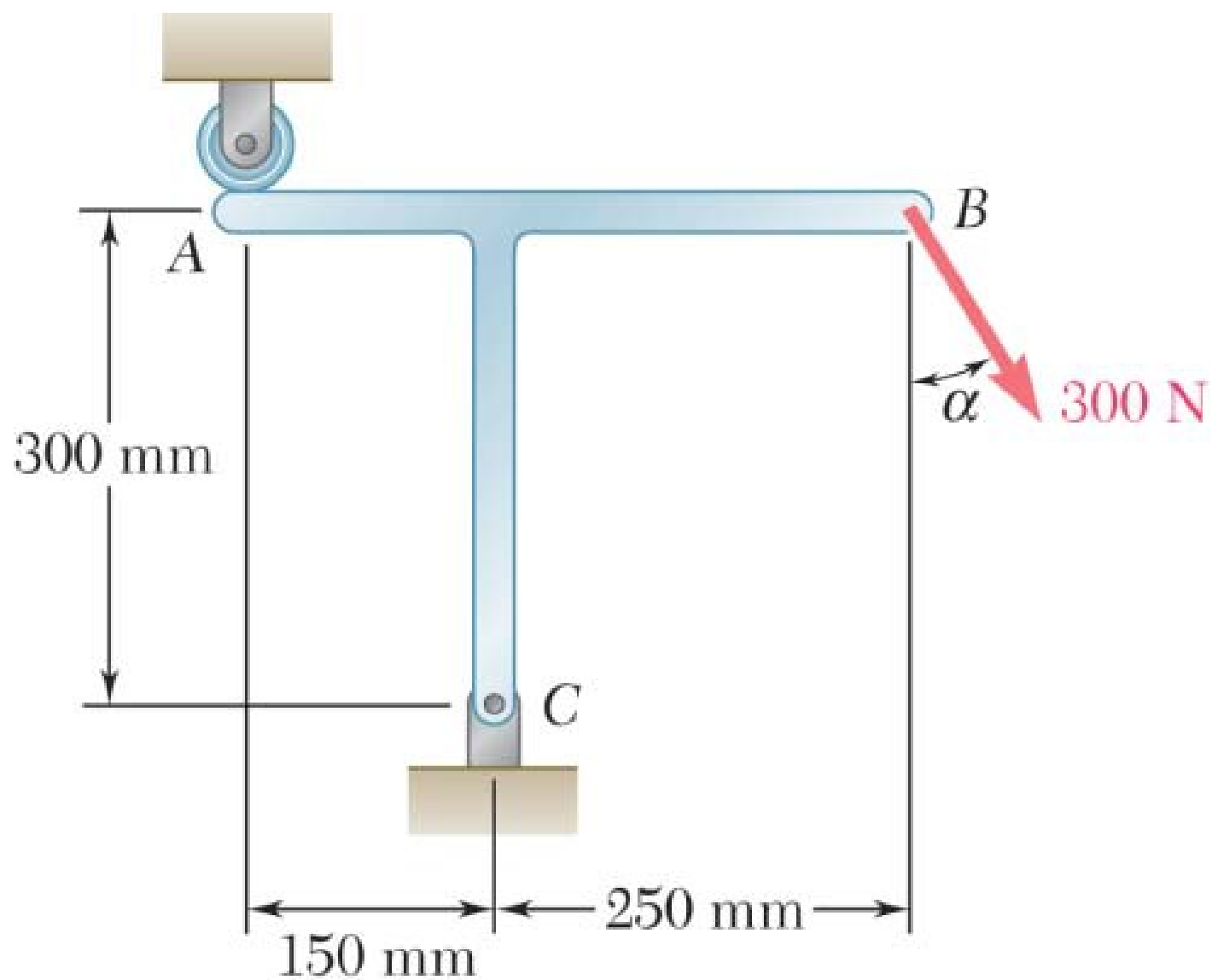
Q20. Determine tension in cables to support the 100-kg crate in equilibrium. (15 marks) [2014–15]



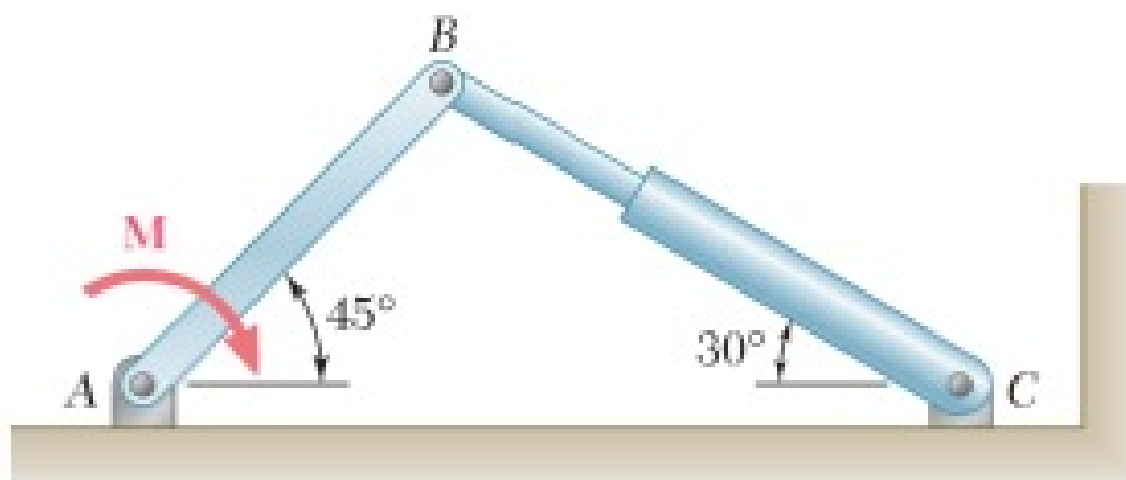
Q21. Three control rods attached to lever ABC exert forces shown. (a) Replace with equivalent force-couple system at B. (b) Determine single equivalent force and its point of application on BC. (20 marks) [2014–15]



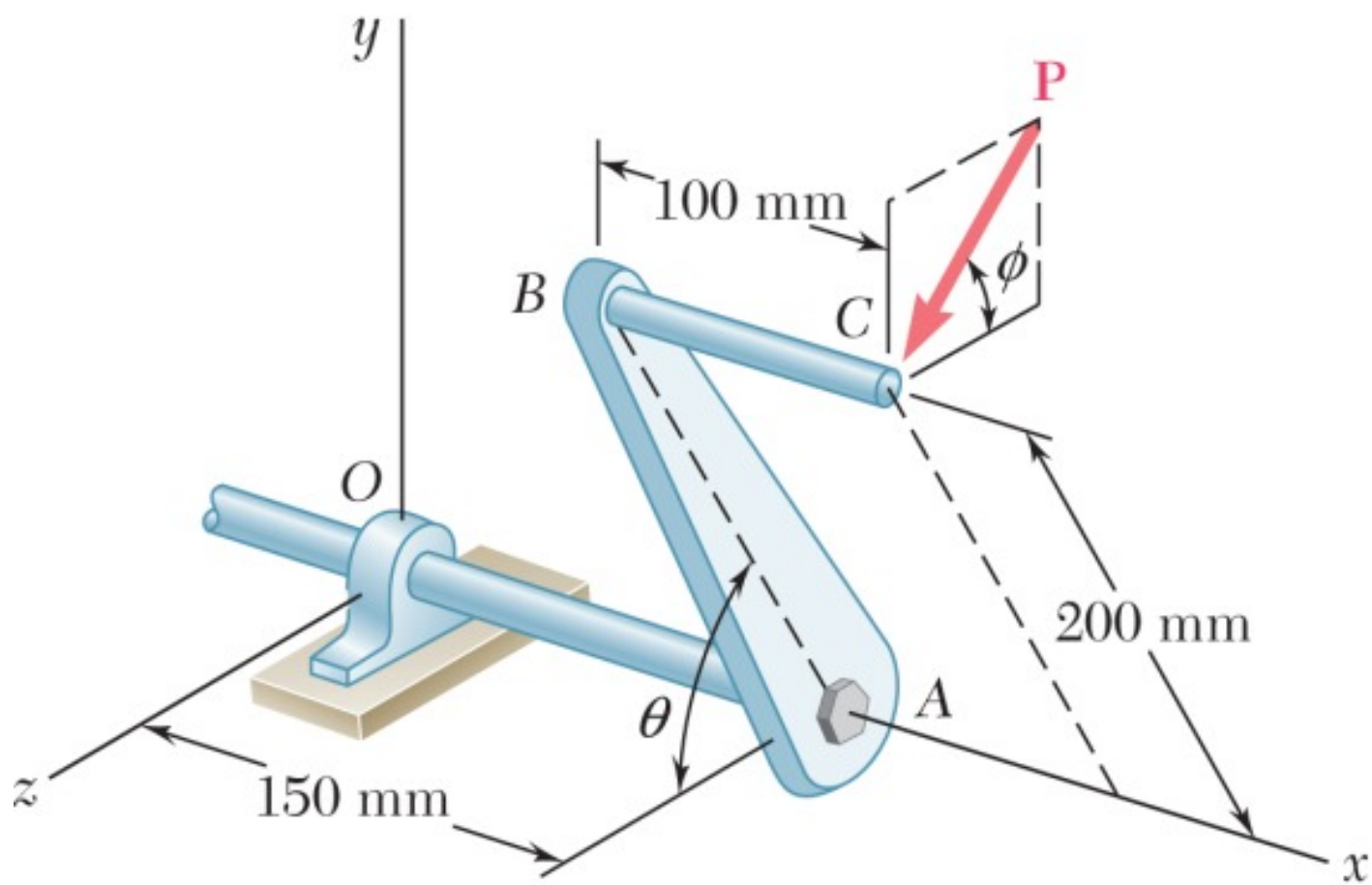
Q22. T-shaped bracket supports a 300-N load. Determine reactions at A and C when $\alpha = 45^\circ$. (15 marks) [2014–15]
 $\hookrightarrow$ Similar: B1-Q21 same bracket, $\alpha = 60^\circ$



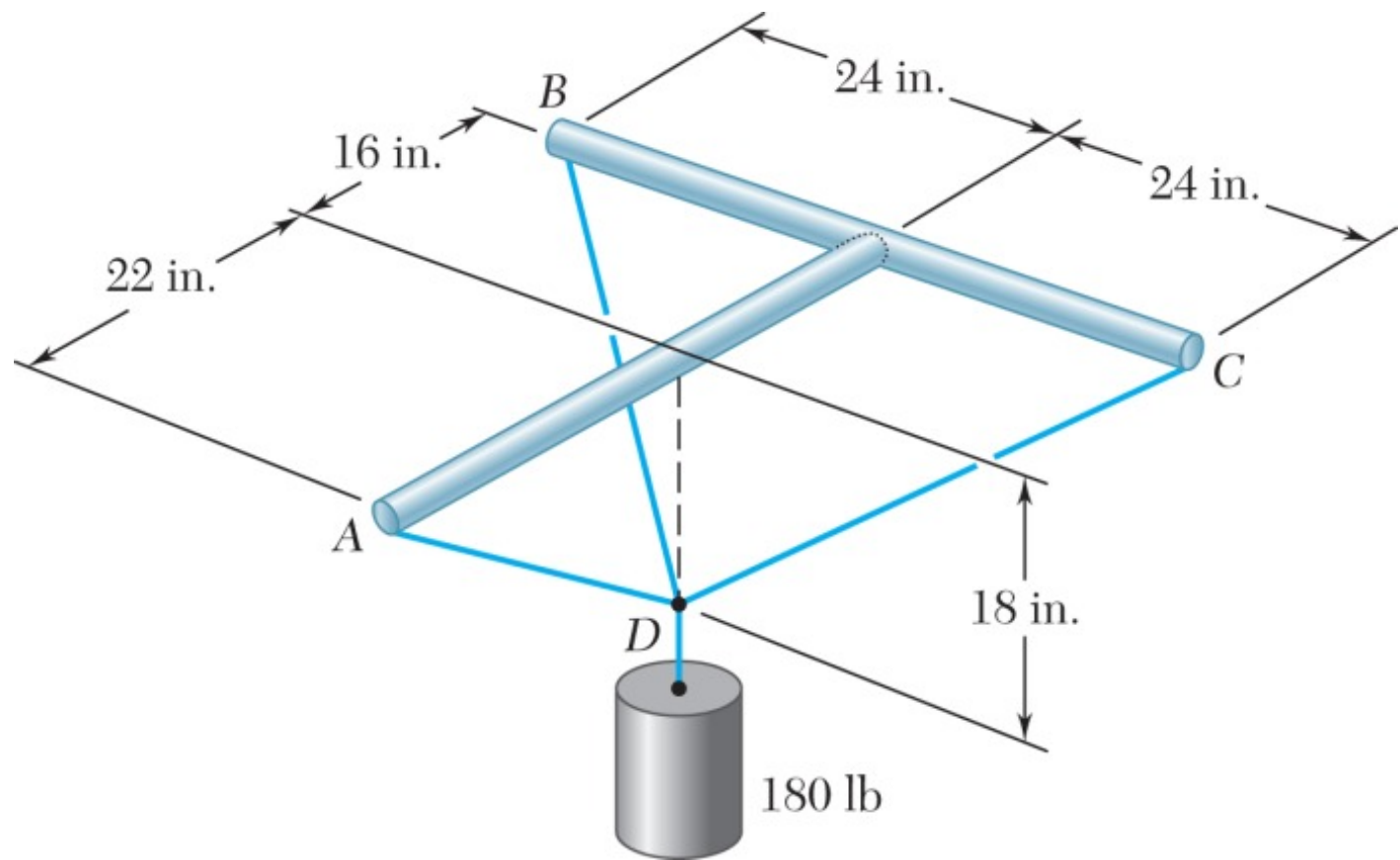
- Q23.** Hydraulic cylinder BC exerts force P on member AB along line BC. P must have a 600-N component perpendicular to AB. Determine:
 (i) magnitude of P , (ii) component along AB. (15 marks) [2013–14]



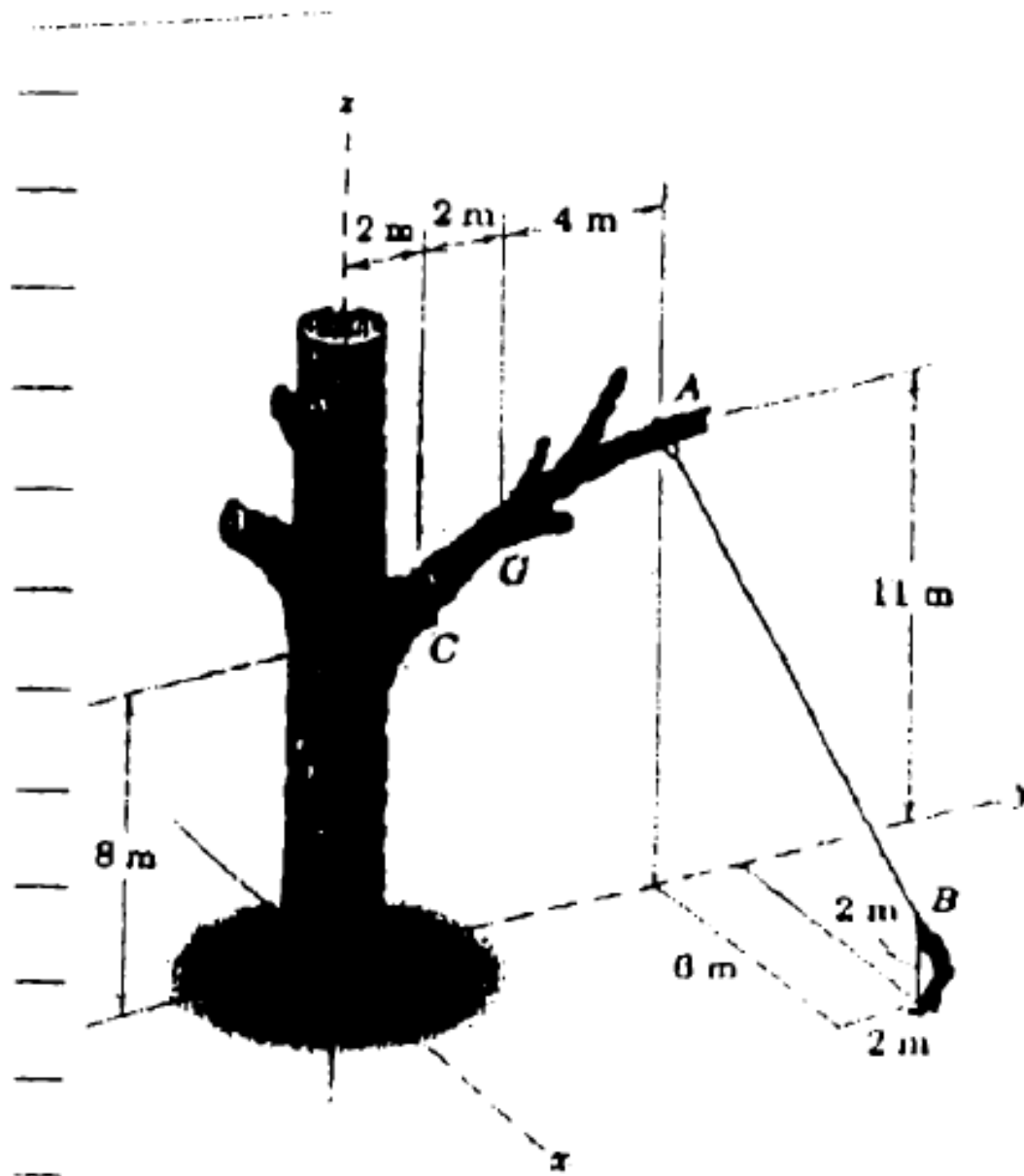
- Q24.** Single force P acts at C perpendicular to handle BC of the crank. $M_x = +20 \text{ N}\cdot\text{m}$, $M_y = -8.75 \text{ N}\cdot\text{m}$, $M_z = -30 \text{ N}\cdot\text{m}$. Determine magnitude of P and angles ϕ and θ . (20 marks) [2013–14]



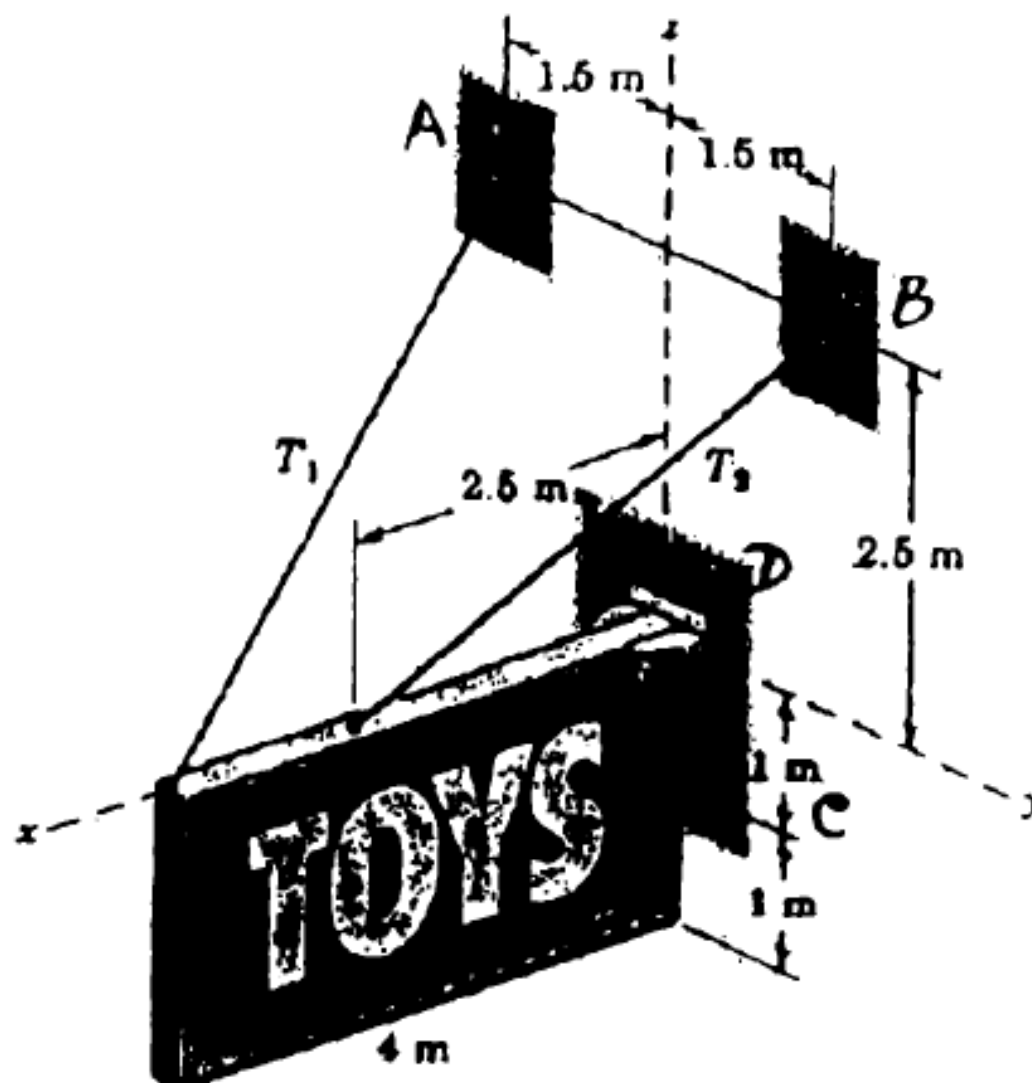
Q25. Three wires connected at point D, 18 in. below T-shaped support ABC. A 180-lb cylinder suspended from D. Determine tension in each wire. **(17 marks)** [2012–13]



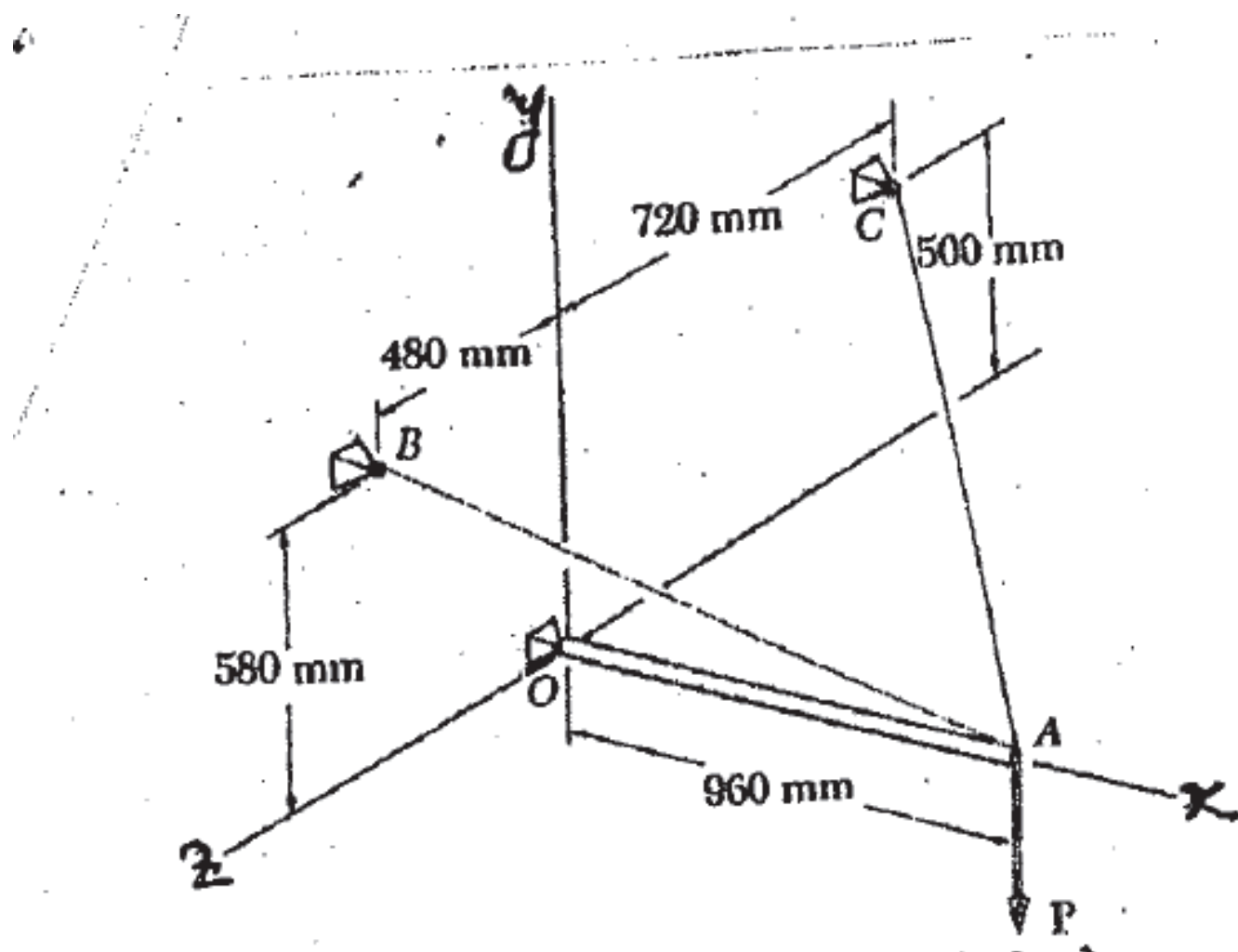
Q26. A tree surgeon exerts a 400-N pull on a line looped around branch at A. Determine moment about C and its magnitude. **(17 marks)** [2012–13]



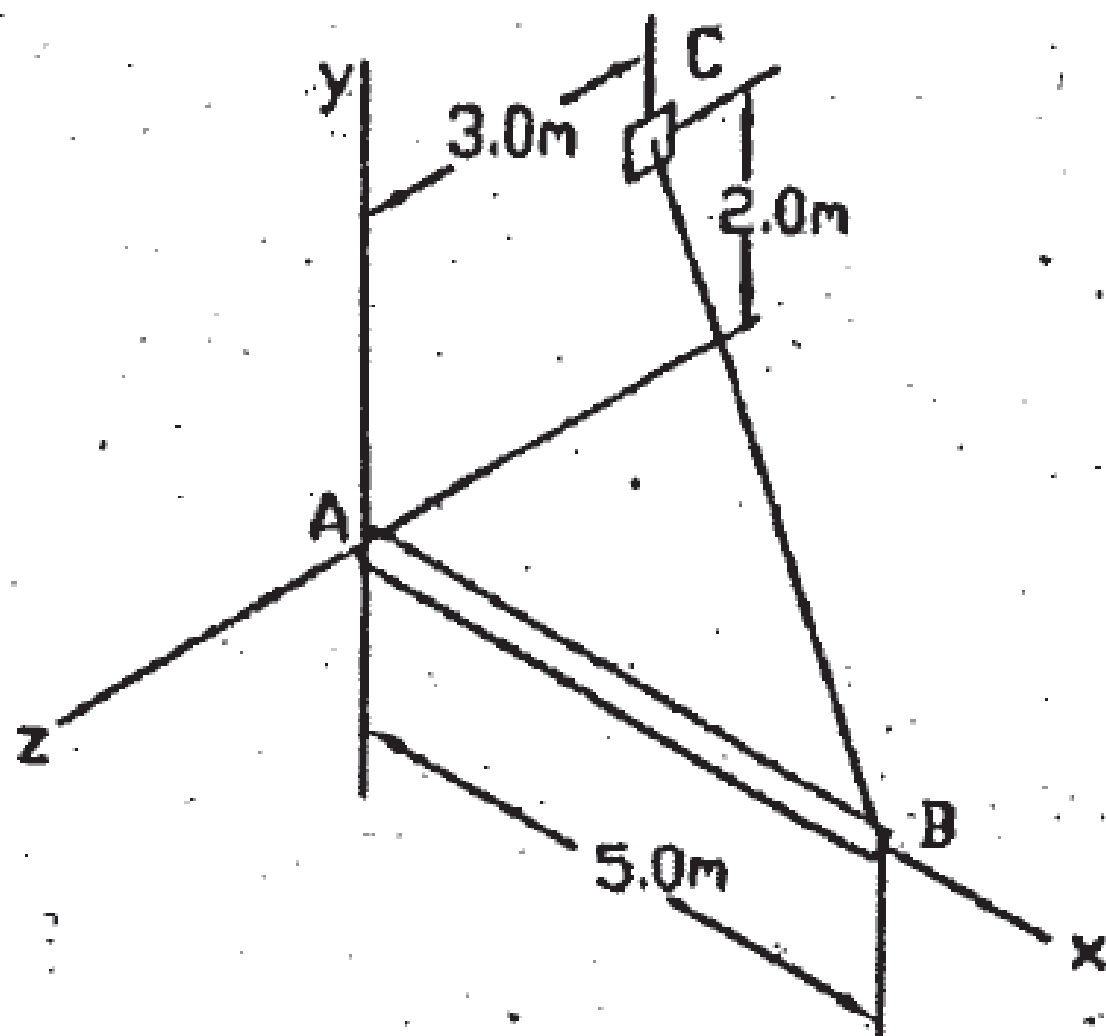
- Q27. Rectangular sign: mass 100 kg, centre of mass at centre. Ball-and-socket at C, y-direction support at D. Calculate T_1 , T_2 , total force at C, and lateral force R at D. (18 marks) [2012–13]



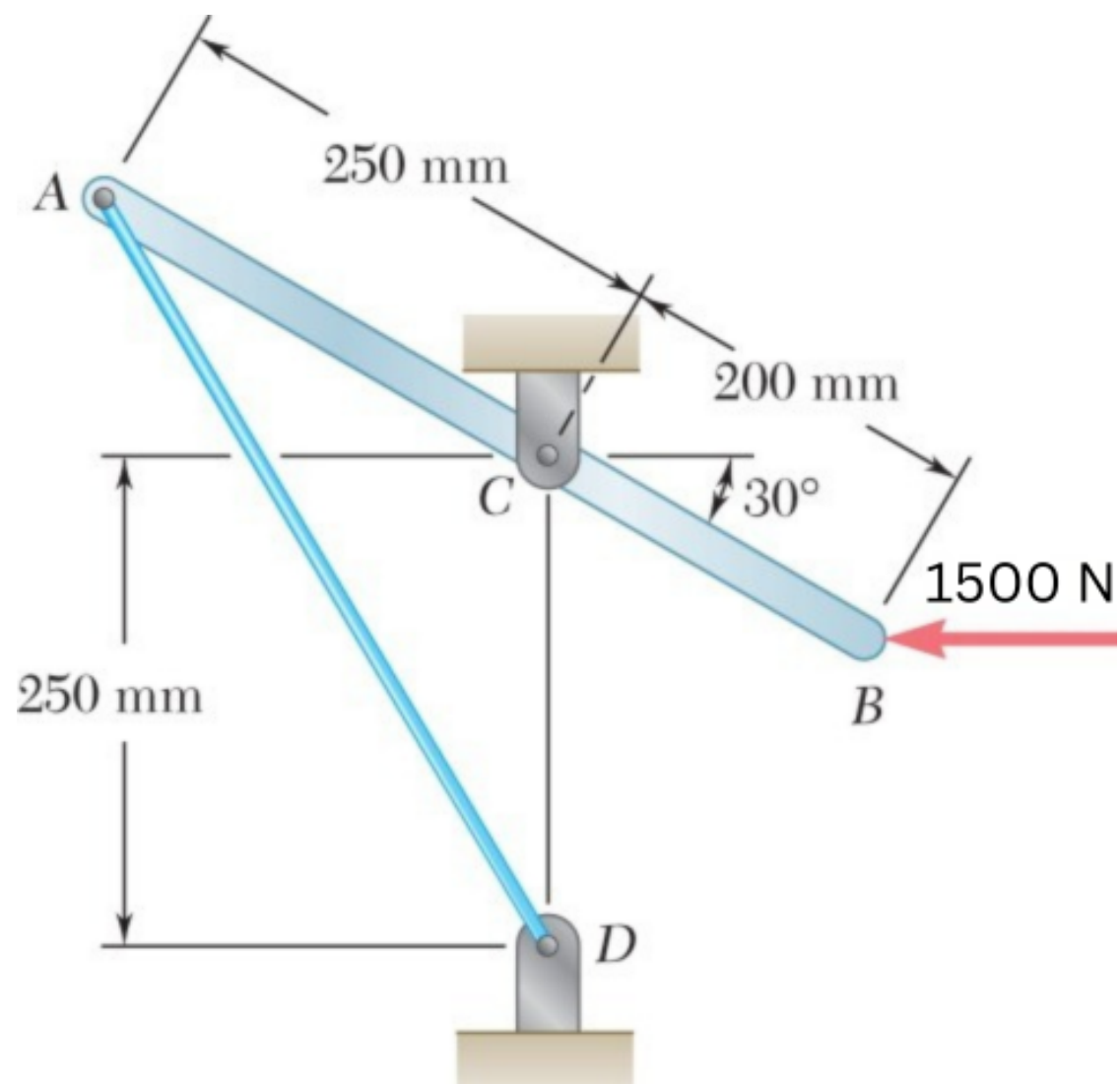
- Q28. Beam OA carries load P, supported by cables AB and AC. Tension in cable AB = 732 N. Resultant of P and forces at A must be along OA. Determine: (i) tension in cable AC, (ii) value of load P. (18 marks) [2011–12]



Q29. 5-m boom AB has fixed end A . Cable from B to point C on vertical wall; tension = 2.5 kN. Determine moment about A and angles of moment with coordinate axes. (17 marks) [2011–12]



Q30. Lever AB hinged at C , attached to control cable at A . A 1.5-kN horizontal force applied at B . Determine: (i) tension in cable, (ii) reaction at C . (17 marks) [2011–12]



Q31. Angle bracket: $\alpha = 50^\circ$. Replace the 600-N force by: (i) equivalent force-couple system at C, (ii) equivalent system of two parallel forces at B and C. (15 marks) [2010–11]

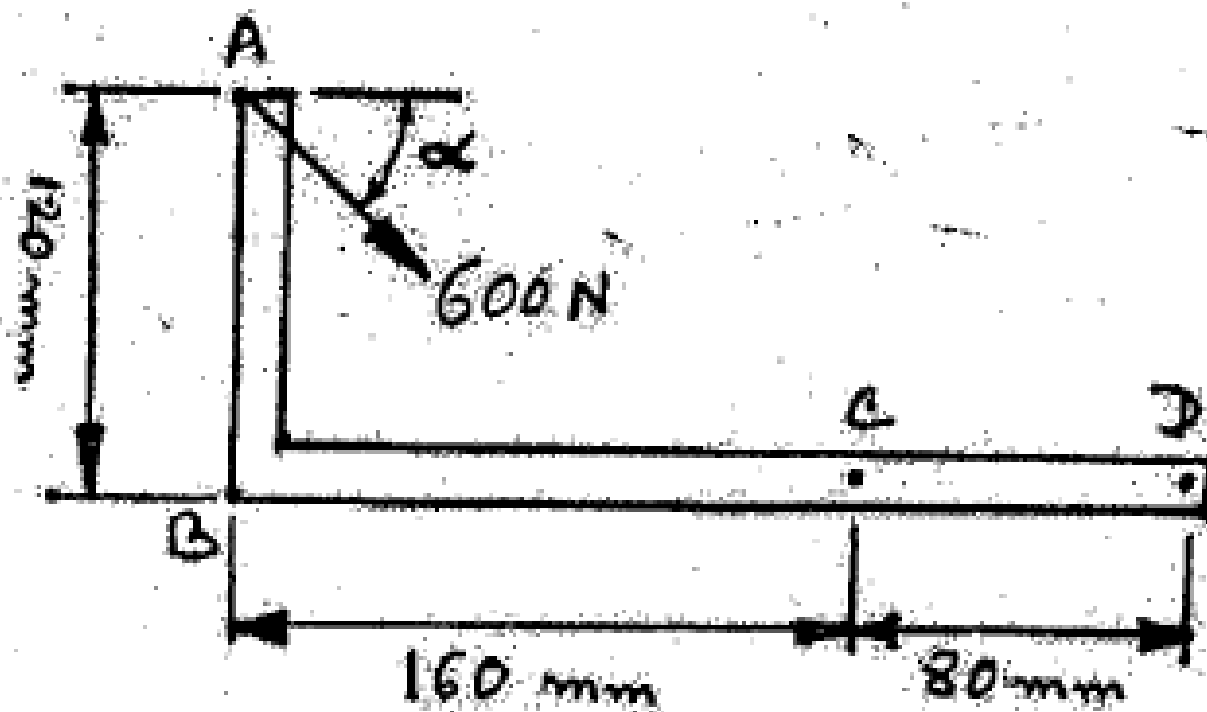
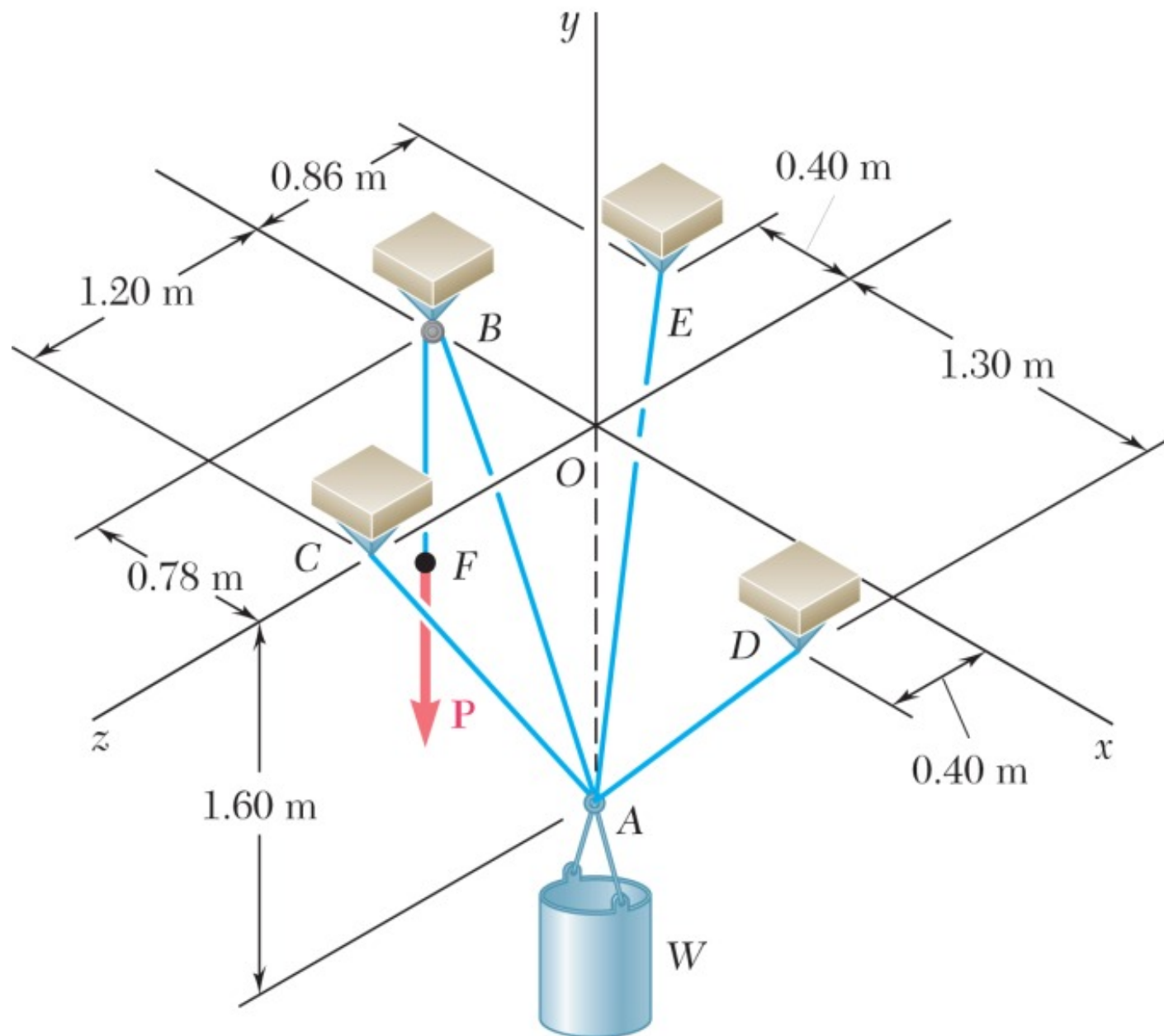
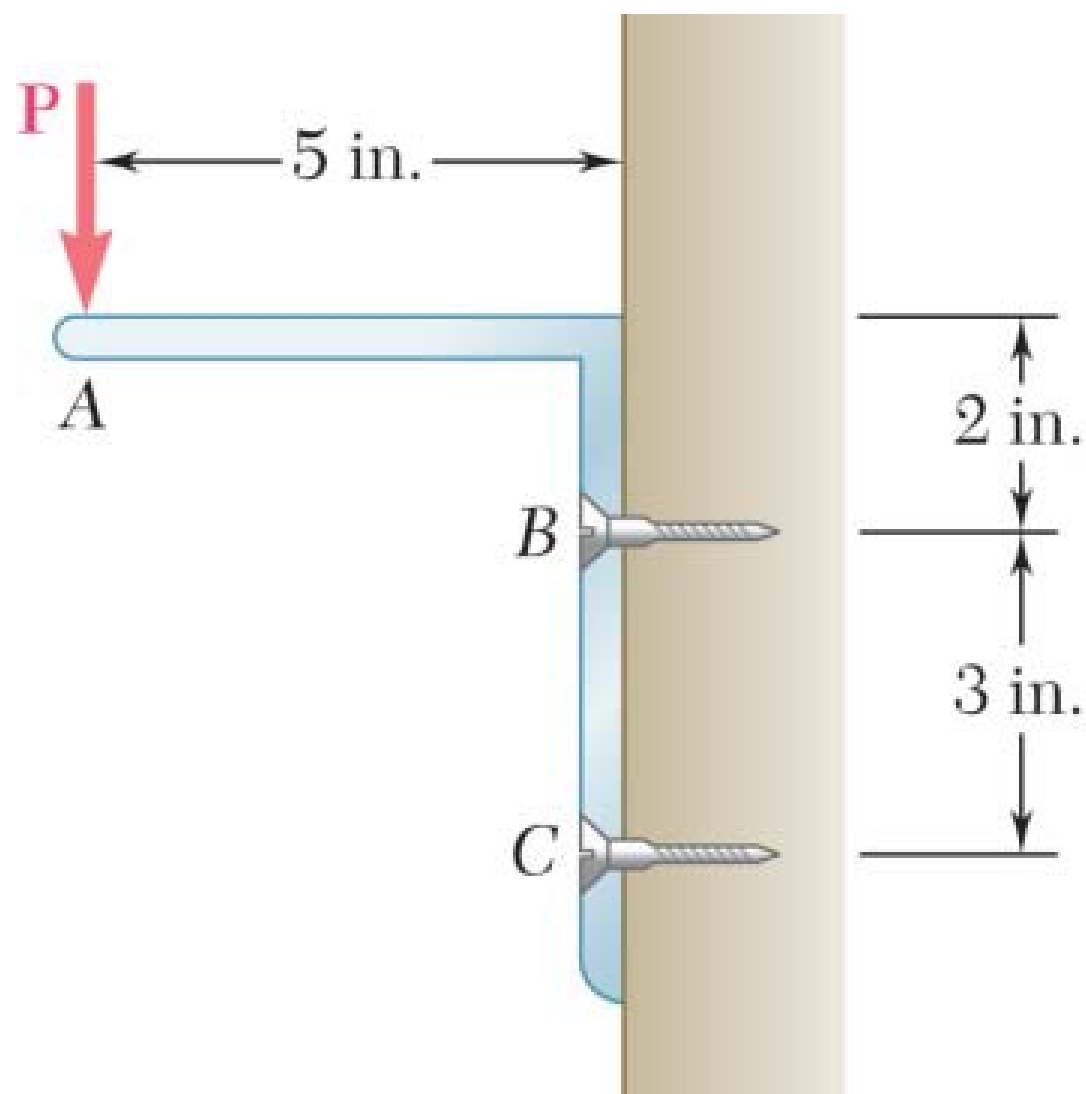


Fig. for Q. No. 5 (a)

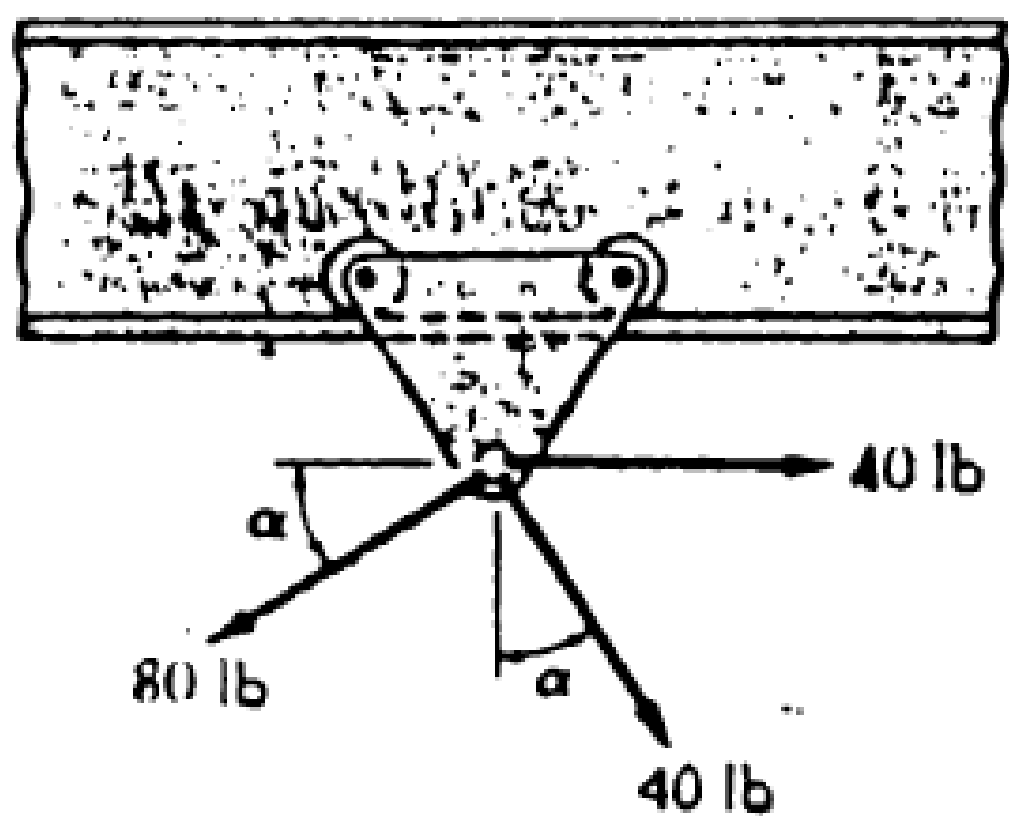
Q32. A container of weight W , is suspended from ring A, to which cables AC and AE are attached. A force P is applied over a pulley at B and through ring A and which is attached to a support at D. Knowing that $W = 1000\text{N}$ determine the magnitude of P . (20 marks) [2010–11]



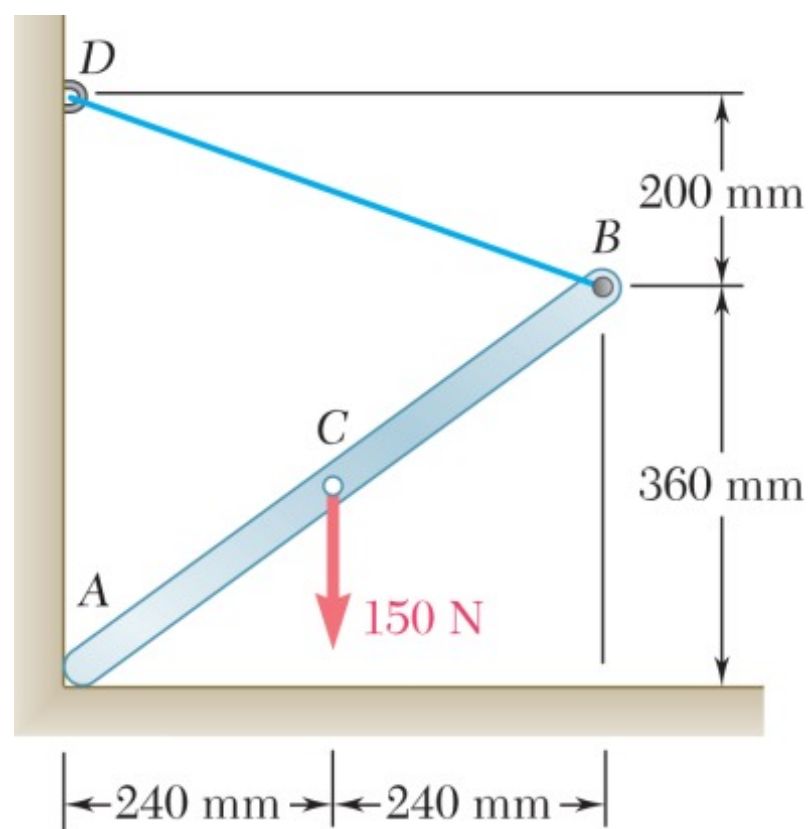
- Q33.** A 30-lb vertical force P is applied at A to the bracket shown, which is held by screws at B and C . (a) Replace P with an equivalent force-couple system at B . (b) Find the two horizontal forces at B and C that are equivalent to the couple obtained in part a. (15 marks) [2008–09]



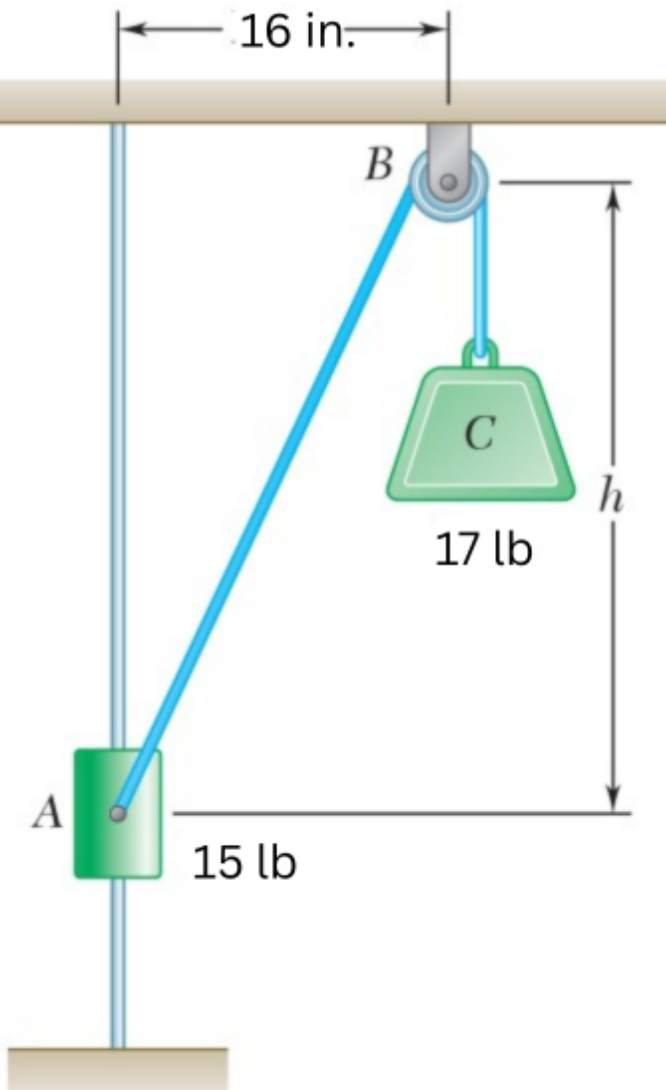
- Q34.** Hoist trolley subjected to three forces. Determine: (i) value of α for which resultant is vertical, (ii) corresponding magnitude of resultant. (17 marks) [2008–09]



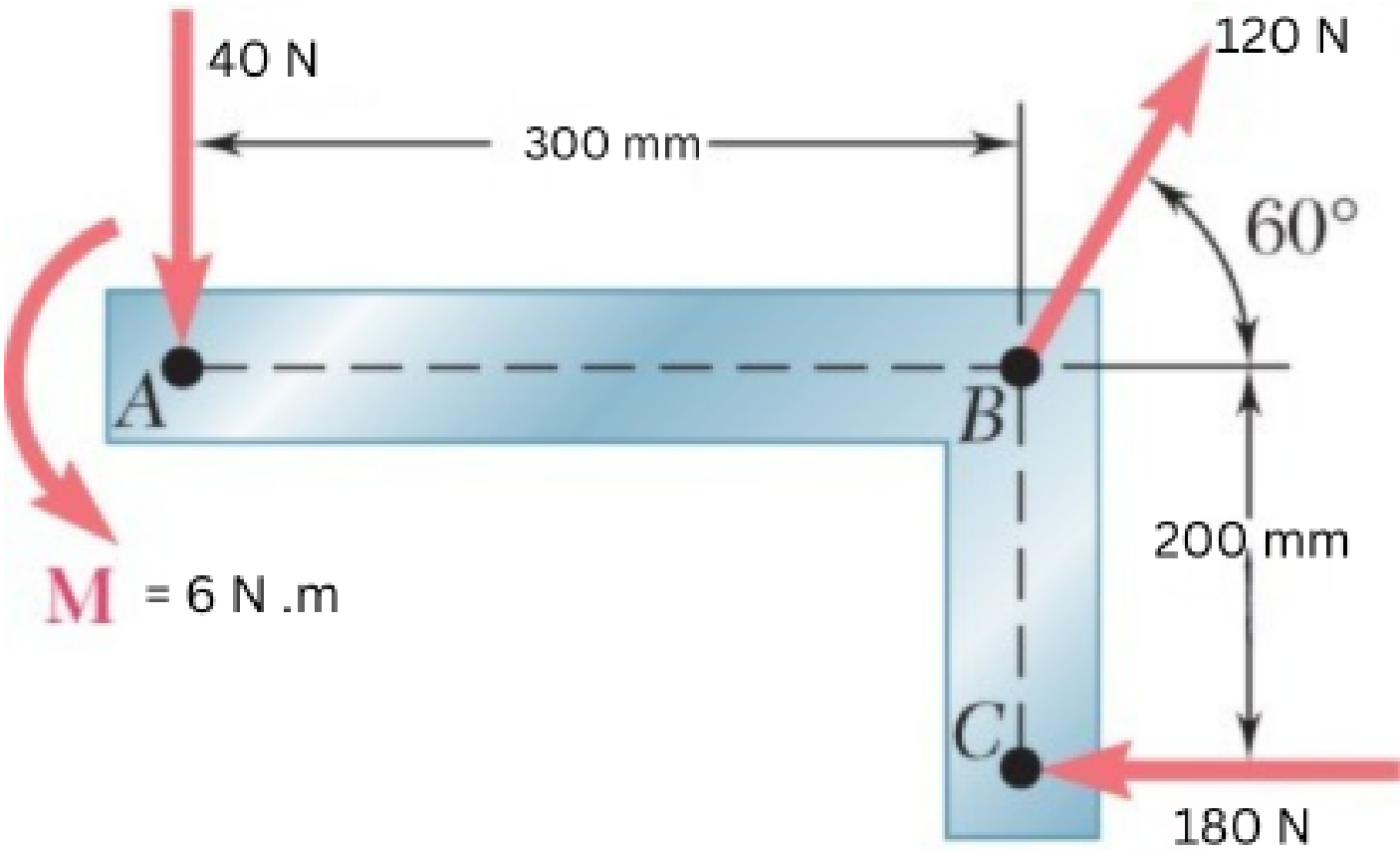
Q35. One end of rod AB rests in corner A; other end attached to cord BD. Rod supports 150-N load at midpoint C. Find reaction at A and tension in cord. (20 marks) [2008–09]



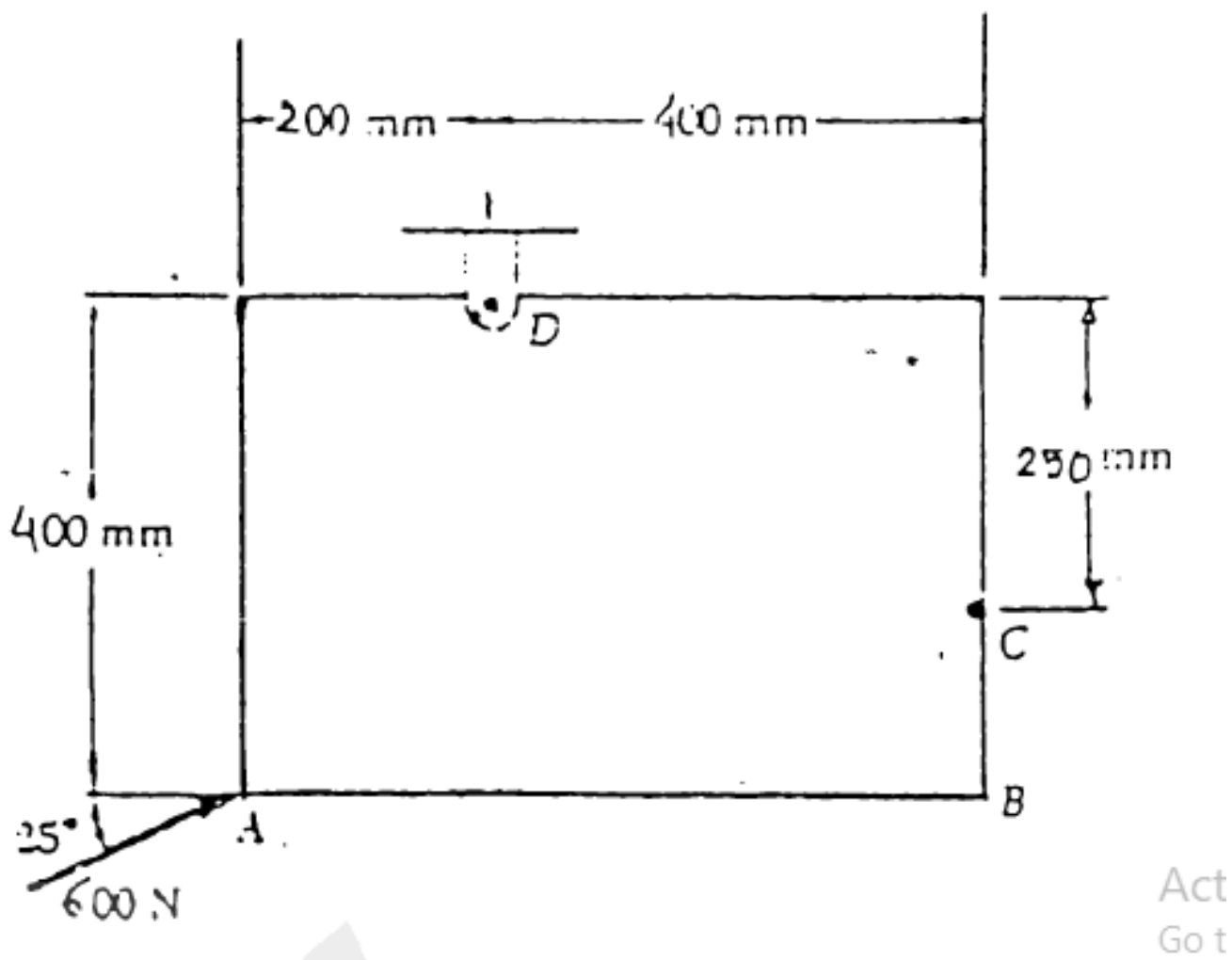
Q36. The 15-lb collar A may slide on a frictionless vertical rod and is connected to a 17-lb counterweight C. Determine value of h for equilibrium. (18 marks) [2008–09]



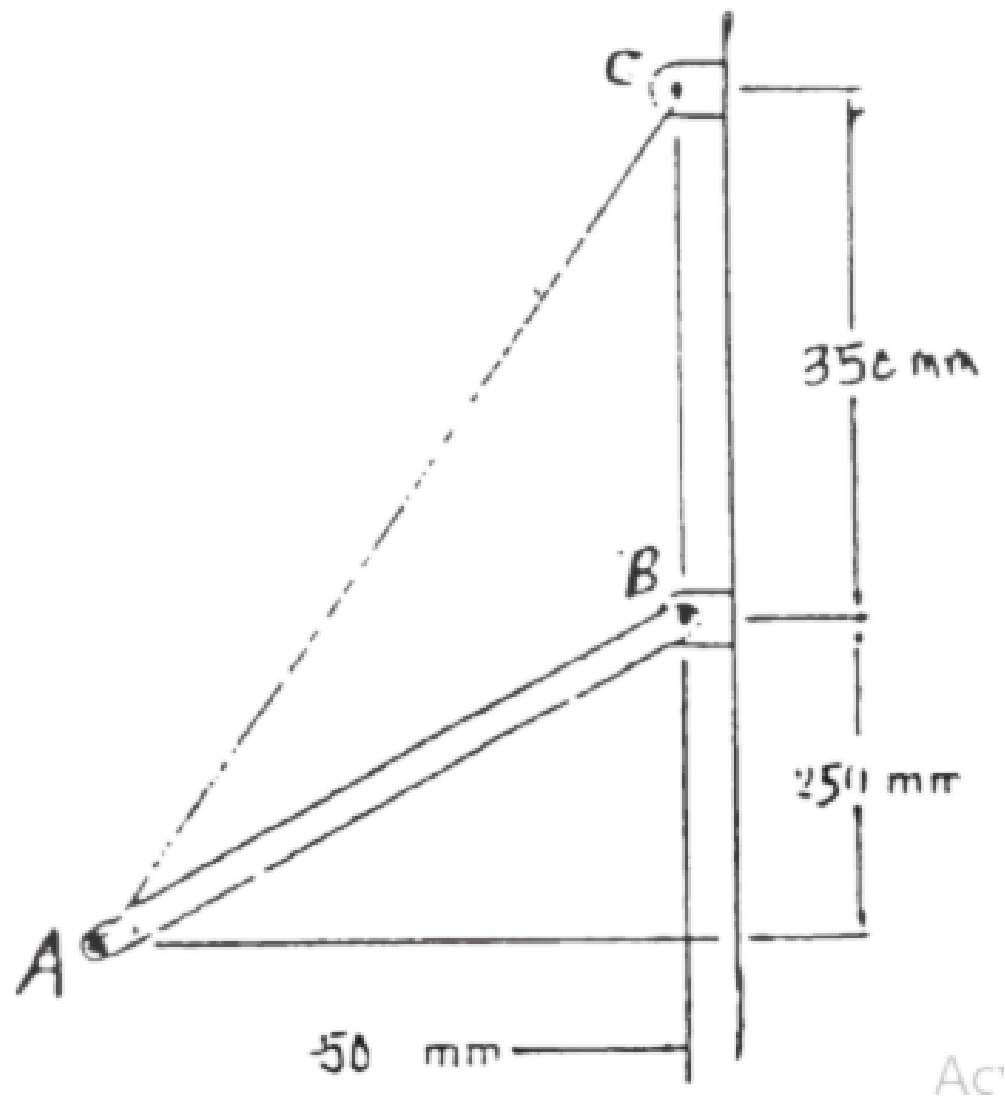
Q37. Three forces and a couple applied to an angle bracket. (i) Find the resultant. (ii) Locate points where line of action intersects lines AB and BC. (13 marks) [2006–07]



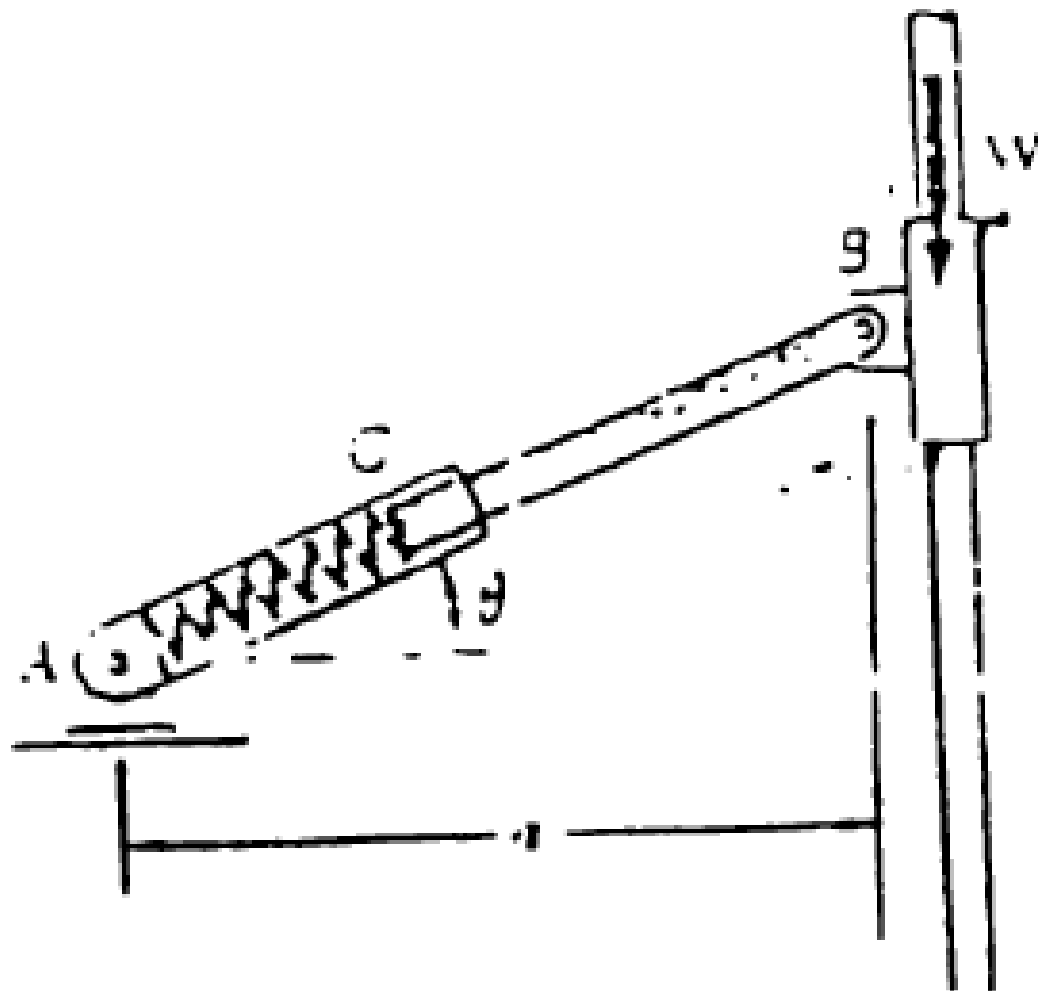
Q38. A 600-N force applied at A. Determine: (a) moment of 600-N force about D, (b) smallest force applied at B that creates the same moment about D. (17 marks) [2005–06]



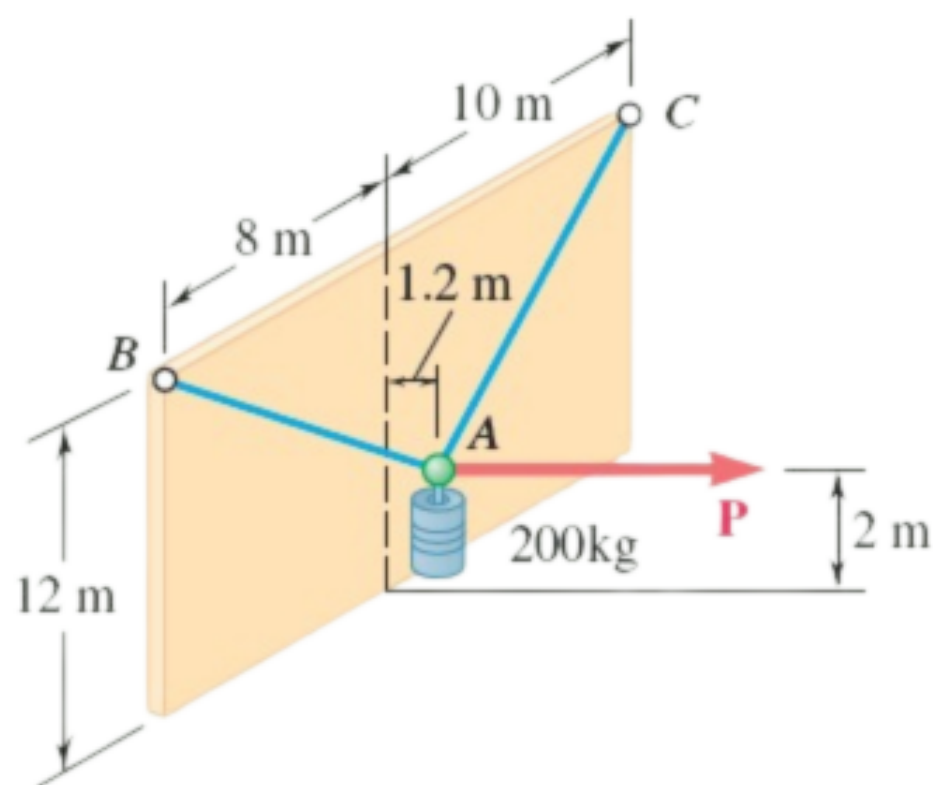
Q39. Rod AB held by cord AC. Tension in cord = 2000 N. Determine moment about B of force at A by resolving into horizontal and vertical components applied: (a) at A, (b) at C. (18 marks) [2005–06]



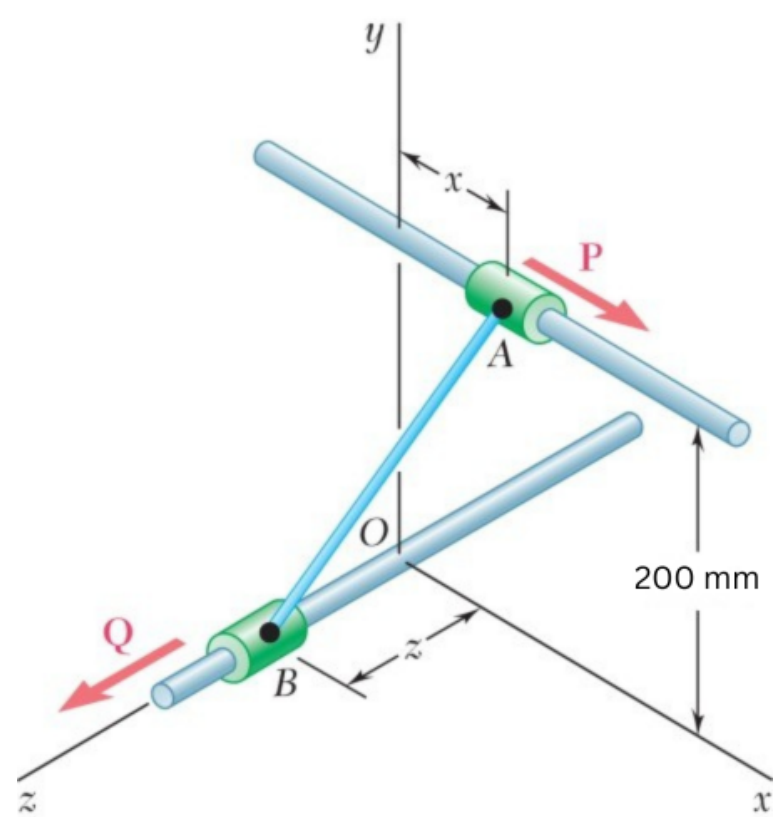
Q40. Internal spring AC has constant K , unstretched at $\theta = 50^\circ$. (a) Derive equilibrium equation in θ , W , a , K . (b) Find K when $W = 75 \text{ N}$, $a = 300 \text{ mm}$, equilibrium at $\theta = 20^\circ$. (18 marks) [2005–06]



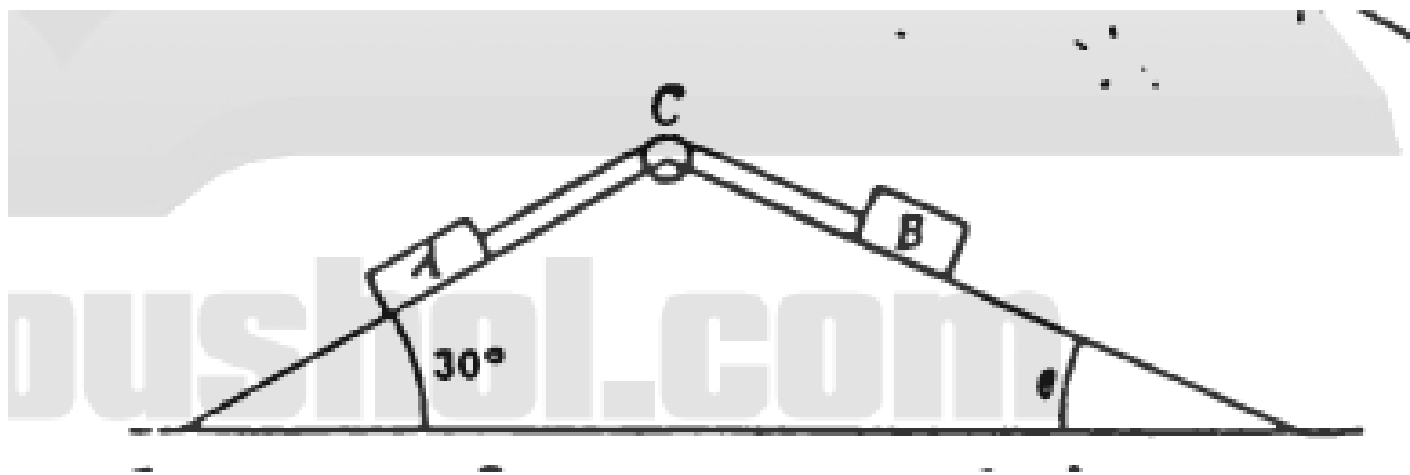
- Q41.** A 200-kg cylinder hung by cables AB and AC attached to top of vertical wall. Horizontal force P holds cylinder in position. Determine P and tension in each cable. (17.5 marks) [2004–05]



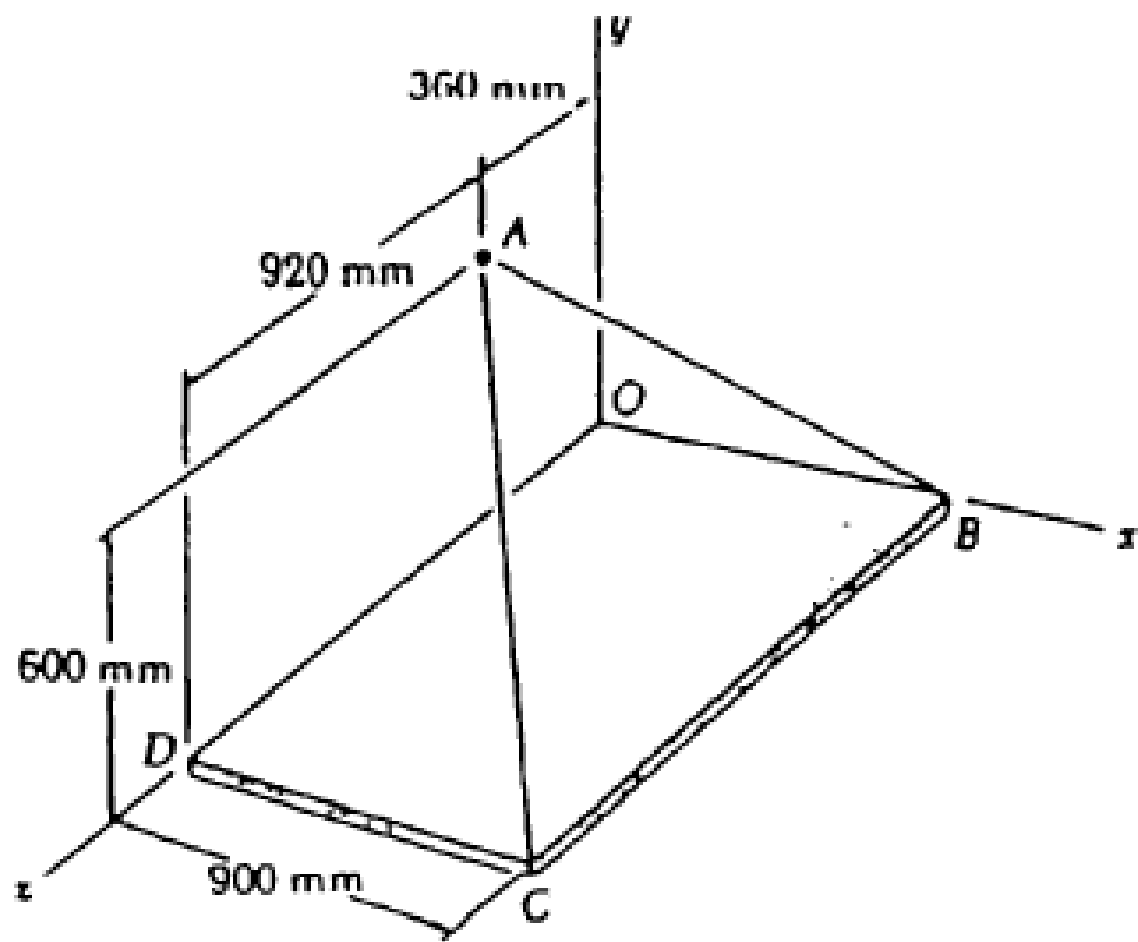
- Q42.** Collars A and B connected by a 250-mm wire; may slide on frictionless rods. Determine distances x and z for equilibrium with $P = 200 \text{ N}$, $Q = 100 \text{ N}$. (17.5 marks) [2004–05]



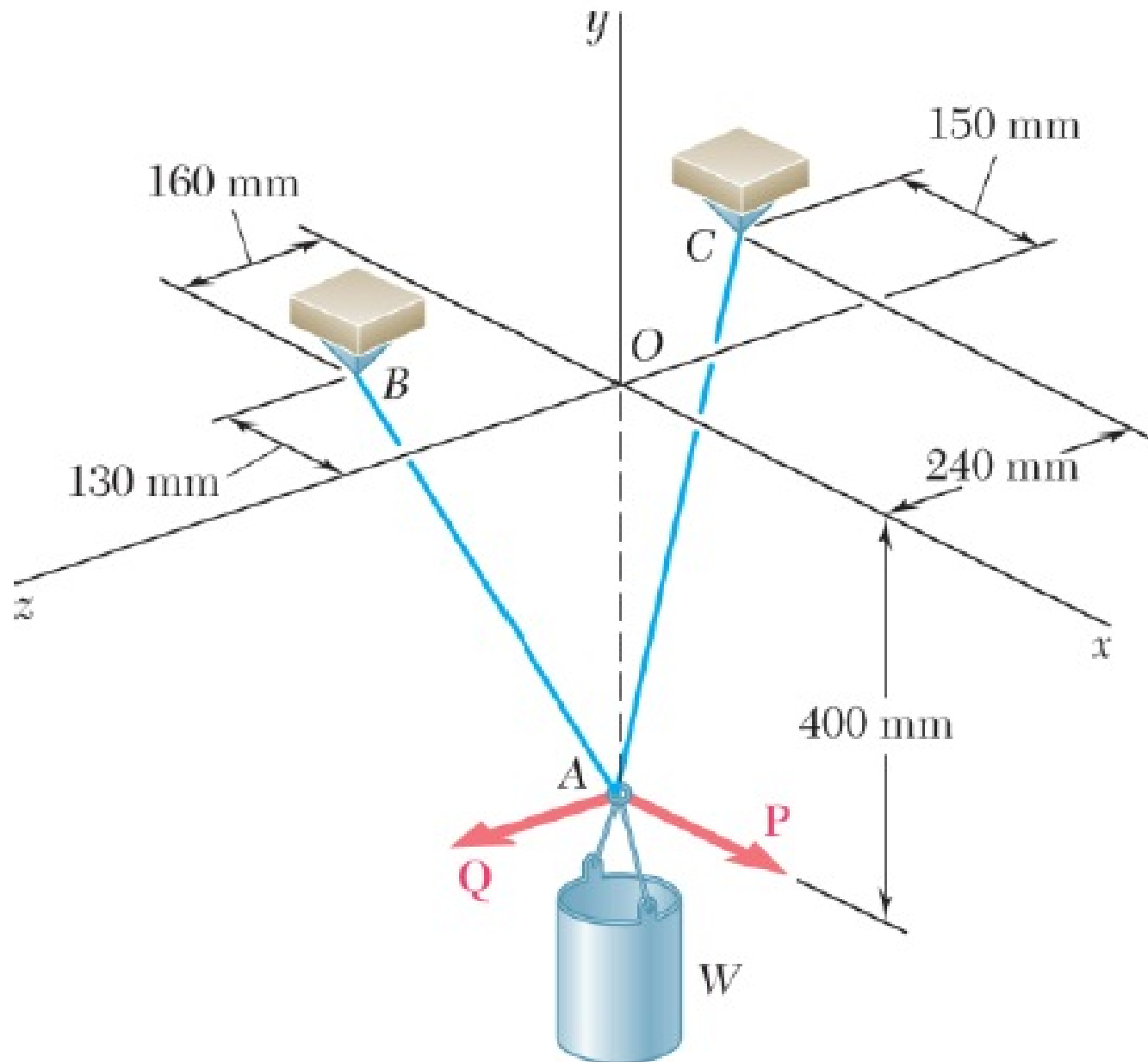
Q43. A and B weigh 40 kg and 30 kg, resting on smooth planes, connected by a cord over frictionless pulley C. Determine angle θ and tension in cord for equilibrium. (17.5 marks) [2004–05]



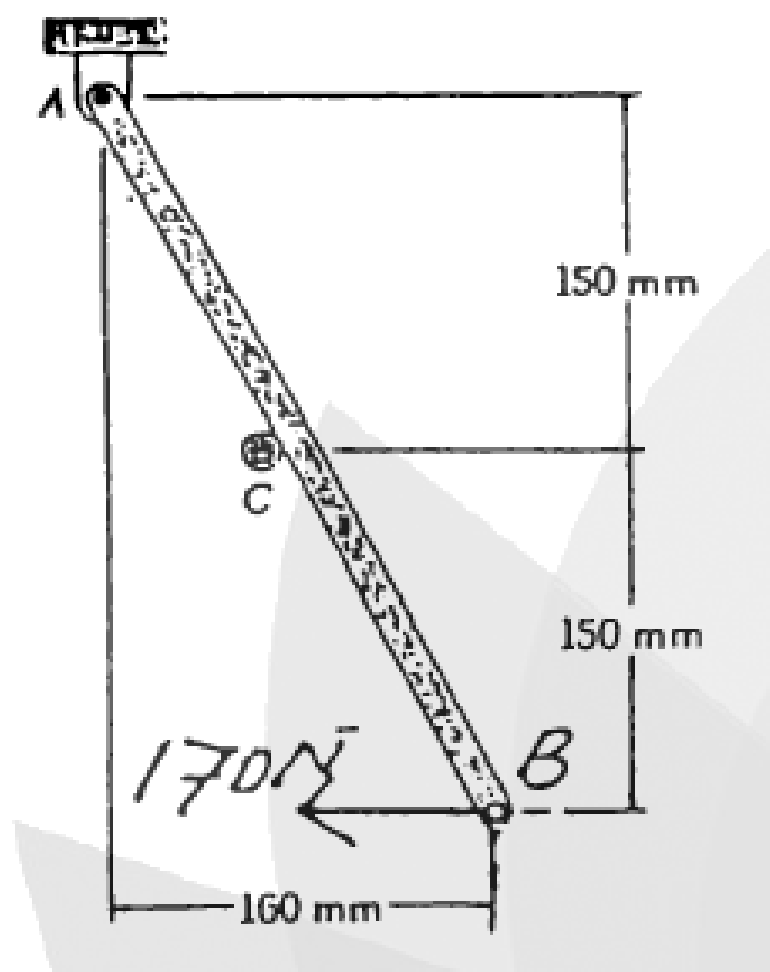
Q44. Tension in cable AB = 570 N. Determine moment about each coordinate axis of the force exerted on the plate at B. (17 marks) [2003–04]



- Q45.** Container of weight $W = 400 \text{ N}$ supported by cables AB and AC at ring A. With $Q = 80 \text{ N}$, determine: (i) magnitude of force P for equilibrium, (ii) tensions in cables AB and AC. **(18 marks)** [2003–04]



- Q46.** Rod AB supported by pin and bracket at A, rests against frictionless peg at C. A 170-N horizontal force applied at B. Determine reactions at A and C. **(17 marks)** [2003–04]



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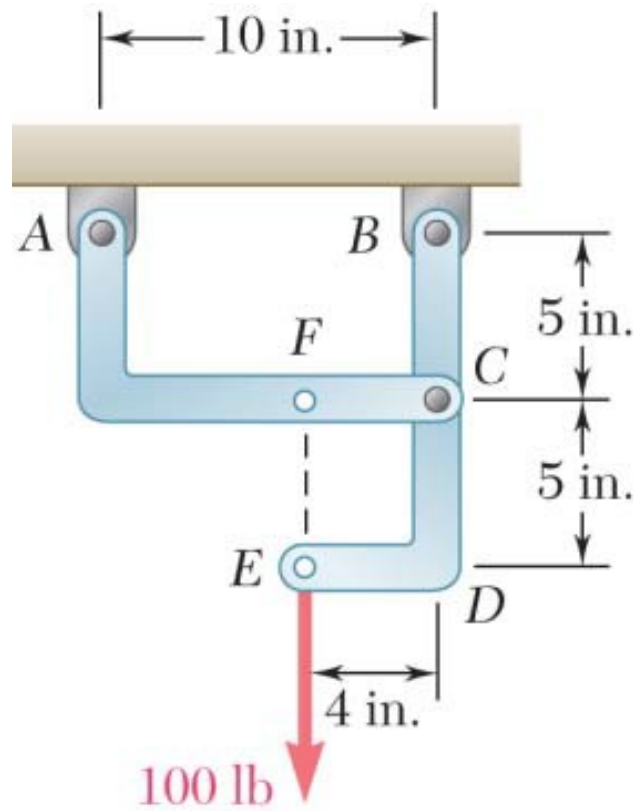
MECHA 165

Section B2

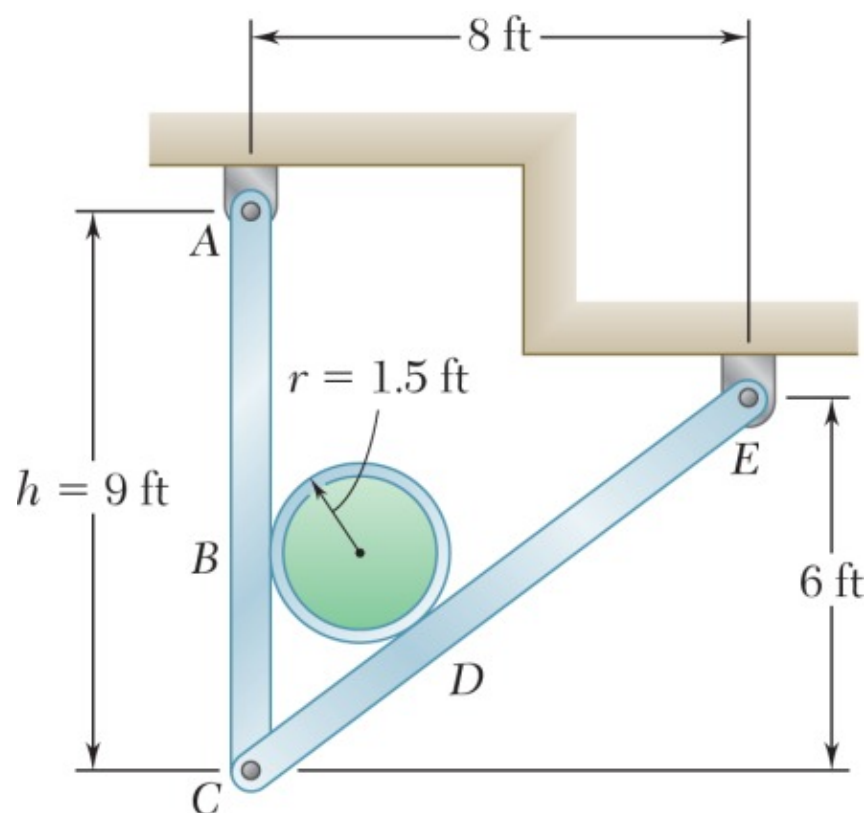
Forces in Trusses & Frames

B2 — Forces in Trusses & Frames

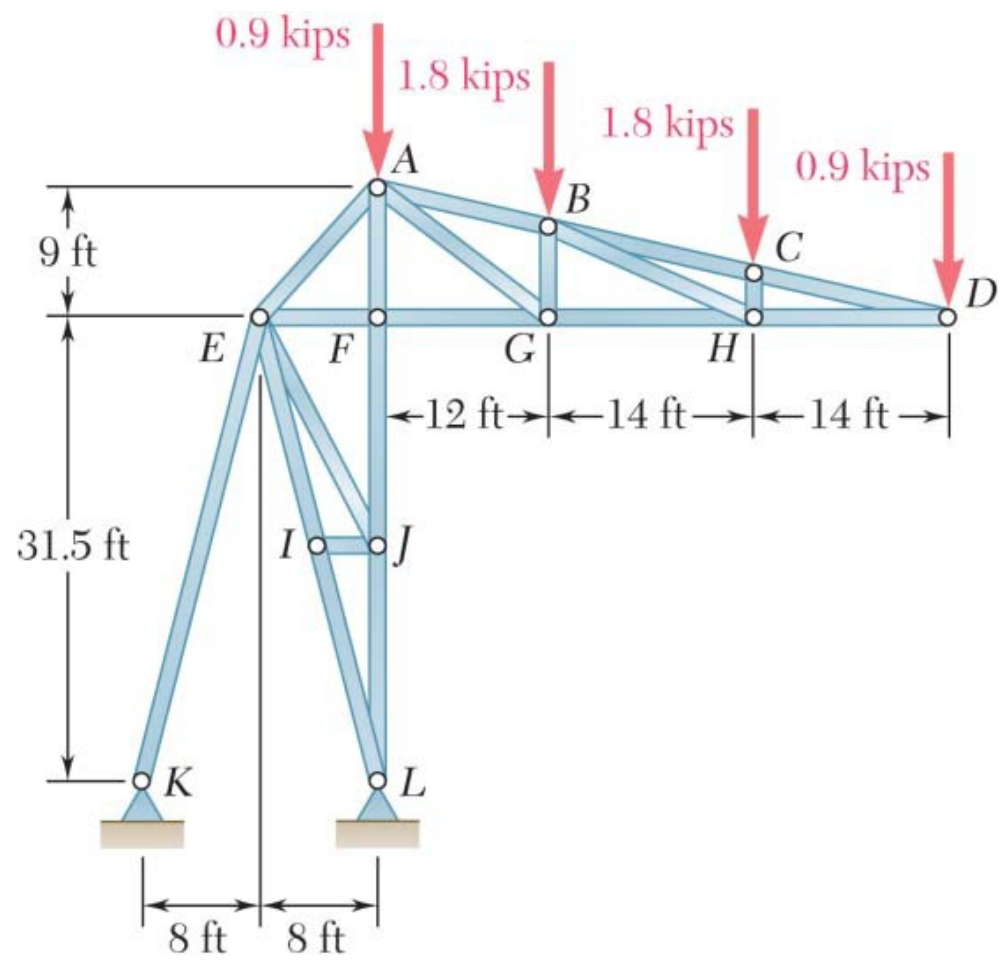
- Q1.** Determine the components of the reactions at A and B: (i) if the 100-lb load is applied as shown, (ii) if the 100-lb load is moved along its line of action to point F. (15 marks) [2023–24]



- Q2.** A 3-ft-diameter pipe supported every 16 ft by a frame. Combined weight of pipe and contents = 500 lb/ft. Frictionless surfaces. Determine: (i) reaction at E, (ii) force at C on member CDE. (15 marks) [2023–24]

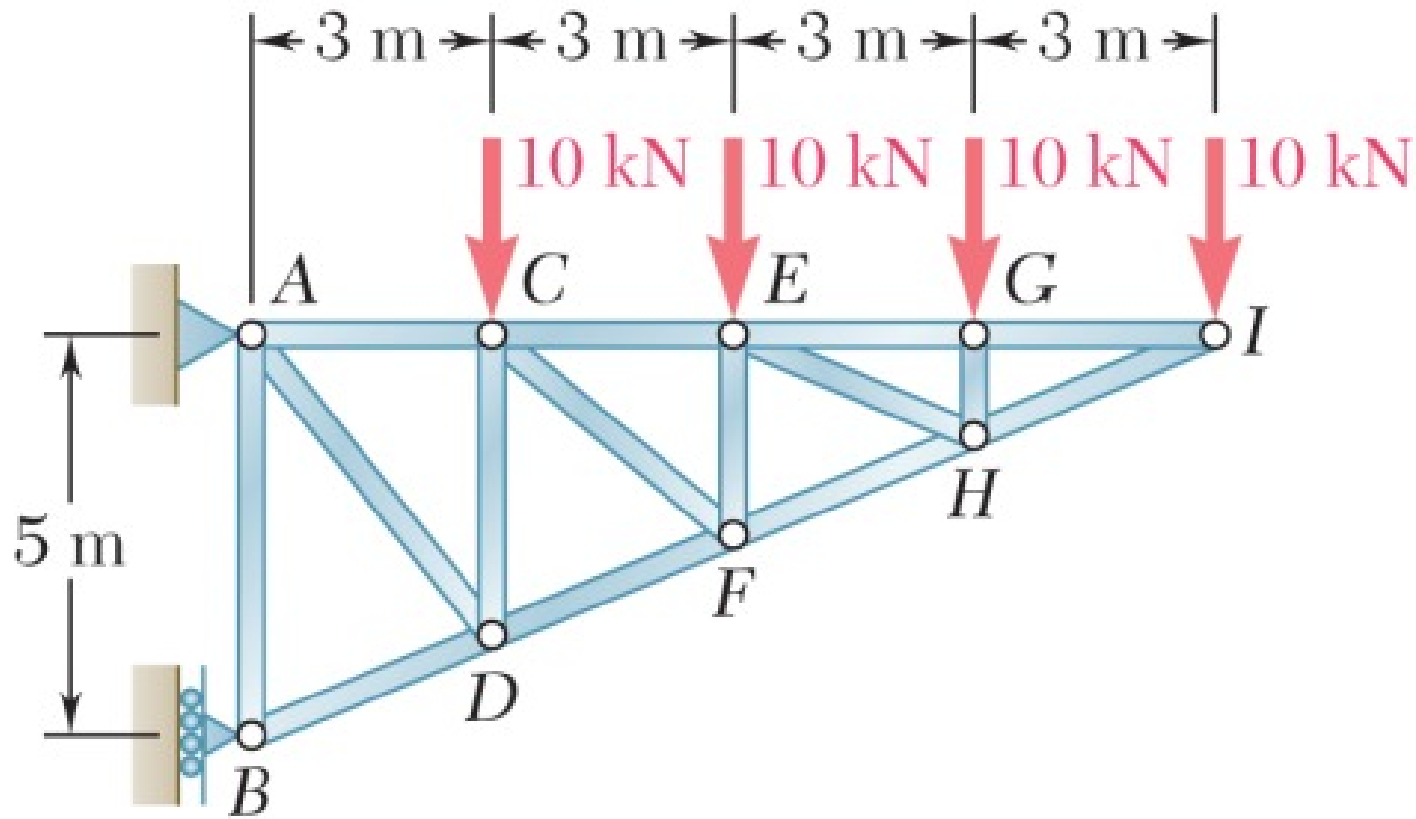


- Q3.** A stadium roof truss is loaded as shown. Determine the force in members AE, EF, and FJ. (15 marks) [2023–24]

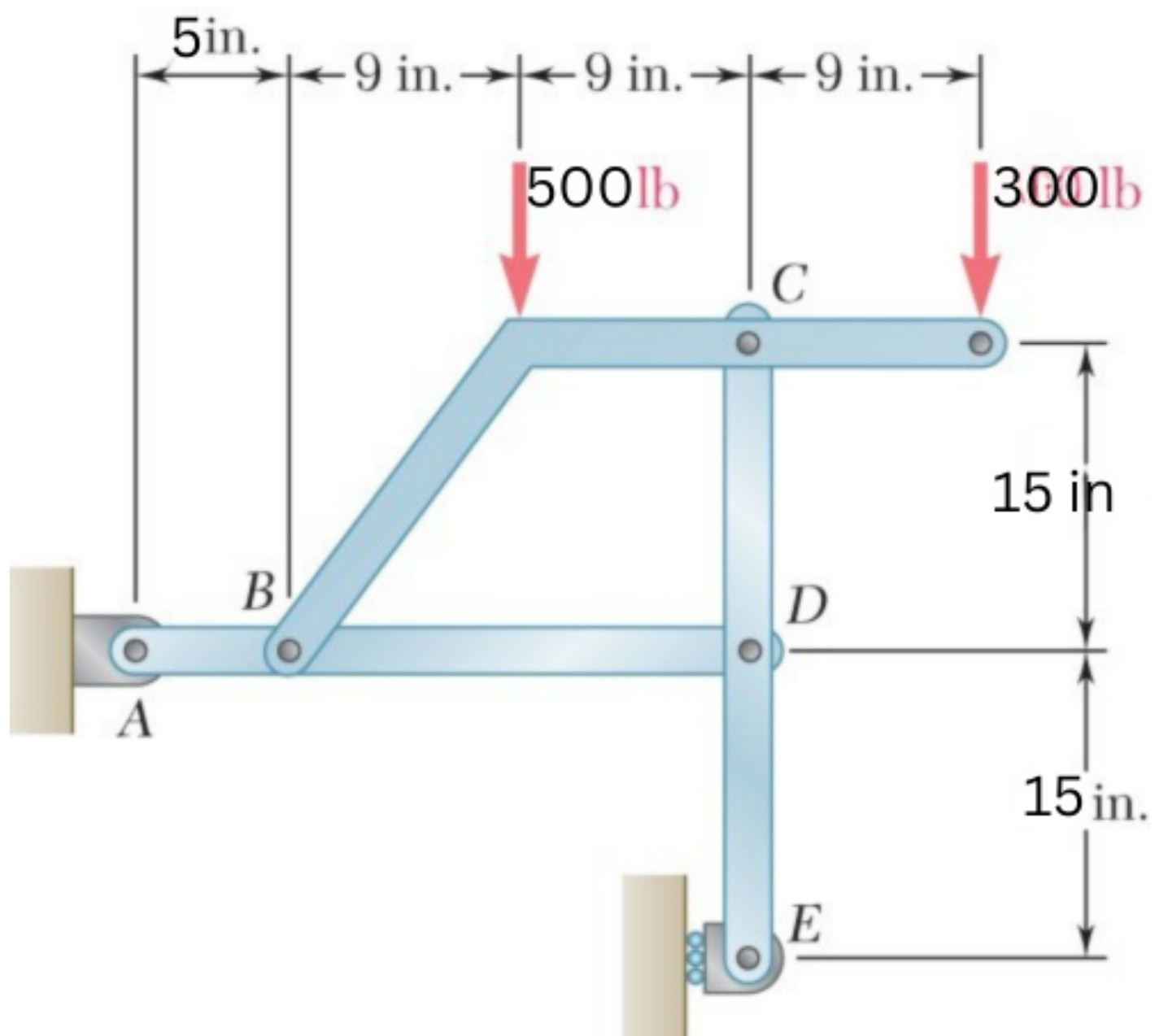


Q4. A stadium roof truss is loaded as shown. Determine the force in members AB, AG, and FG. (15 marks) [2023–24]
↪ Similar: B2-Q1 same truss, different members

Q5. Determine the force in members CE, EF, and FH for the truss. Also determine reactions at A and B. (17 marks) [2021–22 , 2017–18]

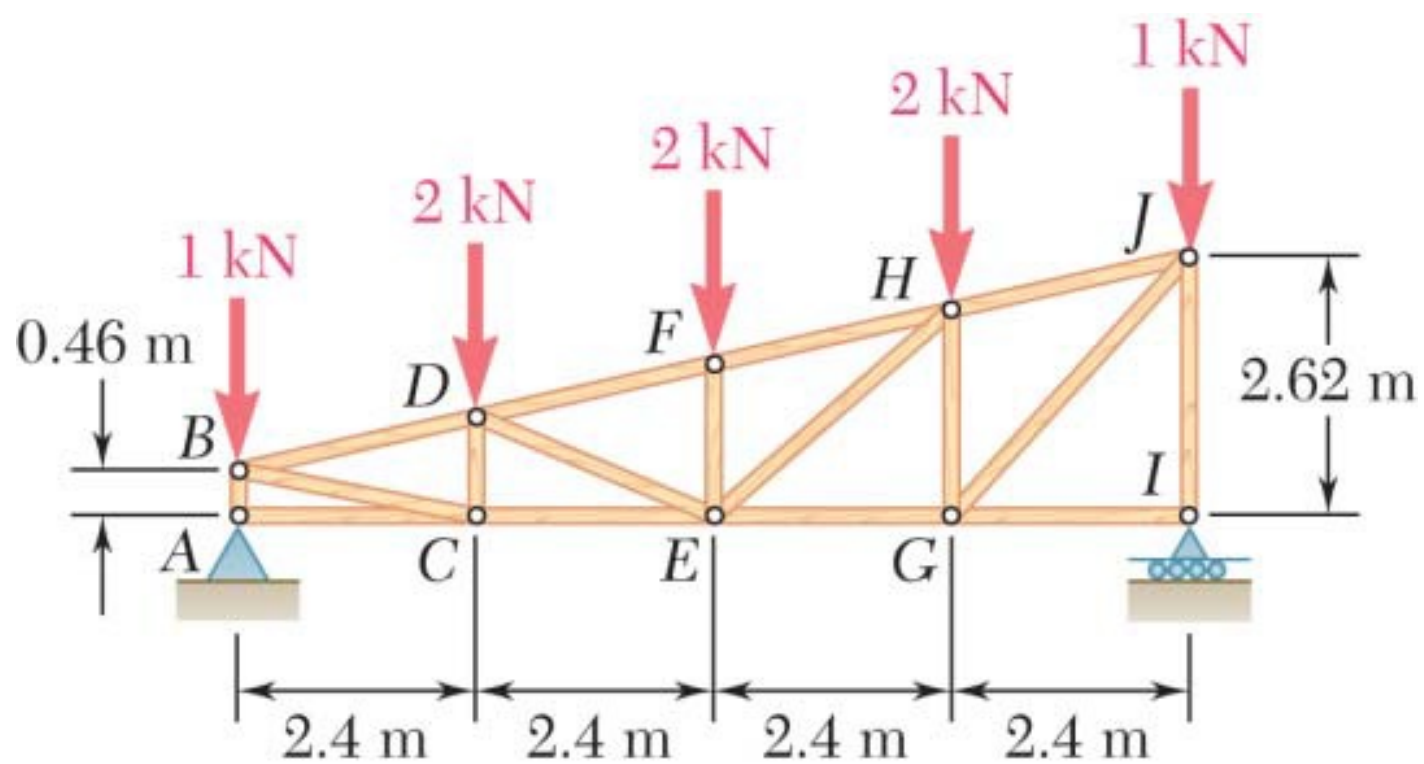


Q6. For the frame and loading, determine the components of all forces acting on members BC and CDE. (18 marks) [2021–22]



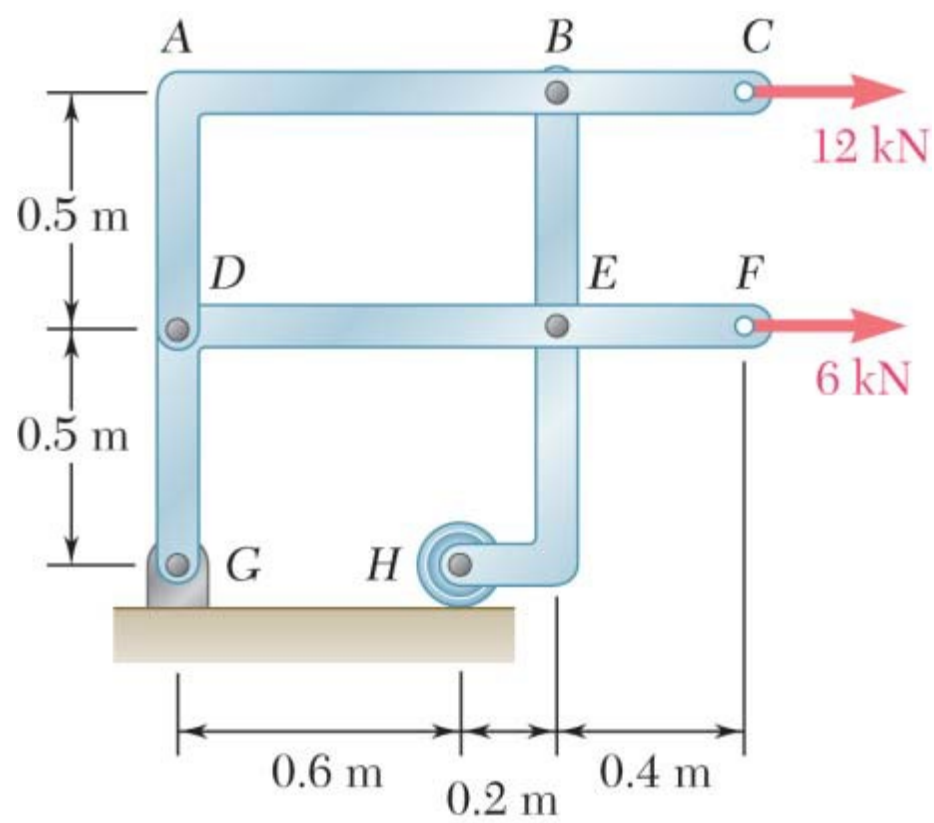
Q7. A pitched flat roof truss. Determine force in members EG, GH, and HJ. **(17 marks)**

[2020–21]

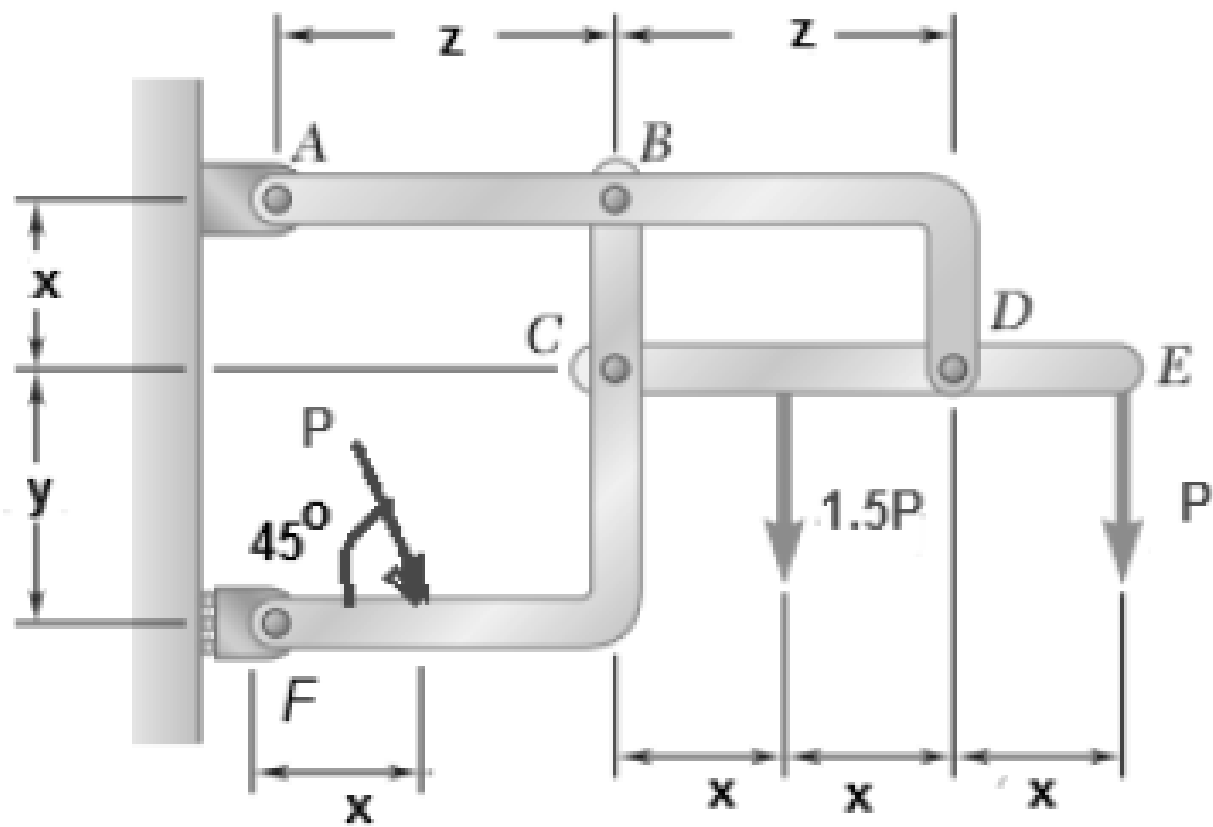


Q8. For the frame and loading, determine the components of forces acting on member DABC at B and D. **(18 marks)**

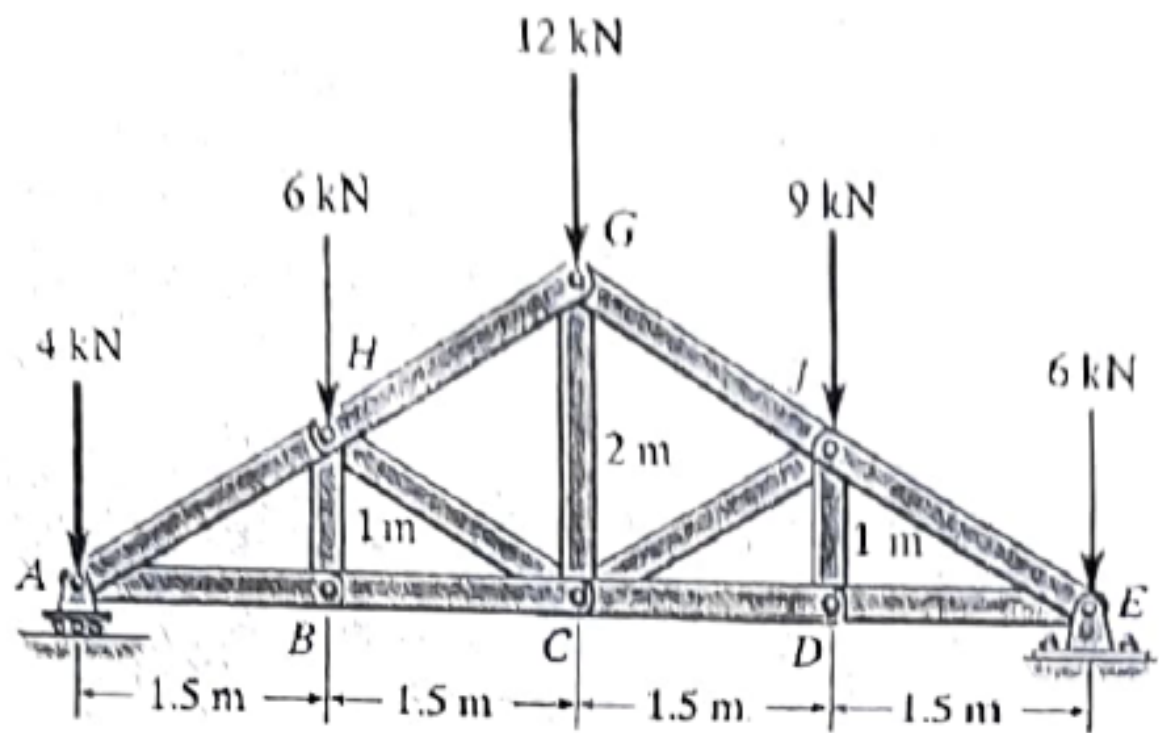
[2020–21]



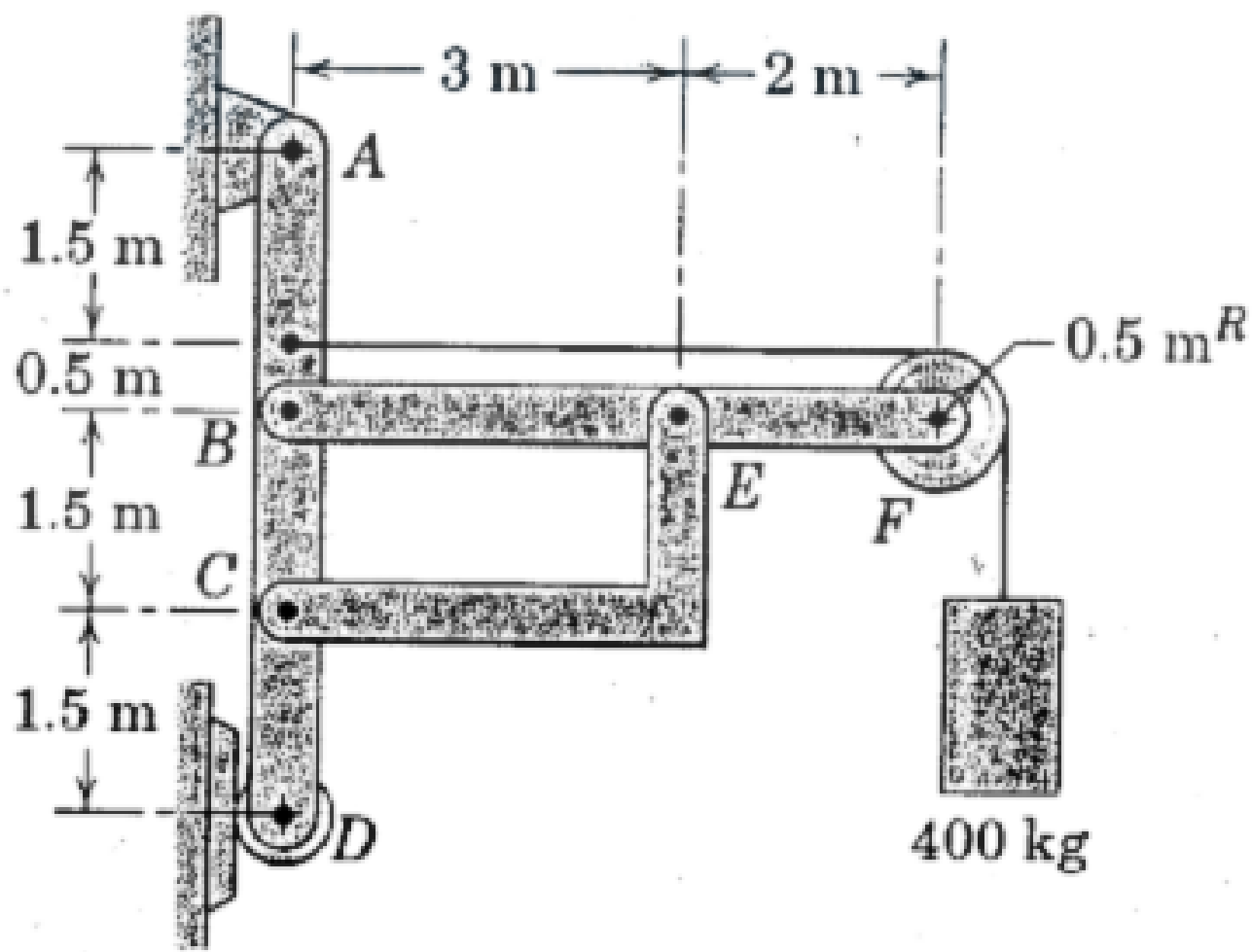
Q9. For the frame and loading, determine the components of all forces acting on member ABD. [Hints: Take: $x=200+2\times(\text{Last three digits of your student ID})$, $y=300+3\times(\text{Last three digits of your student ID})$, $z=400+4\times(\text{Last three digits of your student ID})$ and $P=200+(\text{Last three digits of your student ID})$.] (30 marks) [2019–20]



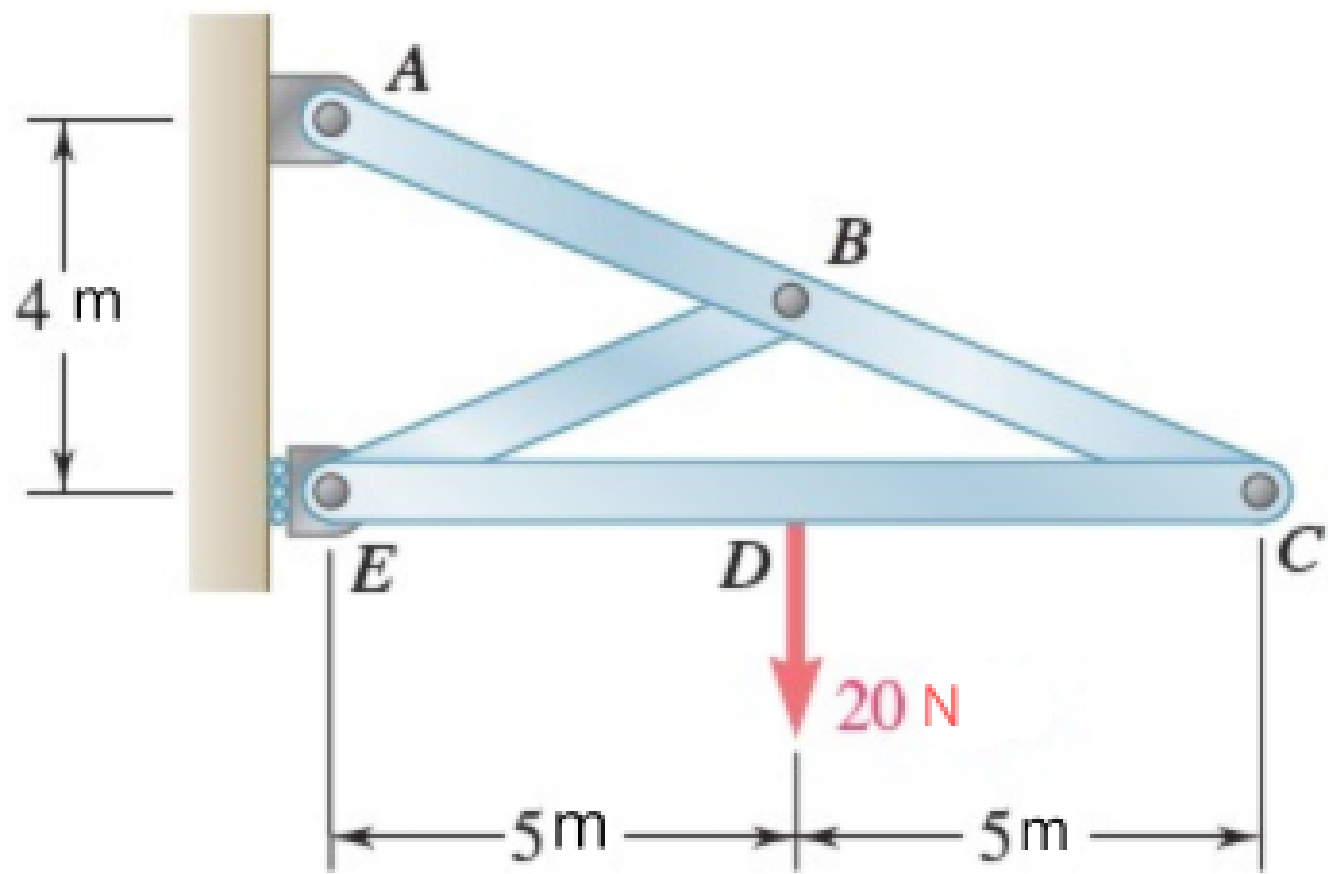
Q10. Determine force in members CD, CJ, and GJ. State tension or compression. (17 marks) [2018–19]



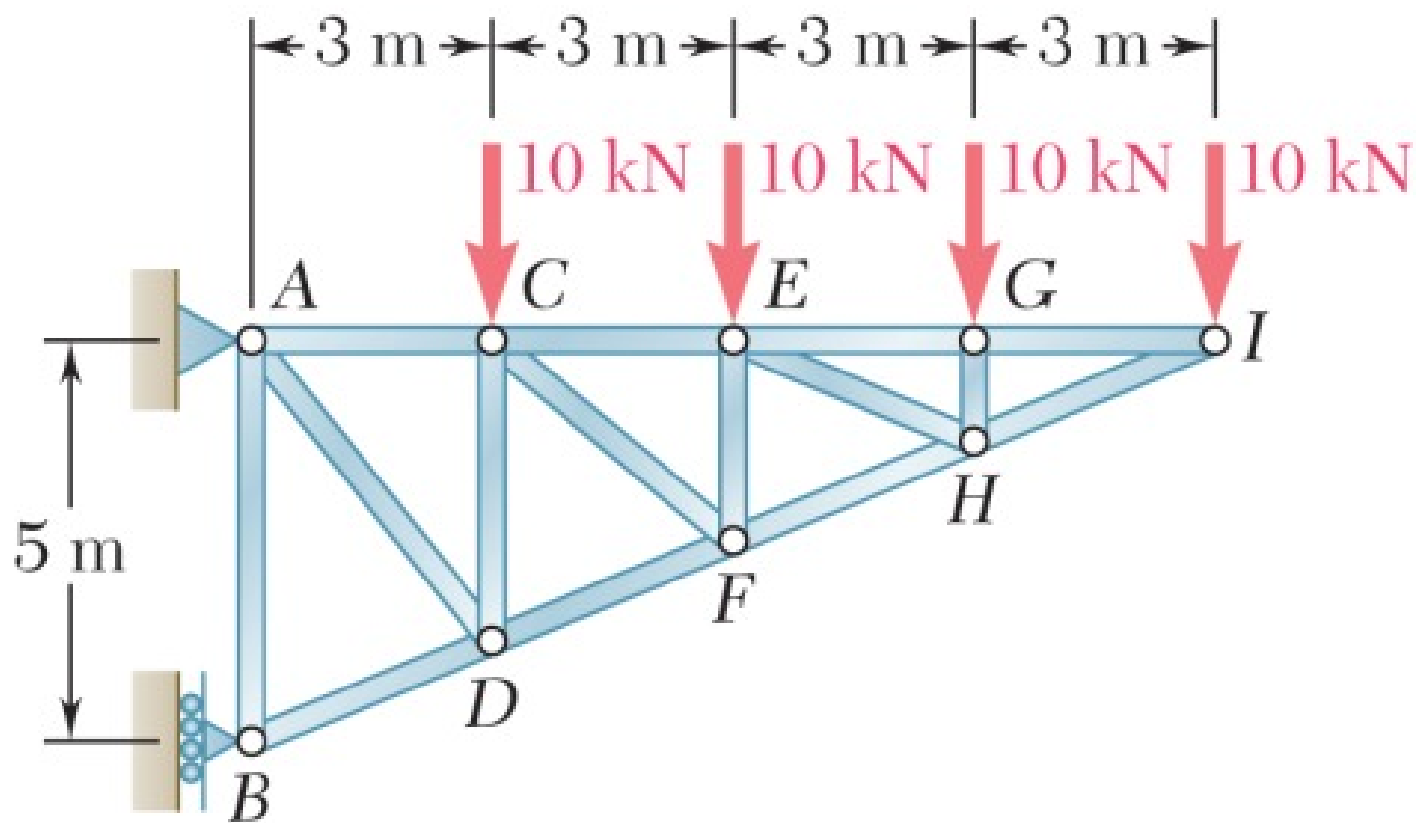
Q11. The frame supports a 400-kg load. Radius of pulley F = 0.5 m. Neglect member weights. Compute all forces on each member. (15 marks) [2016–17]



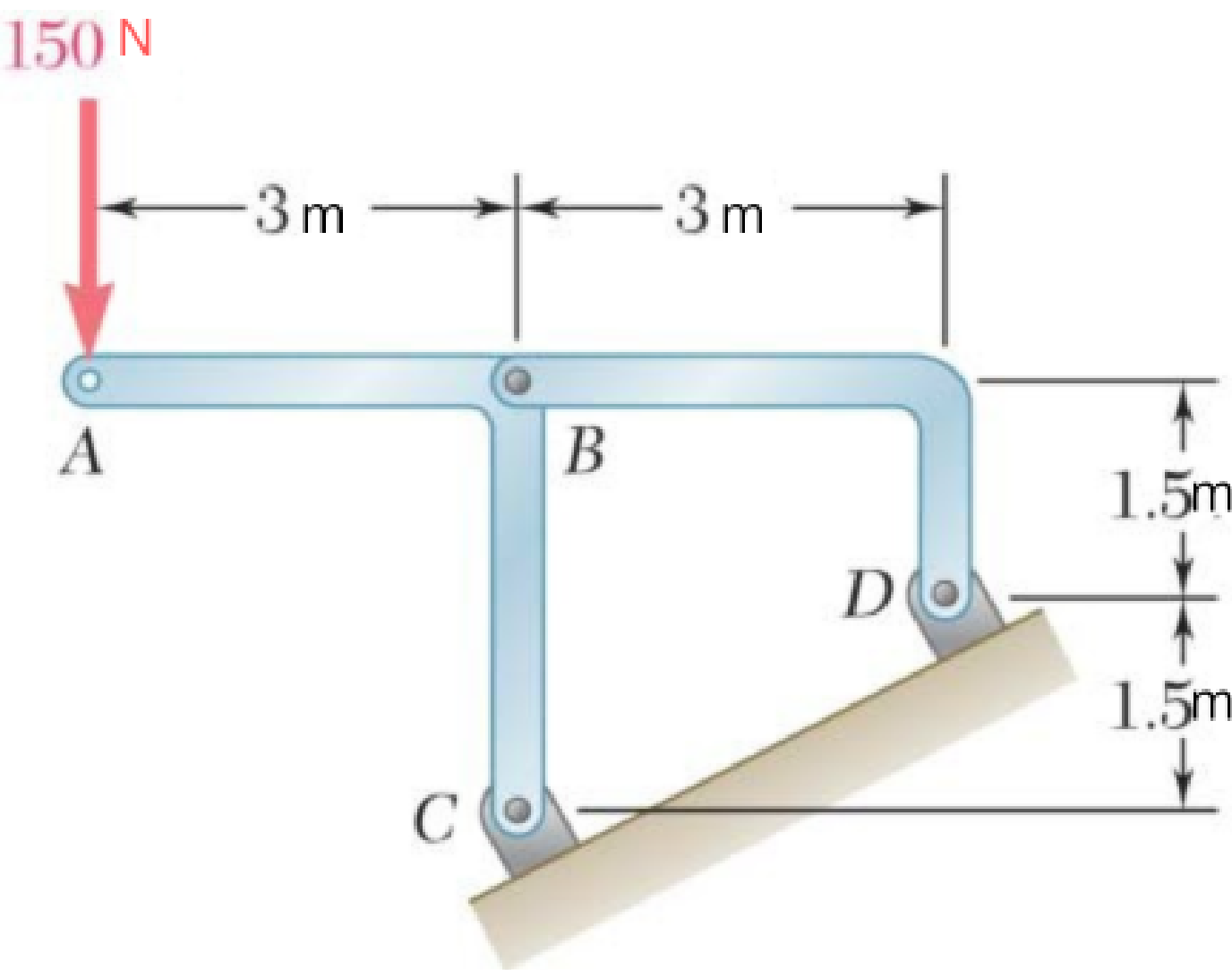
Q12. For the frame and loading, determine the components of all forces acting on member ABC. (15 marks) [2015–16]



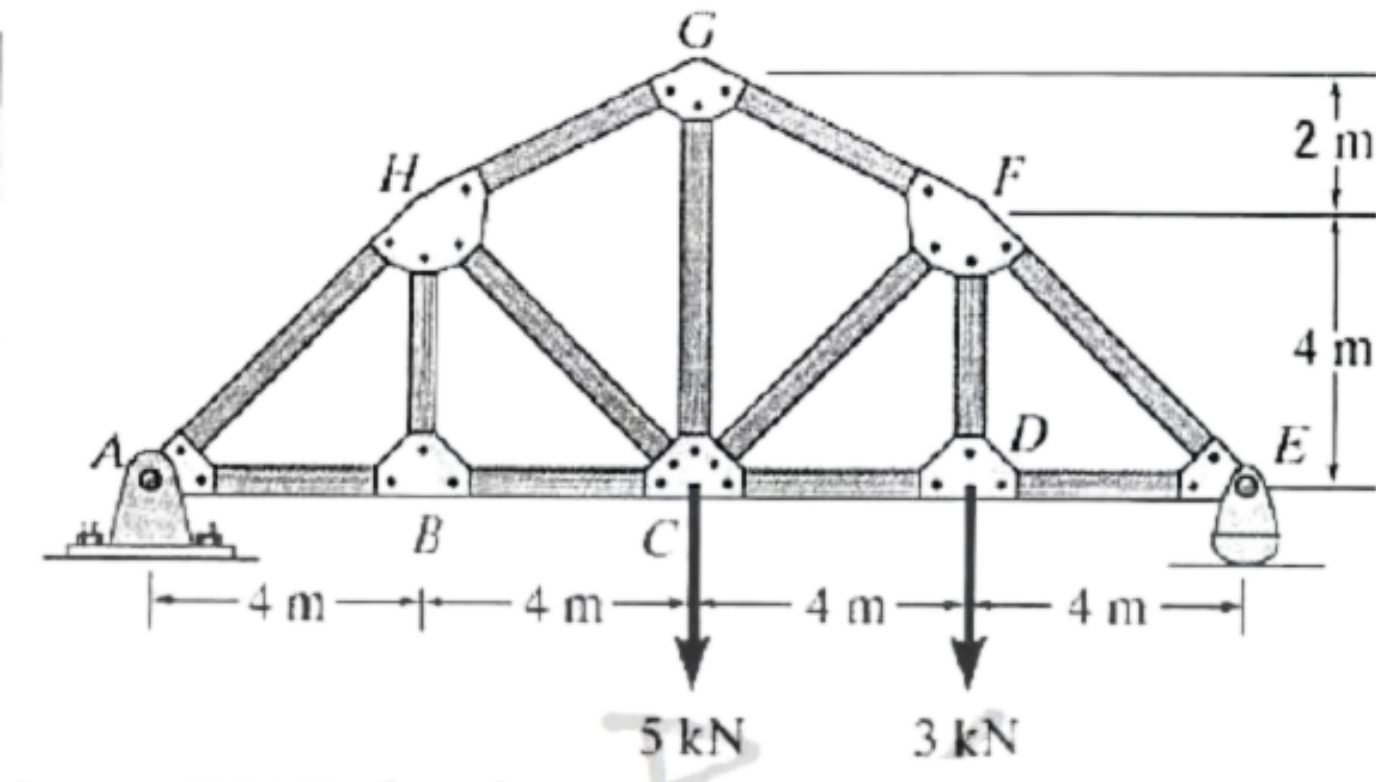
Q13. Determine force in members CD and DF by method of sections. State tension or compression. (20 marks) [2015–16]



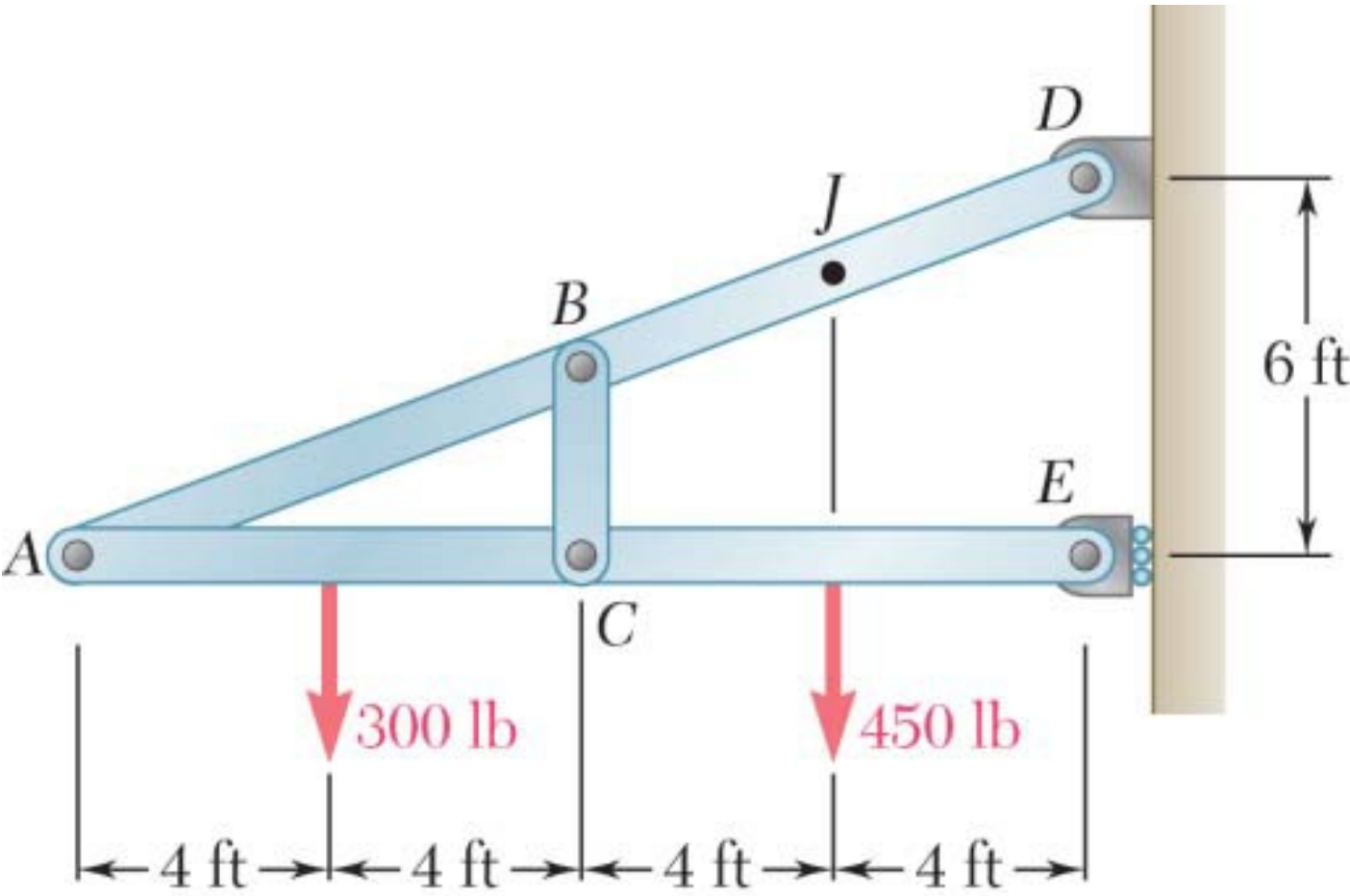
Q14. For the frame and loading, determine the reactions at C and D. (15 marks) [2015–16]



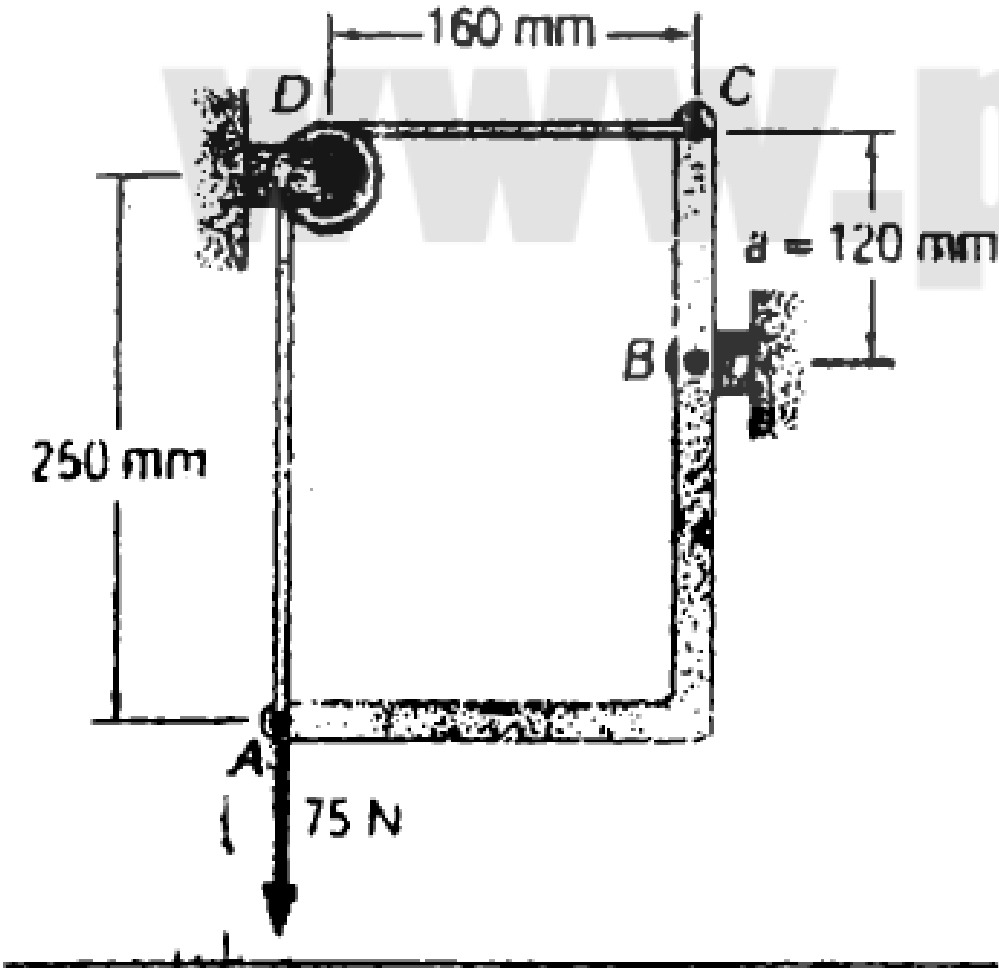
Q15. Determine forces in members CF, CD, and GF of the truss. State tension or compression. (20 marks) [2014–15]



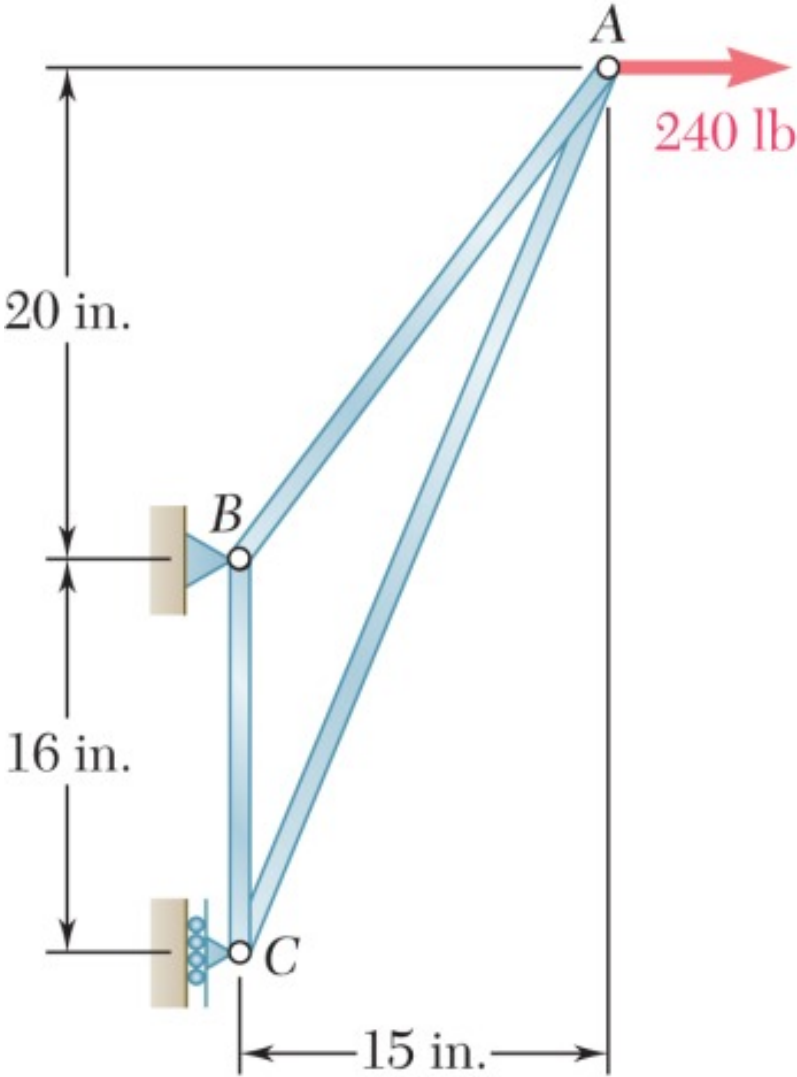
Q16. Determine components of all forces acting on member ABD. (20 marks) [2014–15, 2008–09]



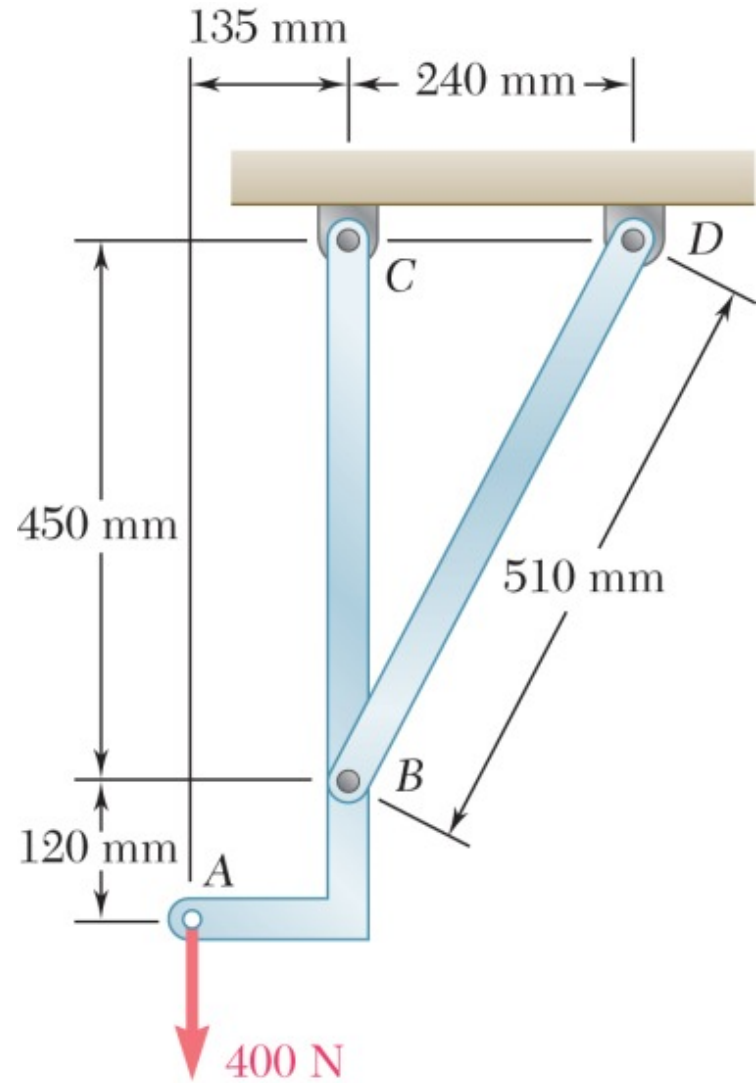
Q17. Member ABC supported by pin bracket at B and inextensible cord over frictionless pulley at D (same tension in AD and CD). Determine tension in cord and reaction at B. (15 marks) [2013–14]



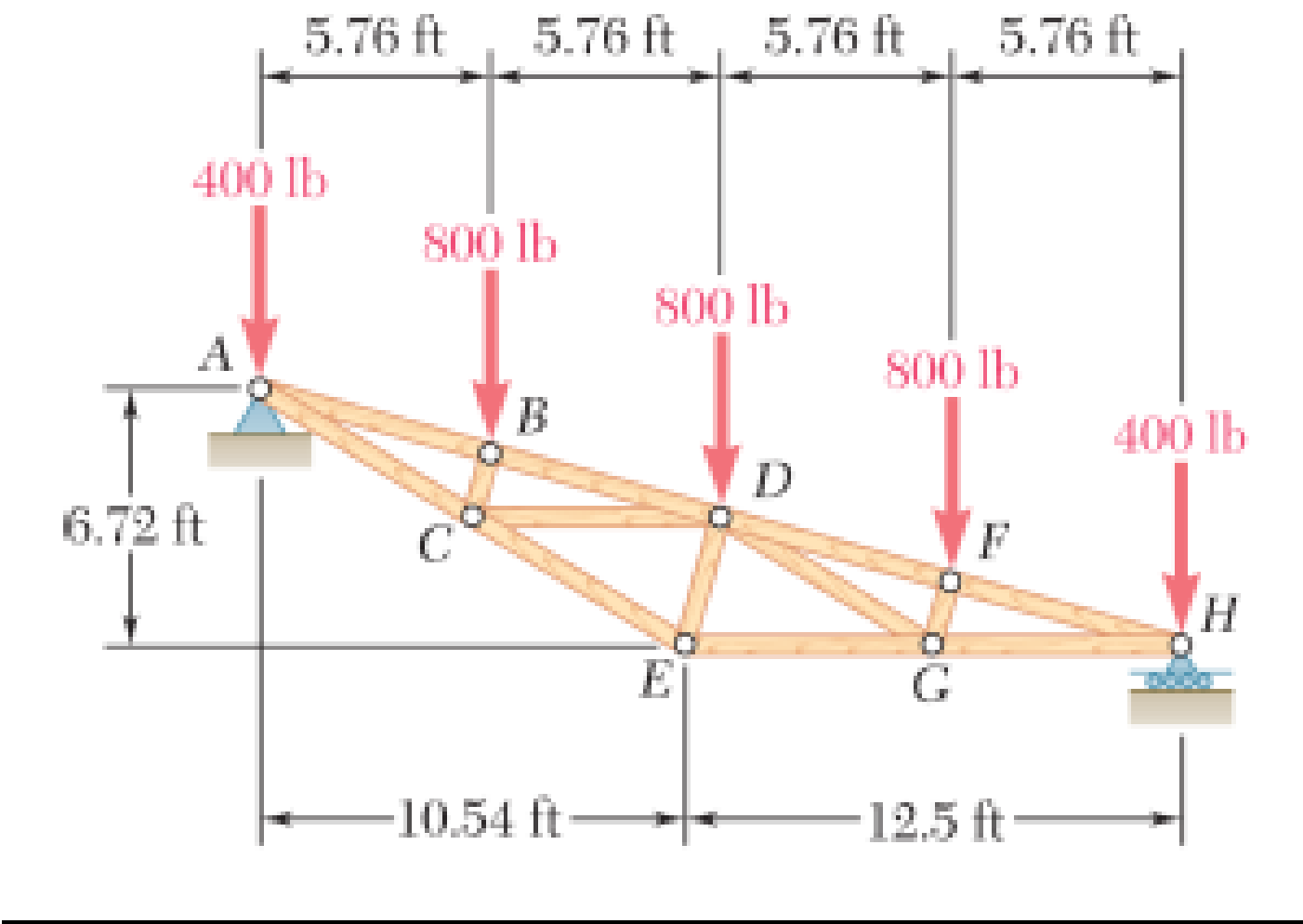
Q18. Using method of joints, determine force in each member of the truss. State tension or compression. (20 marks) [2013–14, 2008–09]



Q19. Determine force in member BD and component of reaction at C. (15 marks) [2013–14]

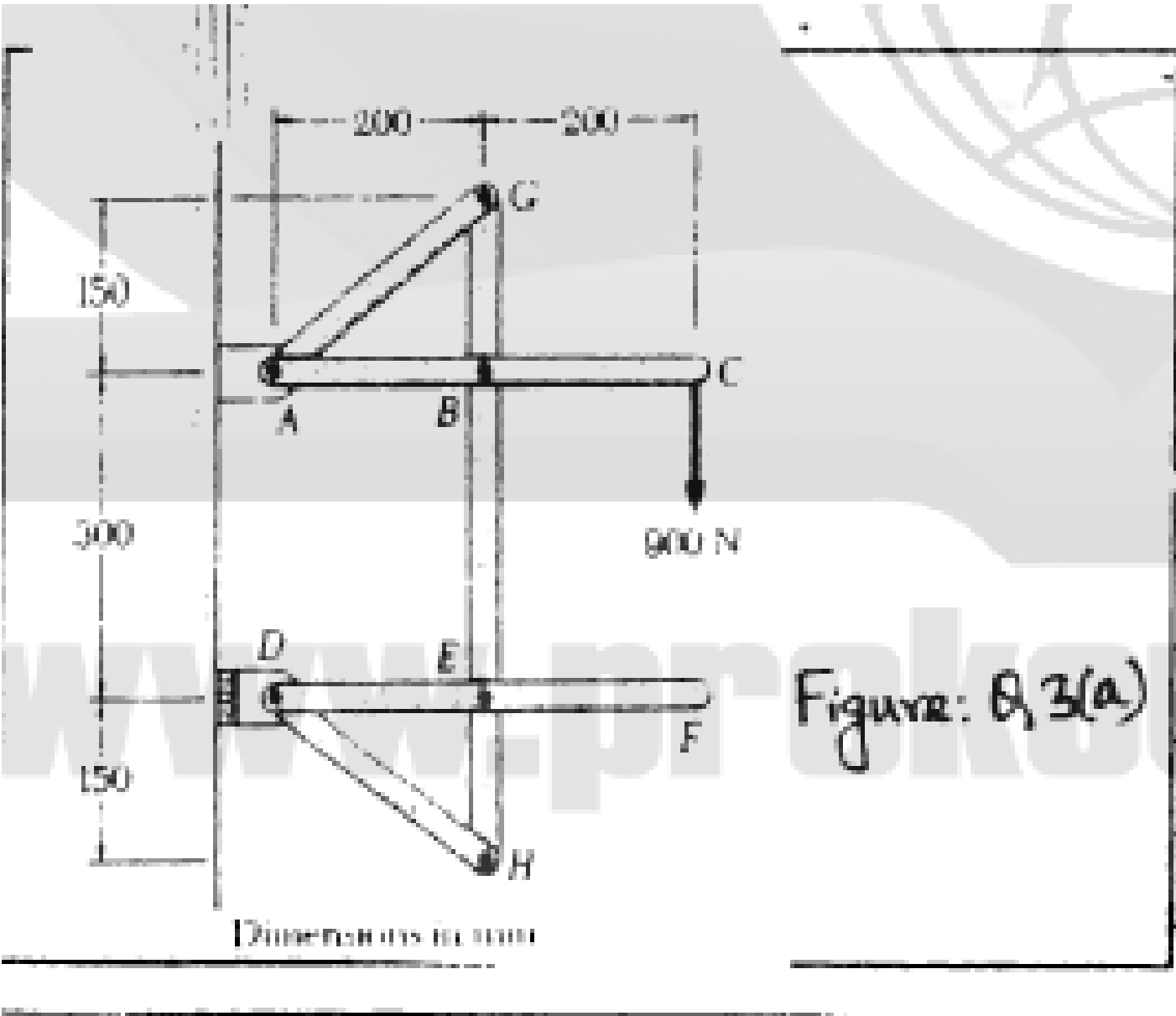


Q20. Determine force in members DE and DC for the inverted Howe roof truss. State tension or compression. (18 marks) [2012–13]



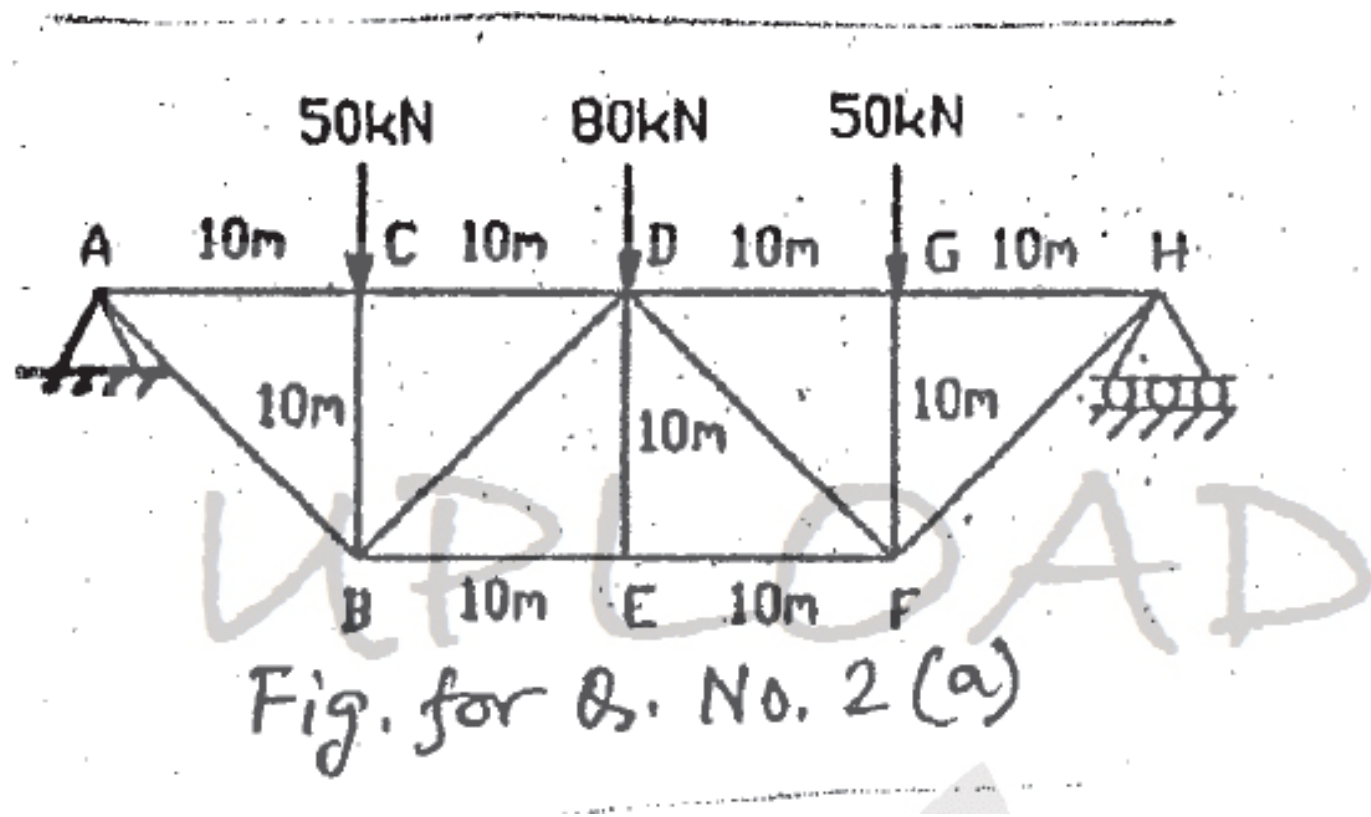
Q21. For the frame and loading, determine components of all forces acting on member GBEH. (18 marks)

[2012–13]

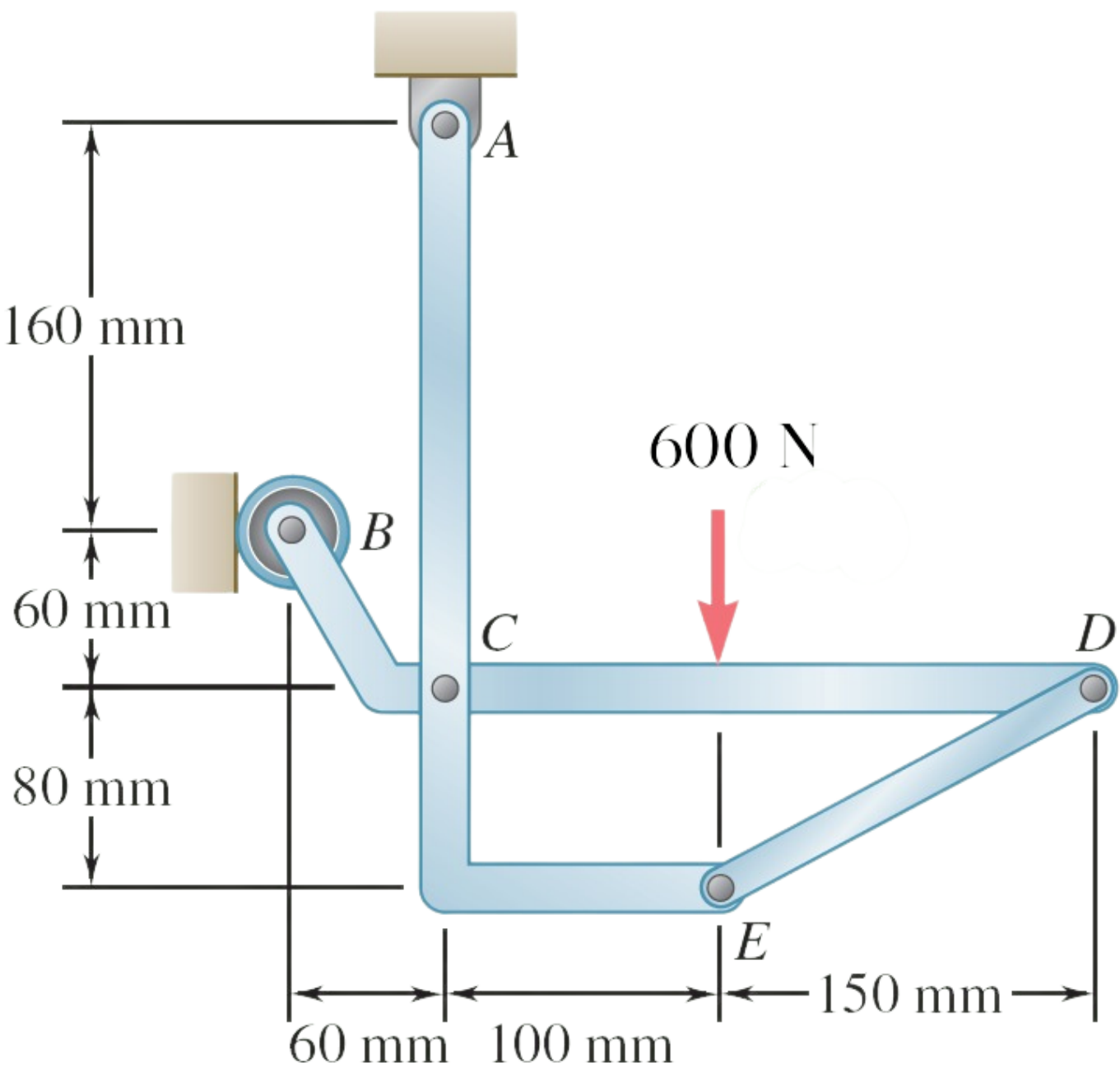


Q22. Find forces in members CD and DB by method of sections. State tension or compression. (17 marks)

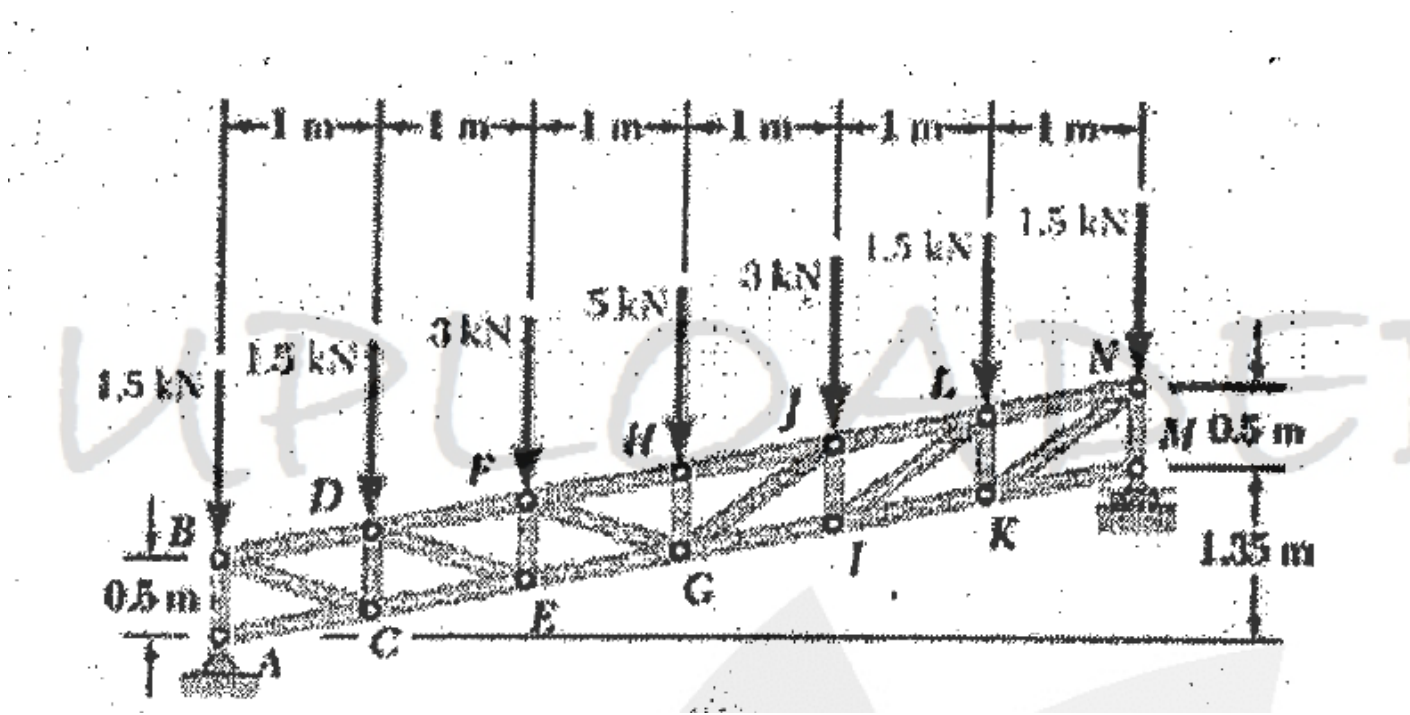
[2011–12]



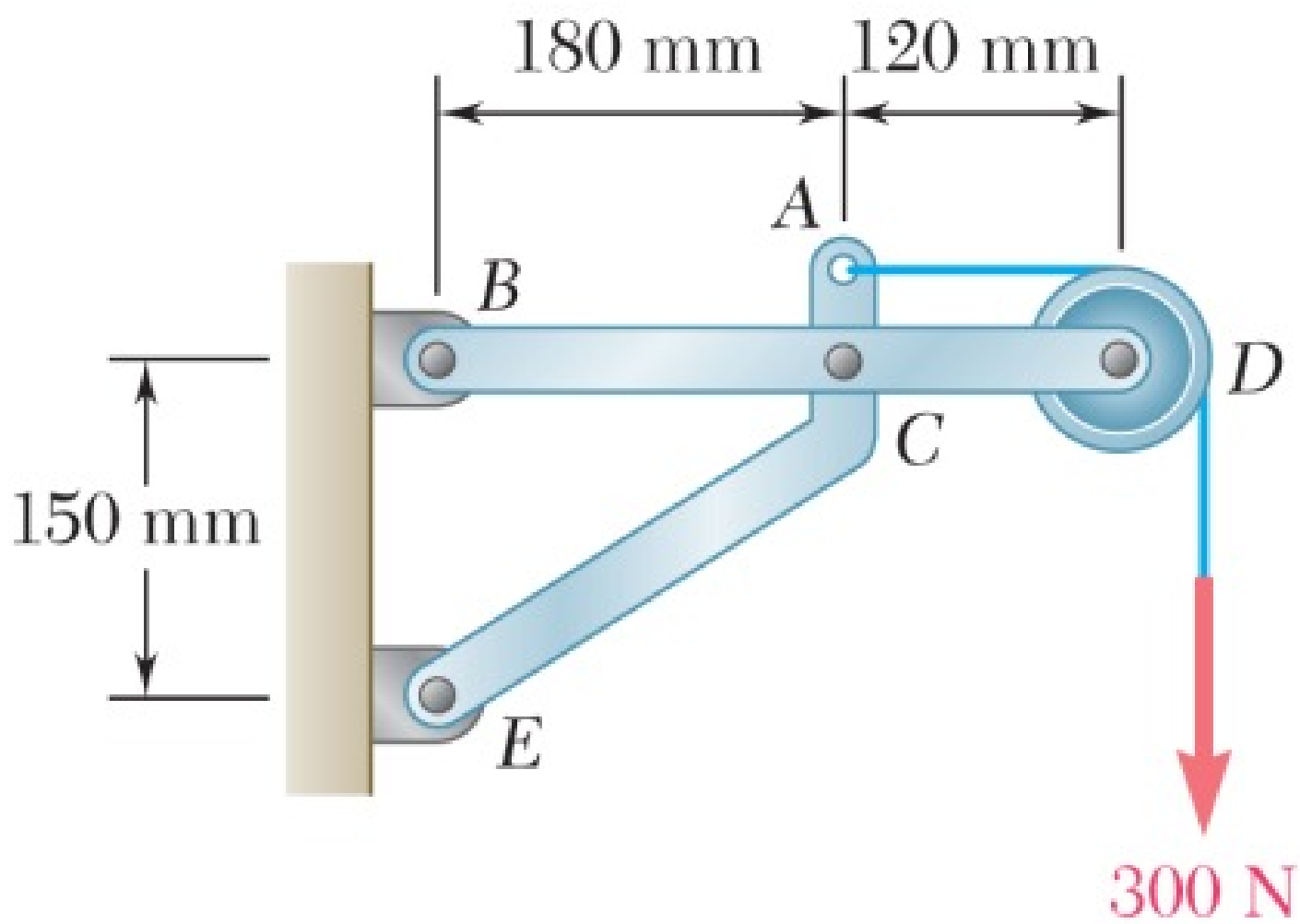
Q23. Members ACE and BCD connected by pin at C and link DE. Determine force in link DE and force at C on member BCD. (18 marks) [2011–12]



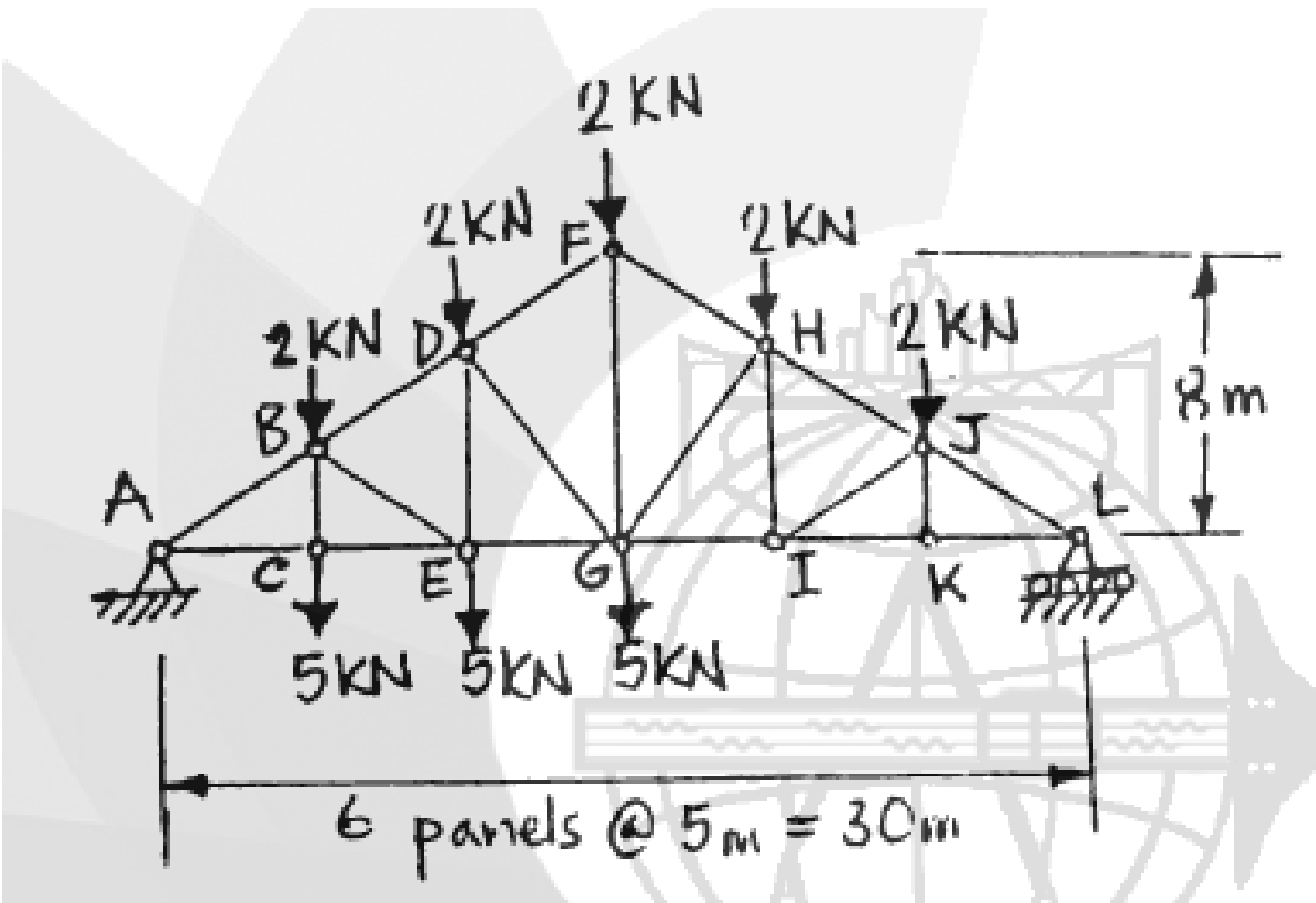
Q24. A parallel chord Pratt truss is loaded. Determine force in members CE, DE, and DF. (18 marks) [2010–11]



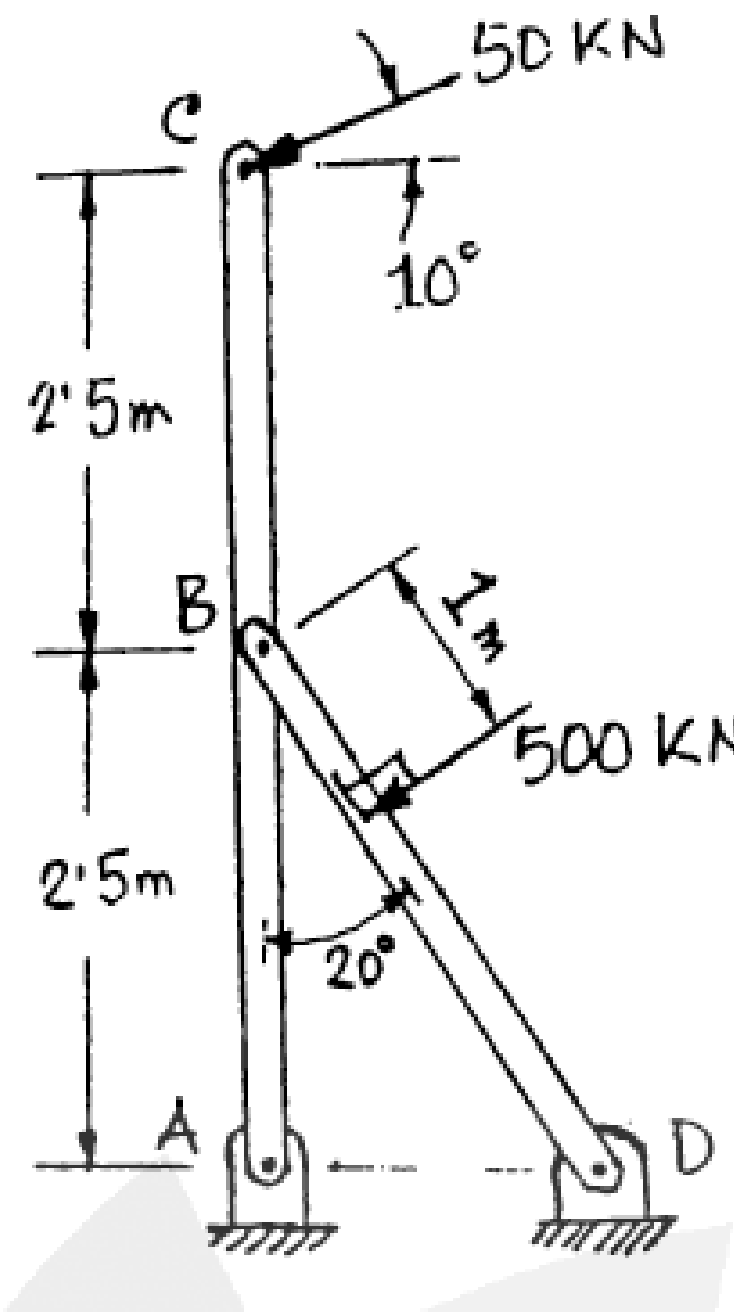
Q25. The pulley has radius 50 mm. Determine components of reactions at B and E. (17 marks) [2010–11]



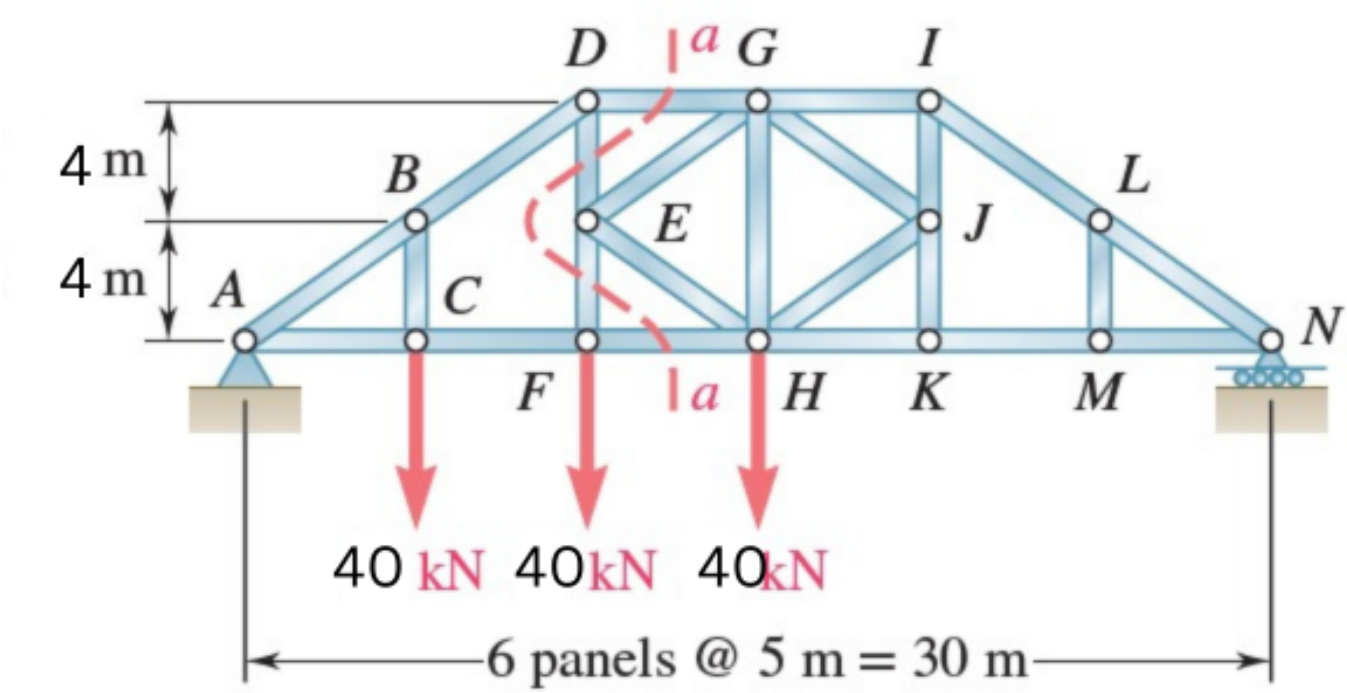
Q26. A roof truss. Determine force in members FH, GH, and GI. (25 marks) [2006–07]



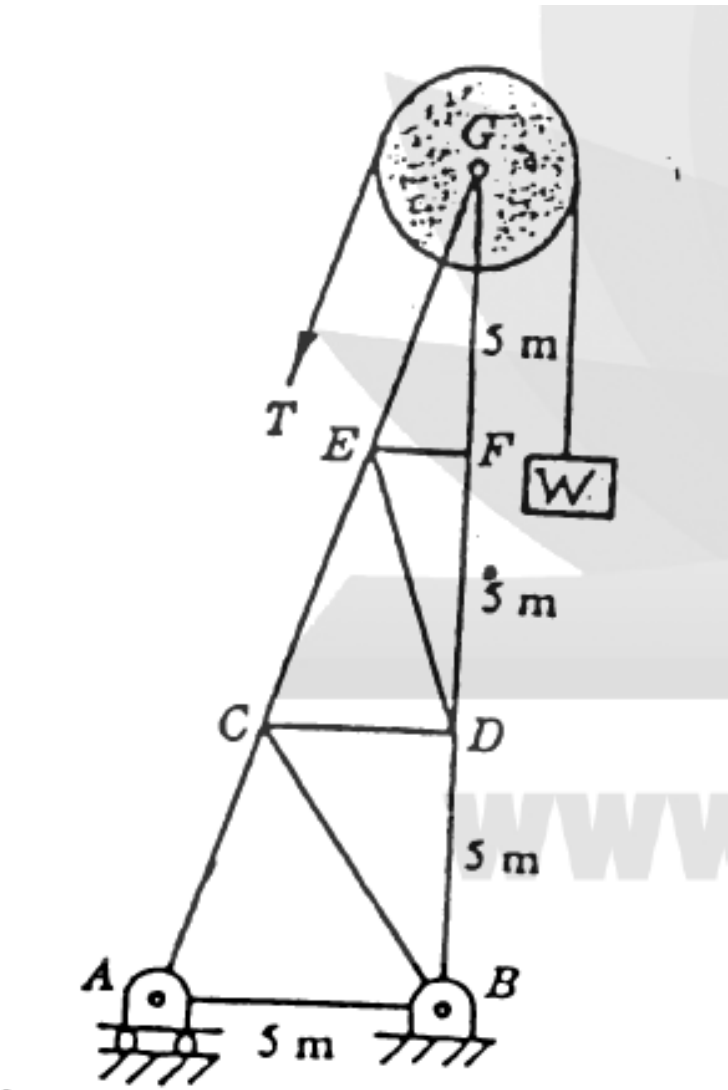
Q27. Simplified model of winch and mast-crutch frame. Calculate forces at points A, B, and D. (22 marks) [2006–07]



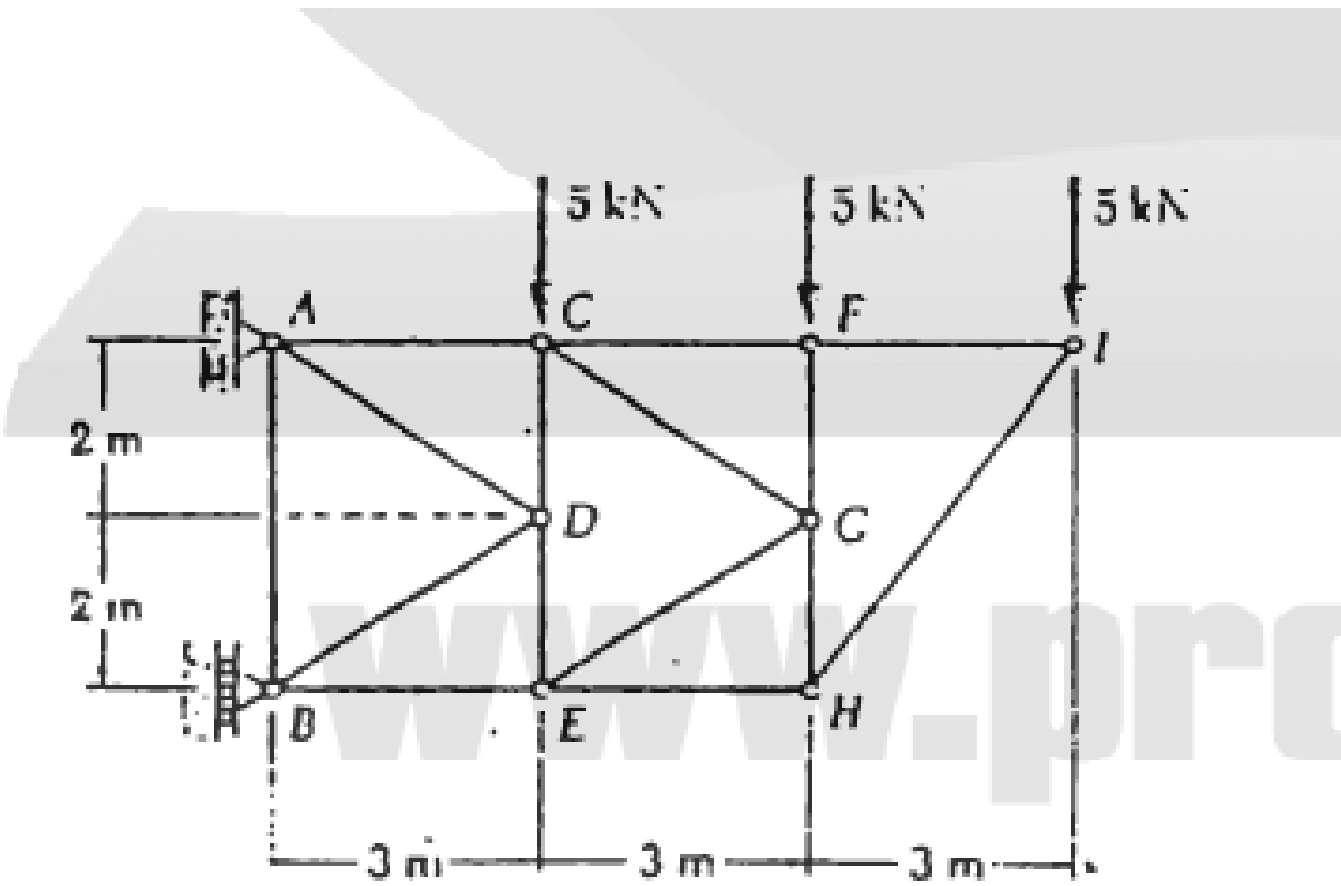
Q28. Determine force in members DG and FH of the truss. (17 marks) [2005–06]



Q29. The cable is hoisting a 2000-kg mass W. Determine forces in AC and BD assuming cable is parallel to the sides. (17.5 marks)
[2004–05]



Q30. Determine force in members AC and BE of the truss. (18 marks) [2003–04]



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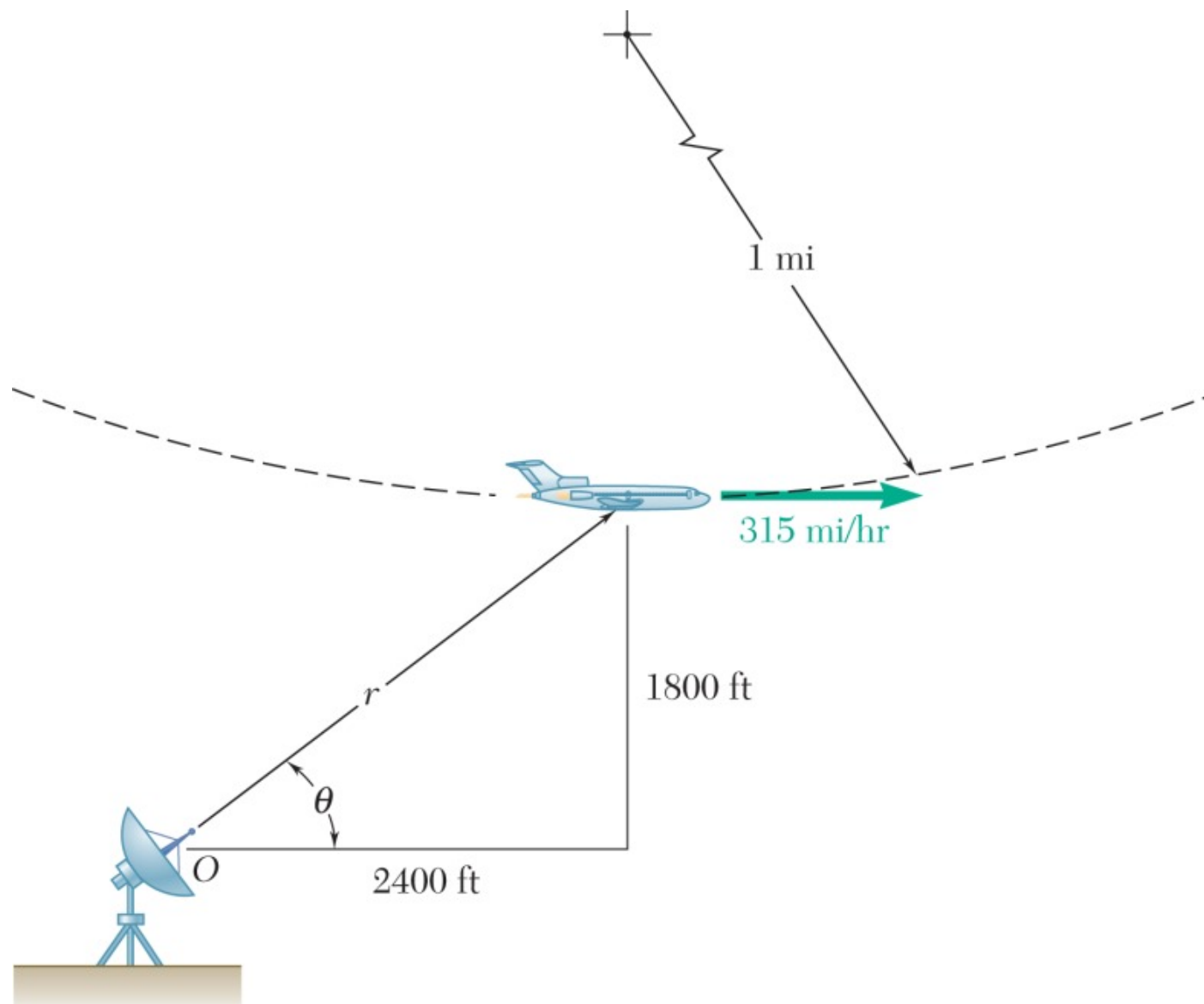
MECHA 165

Section B3

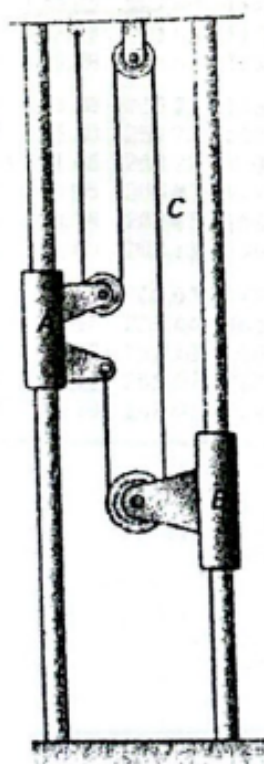
Relative Motion & Kinematics of Particles

B3 — Relative Motion & Kinematics of Particles

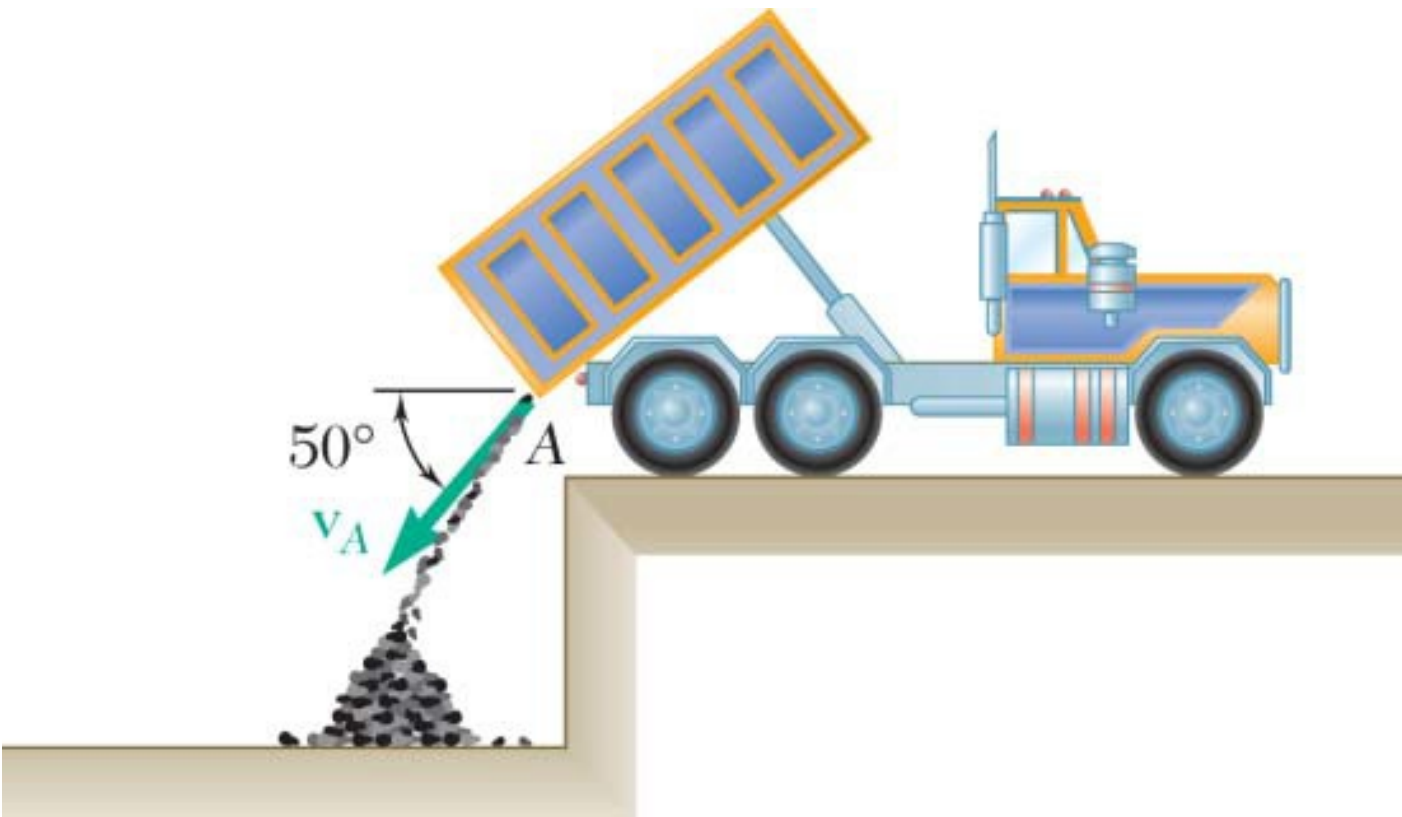
- Q1.** At the bottom of a loop in the vertical plane, an airplane has horizontal velocity 315 mi/h and is speeding up at 10 ft/s^2 . Radius of loop = 1 mi. Tracked by radar at O. Find $\dot{r}$, $\ddot{r}$, $\dot{\theta}$, $\ddot{\theta}$. (15 marks) [2023–24]



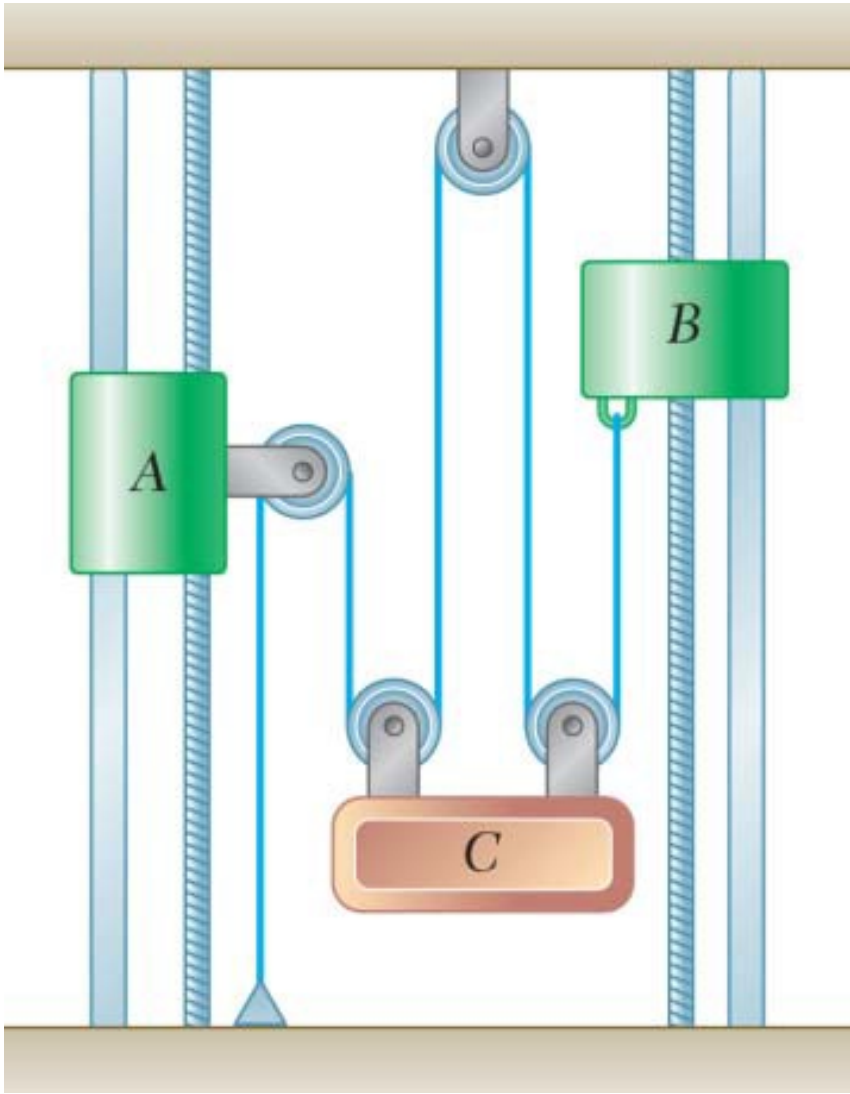
- Q2.** Collar A starts from rest and moves upward with constant acceleration. After 8 s the relative velocity of B w.r.t. A is 24 in./s. Determine: (i) accelerations of A and B, (ii) velocity and change in position of B after 6 s. (15 marks) [2023–24]



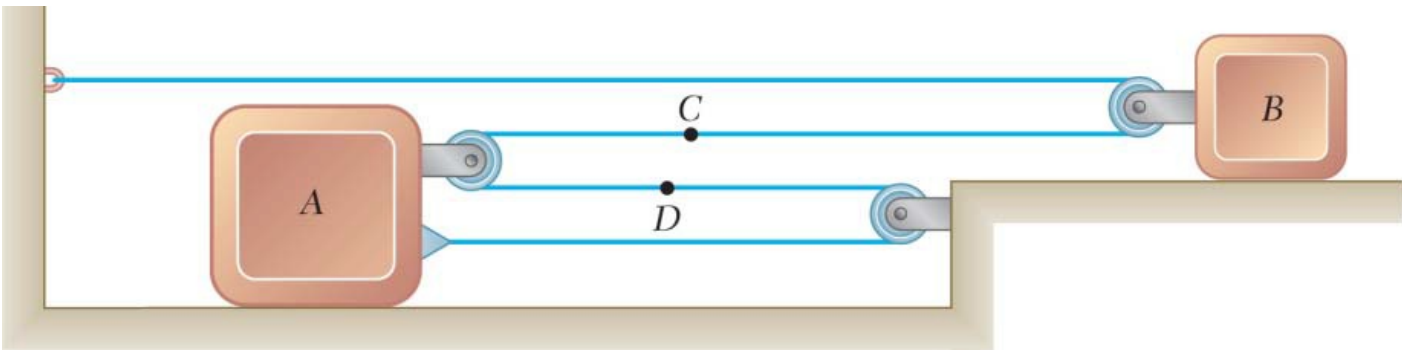
- Q3.** Coal discharged from tailgate A of dump truck with initial velocity $v_A = 2 \text{ m/s}$ at 50° . Determine radius of curvature of trajectory: (i) at A, (ii) at 1 m below A. (15 marks) [2023–24]



Q4. Collar A starts from rest at $t = 0$, moves downward at 7 in./s^2 . Collar B moves upward with constant acceleration and initial velocity 8 in./s ; B moves 20 in. between $t = 0$ and $t = 2 \text{ s}$. Determine: (i) accelerations of B and block C, (ii) time when velocity of C is zero, (iii) distance C has moved at that time. **(20 marks)** [2021–22]



Q5. Slider block B moves with constant acceleration; speed = 150 mm/s . After block A moves 240 mm right, velocity = 60 mm/s . Determine: (i) accelerations of A and B, (ii) acceleration of cable portion D, (iii) velocity and position of B after 4 s . **(17 marks)** [2020–21]



- Q6.** Cars A and B travel at 40 m/s and 30 m/s. B increasing speed at 2 m/s^2 ; A constant speed. Radius of curvature at B: $\rho = 200 \text{ m}$. Determine velocity and acceleration of B w.r.t. A. (18 marks) [2020–21]

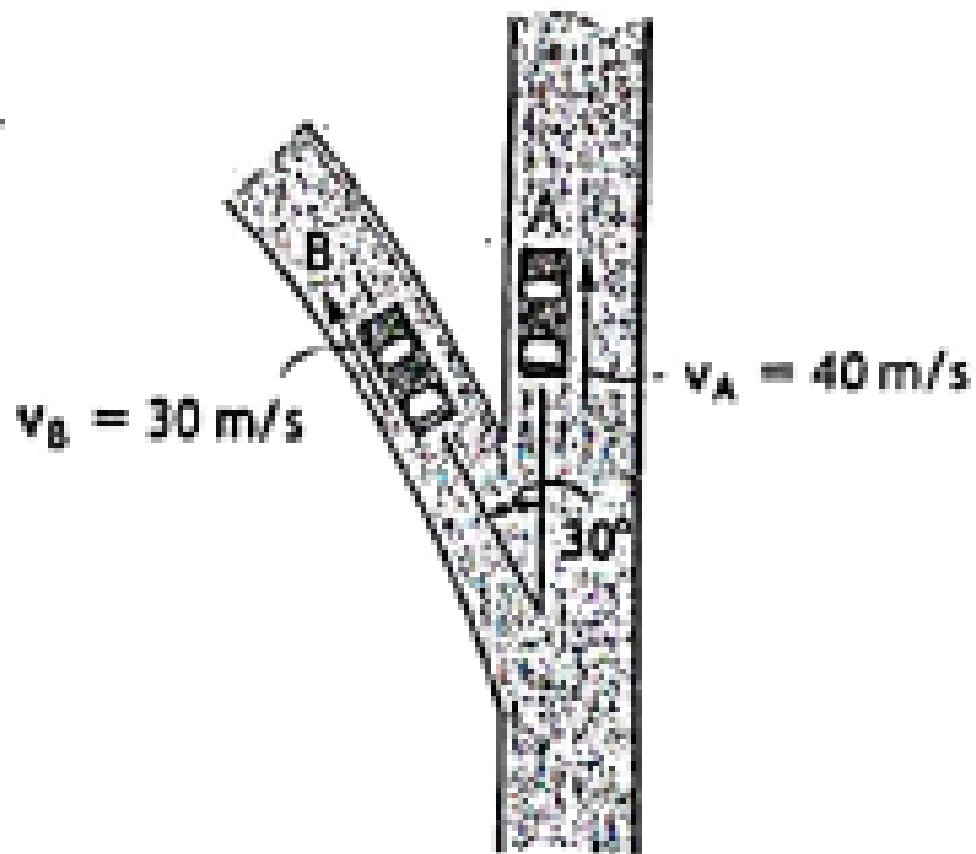


Fig. for Q. 7(b)

- Q7.** Car B travels along curved road at $x \text{ m/s}$ decelerating at $y \text{ m/s}^2$. Car C travels along straight road at $2x \text{ m/s}$ decelerating at $1.5y \text{ m/s}^2$. Determine velocity and acceleration of B relative to C. Hints : Take $[x = 15 + 0.02 \times (\text{student ID}), y = 2 + 0.01 \times (\text{student ID})]$ (30 marks) [2019–20]

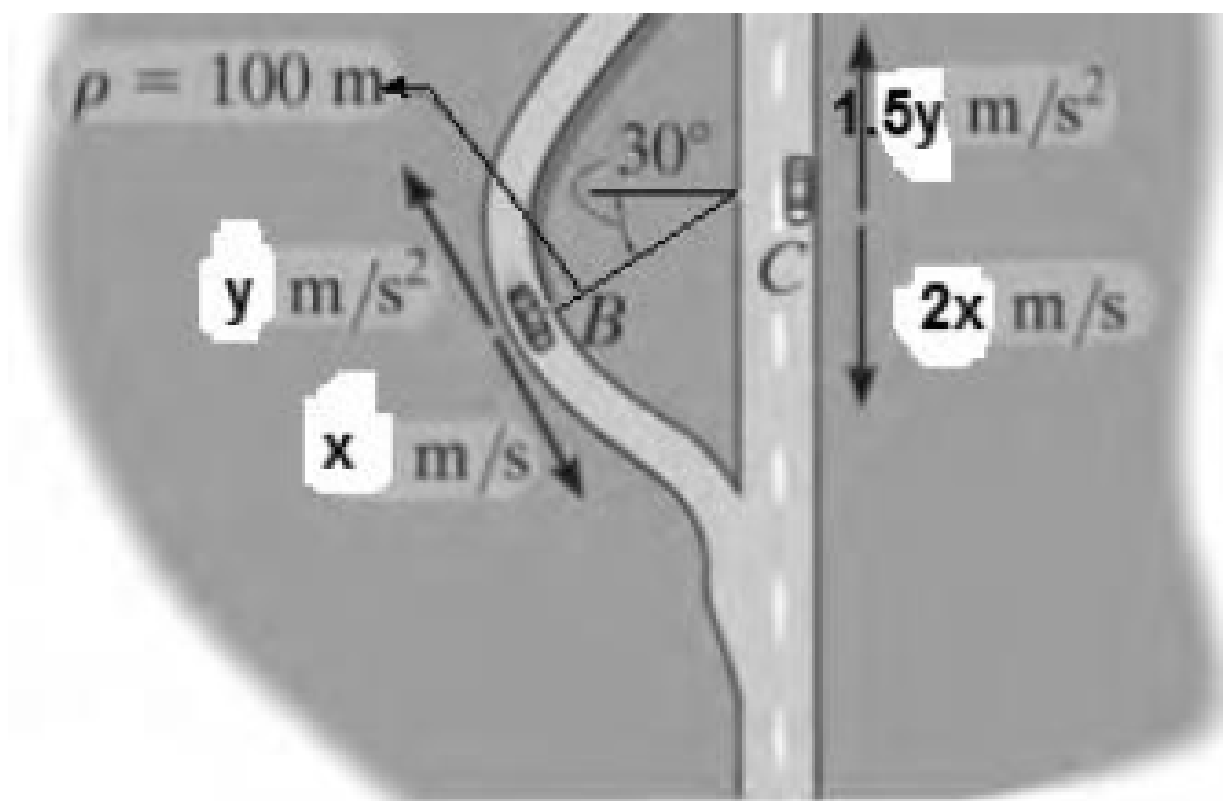
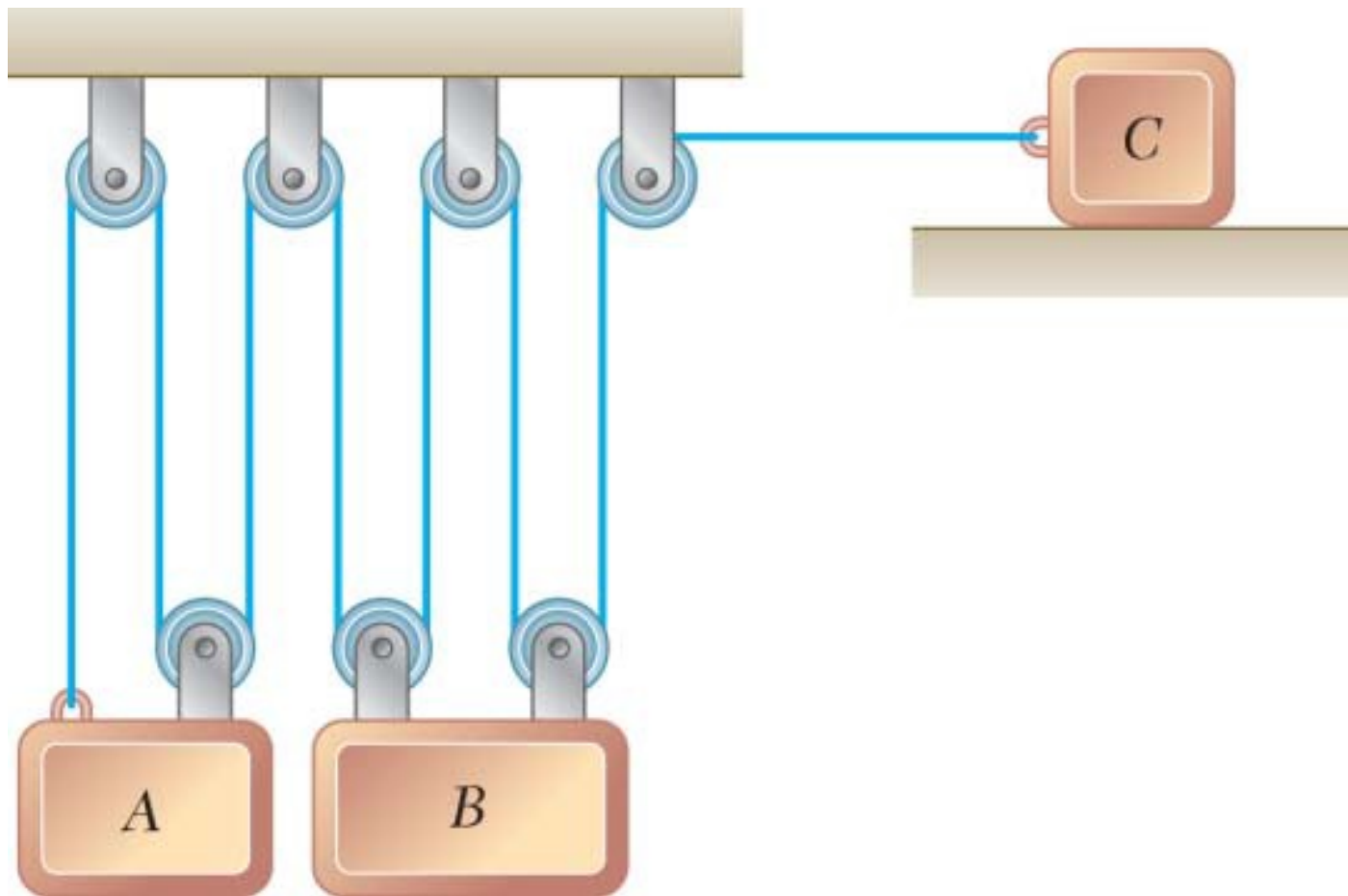
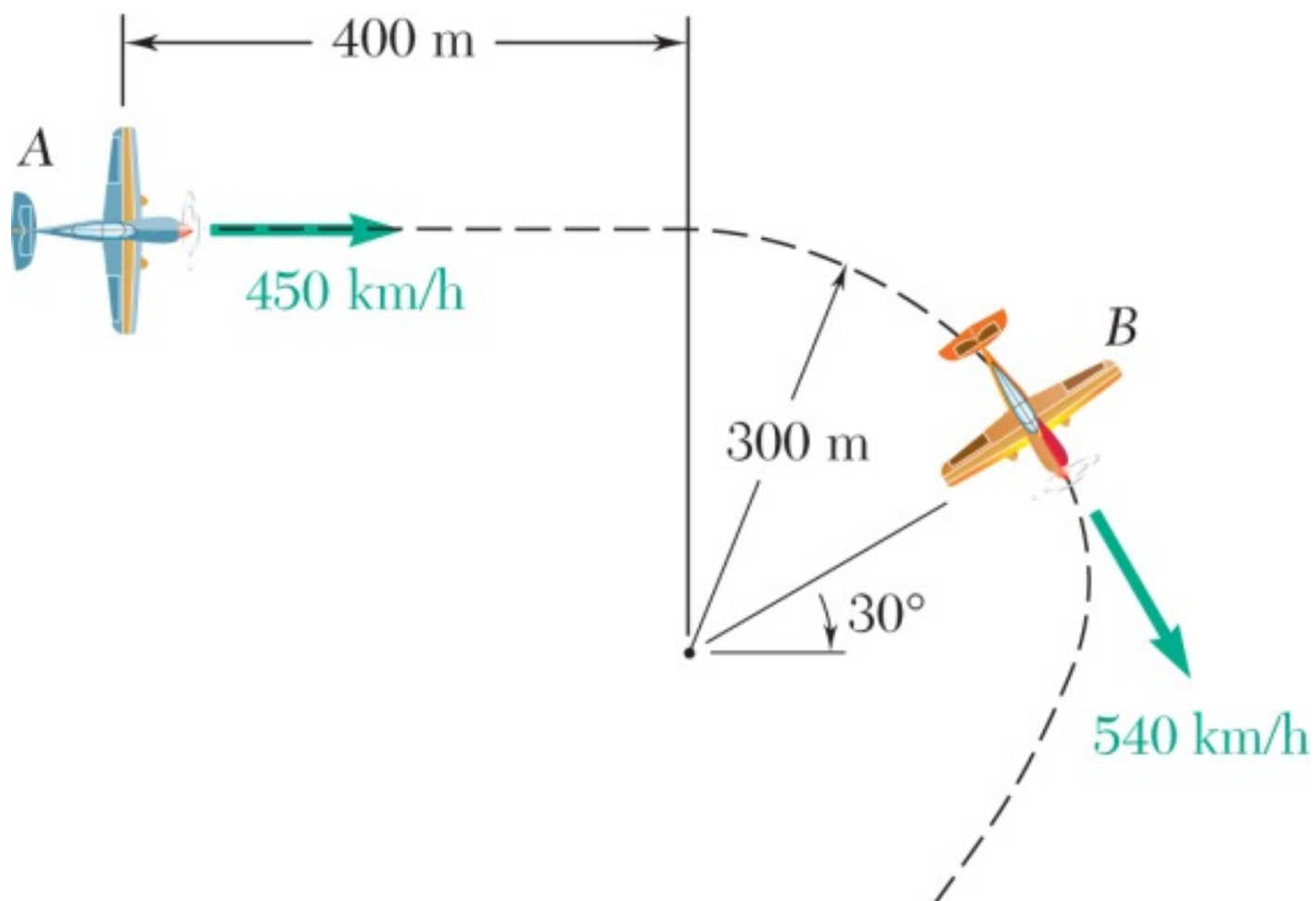


Fig. for Q-7

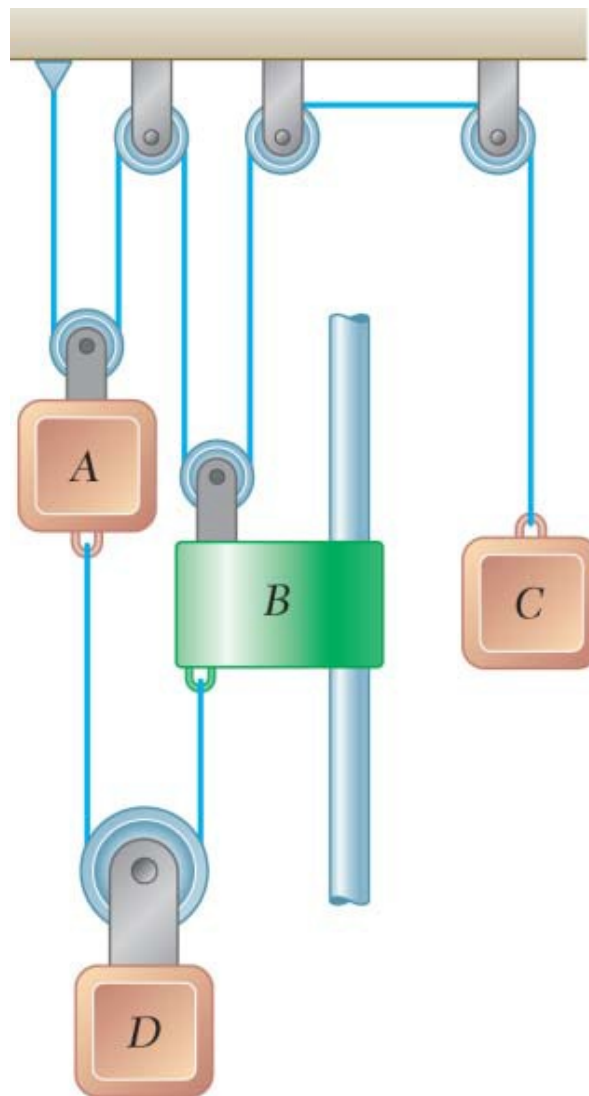
- Q8.** Block B moves downward with a constant velocity of 20 mm/s. At $t = 0$, block A is moving upward with a constant acceleration, and its velocity is 30 mm/s. Knowing that at $t = 3 \text{ s}$ slider block C has moved 57 mm to the right, determine (a) the velocity of slider block C at $t = 0$, (b) the accelerations of A and C, (c) the change in position of block A after 5 s. (17 marks) [2018–19]



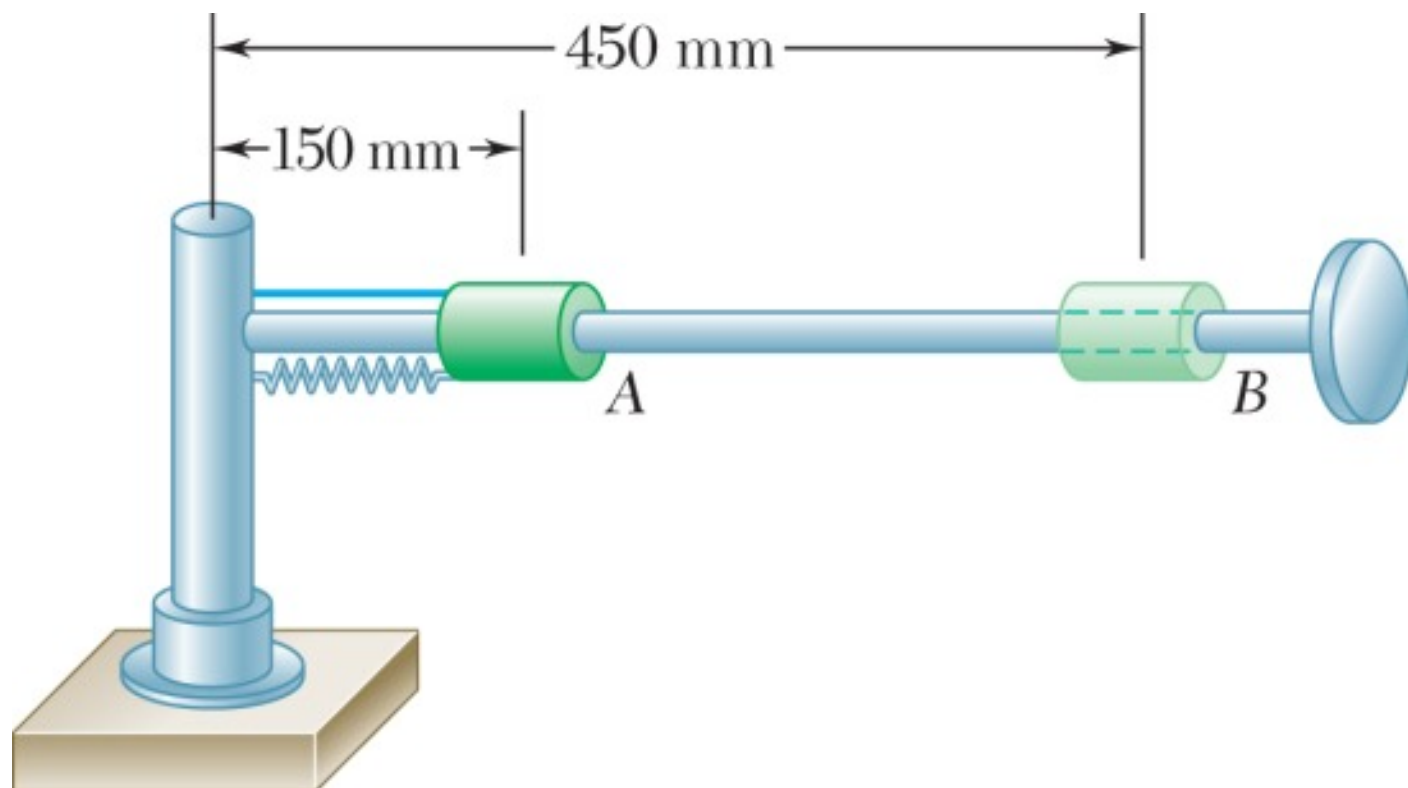
- Q9.** At a given instant in an airplane race, airplane A is flying horizontally in a straight line, and its speed is being increased at the rate of 8 m/s^2 . Airplane B is flying at the same altitude as airplane A and, as it rounds a pylon, is following a circular path of 300-m radius. Knowing that at the given instant the speed of B is being decreased at the rate of 3 m/s^2 , determine, for the positions shown, (a) the velocity of B relative to A, (b) the acceleration of B relative to A. **(17 marks)** [2017–18]



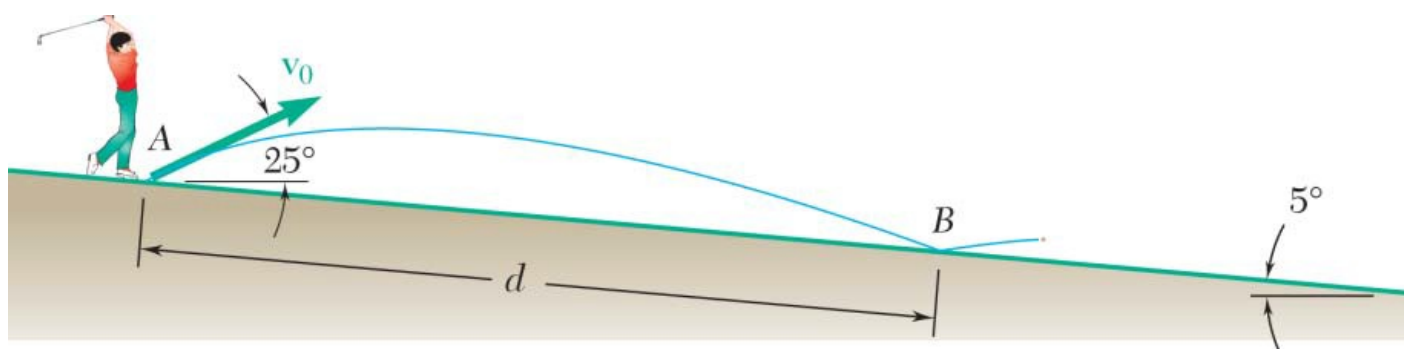
- Q10.** The system shown starts from rest, and each component moves with a constant acceleration. If the relative acceleration of block C with respect to collar B is 60 mm/s^2 upward and the relative acceleration of block D with respect to block A is 110 mm/s^2 downward, determine (a) the velocity of block C after 3 s, (b) the change in position of block D after 5 s. **(20 marks)** [2016–17]



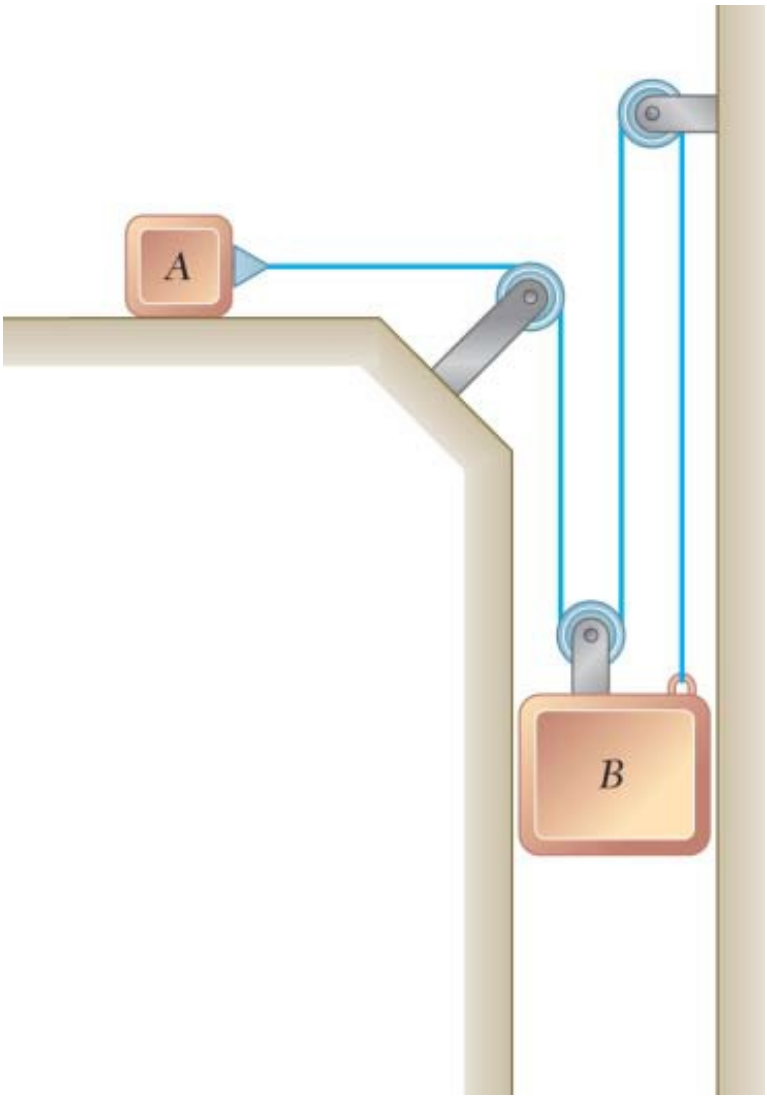
- Q11.** A 1-kg collar slides on a horizontal rod free to rotate about a vertical shaft. Spring constant = 30 N/m, unstretched at A. Rod rotates at 16 rad/s; cord is cut. Determine: (i) radial and transverse accelerations at A, (ii) acceleration of collar relative to rod at A, (iii) transverse velocity at B. **(15 marks)** [2016–17]



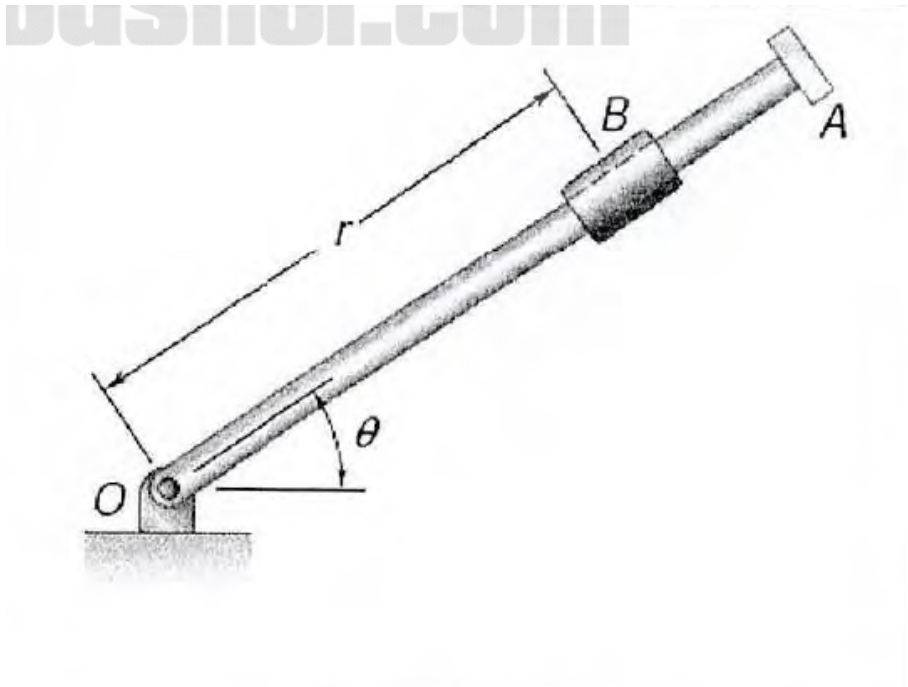
- Q12.** A golfer hits a ball at 50 m/s at 25° above horizontal. Fairway slopes downward at 5° . Determine distance d to point B where ball first lands. **(20 marks)** [2015–16]



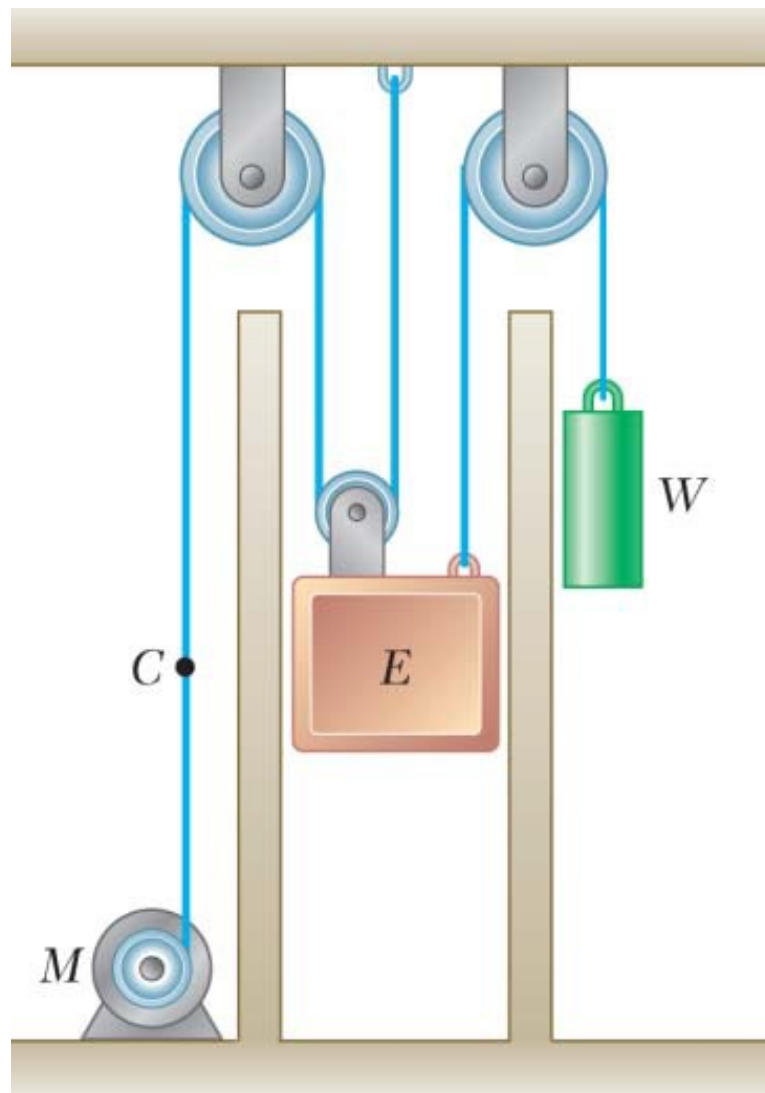
- Q13.** Block B starts from rest and moves downward with constant acceleration. After slider A moves 9 in., velocity = 6 ft/s. Determine: (a) accelerations of A and B, (b) velocity and change in position of B after 2 s. **(20 marks)** [2014–15]



Q14. Rod OA rotates about O in a horizontal plane. Motion of 0.5-lb collar B: $r = 10 + 6 \cos \pi t$, $\theta = \pi(4t^2 - 8t)$. Determine radial and transverse components of force on collar at $t = 0.5$ s. (20 marks) [2014–15]

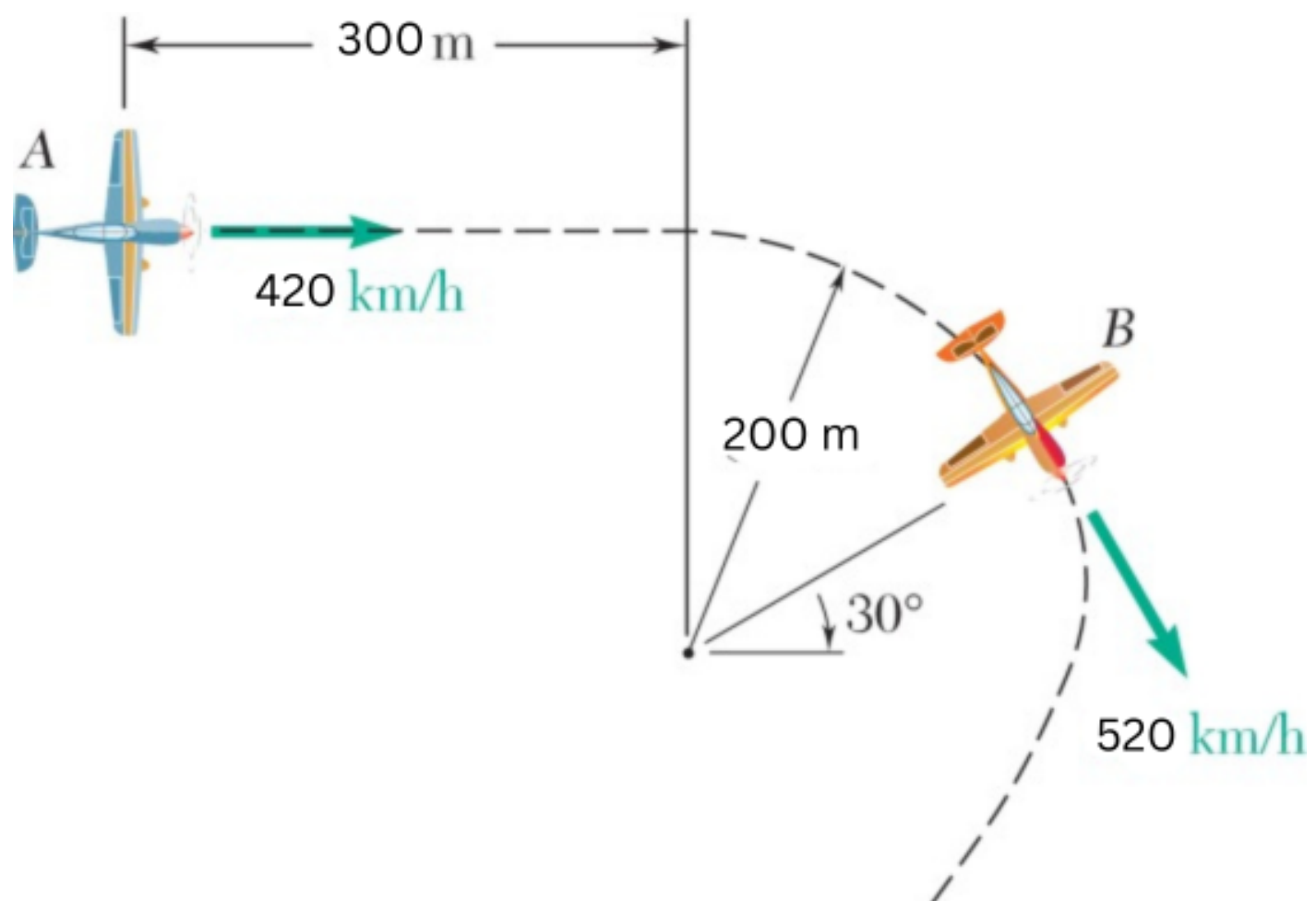


Q15. Elevator moves downward at constant velocity 4 m/s. Determine: (i) velocity of cable C, (ii) velocity of counterweight W, (iii) relative velocity of C w.r.t. elevator, (iv) relative velocity of W w.r.t. elevator. (20 marks) [2013–14]

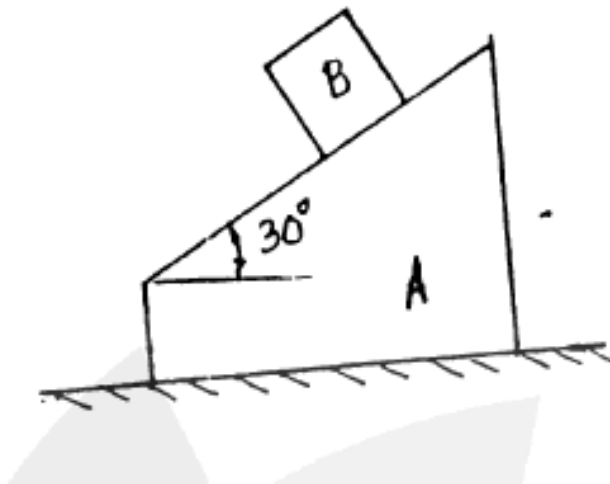


Q16. Airplane A flies horizontally with speed increasing at 6 m/s^2 . Airplane B rounds a pylon on a 200-m circular path with speed decreasing at 2 m/s^2 . Determine: (i) velocity of B relative to A, (ii) acceleration of B relative to A. **(18 marks)** [2012–13]

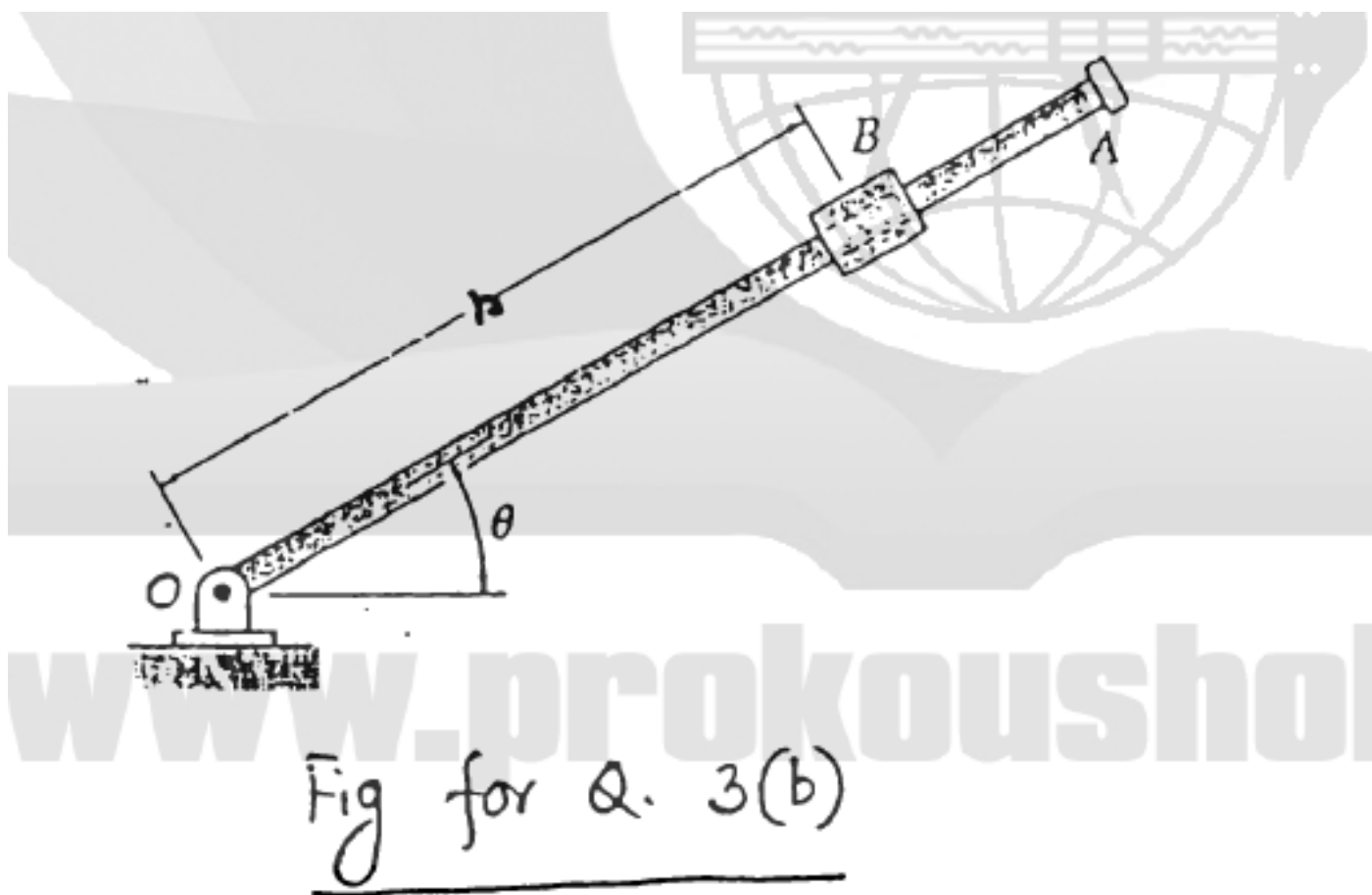
↪ Similar: B3-Q10 same type, different values



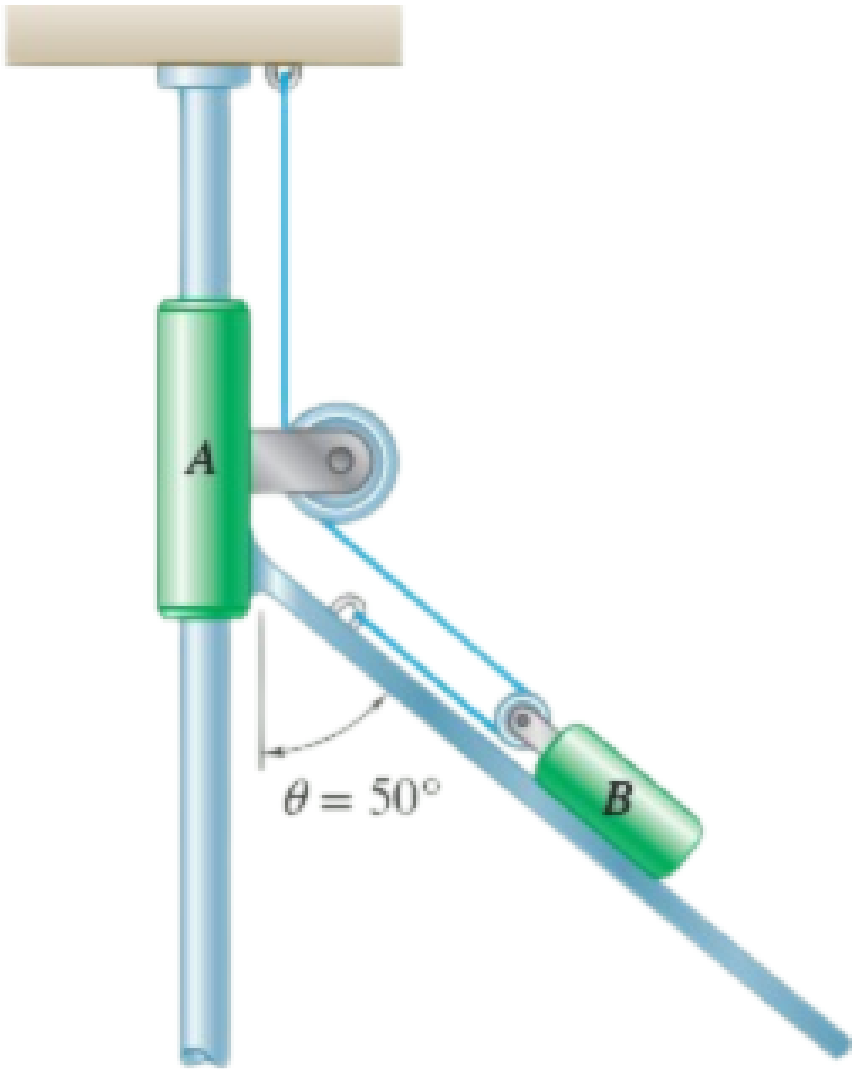
Q17. At $t = 0$, wedge A starts moving right at 100 mm/s^2 and block B starts moving toward left along wedge at 150 mm/s^2 relative to wedge. Determine: (i) acceleration of block B, (ii) velocity of block B at $t = 4 \text{ s}$. **(17 marks)** [2006–07]



- Q18.** The 0.9-m arm OA rotates as $\theta = 0.15t^2$. Collar B slides along arm so that $r = 0.9 - 0.12t^2$. After arm rotates 30° , determine: (i) total velocity, (ii) total acceleration, (iii) relative acceleration of collar w.r.t. arm. **(20 marks)** [2005–06]



- Q19.** Assembly A has velocity 225 mm/s and acceleration 375 mm/s², both downward. Determine: (a) velocity of block B, (b) acceleration of block B. **(17 marks)** [2004–05]



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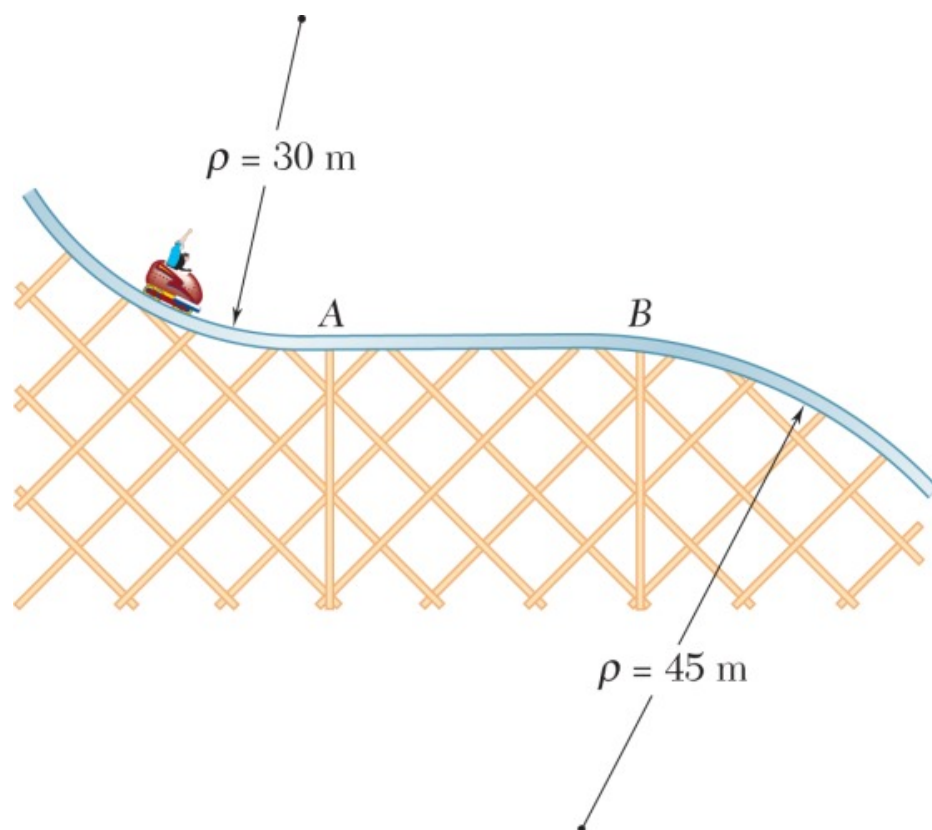
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Section B4

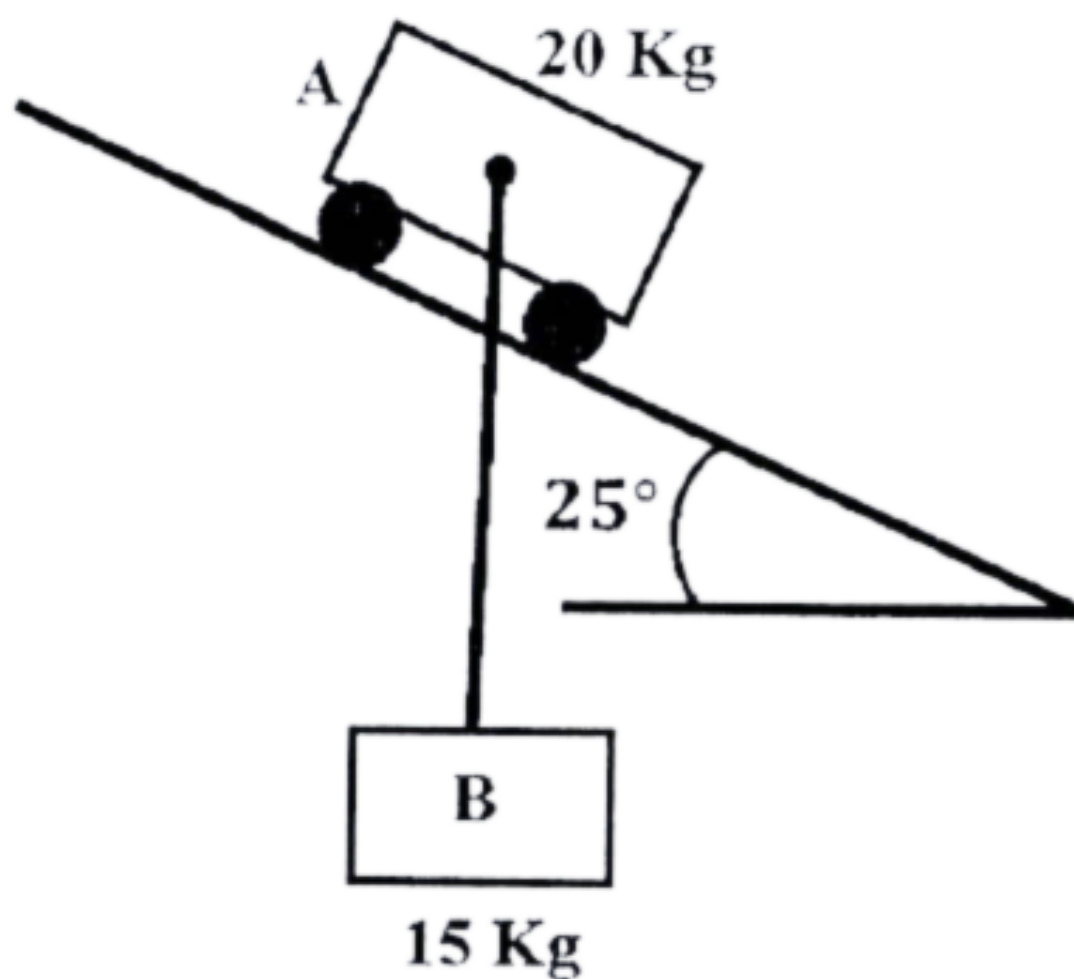
Newton's Second Law of Motion

B4 — Newton's Second Law of Motion

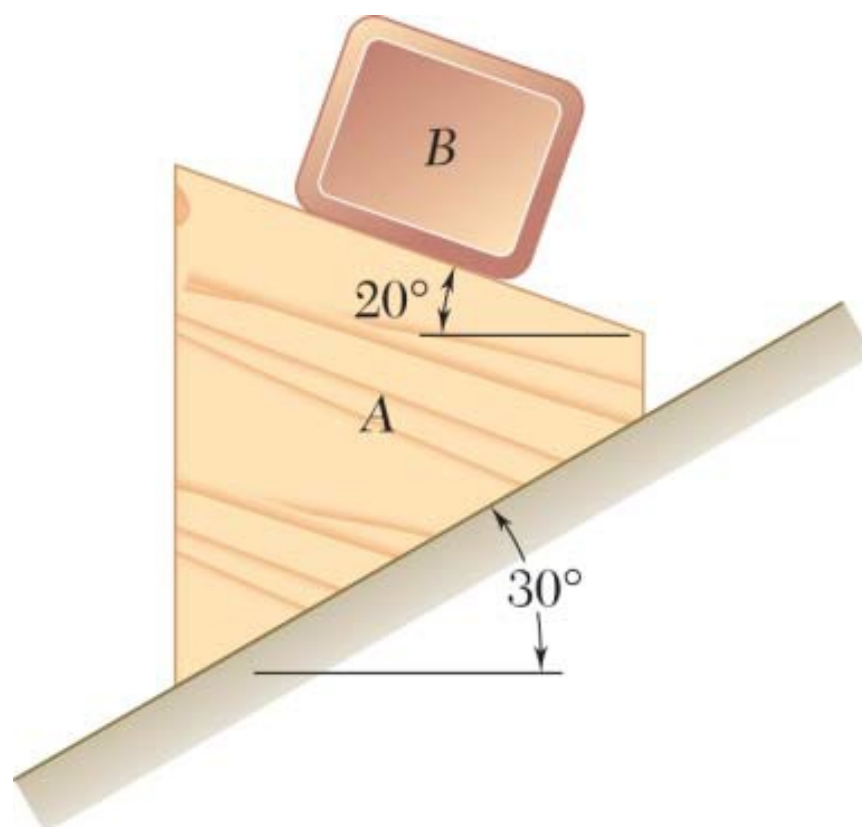
- Q1.** The roller-coaster track is in a vertical plane. Track between A and B is straight and horizontal. Car travels at 72 km/h when brakes are suddenly applied ($\mu_k = 0.20$). Determine initial deceleration if brakes applied as car: (i) almost reached A, (ii) travels between A and B, (iii) just passed B. **(18 marks)** [2021–22, 2010–11]



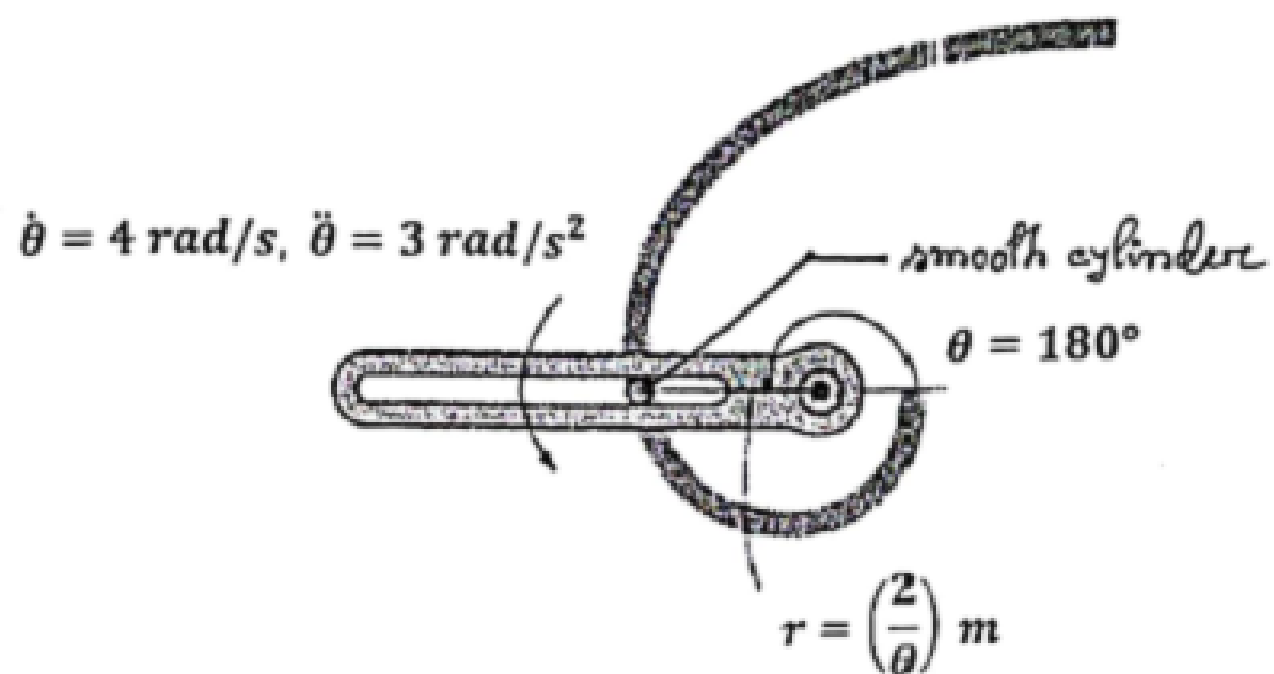
- Q2.** 15-kg block B suspended from 2.5-m cord attached to 20-kg cart A. $\mu_s = 0.24$, $\mu_k = 0.20$. Check static equilibrium. If not in equilibrium, determine: (i) acceleration of cart, (ii) tension in cord, (iii) acceleration of block relative to cart. **(17 marks)** [2021–22]



- Q3.** Block B of mass 10 kg rests on upper surface of 22-kg wedge A. System released from rest. Neglecting friction, determine acceleration of B. **(18 marks)** [2020–21]



- Q4.** Arm rotating at $\dot{\theta} = 4 \text{ rad/s}$ when $\ddot{\theta} = 3 \text{ rad/s}^2$ and $\theta = 180^\circ$. Determine force arm must exert on 0.5-kg smooth cylinder confined to a slotted path in a horizontal plane. (17 marks) [2020–21]



- Q5.** Determine constant force F applied to cord to cause block A (x kg) to have speed y m/s after being displaced z m upward from rest. Neglect pulley weights.
 $[x = 20 + 0.1 \times \text{ID}, y = 2 + 0.01 \times \text{ID}, z = 5 + 0.02 \times \text{ID}]$ (30 marks) [2019–20]

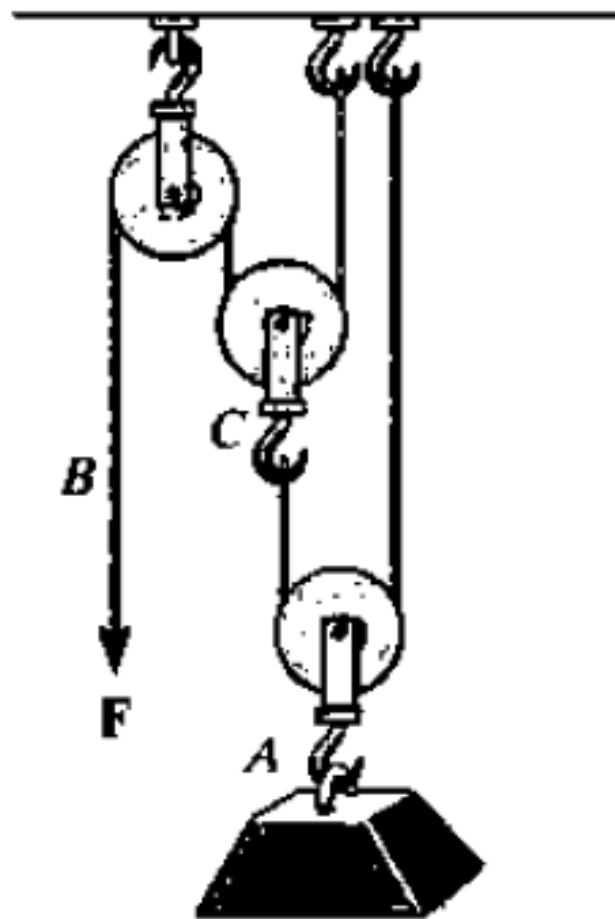
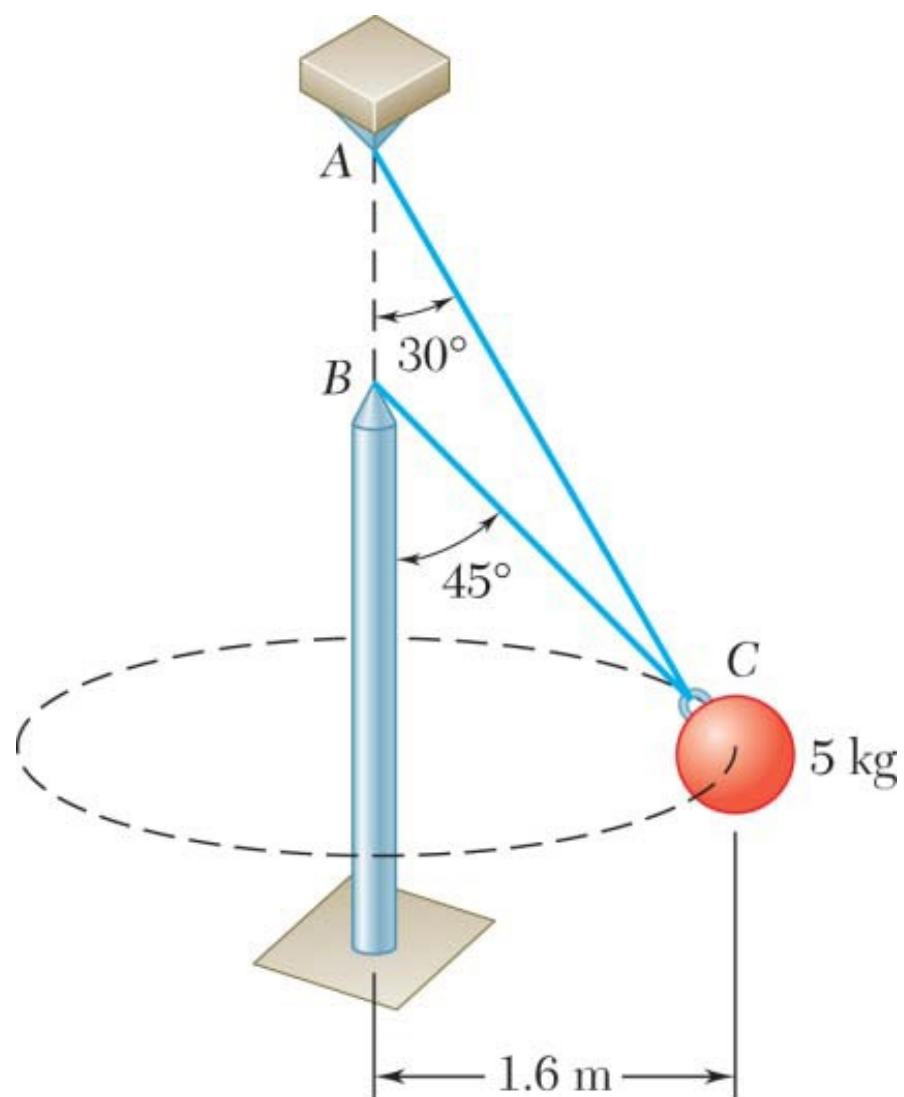
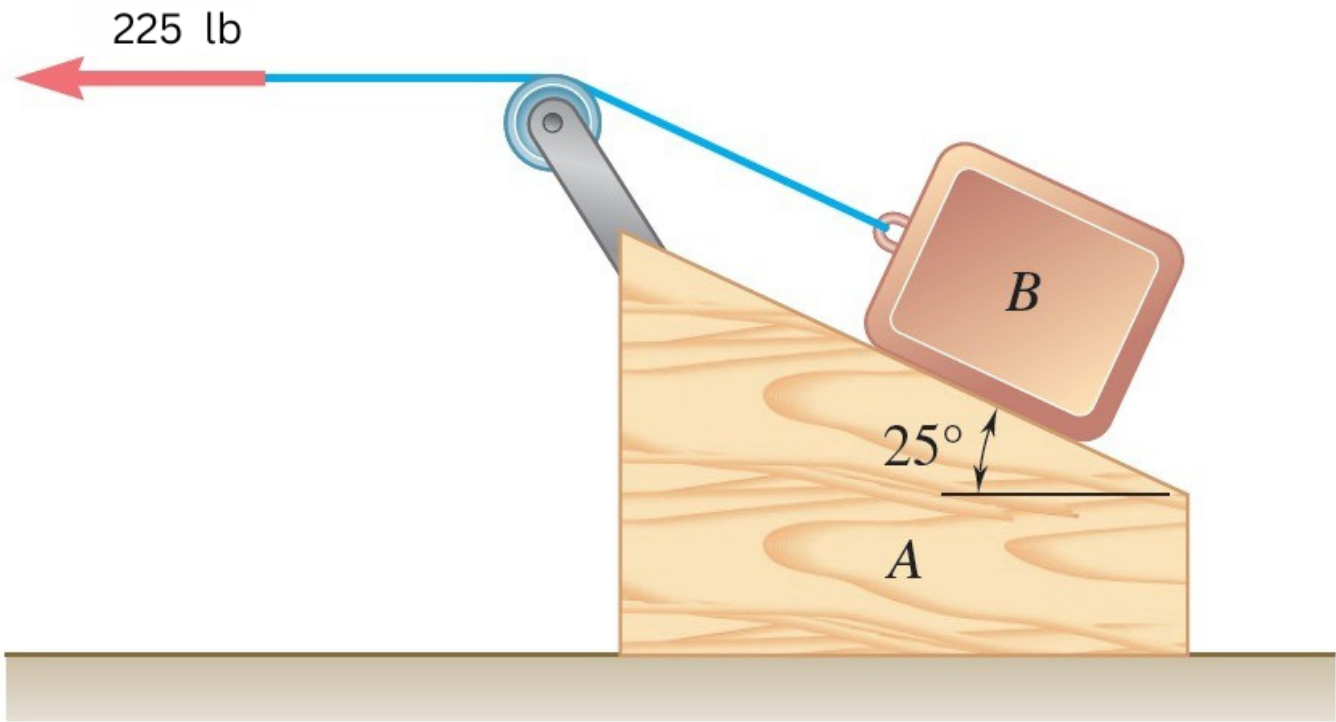


Fig. for Q-8

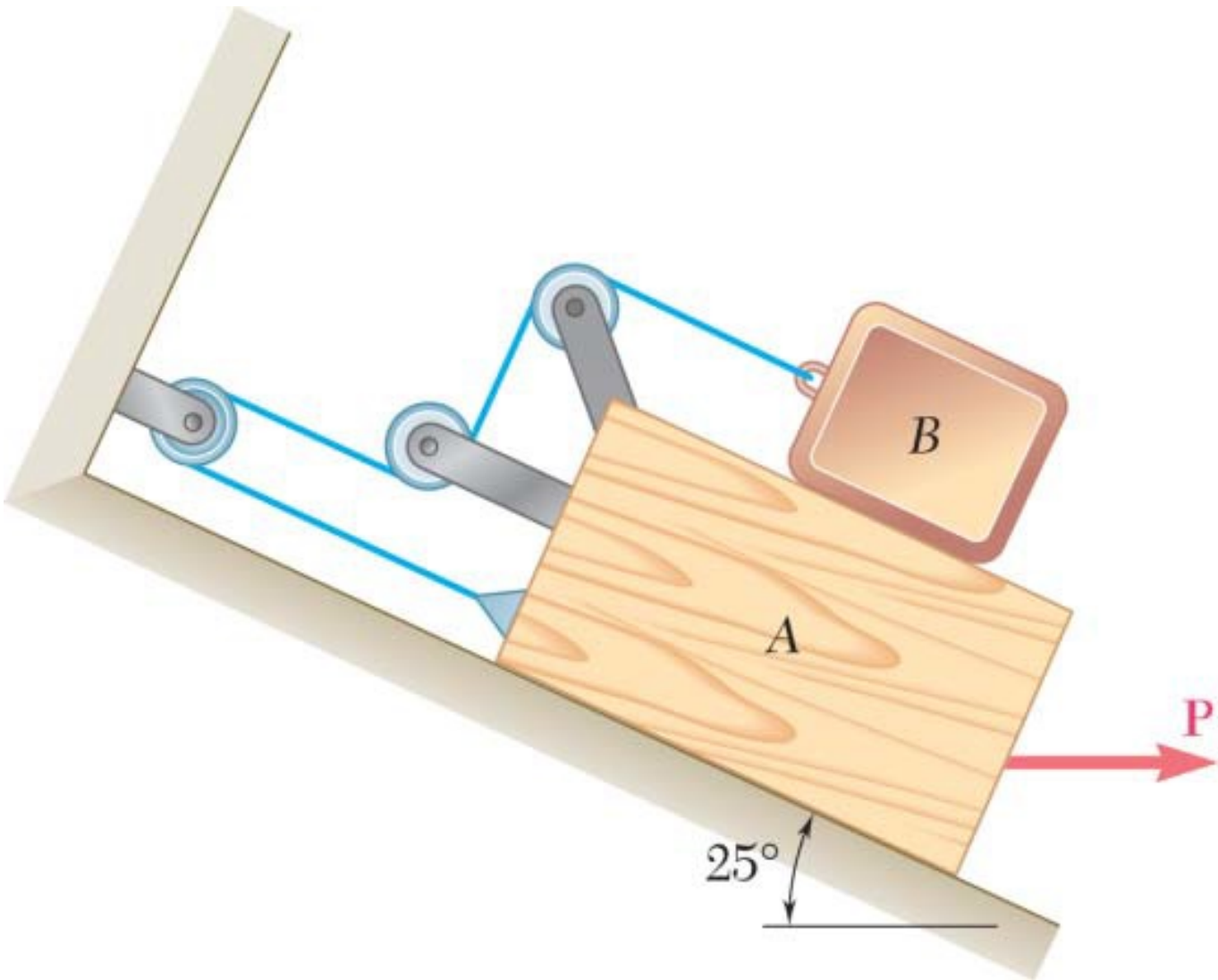
- Q6.** Two wires AC and BC tied at C to a sphere revolving at constant speed v in a horizontal circle. Determine range of allowable values of v if both wires remain taut and tension in BC does not exceed 60 N. (18 marks) [2018–19]



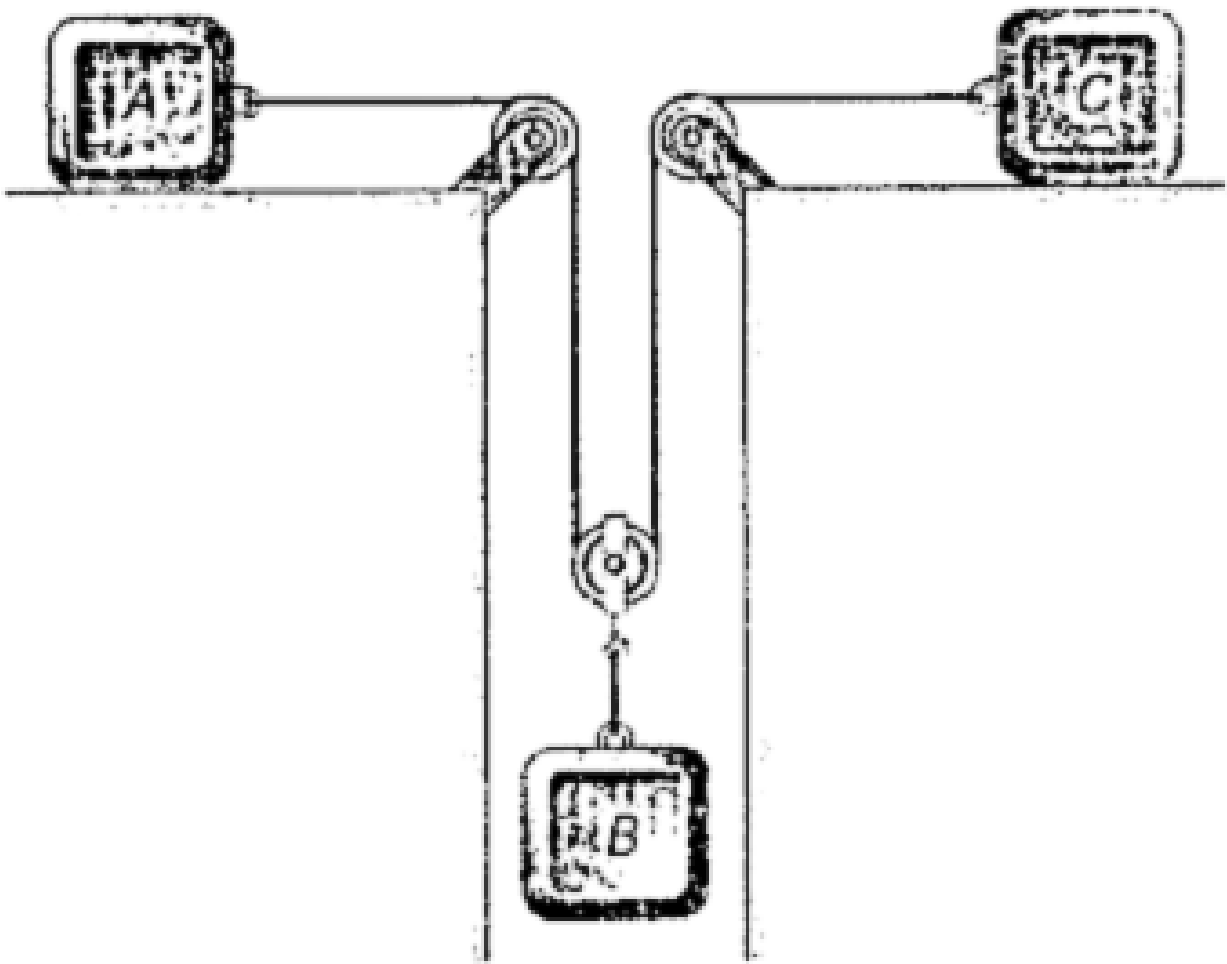
- Q7.** 15-kg block B supported by 25-kg block A, attached to cord with 225-N horizontal force. Neglecting friction, determine: (a) acceleration of block A, (b) acceleration of B relative to A. (18 marks) [2017–18]



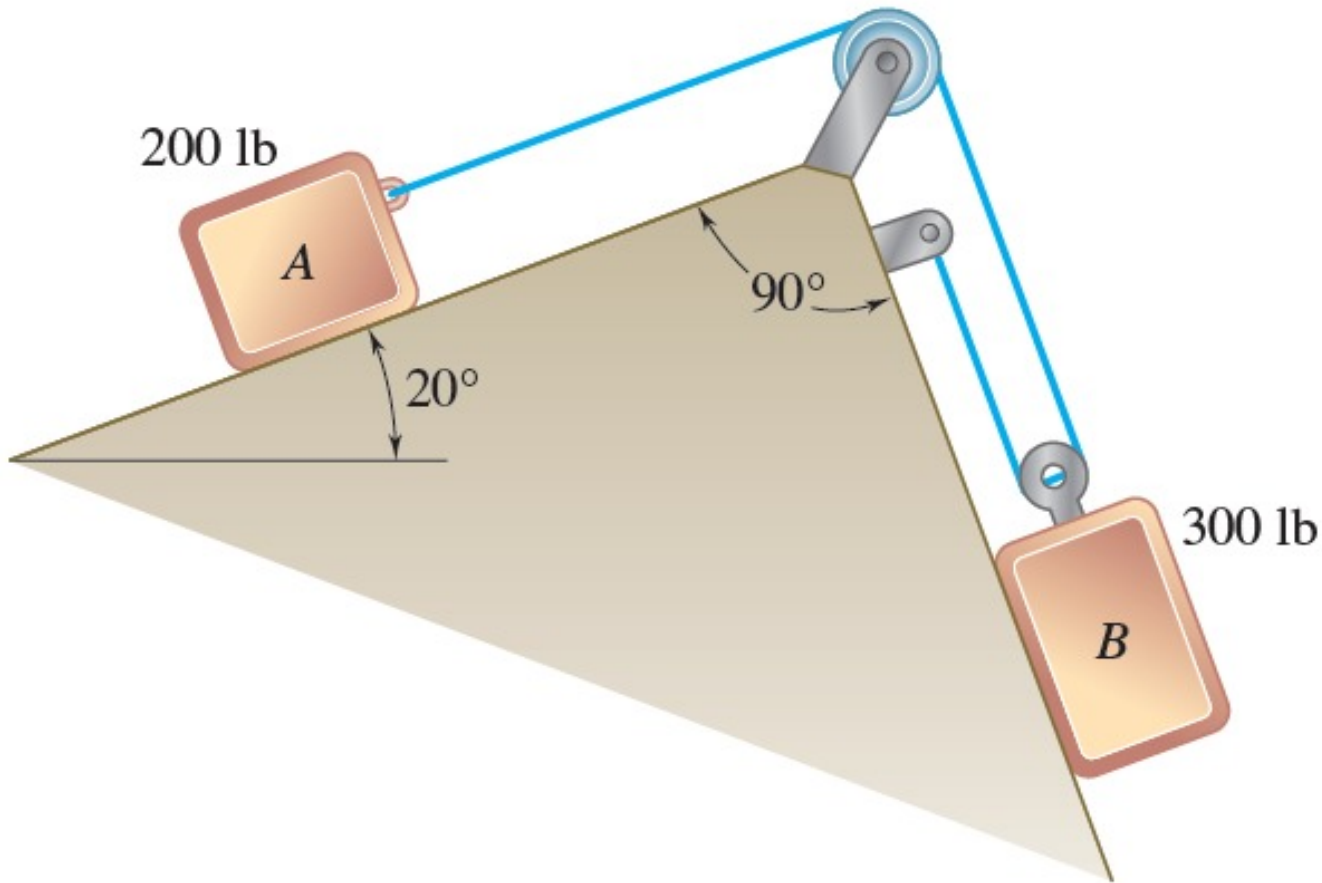
Q8. Block A ($m = 40\text{ kg}$), block B ($m = 8\text{ kg}$), $\mu_s = 0.2$, $\mu_k = 0.15$. With $P = 40\text{ N}$, determine: (i) acceleration of block B, (ii) tension in cord. (20 marks) [2016–17]



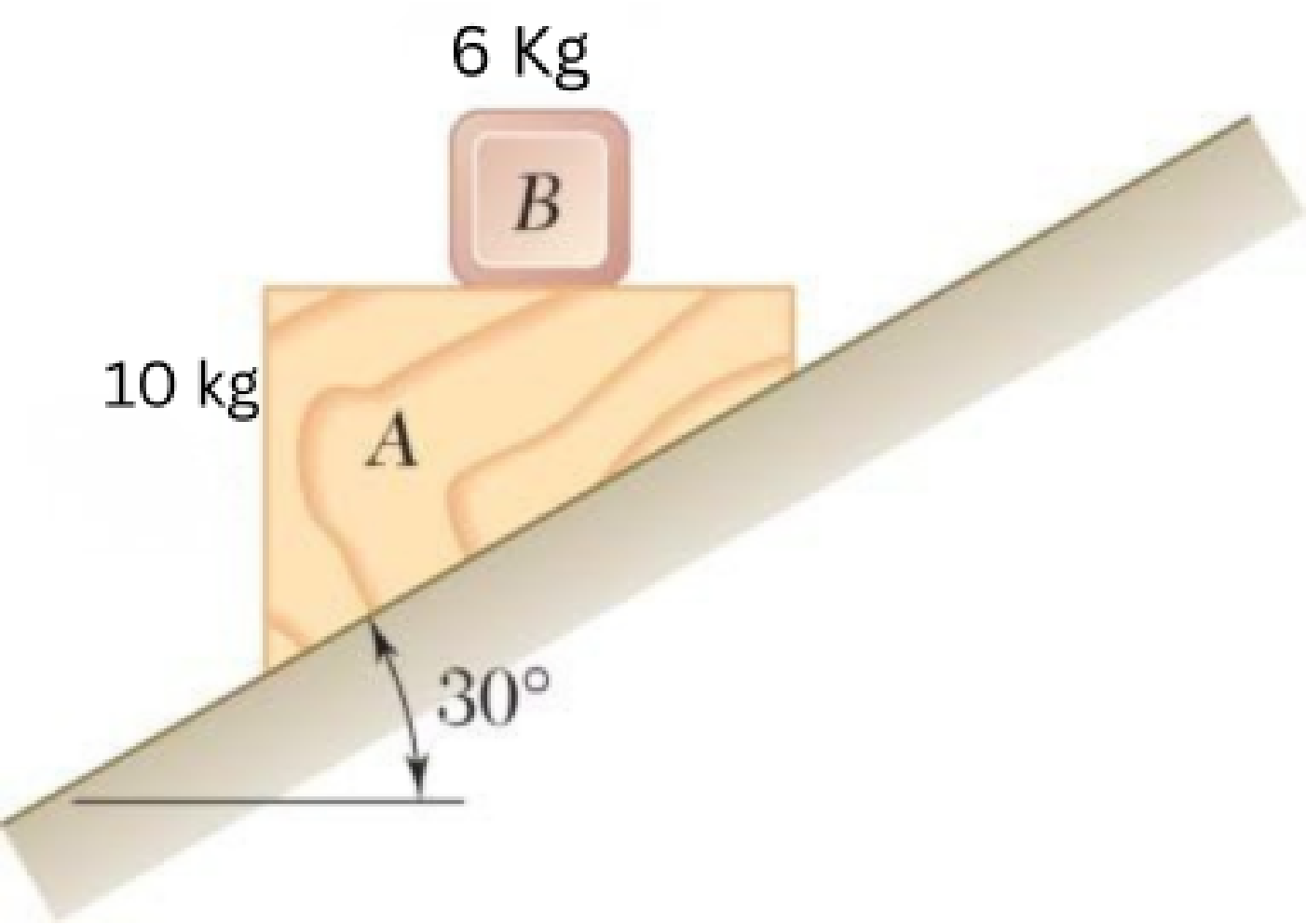
Q9. $m_A = 5\text{ kg}$, $m_B = 10\text{ kg}$, $m_C = 10\text{ kg}$, $\mu_s = 0.24$, $\mu_k = 0.20$. Determine: (i) tension in cord, (ii) acceleration of each block. (20 marks) [2013–14, 2015–16]



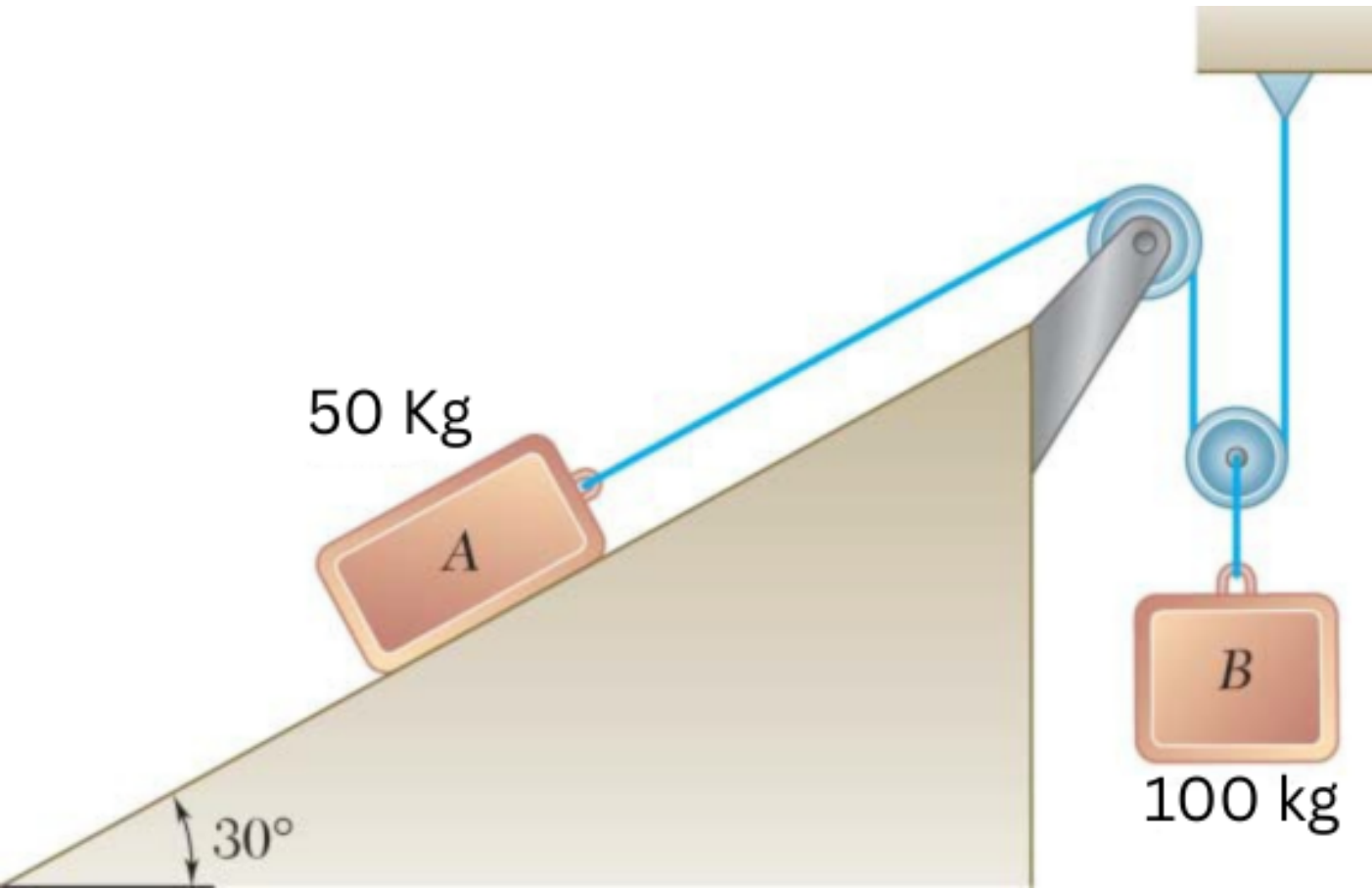
Q10. Two blocks initially at rest. Neglecting pulley masses and friction, determine: (i) acceleration of each block, (ii) tension in cable. (17 marks) [2012–13, 2005–06]



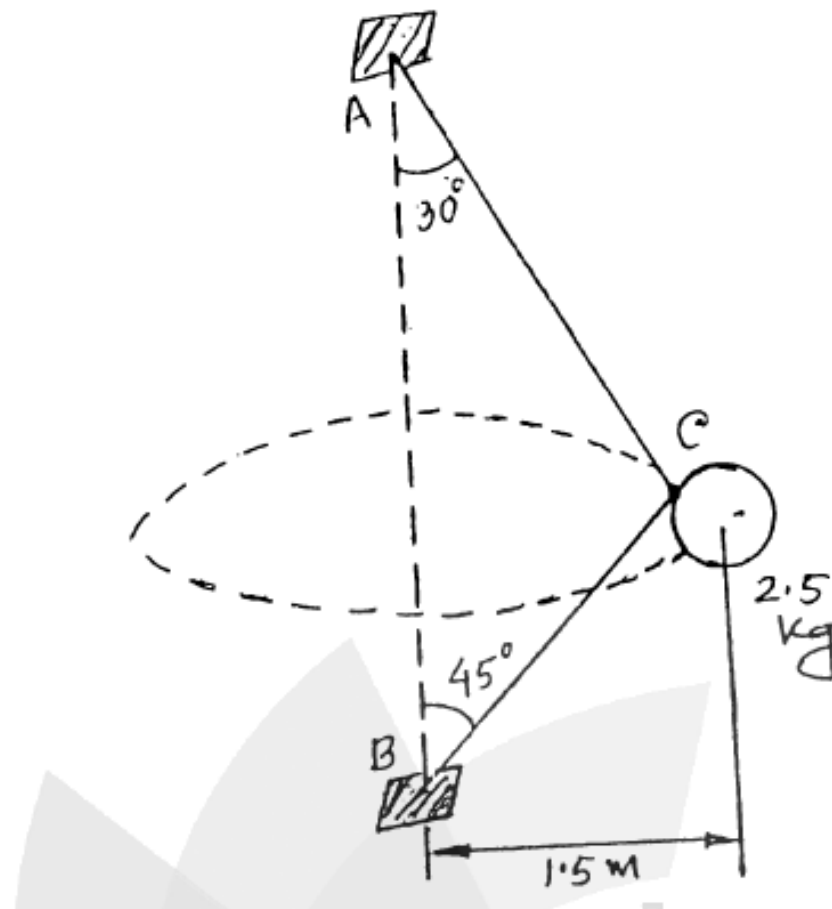
Q11. 6-kg block B rests on upper surface of 10-kg block A. Neglecting friction, immediately after release: (i) acceleration of A, (ii) acceleration of B relative to A. (15 marks) [2010–11]



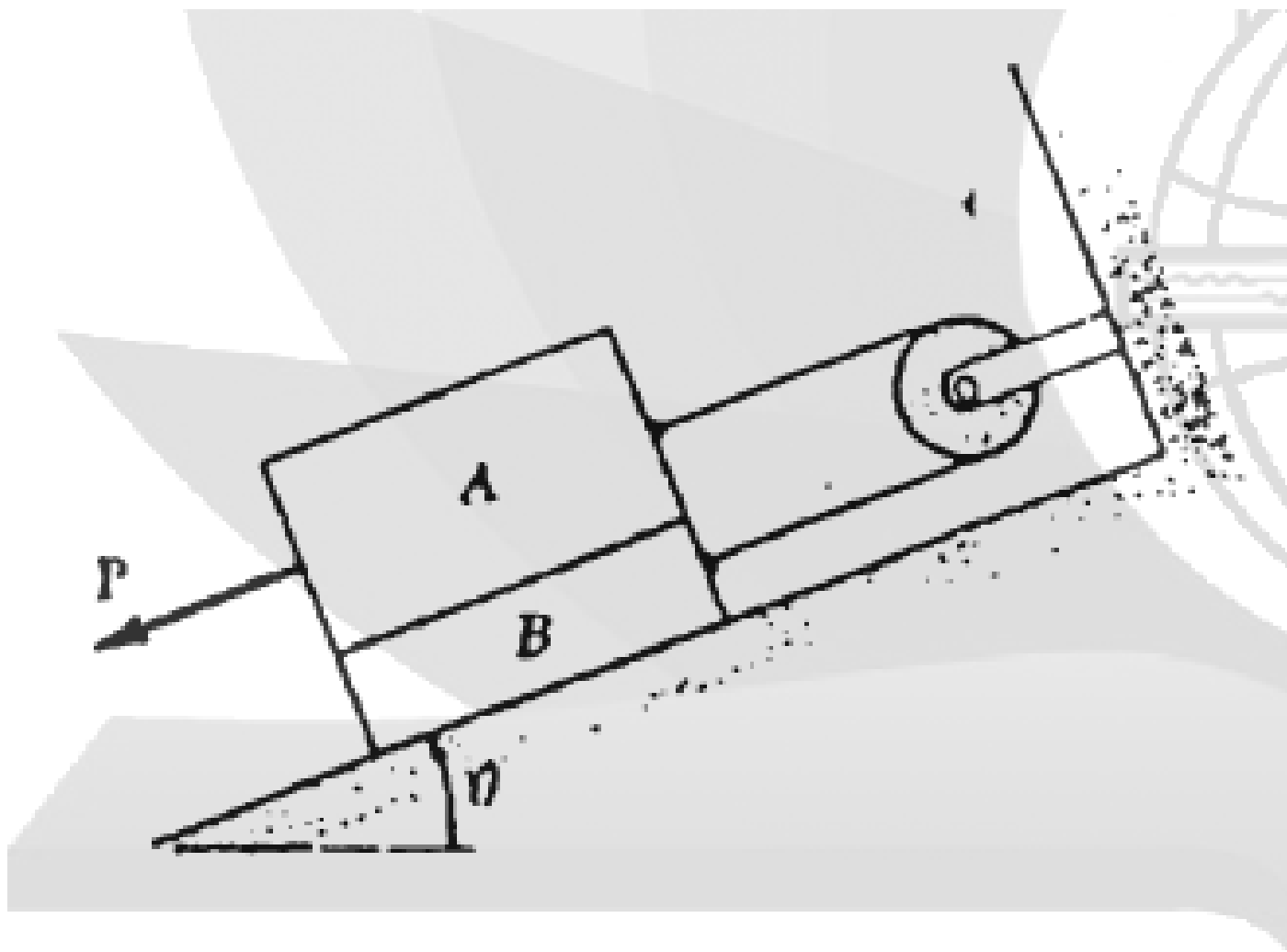
Q12. Two blocks originally at rest. $m_A = 50\text{ kg}$, $m_B = 100\text{ kg}$, $\mu_s = 0.25$, $\mu_k = 0.20$. Determine: (i) acceleration of each block, (ii) tension in cable. **(18 marks)** [2006–07]



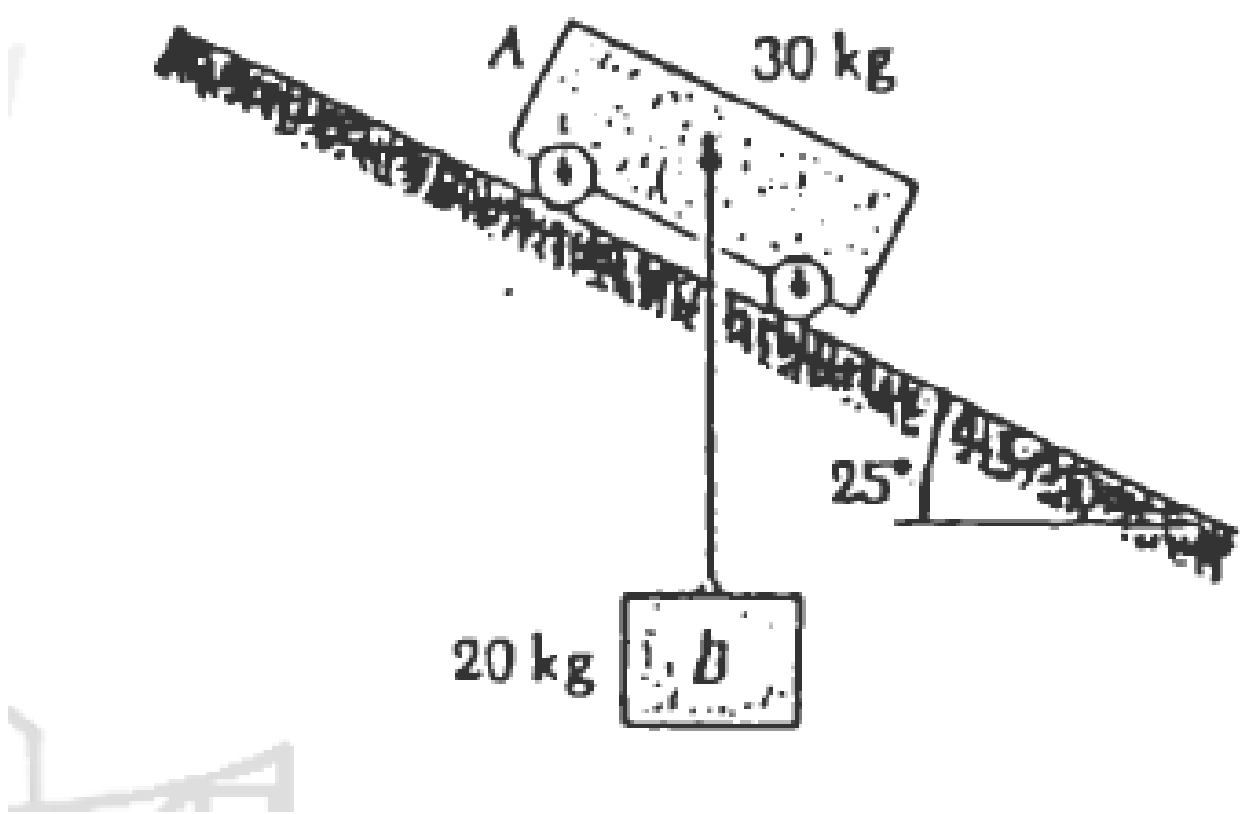
Q13. Single wire ACB passes through ring at C attached to sphere revolving at constant speed v in a horizontal circle. Mass of sphere = 2.5 kg. Tension same in both portions. Determine speed v . **(17 marks)** [2006–07]



- Q14.** Block A ($m = 25 \text{ kg}$), block B ($m = 15 \text{ kg}$), $\mu_s = 0.20$, $\mu_k = 0.15$, $\theta = 25^\circ$, $P = 250 \text{ N}$ applied to A. Determine: (a) acceleration of A, (b) tension in cord. **(18 marks)** [2004–05]



- Q15.** 20-kg block B suspended from 2-m cord attached to 30-kg cart A. Neglecting friction, immediately after release: (i) acceleration of cart, (ii) tension in cord. **(18 marks)** [2003–04]



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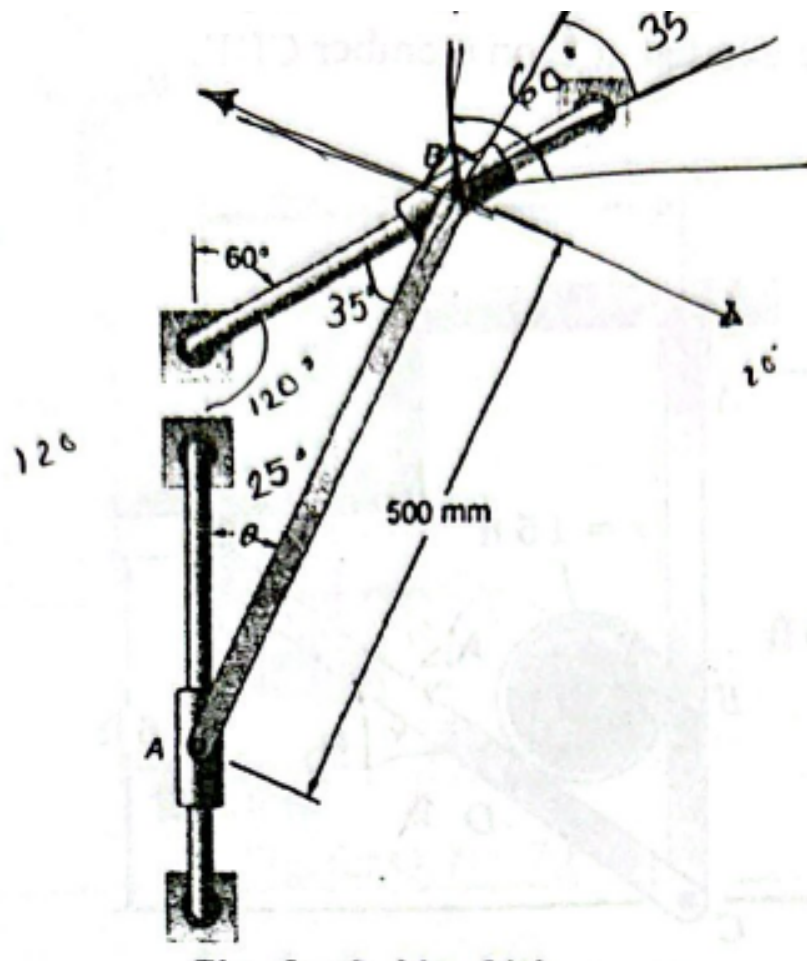
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Section B5

Kinematics of Rigid Bodies

B5 — Kinematics of Rigid Bodies

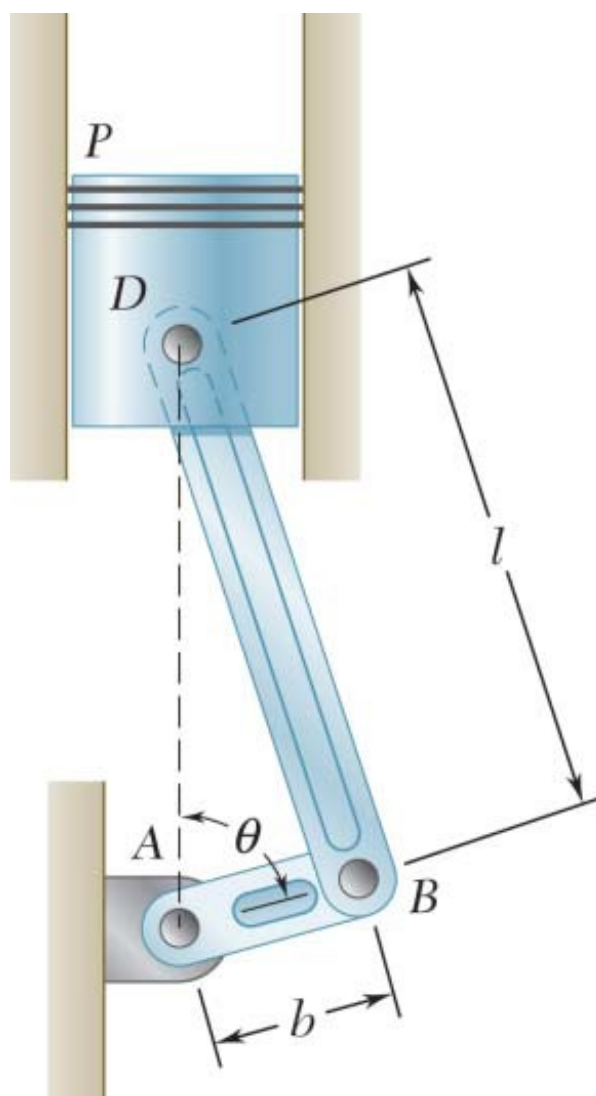
- Q1.** Collar B moves downward to the left at constant velocity 1.6 m/s. At $\theta = 40^\circ$, determine: (a) angular velocity of rod AB, (b) velocity of collar A. **(15 marks)** [2023–24]



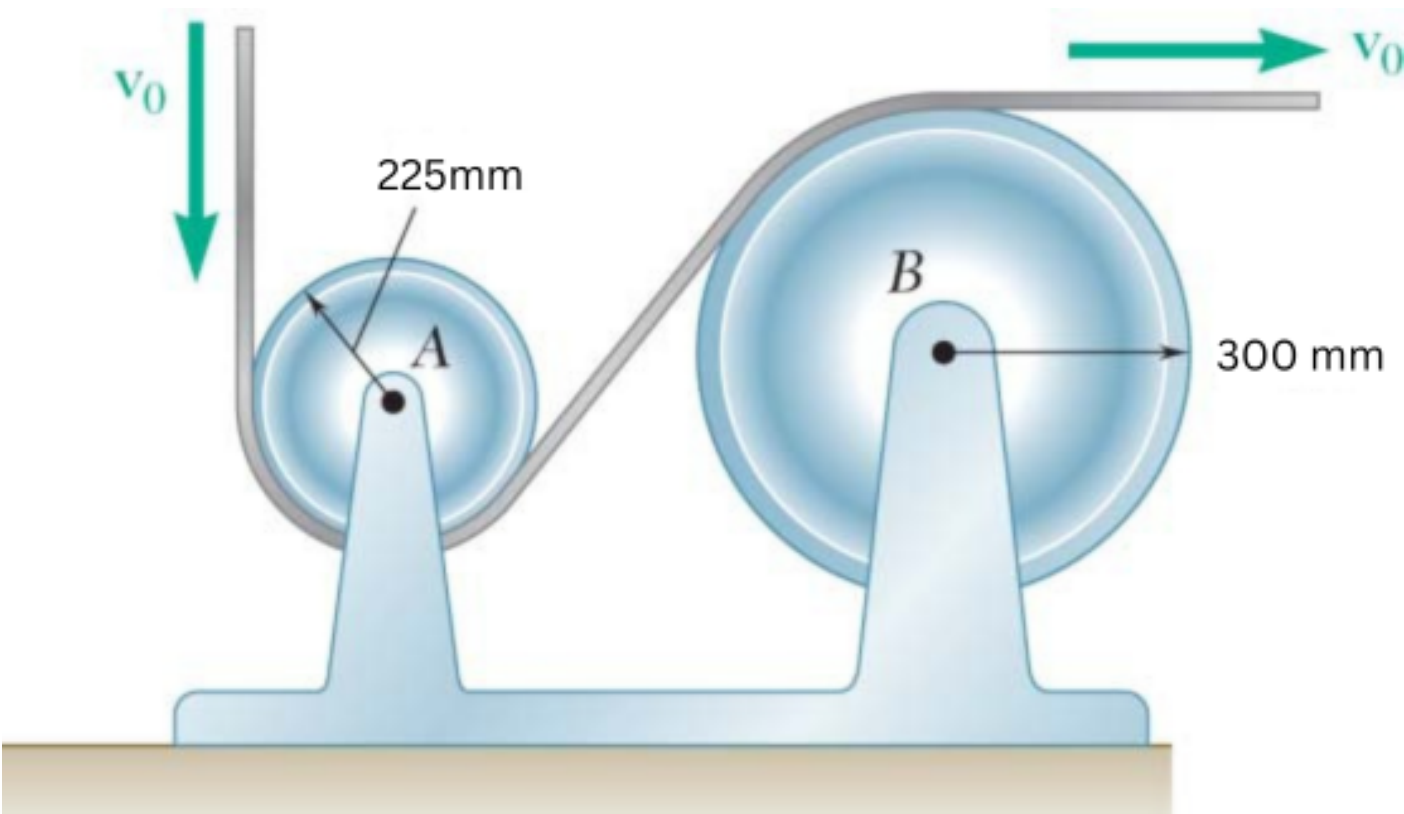
- Q2.** Collar B moves downward to the left at constant velocity 1.6 m/s. At $\theta = 25^\circ$, determine: (i) angular velocity of rod AB, (ii) velocity of collar A. **(15 marks)** [2023–24]

↪ *Similar: B5-Q1 same setup, different angle*

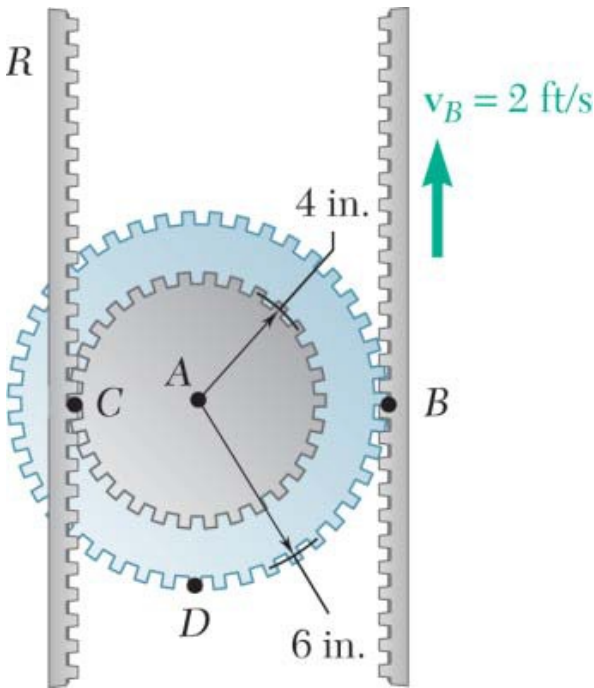
- Q3.** Crank AB clockwise at 1000 rpm, $l = 160$ mm, $b = 60$ mm. At $\theta = 60^\circ$, determine velocity of piston P and angular velocity of connecting rod BD. **(20 marks)** [2018–19]



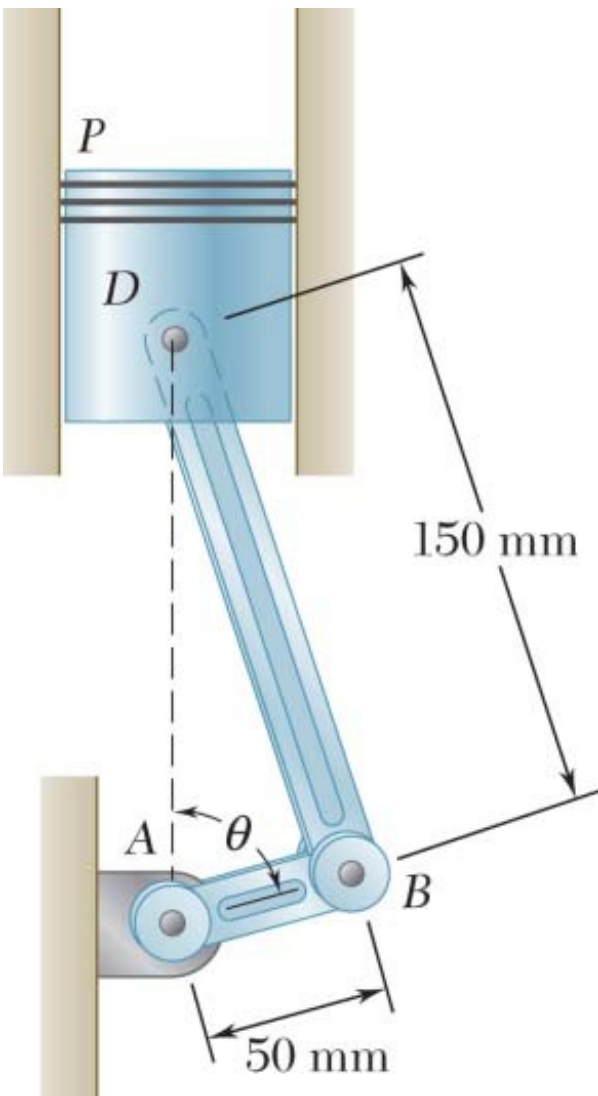
- Q4.** A plastic film moves over two drums. During 4 s, film speed increases uniformly from $V_0 = 1$ m/s to $V_1 = 2$ m/s. Determine: (i) angular acceleration of drum B, (ii) revolutions executed by drum B in 4 s. **(15 marks)** [2015–16]



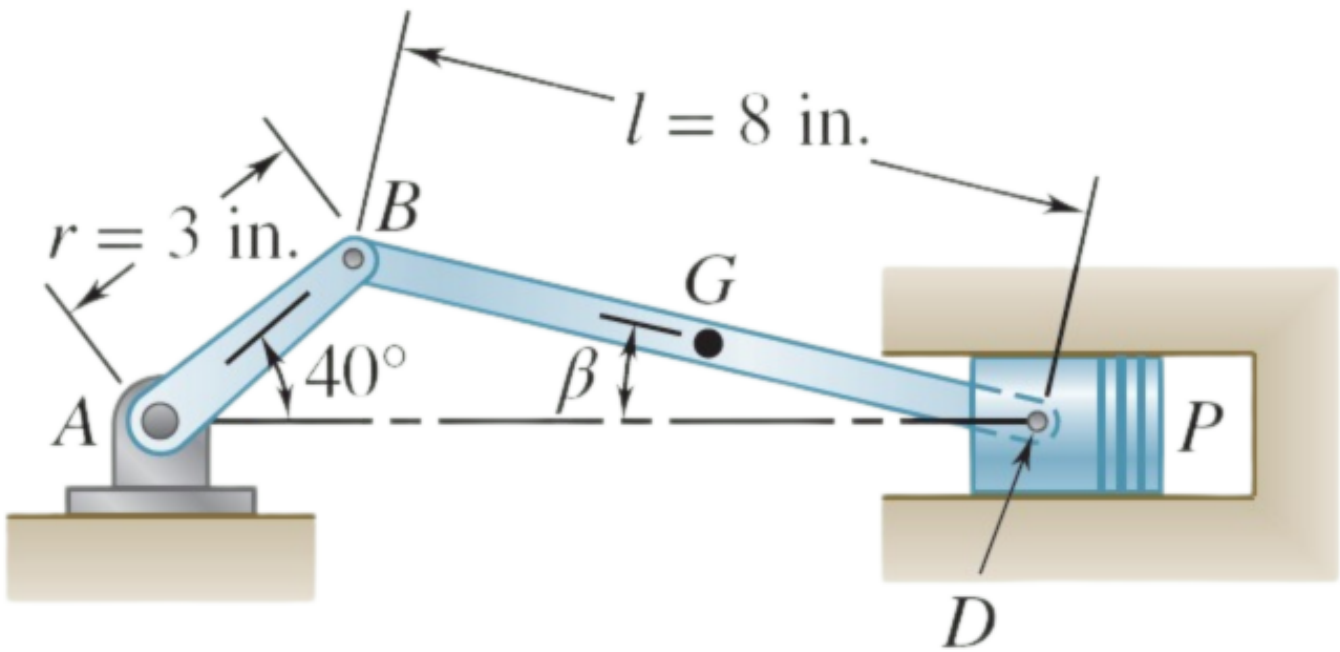
Q5. Double gear rolls on stationary left rack R. Right rack has constant velocity 2 ft/s. Determine: (a) angular velocity of gear, (b) velocities of points A and D. Use instantaneous-centre method. **(15 marks)** [2014–15]



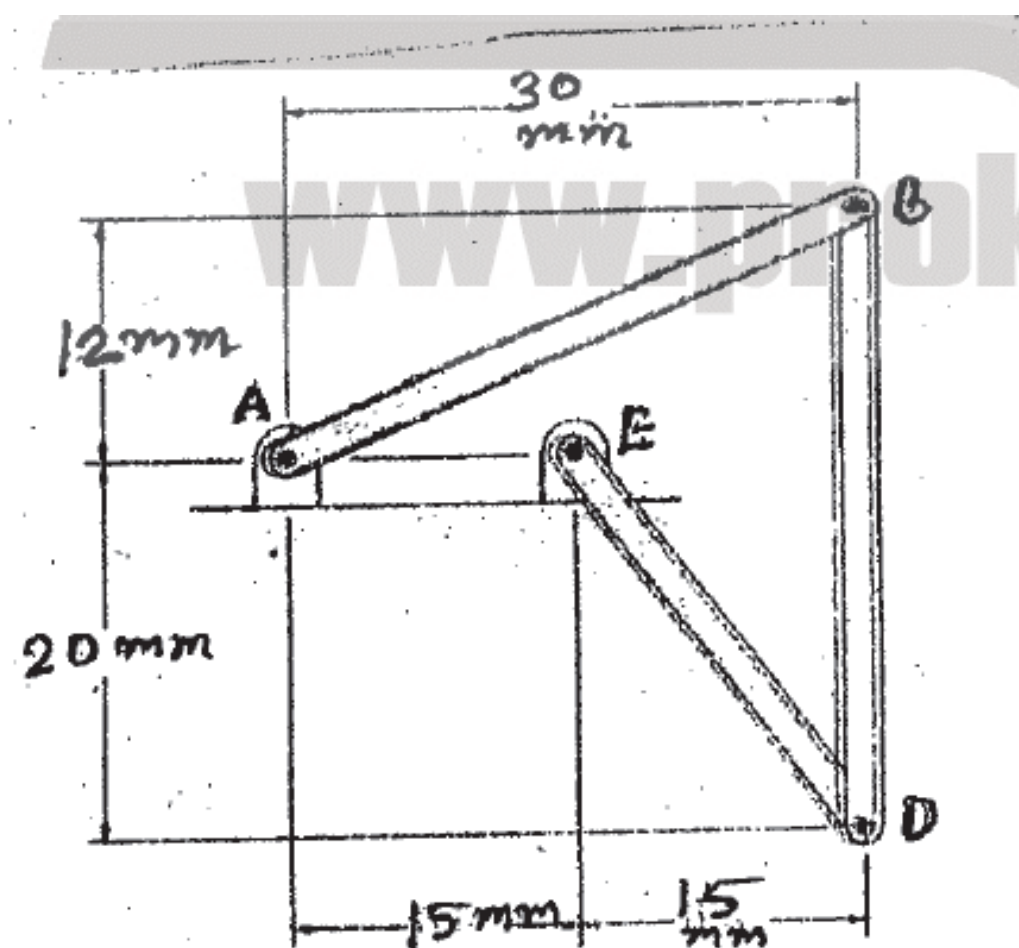
Q6. Crank AB rotates at 900 rpm clockwise about Point A. Determine acceleration of piston P when $\theta = 60^\circ$. **(15 marks)** [2013–14]



Q7. In an engine system, crank AB rotates clockwise at 2000 rpm. Determine: (i) angular velocity of connecting rod BD, (ii) velocity of piston P. (17 marks) [2012–13]

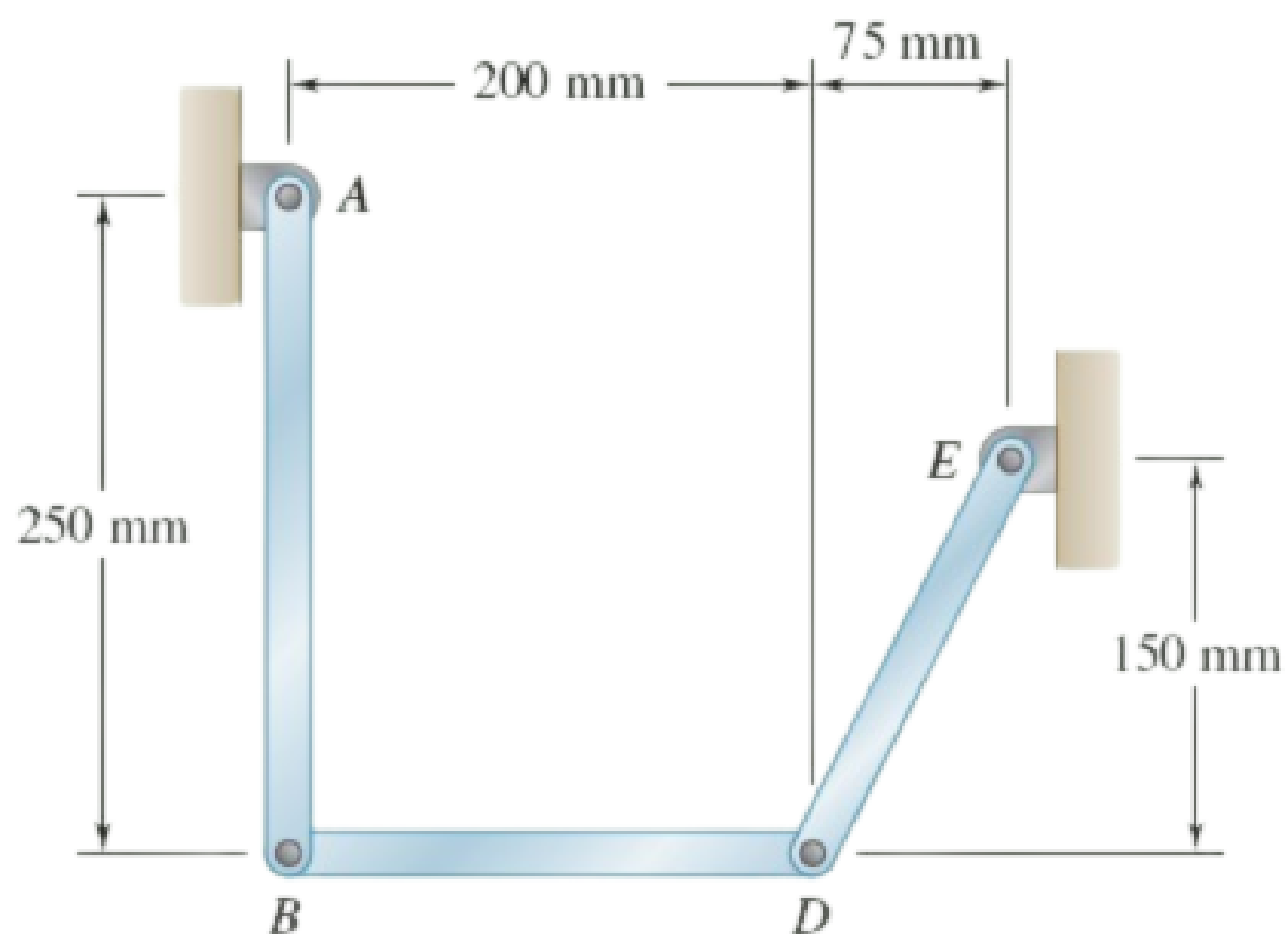


Q8. Bar AB has constant angular velocity 5 rad/s counter-clockwise. Determine angular velocities of bars BD and DE. (17 marks) [2011–12]



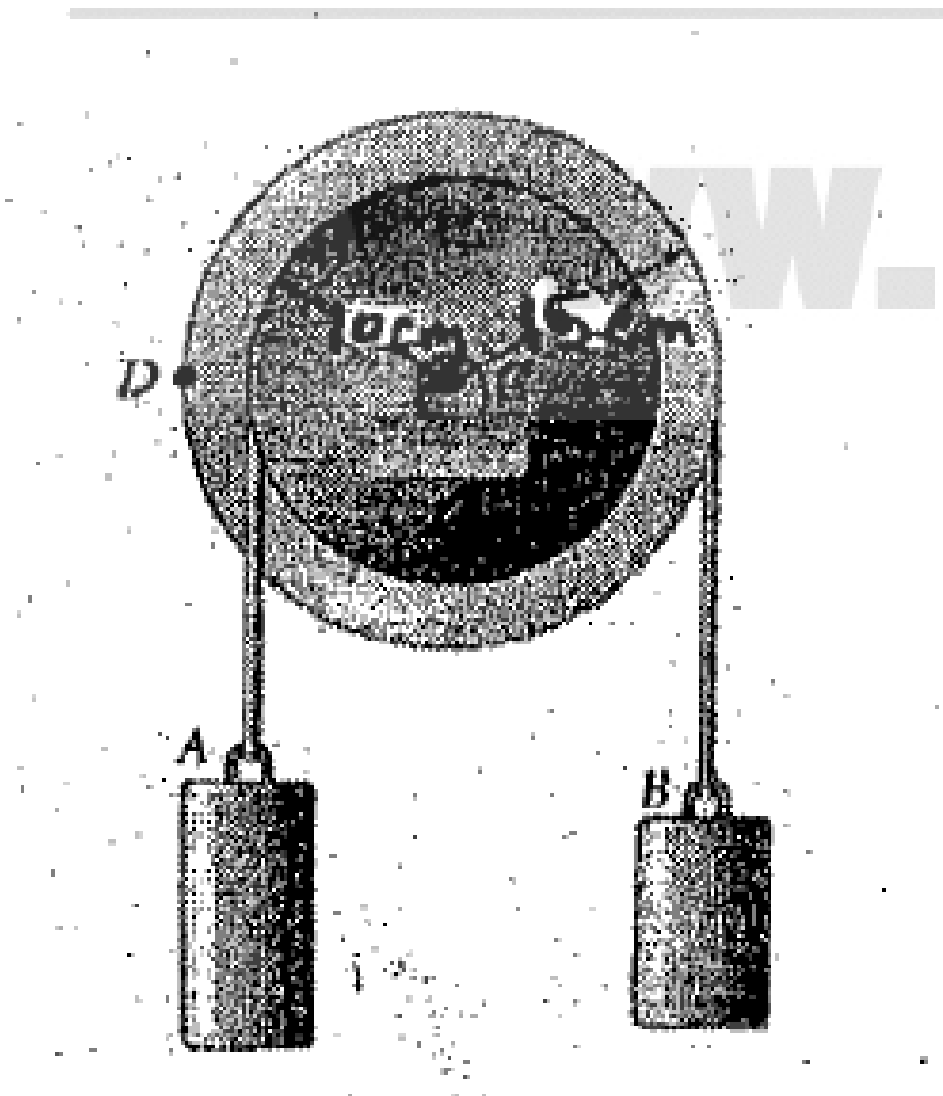
- Q9. Bar AB has angular velocity 4 rad/s clockwise. Determine angular velocities of bars BD and DE. (20 marks)
- Similar: B5-Q7 same linkage, different ω

[2010–11]

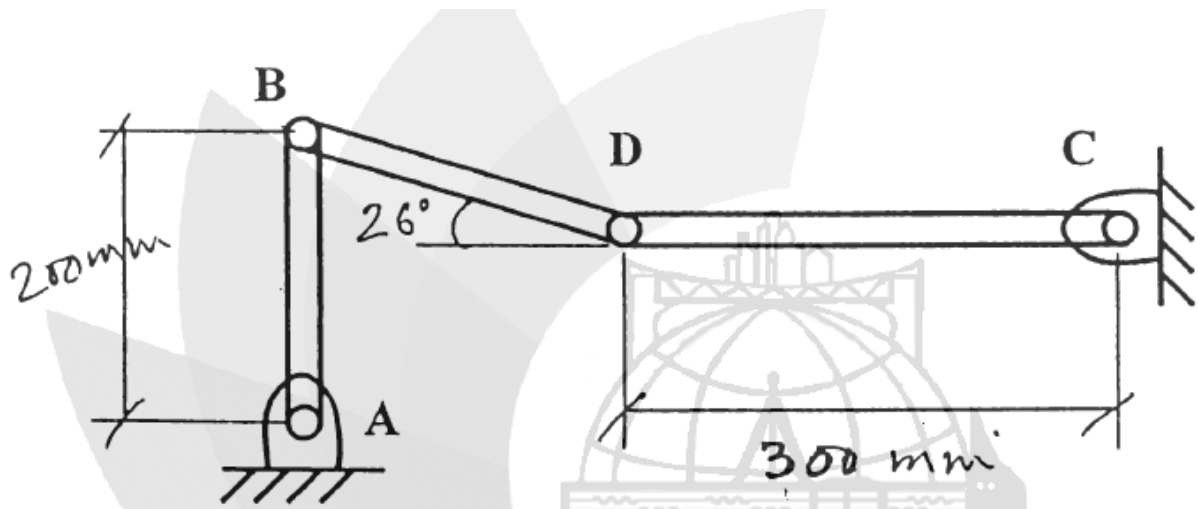


- Q10. A pulley starts from rest at $t = 0$, accelerated at 2.4 rad/s^2 clockwise. At $t = 4 \text{ s}$, determine velocity and position of: (i) load A, (ii) load B. (15 marks)

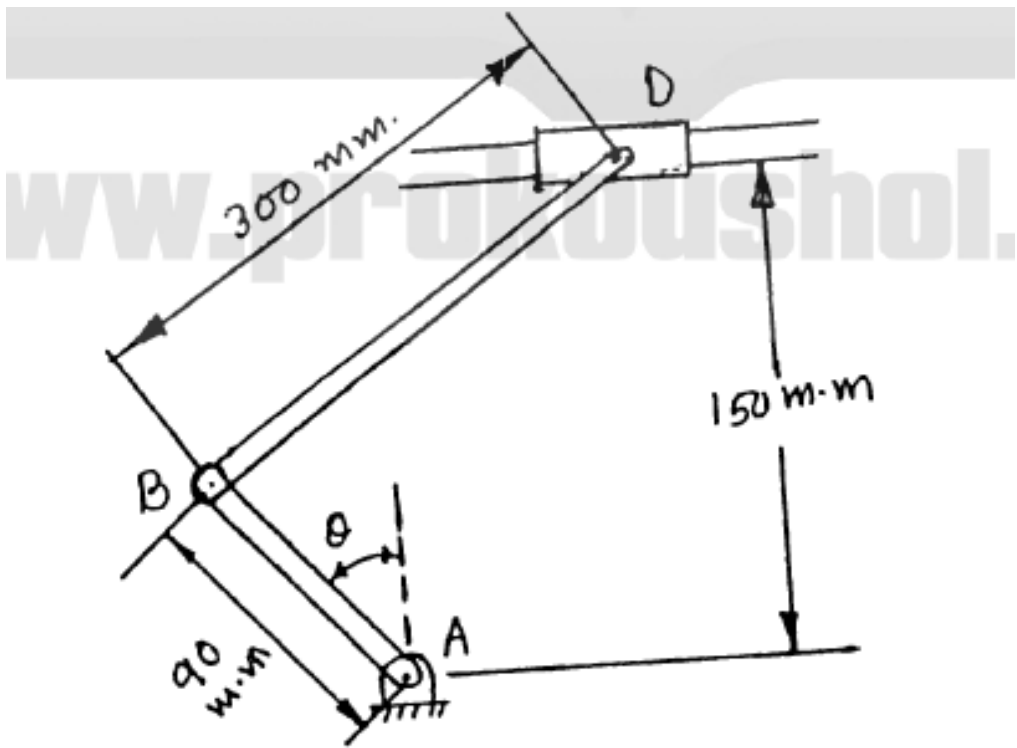
[2010–11]



Q11. Find angular velocity of arm DC if limb AB rotates at 2000 rpm counter-clockwise. (20 marks) [2008–09]



Q12. Crank AB at 200 rpm counter-clockwise. Determine angular velocity of rod BD and velocity of collar D when $\theta = 90^\circ$. (18 marks) [2006–07]



- Q13.** Engine: $l = 200$ mm, $b = 50$ mm, crank AB at 2000 rpm clockwise. At $\theta = 30^\circ$, determine velocity of piston P and angular velocity of connecting rod. (17 marks) [2005–06]

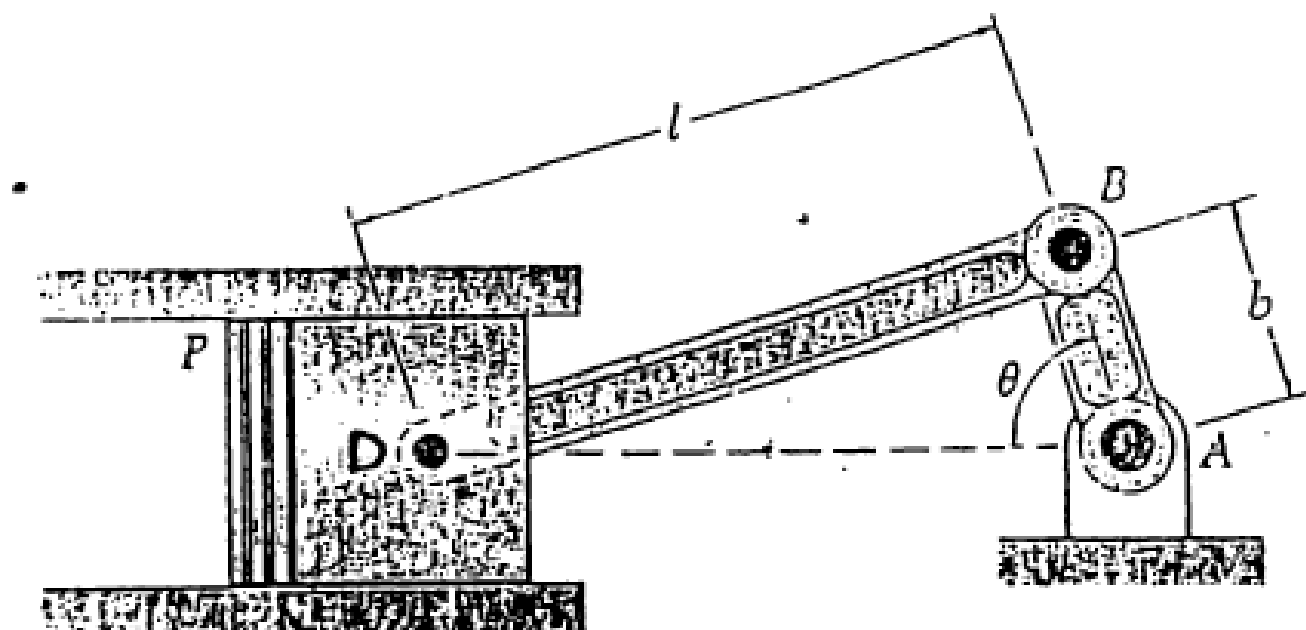
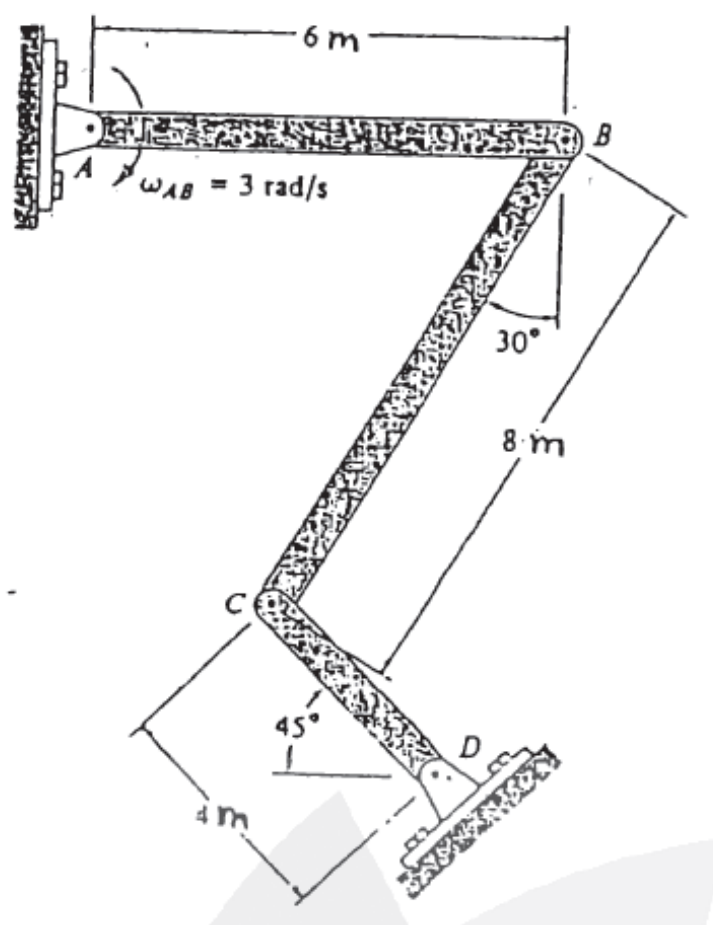
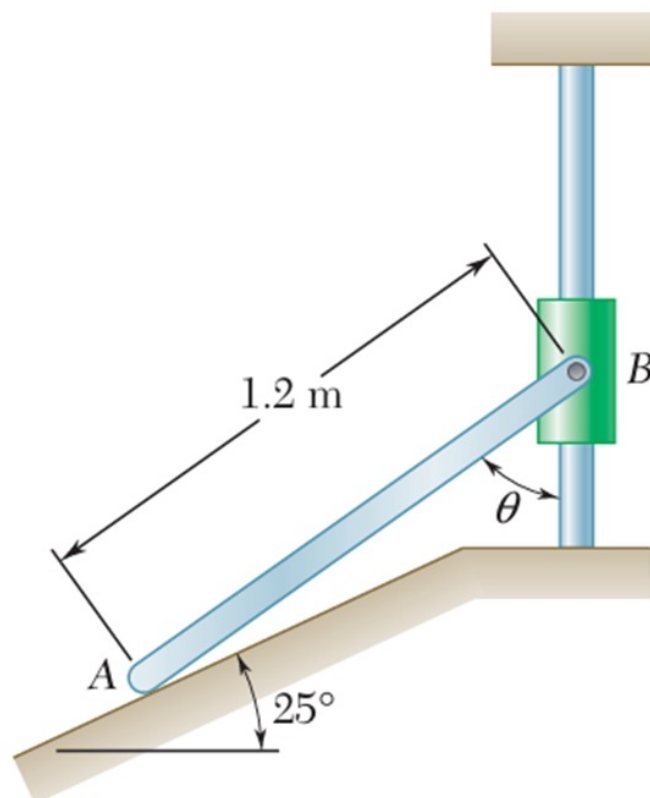


Fig for Q. no 4(a)

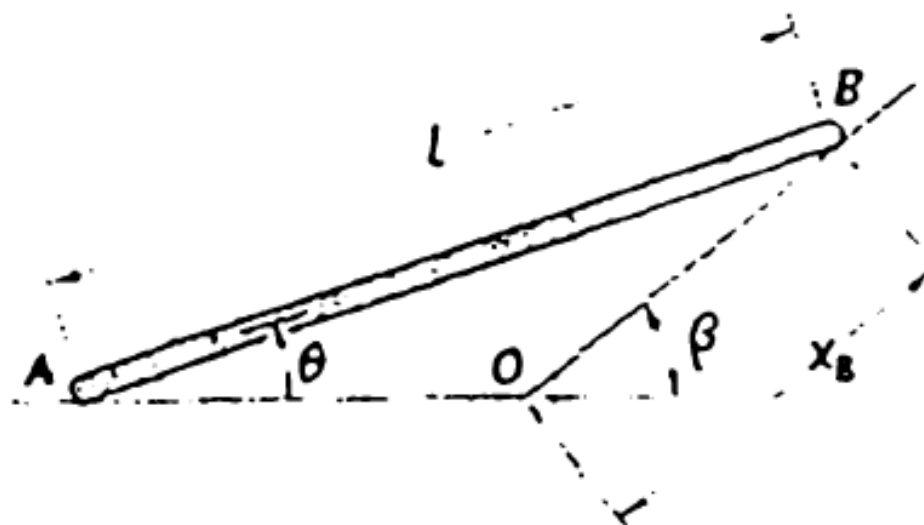
- Q14.** Link AB rotates at 3 rad/s clockwise. Determine angular velocities of links BC and CD at the instant shown. (18 marks) [2005–06]



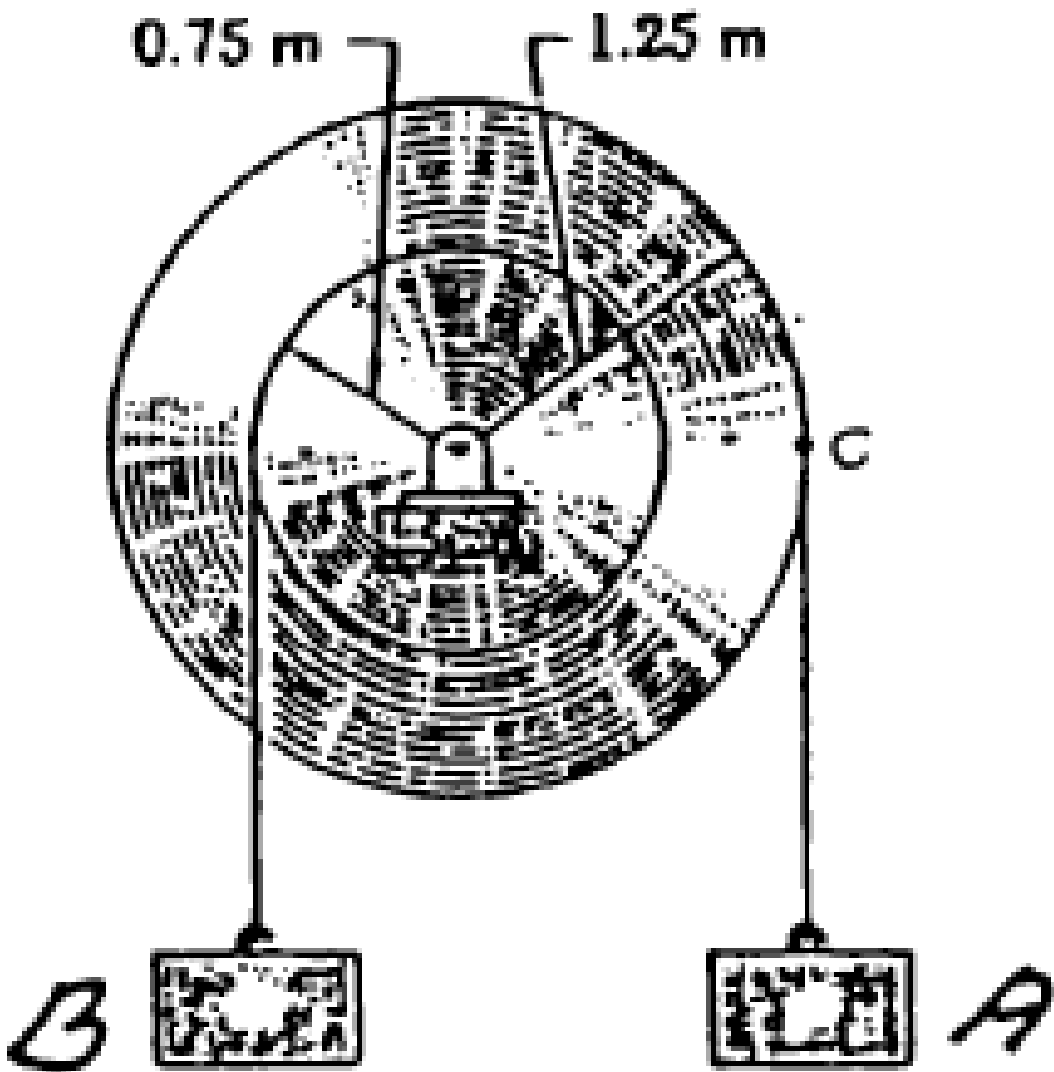
- Q15.** Collar A moves right at constant velocity 900 mm/s. At $\theta = 30^\circ$, determine: (a) angular velocity of rod AB, (b) velocity of collar B. (15 marks) [2004–05]



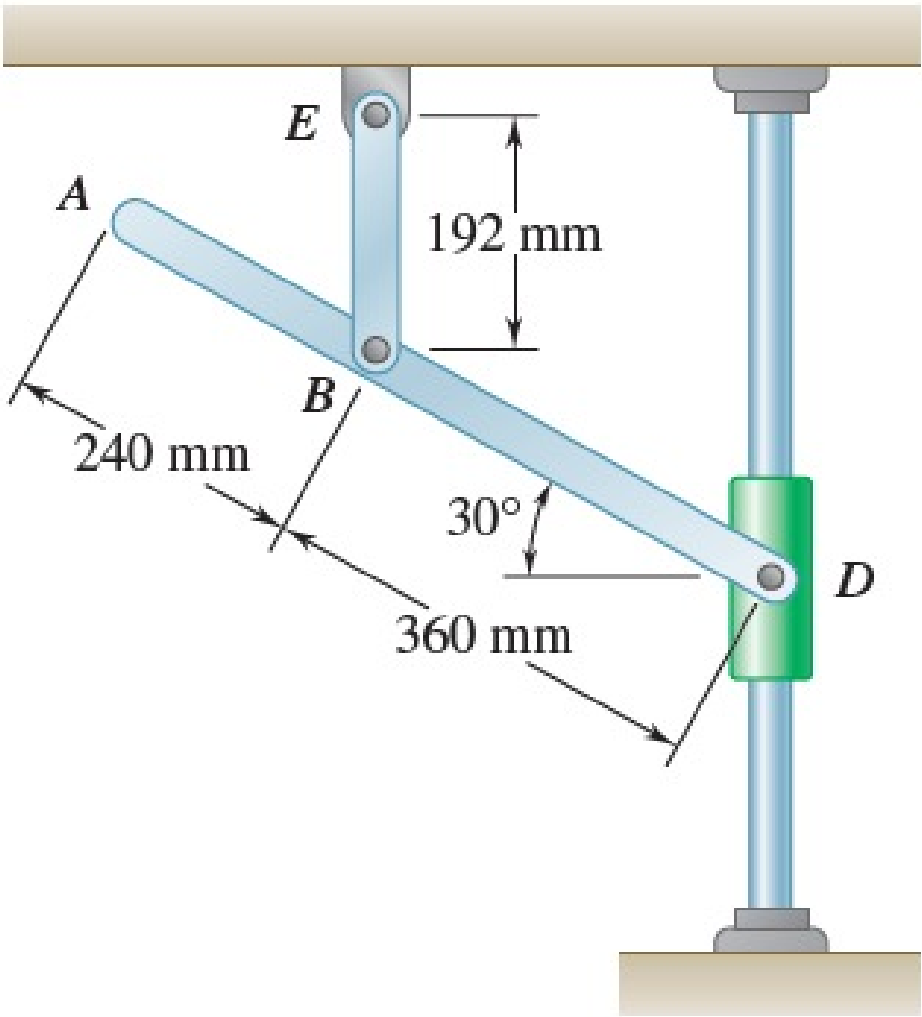
Q16. Derive expression for angular acceleration of rod AB in terms of v_A , θ , l , and β , knowing acceleration of point B is zero. (20 marks) [2004–05]



Q17. Double pulley and two loads connected by inextensible cords. Load A : constant acceleration 2.5 m/s^2 and initial velocity 3.75 m/s , both upward. Determine: (i) revolutions executed by pulley in 3 s , (ii) velocity and position of load B after 3 s . (18 marks) [2003–04]



Q18. Velocity of collar D is 1.8 m/s upward. Determine velocity of point A. (17 marks) [2003–04]



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Section C1

Introduction to Robotics

C1 — Introduction to Robotics

- Q1.** (a) Define joint in the context of robotics. Mention names of any four types of joints and their allowed motions. **(10 marks)**
(b) Write short notes on: (i) Robot software, (ii) Open and closed loop control systems. **(10 marks)** [2021–22]
- Q2.** (a) Write short notes on: (i) Teach programming, (ii) Lead-through programming, (iii) Accuracy and repeatability of a robot. **(15 marks)**
(b) Classify grippers. Draw a schematic diagram of a common two-finger mechanical gripper. **(12 marks)**
(c) In a production chain, a manipulator drills a hole. If the next sheet is placed at twice the distance, how will a parallel manipulator react? How would a serial one react? **(8 marks)** [2020–21]
- Q3.** (a) Write three robot control methods and explain each. **(10 marks)**
(b) What is the most common end-effector that provides the equivalent of a thumb and an opposing finger? **(5 marks)** [2019–20]
- Q4.** (a) Define payload. Briefly classify robots according to the Japanese Industrial Robot Association (JIRA). **(10 marks)**
(b) With schematic diagram, classify and explain different types of actuators. **(10 marks)** [2018–19]
- Q5.** (a) What are the specifications used to describe a robot? What are different joint drive systems used in robotics? **(10 marks)**
(b) What is a robot manipulator? How are they classified? With necessary sketch, discuss the SCARA robot arm. **(10 marks)**
(c) What is robot vision? Discuss how a robot vision system works. **(10 marks)**
(d) What do you understand by “Degeneracy” and “Dexterity”? **(5 marks)** [2017–18]
- Q6.** (a) What is a Robot? Mention the names and functions of robot accessories. **(10 marks)**
(b) How are robot manipulators classified? Discuss the anthropomorphic type with a suitable notation and sketch. **(10 marks)** [2016–17]
- Q7.** Define payload. Briefly classify robots according to JIRA. **(8 marks)** [2015–16]
- Q8.** Briefly classify robots according to the Robotics Institute of America (RIA). **(8 marks)** [2013–14]
- Q9.** Briefly define any two of the following: (i) Payload, (ii) Reach, (iii) Precision. **(6 marks)** [2013–14]
- Q10.** What do you understand by the term “Robot”? Define Payload, Reach, Precision, and Repeatability from the point of view of robot specification. **(10 marks)** [2012–13]
- Q11.** (a) Give a brief description of the main components of a robot. **(10 marks)**
(b) What is meant by robot coordinates? With a neat sketch, explain the working space of a cylindrical coordinate robot. **(8 marks)** [2011–12]
- Q12.** (a) Describe the different parameters used to depict the specification of a robot. **(15 marks)**
(b) Mention some notable applications of robots. **(8 marks)** [2010–11]
- Q13.** (a) What are the steps in Machine Vision Application? **(5 marks)**
(b) Describe the types of gripper used in Industrial Robot Applications. **(10 marks)** [2008–09]
- Q14.** What do you mean by cylindrical coordinate robot? Draw top and side views of workspace for this type. **(5 marks)** [2006–07]
- Q15.** “One machine can do the work of a hundred ordinary men, but no machine can do the work of one extraordinary man.” — Discuss. **(5 marks)** [2004–05, 2005–06]
- Q16.** (a) Draw the basic configurations for: (i) PPP, (ii) RRP, (iii) RYP. **(9 marks)**
(b) How do you compare end effectors and manipulator? **(7 marks)**
(c) Based on the task to be accomplished, how do you recommend the use of hydraulic vs electric drive systems? **(8 marks)** [2003–04]

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Section C2

Plane, Rotational & Spatial Motion; Transformations

C2 — Transformations & Rotation Matrices

- Q1.** (a) Derive a formula by which the description of a point can be changed from one frame to another, where the frames have the same origin but different axis orientations. **(10 marks)**
 (b) Frame $\{B\}$ is rotated relative to $\{A\}$ about z-axis by 45° , then translated 2 units in $\hat{X}_A$ and 3 units in $\hat{Y}_A$. Find ${}^A P$ if ${}^B P = [3 \ 7 \ 0]^T$. **(10 marks)**
 (c) A point $P = [3 \ 0 \ 3]^T$ is subjected to: rotation about y-axis, then translation along x-axis, then rotation about z-axis. Final coordinates are $[4 \ 4 \ 2]^T$. Find possible transformation variables. **(15 marks)** [2023–24]

- Q2.** Frame B is rotated 90° about z-axis, then translated 3 and 5 units relative to n- and o-axes, then rotated 90° about n-axis, then 90° about y-axis.
 (i) Write the equation describing the motion. (ii) Find new location and orientation.

$$B = \begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

(15 marks)

[2023–24]

- Q3.** Derive the matrix that represents a pure rotation about the y-axis of the reference frame. **(10 marks)** [2016–17, 2021–22, 2023–24]

- Q4.** (a) Derive the matrix representation of differential rotation about a general axis k . **(10 marks)**
 (b) Find the effect of a differential rotation of 0.1 rad about the y-axis followed by differential translation of $[0.1, 0, 0.2]$ on the given frame B.

$$\text{where, } B = \begin{bmatrix} 0 & 0 & 1 & 10 \\ 1 & 0 & 0 & 5 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \text{and} \quad \Delta = \begin{bmatrix} 0 & -\delta z & \delta y & dx \\ \delta z & 0 & -\delta x & dy \\ -\delta y & \delta x & 0 & dz \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

(10 marks)

[2017–18]

- Q5.** In a robotic setup, a camera is attached to the fifth link of a robot with six degrees of freedom. The camera observes an object and determines its frames relative to the camera's frame. Using the following information, determine the necessary motion the end effector has to make to get to the object:

$${}^5T_{cam} = \begin{bmatrix} 0 & 0 & -1 & 3 \\ 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 5 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad {}^5T_H = \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^{cam}T_{obj} = \begin{bmatrix} 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 4 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad {}^HT_E = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- Q6.** Derive the matrix that represents a pure rotation about the x-axis of the reference frame. **(15 marks)** [2014–15]

- Q7.** Prove that a rotation matrix is a unitary matrix. **(12 marks)** [2013–14]

- Q8.** (a) Derive the relation $\bar{P}_{xyz} = \bar{R} \bar{P}_{uvw}$ and find $\bar{R}$ if OUVW is rotated by angle ϕ about OY. **(10 marks)**
 (b) A point is subjected to: (i) translation 5 units along +Y, (ii) CW rotation 30° about OX, (iii) CCW rotation 30° about OZ, (iv) translation 3 units along -X. Write the combined transformation equation. **(10 marks)** [2006–07]

- Q9.** (a) Derive the relation $P_{uvw} = Q P_{xyz}$. **(7 marks)**
 (b) Write down the mathematical relation between OXYZ and OUVW for the following motion sequence: (i) CW rotation α about OX, (ii) CW rotation β about OZ, (iii) CCW rotation α about rotated Ou, (iv) CCW rotation β about rotated OV, (v) CCW rotation α about OY, (vi) translation d_y along OY, (vii) CW rotation α about rotated OW, (viii) translation d_x along negative OX. **(20 marks)** [2004–05]

- Q10.** Determine P_{xyz} given: $P_{xyz} = [R_{z,45} \ R_{y,-30} \ T_{z,5}] P_{uvw}$, where P_{uvw} is given. **(8 marks)** [2004–05]

- Q11.** (a) Derive the rotation matrix R transforming coordinates from P_{uvw} to P_{xyz} . Hence find homogeneous transformation matrix T representing: CW rotation α about OX, then CCW rotation θ about OZ, then CW rotation ϕ about OY. **(25 marks)**
 (b) Given $Q_{uvw} = (4, 3, 2)^T$, determine corresponding point after: CCW rotation 45° about OX, then translation 5 units along OZ. **(10 marks)** [2003–04]

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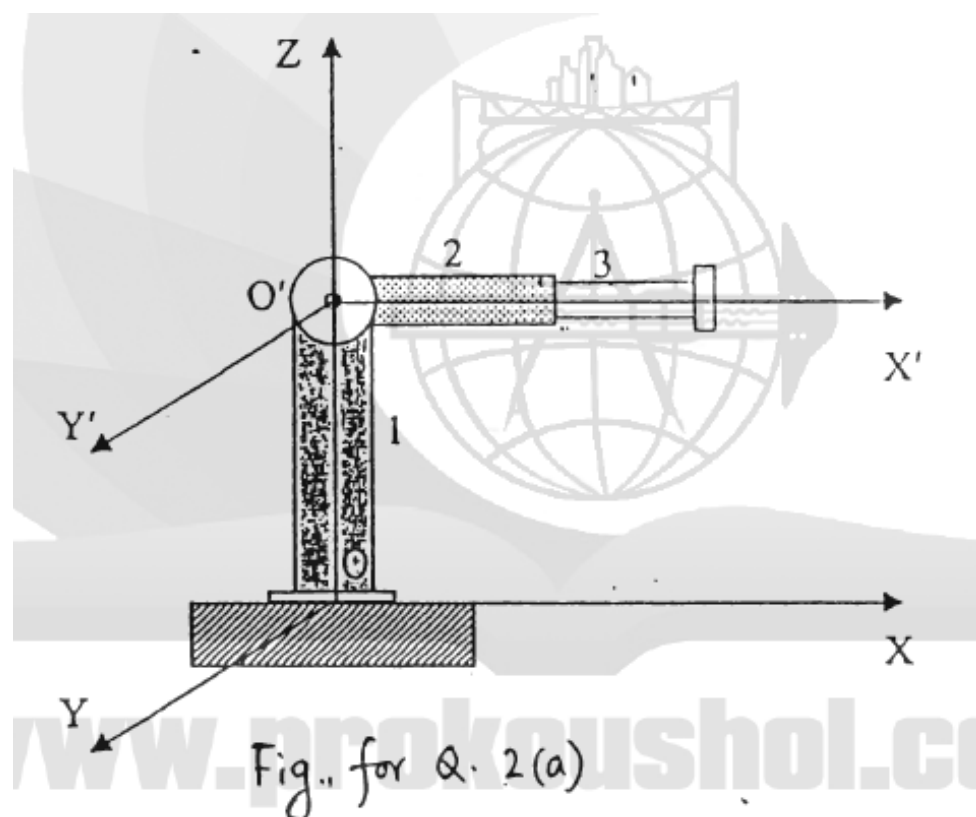
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Section C3

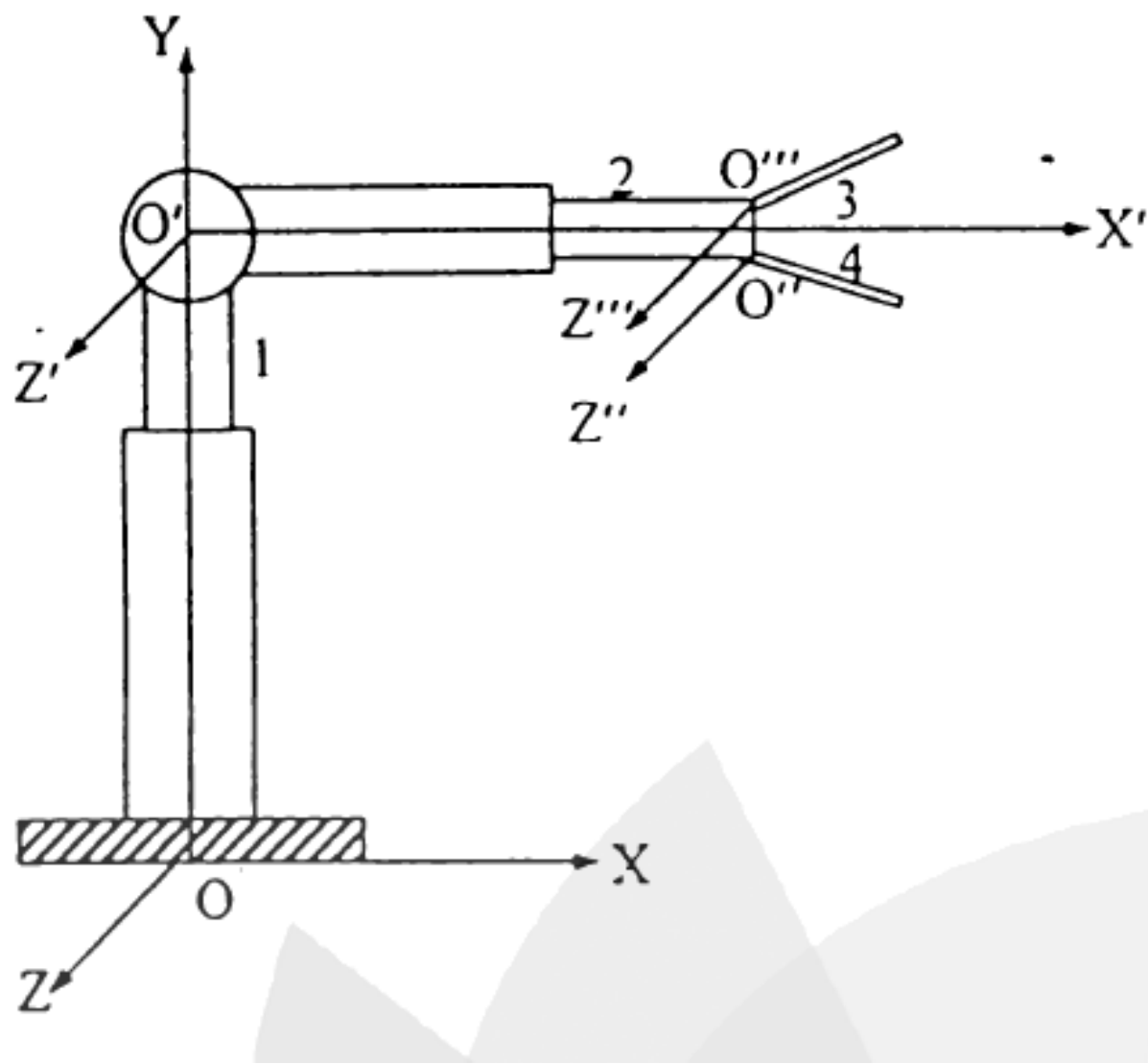
Geometric Configurations: Linkages, Arms, Workspace

C3 — Geometric Configurations & Workspace

- Q1.** (a) Sketch the following manipulator configurations: (i) TR:TLO, (ii) TR:TR. For each, determine the workspace of the wrist portion. **(10 marks)**
 (b) Desired to place origin of hand frame of a cylindrical robot at $p = [-3 \ 4 \ 7]^T$. Calculate the joint variables. **(10 marks)** [2023–24]
- Q2.** A cylindrical robot has three DOF: move along x-axis, rotate about z-axis, move along z-axis. Determine the 4×4 transformation matrix and hence find joint variables to place hand frame origin at $[6, -3, -4]^T$. **(15 marks)** [2021–22]
↪ Similar: C3-Q1 same concept, different target
- Q3.** Briefly classify robots according to various robot coordinates. Derive the single matrix $T_{\text{cyl}}(r, \alpha, l)$ for the cylindrical coordinate system. **(12 marks)** [2015–16, 2021–22]
- Q4.** Cylindrical robot transformation: $T_{\text{cyl}}(r, \alpha, l) = \text{Trans}(0, 0, l) \text{Rot}(z, \alpha) \text{Trans}(r, 0, 0)$. Arm then rotated at $-\alpha$ about z-axis.
 (i) Derive the necessary matrix. (ii) If desired to place hand frame origin at T , calculate joint variables. **(10+5 marks)** [2017–18]
- Q5.** The robotic manipulator is a spherical robot (two revolute + one prismatic joint). Restrictions: Link 1: $0 \leq \text{Rotation about } OZ \leq 120^\circ$; Link 2: $0 \leq \text{Rotation about } O'Y' \leq 60^\circ$; Link 3: Translation along X' . Draw top and side views of work volume. **(8 marks)** [2005–06]



- Q6.** (a) While setting chips and designing circuitry onto a motherboard, what essential factors should an end effector take into consideration? **(6 marks)**
 (b) Is it possible to use an AC power supply for a mobile robot designed to follow a predefined path? Discuss. **(6 marks)** [2004–05]
- Q7.** Draw the schematic diagram of the work volume for a robotic configuration with the following restrictions:
 Link 1: translation along Y-axis = d_y ;
 Link 2: translation along length = d_L ; $0 \leq \text{Rotation about } Y \leq 120^\circ$;
 Link 3: $0 \geq \text{Rotation about } O''Z'' \geq -45^\circ$;
 Link 4: $0 \leq \text{Rotation about } O''Z'' \leq 45^\circ$. **(18 marks)** [2004–05]



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Section C4

Motion Characteristics

C4 — Motion Characteristics

- Q1.** Point $P(3, 0, 3)^T$ subjected to: rotation about y-axis, then translation along x-axis, then rotation about z-axis. Final coordinates are $[4 \ 4 \ 2]^T$. Find possible transformation variables. **(15 marks)** [2023–24]
 ↪ Similar: C2-Q1(c) — identical problem
- Q2.** Point $P(7, 3, 2)^T$ subjected to: (i) rotation 90° about z-axis, (ii) translation $[4, -3, 7]$, (iii) rotation 90° about y-axis. Find final coordinates. **(15 marks)** [2018–19]
- Q3.** Point $P(6, 8, 5)^T$ subjected to: (i) translation $[2, -3, 7]$, (ii) rotation 30° about x-axis, (iii) translation $[-1, 4, 3]$. Find coordinates. **(15 marks)** [2015–16]
- Q4.** Point $P(5, 4, 7)^T$ subjected to: (i) rotation 90° about z-axis, (ii) translation $[4, -3, 2]$, (iii) rotation 45° about y-axis. Find final coordinates. **(20 marks)** [2014–15]
- Q5.** Point $P(4, 5, 3)^T$ subjected to: (i) rotation 90° about z-axis, (ii) translation $[3, -2, 1]$. Find coordinates relative to reference frame. **(15 marks)** [2013–14]
- Q6.** A point T is attached to frame (n, o, a) initially aligned with global frame (x, y, z) . Subjected to: (i) rotation $+30$ about x-axis, (ii) rotation -90 about a-axis, (iii) translation $[0, 4\sqrt{3}, -2\sqrt{3}]$ along (x, y, z) . Find new coordinates. **(25 marks)** [2012–13]
- Q7.** Point $P(3, 3, 0)^T$ subjected to: (i) CW rotation 45° about Y, (ii) CCW rotation 45° about X, (iii) translation $[C, 2, 2]$. Find coordinates relative to OXYZ. **(20 marks)** [2005–06]
- Q8.** (a) Define work envelope. Draw schematic of work envelope for a robot with the given link restrictions. **(11 marks)** [2003–04]

